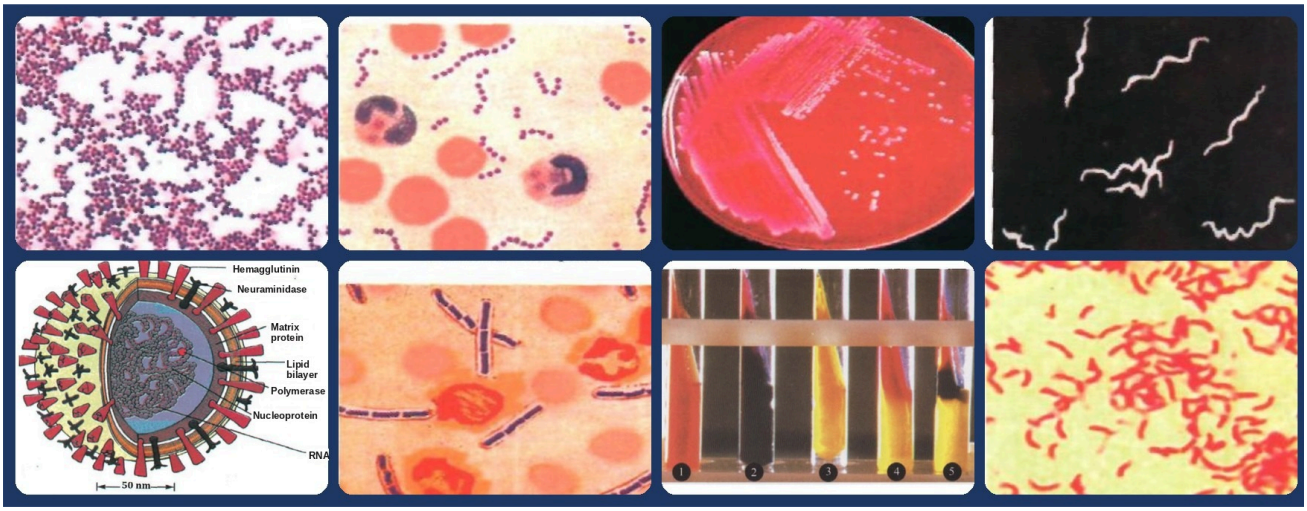
**MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN**
**TASHKENT STATE MEDICAL UNIVERSITY**
**Department of Microbiology, Virology and Immunology**

# MICROBIOLOGY, VIROLOGY AND IMMUNOLOGY

*complete study manual*

**A study manual for practical classes**




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**Tashkent — 2026**

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**Microbiology, Virology and Immunology: Complete Study Manual.** A study manual for practical classes. — Tashkent, 2026.

This manual covers the 30 practical-class topics of the subject “Microbiology, Virology and Immunology” in a single system and consists of two parts. Part I (topics 1–15), based on the calendar-thematic plan of the 2026–2027 academic year, covers general microbiology, virology and immunology: the morphology and structure of microbes, staining methods, physiology and isolation of pure cultures, identification, sanitary microbiology, antibiotics, normal microflora, infection and genetics, general virology, innate and adaptive immunity, and serologic reactions. Part II (topics 16–30), based on the plan of the 2025–2026 academic year, covers systematic bacteriology and virology — the causative agents of purulent, wound, airborne, intestinal, zoonotic, sexually transmitted and vector-borne infections, as well as respiratory, enteric, blood-borne, herpes and pox viruses.

Each topic includes an independent-work topic (TMI), colored boxes (Remember, Clinical significance, Memory aid, Caution), tables, laboratory diagnostic algorithms, review questions, 15 test questions, 3 clinical cases and a quick review table. The answers to the tests are given at the end of the book.

The manual is based on the textbooks of Muhamedov I.M., Aliyev Sh.R. et al. (Microbiology, Virology and Immunology, 2019), Aliyev Sh.R., Muhamedov I.M., Nuruzova Z.A. et al. (A manual for laboratory classes in microbiology, 2013), Zverev V.V., Boychenko M.N. and Borisov L.B., as well as Rapid Review Microbiology and Immunology, Kaplan USMLE Step 1 Immunology and Microbiology and other sources. The photomicrographs and diagrams were taken from these textbooks, with the source given under each; the remaining diagrams were drawn by the compiler.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before applying the clinical information in practice, it is recommended to compare it with the main textbooks.

The manual is intended for students of the general medicine and fundamental medicine faculties of higher medical education institutions.

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# Preface

Microbiology, virology and immunology are among the foundation sciences of medicine. A physician works with a world that cannot be seen but causes disease: bacteria, viruses, fungi, protozoa and the immune system that stands against them. This world can be "seen" only in the laboratory — by staining a smear, plating a colony, observing a biochemical reaction and measuring antibiotic susceptibility. That is why the practical class in this subject is no less important than the textbook: it is here that the student walks, by their own hand, the algorithm for identifying a causative agent — the path from the specimen to the diagnosis.

This manual covers the 30 practical-class topics of the subject in one integrated system and consists of two parts:

- **Part I** — general microbiology, virology and immunology (topics 1–15, based on the calendar-thematic plan of the 2026–2027 academic year): the morphology and structure of microbes, staining methods, physiology and isolation of pure cultures, identification, sanitary microbiology, antimicrobial factors, normal microflora, infection and genetics, general virology and immunology.
- **Part II** — systematic microbiology and virology (topics 16–30, based on the plan of the 2025–2026 academic year): the causative agents of purulent, wound, airborne, intestinal, zoonotic, sexually transmitted and vector-borne infections, as well as respiratory, enteric, blood-borne, herpes and pox viruses.

The two parts complement each other: a method learned in the general part (Gram staining, isolation of a pure culture, a serologic reaction, PCR) is applied to a particular causative agent in the systematic part. For this reason cross-references are given in the text — for example, the link of diphtheria toxin to a phage ties in with the genetics in topic 11, and the antibiogram with topic 9.

Each chapter has the same structure:

- **Learning objectives** — what you should know and be able to do after this class.
- **Theory** — the main concepts of the topic, with tables, diagrams and figures taken from textbooks. Four kinds of colored boxes appear in the text: **Remember** (key information that must be memorized), **Clinical significance** (links theory with practice), **Memory aid** (a mnemonic), **Caution** (a frequent mistake or an important warning).
- **Independent work (TMI)** — the additional topic given in the plan.
- **Practical skill** — a step-by-step algorithm of a laboratory procedure.
- **Review questions**, test questions (15 in each chapter) and clinical cases (with answers).
- **Quick review table** — for revising at a glance before the exam.

Most of the figures are taken from the main textbooks, with the source shown under each; some were drawn by the compiler as diagrams.

The manual does not replace the main textbooks, but it makes preparation for the practical class easier and brings scattered information into one system. The treatment and prophylaxis information in the text is given for general orientation: practical decisions are always made on the basis of current protocols and a physician's orders.

I wish you success.

**The compiler**

# PART I

## *GENERAL MICROBIOLOGY, VIROLOGY AND IMMUNOLOGY*

TOPIC 1

# Subject of medical microbiology. The microbiological laboratory. Morphology of bacteria and microscopic diagnosis

## 1. Learning objectives

In this class the student will master the following:

- to explain the subject, branches and importance of microbiology, virology and immunology in medicine;
- to know the structure, equipment and working rules of a microbiological laboratory and the biosafety requirements;
- to distinguish the methods of microscopy (bright-field, dark-field, phase-contrast, fluorescent, electron) and their applications;
- to identify the main morphological forms, sizes and arrangements of bacteria;
- to acquire the skill of examining a stained smear under the microscope with an immersion objective.

## 2. The science of microbiology: subject, branches and tasks

**Microbiology** (Gr. *mikros* — small, *bios* — life, *logos* — study) is the science that studies the structure, vital activity, heredity and ecology of tiny living organisms that cannot be seen with the naked eye, and their relationship with humans, animals, plants and the environment. The world of microorganisms includes **bacteria, archaea, fungi, protozoa** (eukaryotes) and the acellular forms — **viruses** and **prions**.

**Medical microbiology** studies the microorganisms that cause disease in humans (pathogenic and opportunistic) and the diagnosis, treatment and prevention of the infectious diseases they cause. It is divided into the following branches:

- **bacteriology** — studies bacteria;
- **virology** — viruses;
- **mycology** — fungi;
- **protozoology** — disease-causing protozoa;
- **immunology** — the body’s defense system.

**Table 1.1.** *Practical areas of medical microbiology*

| Area | Task |
|---|---|
| Diagnostic microbiology | Identifying the causative agent: microscopy, culture, biochemical and serological identification, PCR |
| Sanitary microbiology | Assessing the microflora of water, air, soil, food and healthcare facilities |
| Clinical microbiology | Opportunistic microflora, antibiotic susceptibility, hospital-acquired infections |
| Immunology and serology | Immune status, antigen–antibody reactions, vaccines and sera |

**CLINICAL SIGNIFICANCE**

For a physician, microbiology is a tool for identifying the “invisible enemy.” A properly collected specimen, a properly stained smear and a correctly interpreted antibiogram often decide the patient’s life. An improperly collected specimen, on the other hand, makes the entire analysis worthless.

## 3. History of microbiology and the contribution of Uzbek scientists

Several stages are distinguished in the development of microbiology:

1. **Empirical (heuristic) period** — before the discovery of microbes, speculation about the “invisible causes” of diseases (Hippocrates; Avicenna (Ibn Sina) wrote in *“The Canon of Medicine”* about tiny particles that spread disease).
2. **Morphological period** — Antonie van **Leeuwenhoek** (1676) was the first to see “animalcules” (bacteria) in a microscope he built himself.
3. **Physiological period** — Louis **Pasteur** proved that fermentation and putrefaction are the work of microbes and laid the foundations of sterilization and vaccination; Robert **Koch** developed methods of isolating pure cultures, solid culture media, immersion microscopy and “Koch’s postulates.”
4. **Immunological period** — I.I. **Mechnikov** created the theory of phagocytosis and P. **Ehrlich** created the theory of humoral immunity.
5. **Molecular-genetic period** — the discovery of antibiotics (A. Fleming) and the development of virology, genetics and biotechnology.

**MEMORY AID**

Remember the founders: **“LE-PA-KO-ME”** — **Le**euwenhoek (saw), **Pa**steur (fermentation, vaccine), **Ko**ch (pure culture), **Me**chnikov (phagocytosis).

### *Independent work: Scientists from Uzbekistan who contributed to the development of microbiology, virology, immunology and parasitology*

At the beginning of the 20th century, the study of infectious diseases and their causative agents in Turkestan was begun by scientists who worked there — **P.F. Borovsky** (the first to describe, in Tashkent in 1898, the causative agent of cutaneous leishmaniasis), **A.D. Grekov** and **L.M. Isayev**. Under the leadership of L.M. Isayev, **Guinea worm disease (dracunculiasis)** was completely eliminated in the Bukhara oasis (1932) and a system for the fight against malaria was created; the Uzbekistan Institute of Medical Parasitology is named after him.

**Table 1.2.** *Leading scientists of Uzbekistan and their main research areas*

| Scientist | Main scientific contribution |
|---|---|
| T.Kh. Najmiddinov | Causes, prevention and treatment of malaria, brucellosis, leishmaniasis, Guinea worm disease, plague, typhoid–paratyphoid and epidemic hepatitis |
| I.K. Musaboyev (academician) | Advancing the understanding of the pathogenesis and the treatment methods of dysentery, viral hepatitis, cholera and diphtheria; school of intestinal infections |
| V.M. Majidov | Clinical picture, diagnosis and treatment of brucellosis and viral hepatitis |
| O.S. Mahmudov | Intestinal infections in children, typhoid fever; a rational scheme of rehydration therapy |
| T.O. Daminov (academician) | Chronic hepatitis in children, treatment methods for hepatitis B |

| Scientist | Main scientific contribution |
|---|---|
| A.O. Obidov (academician) | Microbial genetics, effects of vaccines on the body, rapid diagnosis of intestinal infections, dysbiosis |
| I.M. Muhamedov, Sh.R. Aliyev and co-authors | The first complete textbook and laboratory manual of “Microbiology, Virology and Immunology” in the Uzbek language |

Today the Department of Microbiology, Virology and Immunology of Tashkent State Medical University continues these traditions.

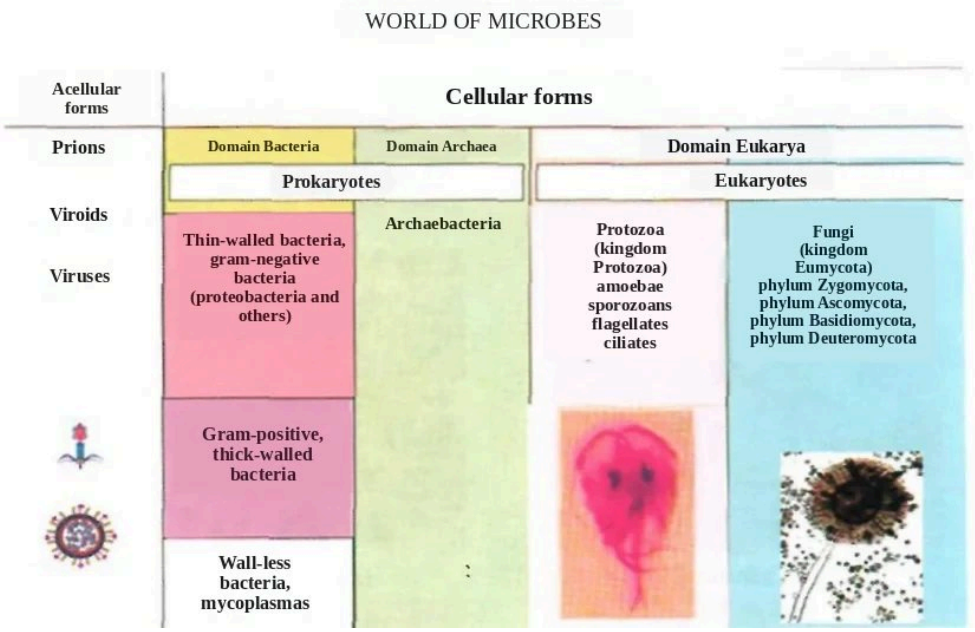



**Figure 1.1.** *The world of microbes: acellular forms (prions, viroids, viruses) and three domains — bacteria, archaea, eukaryotes (fungi, protozoa). Source: Muhamedov I.M. et al., 2019.*

## 4. The microbiological laboratory: structure, equipment and safety

Because a microbiological laboratory works with infectious materials, it has a special regime. It usually consists of the following rooms: specimen reception, preparation (staining, media preparation), inoculation (transfer boxes), incubator room, autoclave (sterilization) and washing room.

**Main equipment:**

- **immersion microscope** — for examining stained smears;
- **incubator** — for growing inoculated media at 37 °C;
- **autoclave** — sterilization under steam pressure (121 °C, 1 atm of excess pressure);
- **dry-heat sterilizer (oven)** — for sterilizing glassware at 160–180 °C;
- **bactericidal lamps (UV)**, **laminar-flow box** — for disinfecting the air and the work area;
- **centrifuge, refrigerator, pH meter, bacteriological loop and Petri dishes.**

> **CAUTION**
> Biosafety is divided into 4 levels (BSL-1 to BSL-4). Work with especially dangerous infections (cholera, plague, anthrax, hemorrhagic fevers) is done only in special laboratories of the highest levels (BSL-3/4). An ordinary diagnostic laboratory is BSL-2.

**Working rules (in brief):** work wearing a lab coat and gloves; treat all inoculated material as potentially infectious; sterilize the loop each time by flaming; after work, disinfect the work area and wash the hands; discard used material only after decontaminating it in an autoclave; eating and touching the face are forbidden in the laboratory.

## 5. Microscopy methods

Because bacteria are very small (0.5–10 µm), they can be seen only under a microscope. The resolving power of the light microscope is ≈ 0.2 µm; the maximum useful magnification is ≈ 1000× (10× eyepiece × 100× immersion objective).

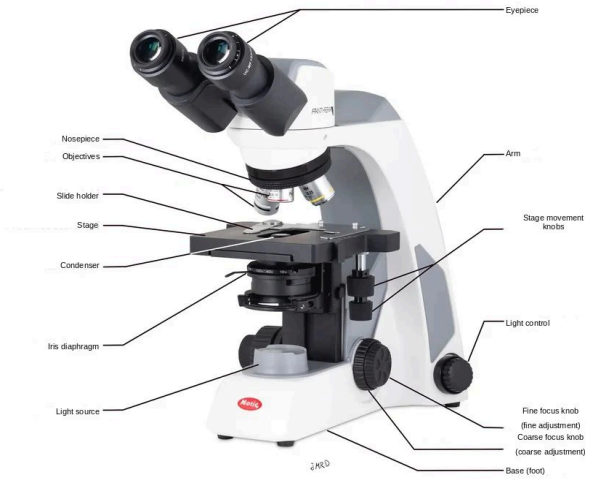



**Figure 1.2.** *Structure and main parts of the light (biological) microscope.*

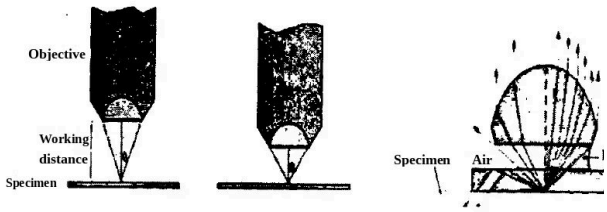



Fig. 5. Path of rays in the immersion system

**Figure 1.3.** *Path of the rays in an immersion system: immersion oil reduces the scattering of rays between the objective and the specimen. Source: Aliyev Sh.R. et al., 2013.*

**Immersion microscopy** is the main method in bacteriology. **Immersion oil** (refractive index close to that of glass, n ≈ 1.515) is placed between the objective and the slide; this reduces the refraction and scattering of the light rays and increases the clarity of the image.

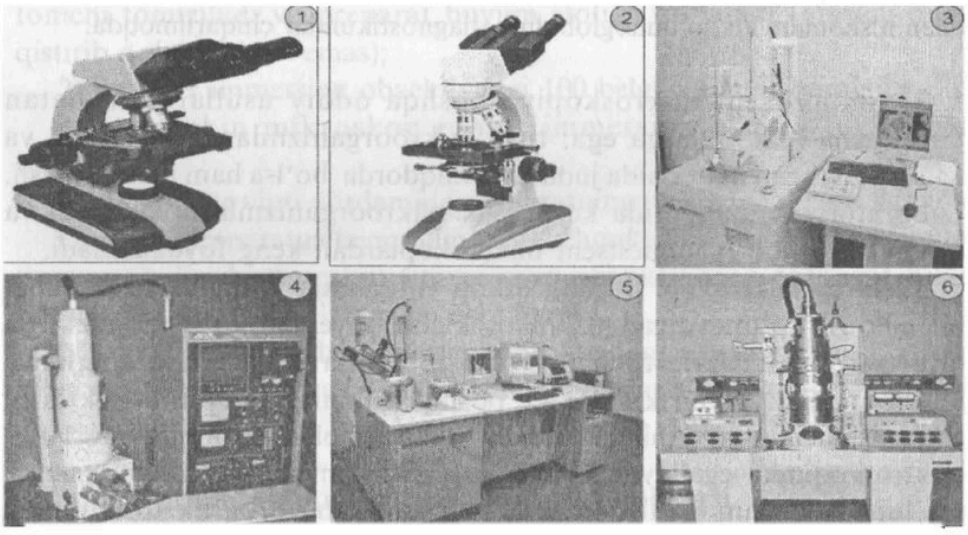



Fig. 6. Microscopes used in microbiological practice: 1, 2 – binocular microscopes; 3, 4 – laser confocal microscope; 5 – interference computer microscope; 6 – electron microscope.

**Figure 1.4.** *Microscopes used in microbiological practice: 1, 2 — binocular microscopes; 3, 4 — laser confocal microscope; 5 — interference computer microscope; 6 — electron microscope. Source: Aliyev Sh.R. et al., 2013.*

**Table 1.3.** *Comparative characteristics of microscopy methods*

| Method | What it shows | Main application |
|---|---|---|
| Bright-field (immersion) | Stained, nonmotile cells | Routine bacterioscopy |
| Dark-field | Live, unstained, delicate cells | Spirochetes (syphilis — *Treponema*) |
| Phase-contrast | Internal structures of the living cell | Observing motility, division, spores |
| Luminescent (fluorescent) | Object glowing with a fluorochrome | Mycobacteria (auramine), immunofluorescence |
| Electron | Ultrastructure, viruses | Viruses, internal cell structure |

## 6. Morphology of bacteria

Bacteria differ in shape, size and arrangement after division. These features are important in the preliminary identification of the causative agent.

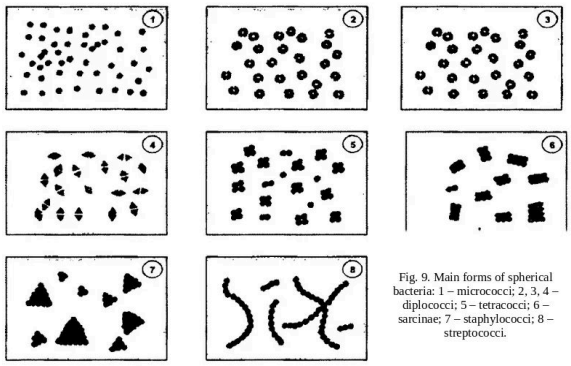


Fig. 9. Main forms of spherical bacteria: 1 – micrococci; 2, 3, 4 – diplococci; 5 – tetracocci; 6 – sarcinae; 7 – staphylococci; 8 – streptococci.

**Figure 1.5.** *Main forms of spherical bacteria: 1-micrococci; 2, 3, 4-diplococci; 5-tetracocci; 6-sarcinae; 7-staphylococci; 8-streptococci (Aliyev Sh.R. et al., 2013)*

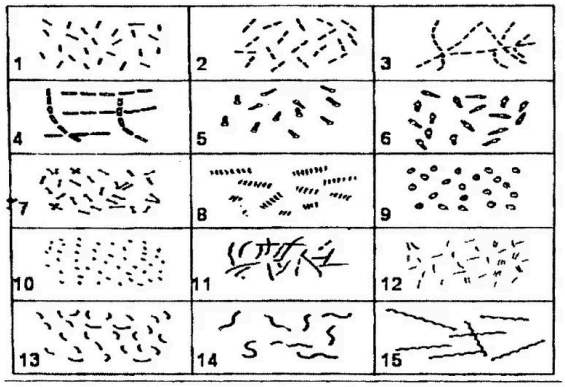


Fig. 10. Main forms of rod-shaped bacteria: 1 – intestinal rod; 2 – diplobacteria; 3 – streptobacteria; 4 – streptobacilli; 5, 6 – clostridia; 7 – causative agent of diphtheria; 8 – diphtheroids; 9 – causative agent of plague; 10 – coccobacteria; 11 – fusobacteria; 12 – mycobacteria; 13 – vibrios; 14 – spirilla; 15 – treponemes.

**Figure 1.6.** *Main forms of rod-shaped and curved bacteria: 1-E. coli; 2-diplobacteria; 3-streptobacteria; 4-streptobacilli; 5, 6-clostridia; 7-causative agent of diphtheria; 8-diphtheroids; 9-causative agent of plague; 10-coccobacilli; 11-fusobacteria; 12-mycobacterium; 13-vibrios; 14-spirilla; 15-treponemes (Aliyev Sh.R. et al., 2013)*

- **Cocci** (spherical): depending on the plane of division and whether the cells separate afterward — micrococcus (single), **diplococcus** (pair: pneumococcus, gonococcus, meningococcus), **streptococcus** (chain), **staphylococcus** (grape-like cluster), tetracoccus (group of four), **sarcina** (packet of 8–16 cells).
- **Rod-shaped**: single (monobacterium), paired (diplobacterium), chain (streptobacterium). The ends may be blunt, pointed or thickened.
- **Curved**: **vibrio** (comma-shaped — causative agent of cholera), **spirillum** (1–3 large coils), **spirochete** (many fine coils — syphilis, leptospirosis).

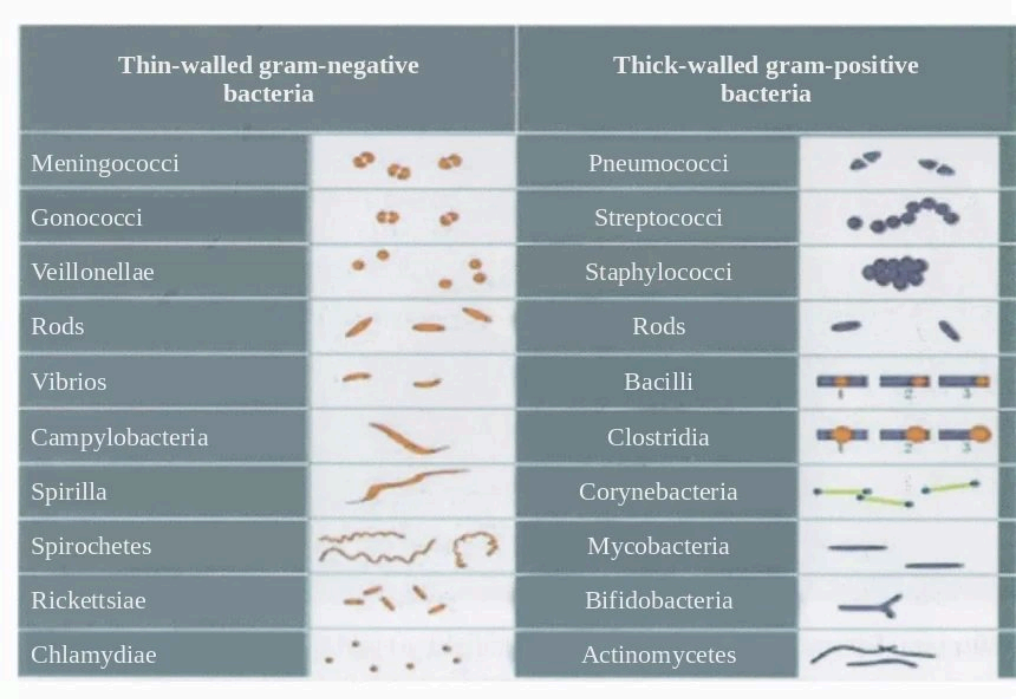



**Figure 1.7.** *Morphological and tinctorial (Gram) groups of bacteria: on the left, gram-negative bacteria with a thin wall; on the right, gram-positive bacteria with a thick wall. Source: Aliyev Sh.R. et al., 2013.*

**REMEMBER**

First guess from the shape: diplococci + in the cerebrospinal fluid → meningococcus; grape-like clusters of cocci + pus → staphylococcus; chains of cocci + throat → streptococcus; comma-shaped + diarrhea → Vibrio cholerae.

## 7. Practical skill: examining a smear under immersion microscopy

Algorithm:

1. Point the microscope toward the light source, raise the condenser to its upper position and open the diaphragm.
2. Secure the stained and dried smear on the stage.
3. Place **one drop of immersion oil** on the smear.
4. Rotate the nosepiece so that the **immersion objective (90–100×)** touches the oil (the objective is immersed in the oil).
5. Looking from the side, carefully bring the objective close to the smear with the coarse focus knob, then looking through the eyepiece bring the image into sharp focus with the **fine focus knob**.
6. Move the field of view and assess the shape, arrangement and staining of the cells.
7. When finished, clean the objective with a soft tissue.

**CAUTION**

When working with the immersion objective, use only the **fine focus knob** — if you turn the coarse focus knob sharply, the objective will break the slide and damage the lens.

## 8. Review questions

1. Into what branches is microbiology divided?

2. What are the main periods in the history of microbiology and who were their founders?
3. What are the rules of work in a microbiological laboratory and the biosafety levels?
4. What is immersion microscopy used for, and what is the function of immersion oil?
5. In which cases are dark-field and fluorescent microscopy used?
6. List the main morphological forms of bacteria and the types of their arrangement.

## Test questions

1. The scientist who was the first to see the world of microbes under the microscope:
   A) Louis Pasteur
   B) Robert Koch
   C) Antonie van Leeuwenhoek
   D) I.I. Mechnikov
   E) Paul Ehrlich

2. The methods of isolating pure cultures, solid culture media and “Koch’s postulates” are associated with the name of the following scientist:
   A) Leeuwenhoek
   B) Robert Koch
   C) Pasteur
   D) Fleming
   E) Ehrlich

3. The founder of the theory of phagocytosis:
   A) P. Ehrlich
   B) L. Pasteur
   C) I.I. Mechnikov
   D) R. Koch
   E) A. Fleming

4. The main function of oil in immersion microscopy:
   A) Cooling the objective
   B) Reducing the refraction and scattering of light rays and increasing the clarity of the image
   C) Staining the specimen
   D) Killing bacteria
   E) Reducing the magnification

5. The most suitable method for viewing live, unstained spirochetes:
   A) Bright-field
   B) Dark-field
   C) Electron microscopy
   D) Immersion-stained smear
   E) Radiography

6. The microscopy used for detecting mycobacteria stained with auramine:
   A) Phase-contrast
   B) Dark-field
   C) Luminescent (fluorescent)
   D) Bright-field
   E) Polarized light

7. Spherical bacteria that form chains:
    A) Staphylococci
    B) Streptococci
    C) Sarcinae
    D) Diplobacteria
    E) Vibrios

8. Cocci that form grape-like clusters:
    A) Streptococci
    B) Tetracocci
    C) Staphylococci
    D) Sarcinae
    E) Micrococci

9. A curved bacterium of comma (cholera) shape:
    A) Spirillum
    B) Spirochete
    C) Vibrio
    D) Streptobacterium
    E) Diplococcus

10. The resolving power of the light microscope is approximately:
    A) 0.2 µm
    B) 2 µm
    C) 0.02 µm
    D) 20 µm
    E) 1 mm

11. Cocci arranged in packets of 8–16 cells:
    A) Tetracocci
    B) Sarcinae
    C) Diplococci
    D) Streptococci
    E) Staphylococci

12. The equipment that sterilizes under steam pressure (121 °C):
    A) Dry-heat oven
    B) Autoclave
    C) Incubator
    D) Centrifuge
    E) Laminar box

13. Microbes that are diplococci in shape and cause meningitis and pneumonia belong to the following group:
    A) Rod-shaped
    B) Spherical (cocci)
    C) Curved
    D) Fungi
    E) Viruses

14. What is used to focus the image sharply when working with the immersion objective:
    A) Coarse focus knob
    B) Fine focus knob
    C) Nosepiece
    D) Diaphragm
    E) Condenser

15. The smallest bacteria that have no cell wall:
    A) Spirochetes
    B) Mycoplasmas
    C) Staphylococci
    D) Vibrios
    E) Sarcinae

## Clinical cases

### *Case 1*

A smear taken from a purulent wound was stained by the Gram method; under the microscope, spherical cells arranged in grape-like clusters and stained dark purple were found.

**Question:** What morphological group and what probable causative agent can be considered?

**Answer:** Spherical (coccus), staphylococcal arrangement, gram-positive cells — probably *Staphylococcus aureus*. A frequent causative agent of purulent infections.

### *Case 2*

A patient is suspected of having syphilis (lues). Fluid taken from the base of an ulcerative lesion must be examined alive, without staining.

**Question:** Which microscopy method is used and why?

**Answer:** Dark-field microscopy. *Treponema pallidum* is a delicate, motile spirochete that takes ordinary dyes poorly; against a dark background it is seen alive as a bright spiral.

### *Case 3*

While working with the immersion objective, a student is looking for the image by sharply turning the coarse focus knob without looking through the eyepiece. As a result, the slide broke.

**Question:** Which rule was violated?

**Answer:** When working with the immersion objective, bring the objective close carefully while watching from the side, and perform fine focusing only with the fine focus knob. Sharp turning of the coarse focus knob damages the lens and the specimen.

## Quick review

**Table 1.4.** *Topic 1 — key concepts*

| Concept | Essence |
|---|---|
| Branches of medical microbiology | Bacteriology, virology, mycology, protozoology, immunology |
| Founders | Leeuwenhoek (saw), Pasteur (fermentation/vaccine), Koch (pure culture), Mechnikov (phagocytosis) |

| Concept | Essence |
|---|---|
| Immersion microscopy | 90–100× objective + immersion oil; useful magnification ≈ 1000× |
| Dark-field | Live spirochetes (syphilis) |
| Luminescent | Mycobacteria (auramine), immunofluorescence |
| Arrangement of cocci | Diplo-, strepto-, staphylo-, tetra-, sarcina |
| Curved forms | Vibrio (cholera), spirillum, spirochete (syphilis, leptospirosis) |
| Sterilization | Autoclave 121 °C (steam); dry oven 160–180 °C |

TOPIC 2

# Structure of the bacterial cell. Gram-positive and gram-negative bacteria. Classification and Bergey's Manual

## 1. Learning objectives

In this class the student will master the following:

- To know the permanent and nonpermanent structures of the bacterial cell, their chemical composition and function;
- to explain the differences in cell wall structure between gram-positive and gram-negative bacteria and the clinical significance of this difference;
- to master the mechanism and technique of Gram staining;
- to know the principles of classification and nomenclature of bacteria and of using Bergey's Manual.

## 2. Permanent structures of the bacterial cell

A bacterium is a **prokaryote**: it has no nucleus enclosed by a nuclear membrane and no membrane-bound organelles (mitochondria, ER, Golgi).

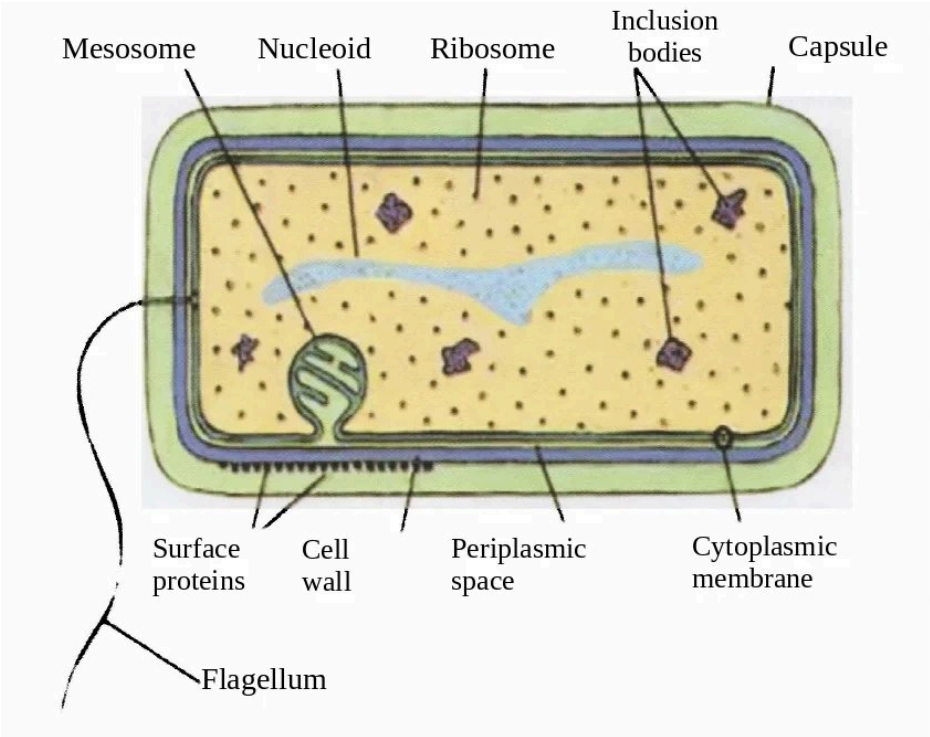



**Figure 2.1.** *Structure of the bacterial cell: capsule, cell wall, periplasmic space and membrane, nucleoid, ribosomes, mesosome, inclusions, flagellum. Source: Aliyev Sh.R. et al., 2013.*

**Permanent structures** (present in almost all bacteria):

- **Cell wall** — determines shape, provides resistance to osmotic pressure, and determines Gram staining and many antigenic properties. Its basis is **peptidoglycan (murein)**.
- **Cytoplasmic membrane (CPM)** — a phospholipid bilayer; selective permeability, transport of nutrients, respiratory enzymes and ATP synthesis (bacteria have no mitochondria — energy is produced in the CPM). Its inward folding is the **mesosome**.

- **Cytoplasm** — a colloidal medium; enzymes, RNA, reserve substances.
- **Nucleoid** — a single circular double-stranded DNA (chromosome), without histones and without a nuclear membrane.
- **Ribosomes** — of the **70S** type (30S + 50S subunits); the site of protein synthesis. Their difference from the eukaryotic ribosome (80S) is the basis for the selective action of many antibiotics.
- **Plasmids** — small circular DNA outside the chromosome that replicates independently (antibiotic resistance, toxin genes — in detail in topic 11).

> **REMEMBER**
> The bacterial ribosome is **70S** (30S + 50S), the human ribosome is **80S**. Aminoglycosides and tetracyclines act on 30S, macrolides and chloramphenicol (levomycetin) on 50S, stopping only bacterial protein synthesis.

**Chemical composition:** water (≈ 80%), proteins, nucleic acids, carbohydrates, lipids and minerals. The composition of the cell wall (peptidoglycan, teichoic acids or LPS) depends on the Gram group.

## 3. The cell wall: gram-positive and gram-negative bacteria

**Peptidoglycan (murein)** — alternately linked chains of N-acetylglucosamine and N-acetylmuramic acid; they are cross-linked by short peptide bridges to form a net-like layer. This cross-linking is carried out by the enzyme **transpeptidase** (penicillin-binding protein) — the target of beta-lactam antibiotics.

**Table 2.1.** *The cell wall of gram-positive and gram-negative bacteria*

| Feature | Gram-positive (Gram +) | Gram-negative (Gram −) |
|---|---|---|
| Peptidoglycan layer | Thick (20–80 layers, ≈90% of the wall) | Thin (1–2 layers, ≈10%) |
| Teichoic acids | Present | Absent |
| Outer membrane | Absent | Present (contains LPS) |
| LPS (endotoxin) | Absent | Present (lipid A — toxic) |
| Periplasmic space | Poorly developed | Well developed |
| Gram staining | Purple (the iodine–dye complex is retained) | Red (decolorized, takes up fuchsin) |
| Susceptibility to antibiotics | More sensitive to penicillin | The outer membrane is a barrier — more resistant |

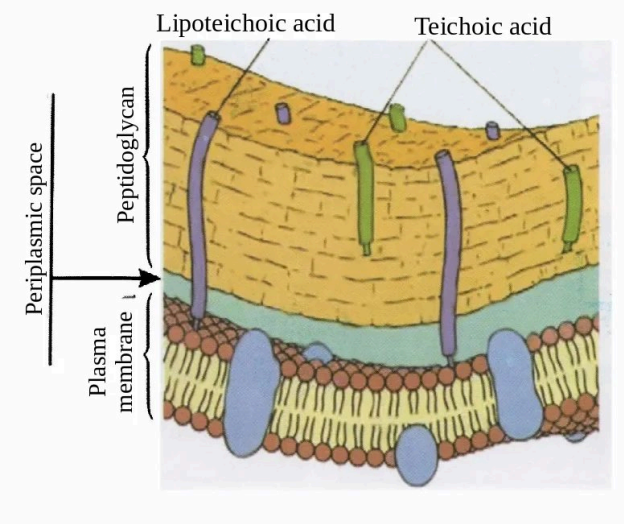



**Figure 2.2.** *Wall of a gram-positive bacterium: thick peptidoglycan, teichoic and lipoteichoic acids (Aliyev Sh.R. et al., 2013)*

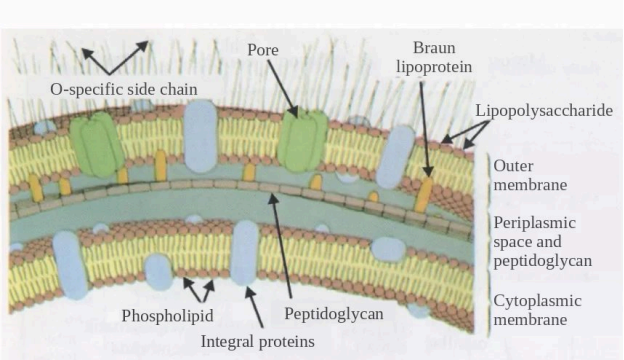



**Figure 2.3.** *Wall of a gram-negative bacterium: outer membrane, LPS, porins, thin peptidoglycan (Aliyev Sh.R. et al., 2013)*

**CLINICAL SIGNIFICANCE**

The **LPS (endotoxin)** of gram-negative bacteria is released when the cell breaks down and causes fever, a fall in blood pressure, DIC syndrome and septic shock. Gram-positive bacteria, in contrast, mainly release exotoxins (proteins).

**Special cases:**

- **Acid-fast** bacteria (mycobacteria — tuberculosis, leprosy): their wall contains a lot of **mycolic acid** and wax; they stain poorly by Gram, so they are stained by the **Ziehl–Neelsen** method (red rods on a blue background).
- **Mycoplasmas** — have no cell wall at all; variable shape, naturally resistant to beta-lactam antibiotics.
- **L-forms** — temporary forms whose wall is partially lost (for example, under the action of penicillin).

## 4. Gram staining

The Gram method (1884, Christian Gram) is the most important differential stain in bacteriology. The difference in staining is based on the structure of the cell wall: thick peptidoglycan is dehydrated by alcohol, its pores narrow, and it retains the **gentian violet–iodine** complex (Gram+); Gram− cells, with thin peptidoglycan and an outer membrane that dissolves in alcohol, lose this complex and then turn red with the counterstain (fuchsin).

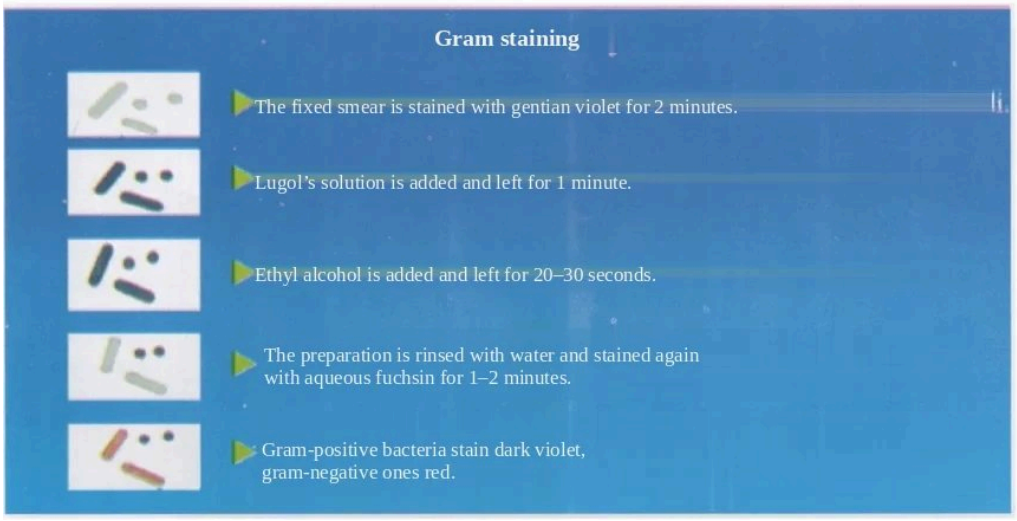



**Figure 2.4.** *Steps of Gram staining (the algorithm from the laboratory manual). Source: Aliyev Sh.R. et al., 2013.*

**Steps:** 1) gentian violet (1–2 min) — all cells turn purple; 2) Lugol's solution (iodine — the fixative, 1 min) — the dye–iodine complex forms; 3) alcohol (decolorizer, 10–20 s) — Gram− cells are decolorized; 4) rinse with water, then fuchsin (counterstain, 1–2 min) — Gram− cells turn red. Result: **Gram+ — purple, Gram− — red (pink).**

> **CAUTION**
> The most important and most commonly mishandled step is **decolorizing with alcohol**. If left in alcohol for too long, Gram+ cells also lose their color and give a false Gram− result; if decolorized too briefly, Gram− cells stay purple and give a false Gram+ result.

## 5. Classification and nomenclature of bacteria and Bergey's Manual

Bacteria are classified by **taxonomic units**: domain → phylum → class → order → **family** → **genus** → **species**.

**Nomenclature is binomial (two-name):** each species is named with two Latin words: the first is the genus name (capitalized), the second is the species name (lowercase), both in italics: for example, *Staphylococcus aureus*, *Escherichia coli*, *Mycobacterium tuberculosis*.

**Bergey's Manual** (*Bergey's Manual of Determinative Bacteriology*) is the main reference that helps identify bacteria by their morphological, tinctorial (Gram), cultural and biochemical features. In practice, bacteria are identified by following this sequence of features.

**Variants within a species.** Although they are not formally part of the classification, in practice bacteria within one species are distinguished by the following features:

**Table 2.2.** *Variants of bacteria within a species*

| Variant | Distinguishing feature |
|---|---|
| Serovar (serotype) | Antigenic structure (by serological reactions) |
| Chemovar | Biochemical properties |
| Fermentovar (biovar) | Enzyme activity, physiological features |
| Phagovar | Sensitivity to certain bacteriophages |
| Resistovar | Antibiotic resistance profile |

**MEMORY AID**

Remember the variants by their root: **SERO**var — **sero**logy (antigen); **PHAGO**var — **phage**; **FERMENT**ovar — **ferment**; **CHEMO**var — **chem**ical.

## 6. Practical skill: the Gram staining algorithm

1. Place a drop of physiological saline on a clean slide, prepare a smear from the culture to be examined with a loop, and air-dry it.
2. **Fix** the smear by passing it 2–3 times over the flame (this fixes the cells to the glass and kills them).
3. Stain with gentian violet for 1–2 minutes.
4. Pour off the dye and apply Lugol's solution for 1 minute.
5. Decolorize with alcohol for 10–20 seconds, until no more color comes off; rinse immediately with water.
6. Counterstain with fuchsin for 1–2 minutes, rinse and dry.
7. Examine with the immersion objective (100×): purple — Gram+, red — Gram−; record the shape and arrangement.

## 7. Review questions

1. What are the permanent structures of the bacterial cell and what is their function?
2. How do prokaryotic and eukaryotic cells differ?
3. What is peptidoglycan, how is it built and which antibiotic acts on it?
4. How does the cell wall of gram-positive bacteria differ from that of gram-negative bacteria?
5. What is the difference between endotoxin (LPS) and exotoxin?
6. Explain the mechanism and steps of Gram staining.
7. What is Bergey's Manual used for? What are serovar, phagovar and fermentovar?

## Test questions

1. Which cell type does a bacterium belong to:
    A) Eukaryotic
    B) Prokaryotic
    C) Virus
    D) Fungus
    E) Protozoan
2. The main component of the bacterial cell wall:
    A) Chitin
    B) Cellulose
    C) Peptidoglycan (murein)
    D) Cholesterol
    E) Creatine

3. The type of the bacterial ribosome:
    A) 80S
    B) 70S
    C) 60S
    D) 40S
    E) 90S

4. Present in the cell wall of gram-positive bacteria but absent in gram-negative ones:
    A) Outer membrane
    B) LPS
    C) Teichoic acids
    D) Periplasmic space
    E) Porins

5. The endotoxin of gram-negative bacteria is the following substance:
    A) Peptidoglycan
    B) Teichoic acid
    C) Lipopolysaccharide (lipid A)
    D) Mycolic acid
    E) Flagellin

6. What color do gram-positive bacteria stain with the Gram method:
    A) Red
    B) Purple
    C) Blue
    D) Green
    E) Yellow

7. The most important (differentiating) step in Gram staining:
    A) Staining with gentian violet
    B) Applying Lugol’s solution
    C) Decolorizing with alcohol
    D) Staining with fuchsin
    E) Fixation

8. Bacteria that have no cell wall and are naturally resistant to beta-lactam antibiotics:
    A) Staphylococci
    B) Mycoplasmas
    C) Streptococci
    D) Mycobacteria
    E) Spirochetes

9. The method used for staining mycobacteria:
    A) Gram
    B) Ziehl–Neelsen
    C) Burri
    D) Ozheshko
    E) Neisser

10. The structure of the bacterium in which energy (ATP) is produced:
    A) Mitochondrion
    B) Cytoplasmic membrane
    C) Golgi apparatus
    D) Lysosome
    E) Nucleus

11. Small circular DNA outside the chromosome that carries antibiotic resistance genes:
    A) Nucleoid
    B) Plasmid
    C) Ribosome
    D) Mesosome
    E) Capsule

12. In the binomial nomenclature of a bacterium, the first word denotes:
    A) Species
    B) Genus
    C) Family
    D) Serovar
    E) Strain

13. The variant distinguished by the antigenic structure of bacteria:
    A) Phagovar
    B) Chemovar
    C) Serovar
    D) Fermentovar
    E) Resistovar

14. The reference used to identify bacteria by their morphological, tinctorial and biochemical features:
    A) Pharmacopoeia
    B) Bergey’s Manual
    C) ICD-10
    D) Vidal table
    E) Graham table

15. Drugs that act on the 70S ribosome and stop bacterial protein synthesis but do not act on the human (80S) ribosome:
    A) Beta-lactams
    B) Aminoglycosides and macrolides
    C) Sulfonamides
    D) Fluoroquinolones
    E) Polymyxins

## Clinical cases

### *Case 1*

A smear was stained by the Gram method, and red (pink) rods were seen under the microscope. The patient has signs of an intestinal infection.

**Question:** Which Gram group does the bacterium belong to and what is its wall structure?

**Answer:** A gram-negative (Gram−) rod — it has thin peptidoglycan, an outer membrane and LPS (endotoxin). Enteric rods (Enterobacteriaceae — for example, *E. coli*) are gram-negative.

### *Case 2*

While performing Gram staining, a student kept the alcohol decolorization step far too long (1 minute). As a result, a known gram-positive culture was also stained red.

**Question:** What error occurred and how is the result assessed?

**Answer:** Over-decolorization with alcohol — the thick peptidoglycan also lost the gentian violet–iodine complex, and the gram-positive cells looked falsely gram-negative (red). The result is incorrect; the staining must be repeated.

### *Case 3*

A patient was prescribed an antibiotic of the penicillin group, but it had no effect on the causative agent (a mycoplasma).

**Question:** Why was penicillin ineffective?

**Answer:** Mycoplasmas have no cell wall (peptidoglycan), while penicillin blocks peptidoglycan synthesis (transpeptidase). With no wall, the drug has no target — which is why mycoplasmas are naturally resistant to beta-lactams (macrolides and tetracyclines act on them).

## Quick review

**Table 2.3.** *Topic 2 — key concepts*

| Concept | Essence |
|---|---|
| Permanent structures | Cell wall, CPM, cytoplasm, nucleoid, ribosomes (70S) |
| Sign of a prokaryote | No nuclear membrane, mitochondria or ER; the DNA is circular |
| Peptidoglycan | Murein; transpeptidase cross-links it — the target of beta-lactams |
| Gram+ | Thick peptidoglycan + teichoic acid; purple; exotoxin |
| Gram− | Thin peptidoglycan + outer membrane + LPS; red; endotoxin |
| Steps of Gram staining | Gentian violet → Lugol → alcohol (decolorization) → fuchsin |
| Wall-less / acid-fast | Mycoplasma (wall-less); mycobacterium (Ziehl–Neelsen) |
| Nomenclature | Binomial: *Genus species* (italic) |
| Variant within a species | Serovar (antigen), phagovar (phage), fermentovar, chemovar |

TOPIC 3

# Nonpermanent structures of bacteria: spore, capsule, flagella, inclusions. Complex staining methods. Fungi

## 1. Learning objectives

In this class the student will master the following:

- To know the structure and function of the nonpermanent (optional) structures of bacteria — spore, capsule, flagellum, pili and inclusions;
- to explain the importance of these structures as differential features in identifying the species;
- to master the complex (differential) staining methods — Ozheshko, Burri–Hiss, Neisser, Ziehl–Neelsen — and the methods of determining motility;
- Independent work: to know the morphology and structure of fungi and the methods of studying them.

## 2. General concept of nonpermanent structures

Permanent structures (cell wall, cytoplasmic membrane, cytoplasm, nucleoid, ribosomes) are present in every bacterium. **Nonpermanent structures**, by contrast, occur only in some species or only under certain conditions. They are not essential for the bacterium’s survival, but they serve as an **important differential feature in identifying the species**, and for this reason they are studied separately in the practical class.

**Table 3.1.** *Nonpermanent structures and their detection*

| Structure | Chemical basis | Task | In which bacteria (example) | Method of detection |
|---|---|---|---|---|
| Spore | Dipicolinic acid–calcium, multilayered coat | Preservation of the species | Genera *Bacillus*, *Clostridium* | Ozheshko method |
| Capsule | Polysaccharide (in most), polypeptide (anthrax bacillus) | Resistance to phagocytosis, virulence | Pneumococcus, Klebsiella, meningococcus | Burri–Hiss |
| Flagellum | Flagellin protein | Motility, chemotaxis | E. coli, Salmonella, vibrio | Hanging drop, semisolid agar, Löffler |
| Pili (fimbriae) | Pilin protein | Adhesion, conjugation | Gonococcus, E. coli | Electron microscopy |
| Inclusions | Volutin, glycogen, lipid | Reserve substance | Diphtheria bacillus (volutin) | Neisser, Löffler |

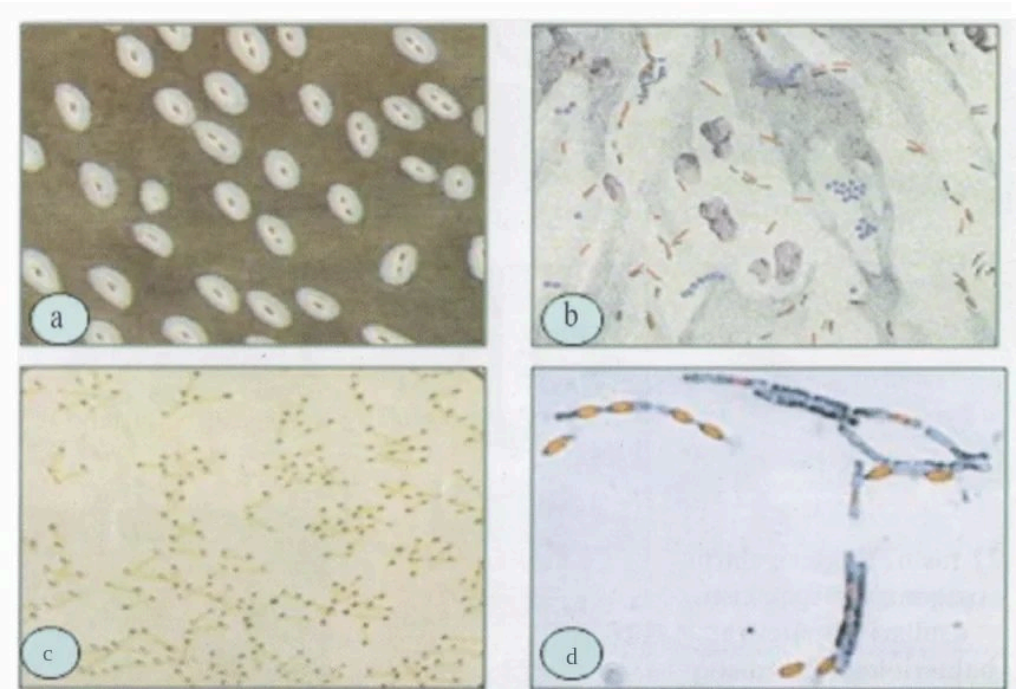



**Figure 3.1.** *Preparations stained by complex methods: a) Burri–Hiss (capsule), b) Ziehl–Neelsen (acid-fast rods), c) Neisser (volutin granules), d) Ozheshko (spore). Source: Aliyev Sh.R. et al., 2013.*

## 3. Spore

**Spore** — a resting form produced under unfavorable conditions by some gram-positive rods (*Bacillus* — aerobes, *Clostridium* — anaerobes). It is a form of **preservation, not reproduction**: one spore forms from one cell, and one vegetative cell grows back from the spore.

**Structure.** In the center are dehydrated cytoplasm and the nucleoid, surrounded by the cortex, a multilayered coat and sometimes an exosporium. There is almost no free water, and there is a lot of the **calcium salt of dipicolinic acid** — because of this, metabolism in the spore stops and it is much more resistant than the vegetative cell. That is why in medicine the reliability of sterilization is assessed precisely by the **destruction of spores** (see topic 9: autoclave and biological indicators).

**Spore location** — a morphological feature used in identifying the species:

- **central** — in the middle of the cell, does not change the shape of the cell (anthrax bacillus);
- **subterminal** — near the end, giving a “tennis racket” appearance (the causative agent of botulism);
- **terminal** — at the very end, giving a **“drumstick bacillus”** appearance (the causative agent of tetanus).

If the spore diameter is greater than the width of the cell, the cell swells and takes the **clostridial (spindle-shaped)** form; if smaller, the **bacillus** form is retained.

**REMEMBER**
A spore does not take up an ordinary dye: in an ordinary smear it appears as an **unstained white space**. With the Ozheshko method, however, the **spore is red and the vegetative cell is blue**.

**MEMORY AID**
Remember the location by the shape: **terminal — drumstick** (tetanus), **subterminal — racket** (botulism), **central — the shape does not change** (anthrax).

### *The Ozheshko method (spore staining) — procedure*

1. A smear is prepared and air-dried.
2. A 0.5% solution of hydrochloric acid is dropped onto the preparation and heated for 1–2 minutes — the spore coat becomes permeable to the dye.
3. The acid is washed off with water and the smear is fixed.
4. It is stained with carbol fuchsin, heated until steam rises (5–7 minutes), and rinsed.
5. It is counterstained with methylene blue (3–5 minutes), rinsed and dried.
6. It is examined under the immersion objective: **red spore + blue vegetative cell**.

## 4. Capsule

**Capsule** — a mucous layer with a distinct border on top of the cell wall. If thicker than 0.2 µm it is a **macrocapsule** (visible with a light microscope); a thinner one is called a **microcapsule**. A layer without a distinct border that is easily removed is the **slime layer (glycocalyx)**; it forms the basis of the biofilm and is important in infections associated with catheters, prostheses and implants.

**Chemical composition:** a polysaccharide in most bacteria (pneumococcus, Klebsiella, meningococcus, *Haemophilus influenzae*), a polypeptide in the anthrax bacillus.

**Significance:**

- resists phagocytosis — a **virulence factor** (unencapsulated strains are not virulent);
- determines the **serovars within a species**: pneumococcus has more than 90 capsular serotypes, and vaccines are made precisely from these polysaccharides;
- protects against desiccation and immune factors.

Most bacteria form the capsule **in the body** or on media with added serum, and lose it on ordinary artificial media. That is why a smear prepared from **pathological material** (sputum, CSF, blood) is usually used to see the capsule.

**Burri–Hiss method:** India ink is added to the material and smeared (the background becomes black); after fixation, it is stained with fuchsin. As a result, a **red cell** on a **black background** is seen, and around it a **colorless halo — the capsule**.

> **CLINICAL SIGNIFICANCE**
> In a cerebrospinal fluid smear, encapsulated gram-positive diplococci raise suspicion of pneumococcal meningitis; in sputum, large encapsulated gram-negative rods (a slimy, dome-shaped colony) raise suspicion of Klebsiella pneumonia.

## 5. Flagella and motility

**Flagellum** — a filamentous appendage made of flagellin protein and anchored to the cytoplasmic membrane; it gives the bacterium motility and **chemotaxis** (toward nutrients, away from harmful substances). The flagellum is used as the **H-antigen** in serological identification.

**Table 3.2.** *Types of flagella by arrangement*

| Type | Arrangement | Example |
|---|---|---|
| Monotrichous | A single flagellum at one end | Vibrio cholerae |
| Lophotrichous | A tuft of flagella at one end | Some species of *Pseudomonas* |
| Amphitrichous | Flagellum (flagella) at both ends | *Spirillum* |

| Type | Arrangement | Example |
|---|---|---|
| Peritrichous | Over the entire surface | E. coli, Salmonella, Proteus |
| Atrichous | No flagella | Cocci, Klebsiella |

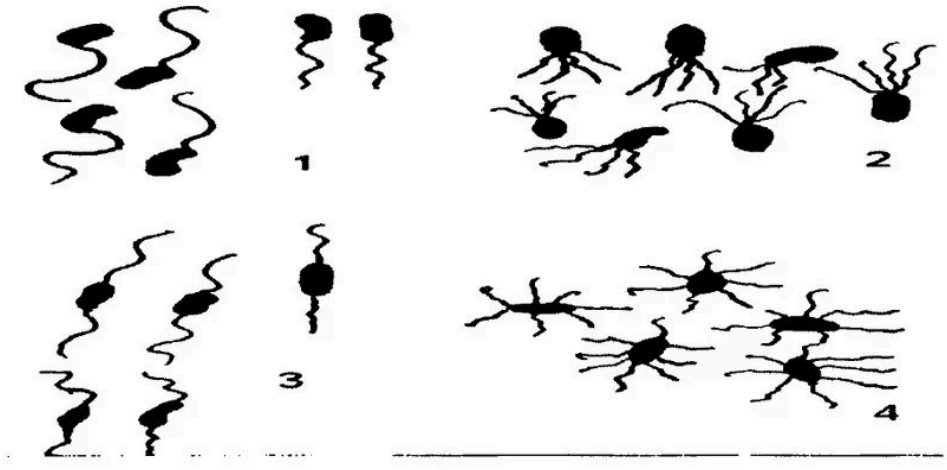



Fig. 15. Bacterial flagella (diagram). 1 – monotrichous; 2 – lophotrichous; 3 – amphitrichous; 4 – peritrichous.

**Figure 3.2.** *Types of bacteria by arrangement of flagella (scheme): 1-monotrichous; 2-lophotrichous; 3-amphitrichous; 4-peritrichous. Source: Aliyev Sh.R. et al., 2013.*

**Methods of determining motility:**

1. **Hanging drop** — living cells in a drop between a concave slide and a coverslip are observed by phase-contrast or dark-field microscopy.
2. **Crushed (wet-mount) drop** — a coverslip is placed over a drop on an ordinary slide; simpler, but the drop dries quickly.
3. **Stab inoculation into semisolid agar** — motile bacteria spread sideways from the stab line and make the medium cloudy; nonmotile ones grow only along the stab line. This method is the most widely used in routine practice.

> **CAUTION**
> In a hanging drop, **true motility** must be distinguished from **Brownian motion**: in Brownian motion the cells merely vibrate in place, whereas in true motility they cross the field of view and change position.

## 6. Pili and inclusions

**Pili (fimbriae)** — numerous protein appendages that are shorter and thinner than flagella. They are divided into two groups:

- **common-type pili (adhesins)** — ensure attachment of the cell to the epithelium; this is the first step of infection (for example, the P pili of E. coli in urinary tract infection, gonococcal pili);
- **sex (F) pili** — ensure the transfer of DNA between donor and recipient cells during conjugation (topic 11).

**Inclusions** — reserve substances in the cytoplasm: **volutin (metachromatic granules)**, glycogen, starch, lipid droplets, sulfur granules. The most important for diagnosis is **volutin**: it is located at the ends of the diphtheria bacillus, and with the **Neisser** stain the cell appears yellow-brown while the volutin granules appear **dark blue-black**. This is the classic sign in the bacterioscopic diagnosis of diphtheria.

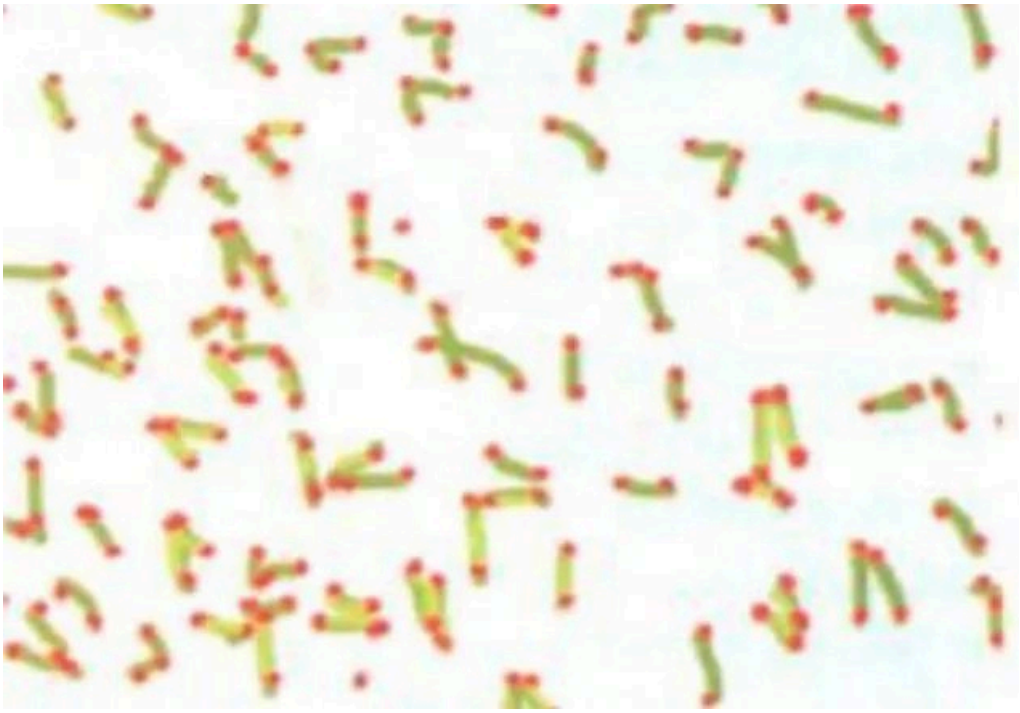


**Figure 3.3.** *Diphtheria bacillus stained by the Neisser method: dark volutin granules at the ends of the cells. Source: Muhamedov I.M. et al., 2019.*

**MEMORY AID**

**NEI**sser — **NEI**ther end is clear: the tips are black, because of **volutin**. Diphtheria + granules at the ends = a “matchstick” appearance, and the cells lie at an angle to one another (in the shape of the **Latin letters V and X**).

## 7. Independent work: Fungi — morphology, structure and methods of study

**Fungi** — **eukaryotic**, chlorophyll-free, heterotrophic organisms. They differ from bacteria in having a formed nucleus, membrane-bound organelles, **80S** ribosomes, a cell wall made of **chitin and glucan**, and **ergosterol** in the membrane (the target of antifungal drugs). That is why antibacterial antibiotics do not act on fungi.

**Table 3.3.** *Main morphological forms of fungi*

| Form | Structure | Example |
|---|---|---|
| Yeast-like | Single round cells, reproduce by budding | *Candida*, *Cryptococcus* |
| Mycelial (filamentous) | A network of hyphae forms the mycelium | *Aspergillus*, *Penicillium*, dermatophytes |
| Dimorphic | Yeast-like in the body, mycelial in the environment | *Histoplasma*, *Blastomyces* |

**Mycelium** may be septate or nonseptate. Reproduction: **asexual** — conidia, sporangiospores, blastospores, chlamydospores, arthrospores; **sexual** — asco-, basidio- and zygospores.

**Methods of study:**

1. **Microscopic:** examining the material (hair, nail, skin scrapings) cleared in 10–20% KOH solution; staining smears with Gram, Romanowsky–Giemsa, PAS or calcofluor white.
2. **In culture:** inoculation onto **Sabouraud** medium and growth at 22–28 °C (and 37 °C) for several days to several weeks; identification by colony appearance, pigment and microscopic

structure.

3. **Serological and molecular:** antigen detection (cryptococcal capsular antigen, galactomannan), ELISA, PCR.

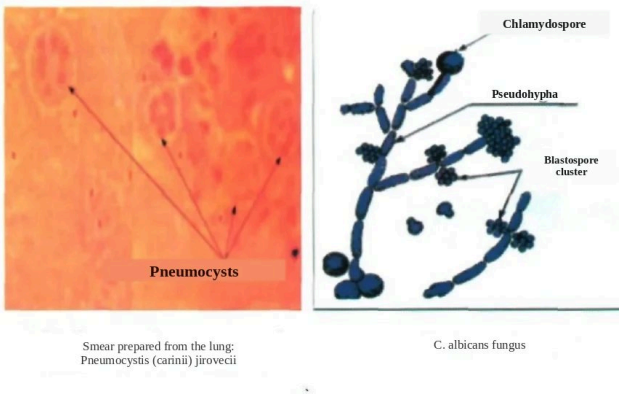



**Figure 3.4.** *Microscopic appearance of fungi: a) Pneumocystis in a lung smear; b) Candida albicans (Muhamedov I.M. et al., 2019)*

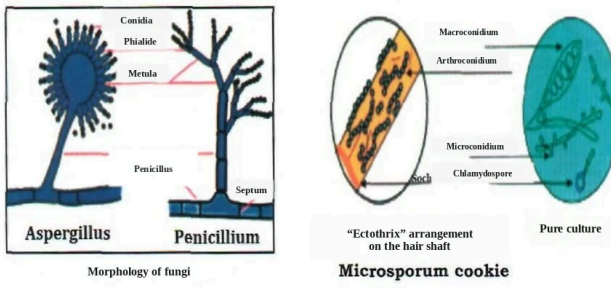



**Figure 3.5.** *Conidial heads of Aspergillus and Penicillium; elements of a dermatophyte (Microsporum) (Muhamedov I.M. et al., 2019)*

**CLINICAL SIGNIFICANCE**

*Candida albicans* is a member of the normal microflora; against a background of antibiotics, diabetes mellitus or reduced immunity it turns into candidiasis (oral cavity — thrush, esophagus, urinary tract, generalized form). That is why finding a fungus in the material is not enough by itself — its quantity is assessed together with the clinical picture.

## 8. Practical skill: choosing complex staining methods

In the laboratory, the staining method is chosen according to the **structure being sought**:

1. Shape, arrangement and Gram group needed → **Gram method**.
2. Acid-fast rods (tuberculosis, leprosy) → **Ziehl–Neelsen** (red rods, blue background) or the fluorescent method with auramine.
3. Spore → **Ozheshko** (red spore, blue cell).
4. Capsule → **Burri–Hiss** (black background, colorless halo).
5. Volutin granules (diphtheria) → **Neisser**.
6. Spirochetes → **Romanowsky–Giemsa**, **Morozov** (silver impregnation) or dark-field.
7. Fungi → **KOH preparation**, PAS, calcofluor.

**CAUTION**

In complex methods, the order of steps and the timing must be strictly observed. In the Ozheshko method, the spore is not stained unless it is treated with acid; in the Neisser method, if the counterstain is held too long, the volutin granules merge with the background and become invisible.

## 9. Review questions

1. How do permanent and nonpermanent structures differ? What is their diagnostic significance?
2. How is a spore formed? Describe its structure and the types of its location.
3. By which method is a spore stained and what is the result?
4. Explain the chemical composition, function and method of detection of the capsule.
5. Into what types are flagella divided by arrangement? How is motility determined?

6. How do you distinguish true motility from Brownian motion?
7. What are volutin granules, in which bacterium and by what method are they detected?
8. By what features do fungi differ from bacteria? What is the function of Sabouraud medium?

## Test questions

1. The main biological meaning of spore formation:
   A) Reproduction
   B) Preservation of the species under unfavorable conditions
   C) Movement
   D) Attachment
   E) Toxin production

2. The substance in the spore that provides resistance:
   A) Teichoic acid
   B) Calcium salt of dipicolinic acid
   C) Mycolic acid
   D) Lipopolysaccharide
   E) Peptidoglycan

3. Spore location and cell appearance typical of the causative agent of tetanus:
   A) Central, the shape does not change
   B) Subterminal, “tennis racket”
   C) Terminal, “drumstick bacillus”
   D) No spore
   E) Central with an exosporium

4. The method of spore staining:
   A) Neisser
   B) Burri–Hiss
   C) Ozheshko
   D) Romanowsky–Giemsa
   E) Löffler

5. How does the capsule appear in the Burri–Hiss method:
   A) A dark blue granule
   B) A colorless halo on a black background
   C) A red rod
   D) A green glowing cell
   E) A purple coccus

6. The chemical nature of the capsule of the anthrax bacillus:
   A) Polysaccharide
   B) Polypeptide
   C) Lipid
   D) Nucleic acid
   E) Mycolic acid

7. The main function of the capsule:
   A) Movement
   B) Resistance to phagocytosis and virulence
   C) Protein synthesis
   D) Conjugation
   E) Energy production

8. The type of bacterium with flagella over its entire surface:
   A) Monotrichous
   B) Lophotrichous
   C) Amphitrichous
   D) Peritrichous
   E) Atrichous

9. The flagellum corresponds to which antigen:
   A) O-antigen
   B) H-antigen
   C) K-antigen
   D) Vi-antigen
   E) Lipid A

10. The most commonly used cultural method for determining motility:
    A) Inoculation onto blood agar
    B) Stab inoculation into semisolid agar
    C) Inoculation onto Endo medium
    D) Inoculation onto Sabouraud medium
    E) Inoculation onto Kligler medium

11. In which bacterium do volutin granules have diagnostic significance:
    A) E. coli
    B) Diphtheria bacillus
    C) Vibrio cholerae
    D) Meningococcus
    E) Klebsiella

12. What color do volutin granules stain in the Neisser method:
    A) Red
    B) Dark blue-black
    C) Green
    D) Yellow
    E) Colorless

13. The function of the sex (F) pilus:
    A) Movement
    B) Adhesion
    C) Transfer of DNA in conjugation
    D) Protection from phagocytosis
    E) Accumulation of reserve substances

14. The medium used for growing fungi:
    A) Endo
    B) Sabouraud
    C) Kligler
    D) Ploskirev
    E) Löwenstein–Jensen

15. The main component of the fungal cell wall:
    A) Peptidoglycan
    B) Chitin and glucan
    C) Mycolic acid
    D) Teichoic acid
    E) Lipopolysaccharide

## Clinical cases

### *Case 1*

In a smear of material taken from a deep, soil-contaminated wound, gram-positive rods were found; at their ends there is a thick round structure, and the cell has a “drumstick bacillus” appearance.

**Question:** What is this structure and which causative agent should be considered?

**Answer:** A terminally located spore; the morphological appearance is typical of the causative agent of tetanus (*Clostridium tetani*). To confirm the diagnosis, culture under anaerobic conditions and toxin detection are needed (topic 17).

### *Case 2*

In a preparation made with India ink from the sputum smear of a patient with pneumonia, red rods on a black background and wide colorless halos around them were seen.

**Question:** Which method was used, what is the colorless halo and which causative agent is likely?

**Answer:** The Burri–Hiss method; the colorless halo is the macrocapsule. Large encapsulated gram-negative rods are typical of *Klebsiella pneumoniae*. Culture and identification are needed for a final answer.

### *Case 3*

A student, seeing small vibrations of cells in place in a hanging drop preparation, judged the culture “motile.” But when it was stab-inoculated into semisolid agar, growth occurred only along the stab line.

**Question:** What is the error?

**Answer:** The student mistook Brownian motion for true motility. In true motility, the cell crosses the field of view. The result in semisolid agar confirms that the culture is nonmotile (atrichous).

## Quick review

**Table 3.4.** *Topic 3 — key concepts*

| Concept | Essence |
|---|---|
| Spore | Preservation form; 1 cell → 1 spore; dipicolinic acid–Ca |

| Concept | Essence |
|---|---|
| Spore location | Central (anthrax), subterminal (botulism), terminal (tetanus) |
| Ozheshko | Spore — red, cell — blue |
| Capsule | Polysaccharide (polypeptide — anthrax); resistance to phagocytosis |
| Burri–Hiss | Black background, red cell, colorless halo |
| Types of flagella | Mono-, lopho-, amphi-, peritrichous, atrichous; H-antigen |
| Detecting motility | Hanging/crushed drop, semisolid agar |
| Volutin | At the ends of the diphtheria bacillus; Neisser — dark blue granules |
| Pili | Adhesion (common type), conjugation (F pili) |
| Fungi | Eukaryotic, chitin + glucan wall, ergosterol, 80S; Sabouraud medium |

TOPIC 4

# Morphology of spirochetes, mycoplasmas, chlamydiae, rickettsiae, actinomycetes and fungi. Protozoa

## 1. Learning objectives

In this class the student will master the following:

- To know the morphology and biological features of the groups that differ in structure from ordinary bacteria — spirochetes, mycoplasmas, chlamydiae, rickettsiae and actinomycetes;
- to explain the special methods of studying these groups (dark-field, silver impregnation, cell culture);
- to be able to name the main diseases caused by each group and the principle of their laboratory diagnosis;
- Independent work: to know the morphology and structure of protozoa and the methods of studying them.

## 2. General description: why these groups are studied separately

Although the groups listed above are prokaryotes (apart from fungi and protozoa), each of them has a **feature that differs markedly from an ordinary bacterium**: one is extremely thin and coiled, one has lost its cell wall, one multiplies only inside cells. These differences directly affect practice — they do not grow on ordinary media, they stain poorly by Gram, and beta-lactam antibiotics do not always act on them.

**Table 4.1.** *Comparative characteristics of the special groups*

| Group | Cell wall | Grows on ordinary media? | Main diseases | Method of choice |
|---|---|---|---|---|
| Spirochetes | Present, delicate; axial fibrils | Some (Leptospira) — on special media | Syphilis, relapsing fever, leptospirosis, Lyme disease | Dark-field, silver impregnation, serology |
| Mycoplasmas | **Absent** (sterols in the membrane) | Special medium, “fried-egg” colony | Atypical pneumonia, urogenital infection | PCR, special medium, serology |
| Chlamydiae | Present (little peptidoglycan) | **No** — only inside cells | Trachoma, urogenital chlamydiosis, ornithosis | Cell culture, PCR, ELISA |
| Rickettsiae | Present (gram-negative type) | **No** — only inside cells | Epidemic typhus, Q fever | Serology, PCR, chick embryo |
| Actinomycetes | Present (gram-positive type) | Yes, they grow slowly | Actinomycosis, nocardiosis | Microscopy (druses), culture |

**REMEMBER**

Prokaryotes that **multiply only inside cells** — **chlamydiae and rickettsiae** (they do not grow at all on artificial media). **No cell wall** — **mycoplasmas** (and therefore naturally resistant to penicillin).

## 3. Spirochetes

**Spirochetes** — thin, long, spiral-shaped, motile bacteria. Their structure differs from ordinary bacteria: in the center is the **protoplasmic cylinder** (cytoplasm + nucleoid, surrounded by the cell

wall), around it are the **axial fibrils** (axial filaments), and the **outer sheath** covers the whole cell. It is the contraction of the axial fibrils that allows the spirochete to move by rotating and flexing.

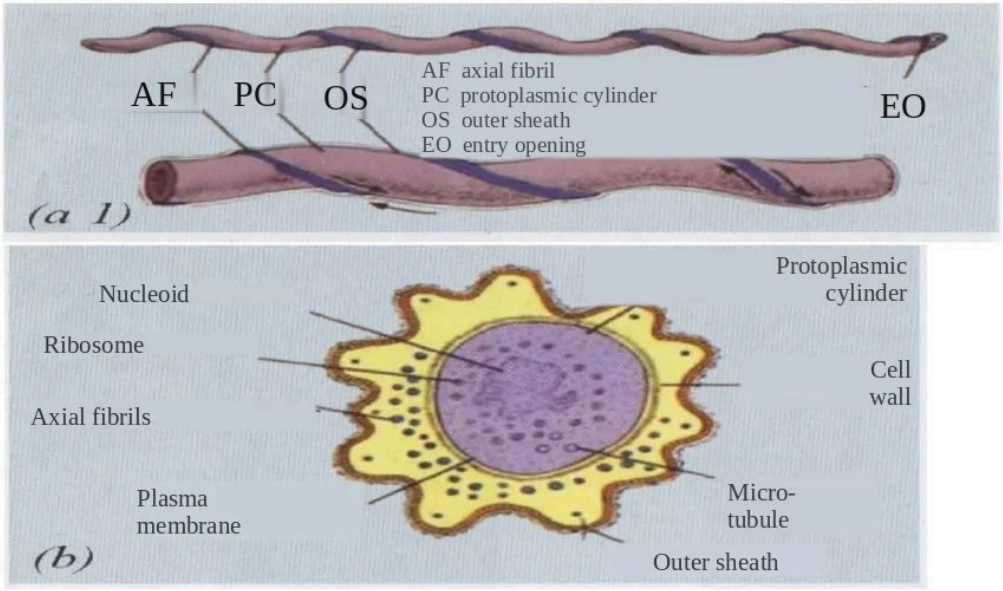



**Figure 4.1.** *Structure of a spirochete: a) external appearance; b) cross-section — protoplasmic cylinder, axial fibrils, outer sheath. Source: Aliyev Sh.R. et al., 2013.*

**Table 4.2.** *Spirochetes of medical importance*

| Genus | Coils | Disease | Staining/visualization method |
|---|---|---|---|
| *Treponema* | 8–14, small and regular | Syphilis | Dark-field, Morozov (silver impregnation), Romanowsky–Giemsa (pale pink) |
| *Borrelia* | 3–8, large and irregular | Relapsing fever, Lyme disease | Romanowsky–Giemsa (blue-violet), blood smear |
| *Leptospira* | Very many, small; hooked ends (S or C shape) | Leptospirosis | Dark-field, silver impregnation, special liquid medium |

**MEMORY AID**
By the number of coils: **B**orrelia — **b**ig, loose coils (few and large); **T**reponema — **t**ight and tiny (regular, fine coils); **L**eptospira — **l**ots of coils + hooked ends.

**CLINICAL SIGNIFICANCE**
*Treponema pallidum* takes ordinary dyes poorly (hence its name — “pale”). Therefore in primary syphilis a preparation made from the discharge of the hard chancre is examined **alive, in dark field**. In late stages, diagnosis relies mainly on **serological** methods (RPR, TPHA, ELISA) (topic 25).

## 4. Mycoplasmas

**Mycoplasmas** — the smallest self-replicating prokaryotes (0.1–0.3 µm). Their main feature is the **complete absence of a cell wall**; they are surrounded only by a three-layered cytoplasmic membrane, and, unlike other bacteria, the membrane contains **sterols**.

**Consequences:**

- the shape is not constant (**pleomorphism**): round, pear-shaped, filamentous forms;
- **practically not stained by the Gram method** (Romanowsky–Giemsa is used);

- **naturally resistant to beta-lactam antibiotics** (penicillin stops cell wall synthesis, and there is no wall) — macrolides, tetracyclines and fluoroquinolones are used in treatment;
- can pass through bacterial filters;
- grows slowly on specially enriched medium and forms tiny colonies with a depressed center, a **“fried-egg”** appearance.

Medically important: *Mycoplasma pneumoniae* (atypical pneumonia), *Mycoplasma hominis* and *Ureaplasma urealyticum* (urogenital infections).

## 5. Chlamydiae

**Chlamydiae** — gram-negative prokaryotes that have a cell wall but little peptidoglycan; **obligate intracellular parasites**, because they cannot synthesize ATP themselves (“energy parasites”). Their developmental cycle consists of alternation between two forms:

1. **Elementary body (EB)** — a small, dense, metabolically inactive, **infectious** form that lives outside the cell; it enters the cell.
2. **Reticulate body (RB)** — a large, metabolically active form that divides inside the cell, but is not infectious.

Cycle: EB enters the cell → prevents phagolysosome formation → turns into RB and multiplies → turns into new EBs → the cell breaks down and EBs are released. The whole cycle lasts 48–72 hours.

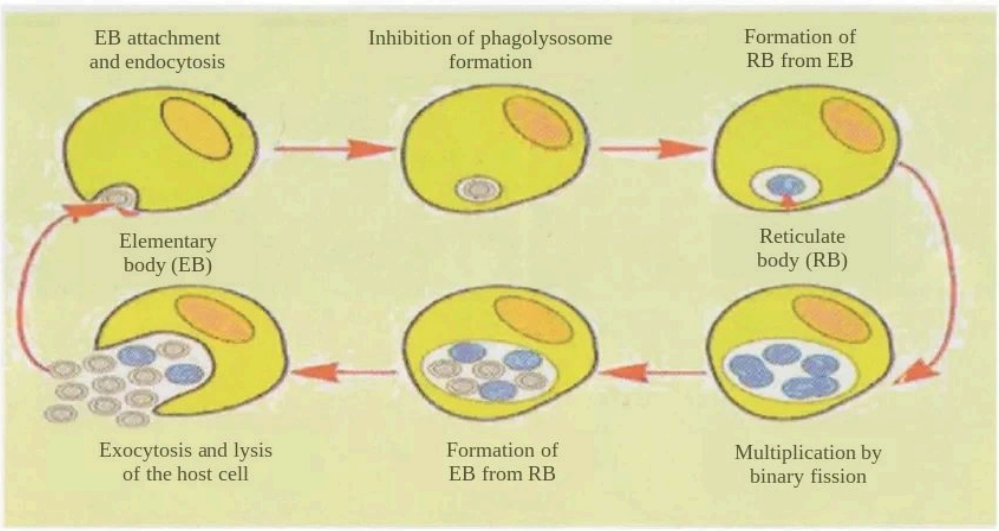



**Figure 4.2.** *Developmental cycle of Chlamydia trachomatis: alternation of elementary and reticulate bodies. Source: Aliyev Sh.R. et al., 2013.*

**Main species and diseases:** *C. trachomatis* — trachoma, urogenital chlamydiosis, neonatal conjunctivitis; *C. psittaci* — ornithosis (from birds); *C. pneumoniae* — respiratory infection.

**Diagnosis:** inoculation into cell culture (the reference method, rarely used in practice), **PCR** (the main method today), direct immunofluorescence, antibody detection by ELISA; seeing intracellular inclusions in a smear with Romanowsky–Giemsa.

> **CAUTION**
> Chlamydiae and rickettsiae **cannot be cultured on ordinary nutrient media**. If a report states “chlamydia was grown on blood agar,” this is an error. They are propagated in living cell culture, chick embryo or laboratory animals.

## 6. Rickettsiae

**Rickettsiae** — small, polymorphic (coccoid or rod-shaped) gram-negative bacteria, **obligate intracellular parasites**. Most are transmitted by **arthropods** (lice, ticks, fleas) — which is why the infections they cause are called **vector-borne** (topic 26).

Main ones: *Rickettsia prowazekii* — epidemic typhus (via lice); *R. typhi* — endemic (murine, rat) typhus (via fleas); *Coxiella burnetii* — Q fever (also transmitted by the airborne route).

**Staining:** Romanowsky–Giemsa, the **Zdrodovsky** method (rickettsiae red, cell cytoplasm blue). **Diagnosis** is made mainly by serological methods (CFT, PHA, ELISA) and PCR.

## 7. Actinomycetes

**Actinomycetes** — gram-positive bacteria that form branching filamentous outgrowths (mycelium). They were formerly considered fungi, but their prokaryotic structure (a wall with peptidoglycan, 70S ribosomes, no nucleus) places them among the bacteria; they are sensitive to antibacterial antibiotics.

- *Actinomyces israelii* — a member of the normal oral microflora; when immunity falls or after injury it causes **actinomycosis** (chronic granulomatous inflammation, fistulas). Yellow **druses** (sulfur granules; clumps of mycelium) visible to the naked eye are found in the pus — this is the classic diagnostic sign.
- *Nocardia* — partially acid-fast; pulmonary and cutaneous nocardiosis, often in patients with reduced immunity.

> **CLINICAL SIGNIFICANCE**
> A chronic, indurated infiltrate with fistulous openings in the maxillofacial region + yellow granules (druses) in the pus — a picture typical of actinomycosis. Seeing the druses on ordinary microscopy speeds up the diagnosis.

## 8. Independent work: Protozoa

**Protozoa** — unicellular **eukaryotic** organisms: a formed nucleus, membrane-bound organelles, 80S ribosomes. Most exist in two forms: the **vegetative (trophozoite)** — the active, multiplying form, and the **cyst** — the form resistant to the external environment and infectious.

**Table 4.3.** *Protozoa of medical importance*

| Group | Representative | Disease | Main material / method |
|---|---|---|---|
| Sarcodina (amoebae) | *Entamoeba histolytica* | Amebic dysentery, liver abscess | Stool (trophozoite/cyst), Lugol staining |
| Flagellates | *Giardia lamblia* | Giardiasis | Stool, duodenal juice |
| | *Trichomonas vaginalis* | Trichomoniasis | Urogenital discharge, native preparation |
| | *Leishmania* | Leishmaniasis | Smear from the edge of the ulcer, Romanowsky–Giemsa |
| | *Trypanosoma* | Sleeping sickness, Chagas disease | Blood smear |
| Sporozoa | *Plasmodium* | Malaria | Thick and thin blood smears |
| | *Toxoplasma gondii* | Toxoplasmosis | Serology, PCR |
| Ciliates | *Balantidium coli* | Balantidiasis | Stool |

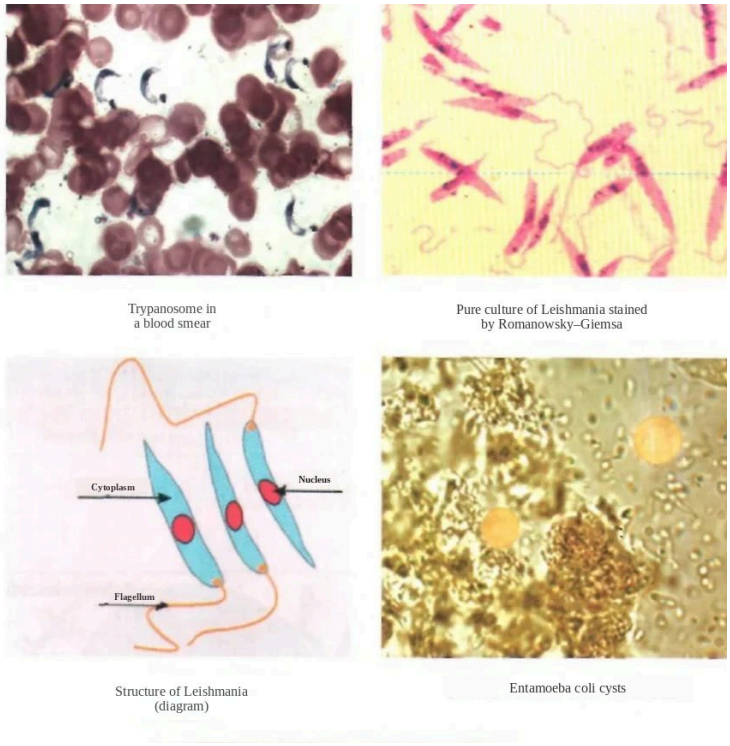



**Figure 4.3.** *Protozoa: trypanosome in a blood smear; Leishmania (pure culture and structural scheme); Entamoeba cysts. Source: Muhamedov I.M. et al., 2019.*

**Methods of study:** native (unstained) preparation — to see motility; staining with Lugol's solution — to see the cyst nuclei; **Romanowsky–Giemsa** — the main method for blood parasites (malaria, Leishmania, trypanosomes); enrichment methods (flotation/sedimentation of stool); serology and PCR.

> **REMEMBER**
> In malaria, the gold standard of diagnosis is the **thick smear** (to find the parasite) and the **thin smear** (to identify the species), stained with Romanowsky–Giemsa.

## 9. Practical skill: choosing the right material and method

1. **Suspected primary syphilis** → ulcer discharge → **dark-field** microscopy (live, unstained preparation).
2. **Suspected relapsing fever** → **blood smear** taken at the height of fever → Romanowsky–Giemsa.
3. **Suspected atypical pneumonia** → throat–nasal swab/sputum → **PCR** (mycoplasma, chlamydia).
4. **Suspected urogenital chlamydiosis** → urethral/cervical canal **scraping** (it must contain epithelial cells, not the discharge alone) → PCR or IF.
5. **Suspected actinomycosis** → fistula discharge → look for druses + culture.
6. **Suspected trichomoniasis** → **warm native preparation** of the discharge → look for motile flagellated cells.

> **CAUTION**
> The Trichomonas preparation must be examined **immediately and warm**: in a cooled preparation the parasite stops moving and it becomes difficult to distinguish it from an epithelial cell.

## 10. Review questions

1. How does the structure of spirochetes differ from that of ordinary bacteria?
2. How do you distinguish Treponema, Borrelia and Leptospira by their morphological features?
3. What is the main feature of mycoplasmas and how does it affect treatment?
4. Explain the developmental cycle of chlamydiae. What is the difference between EB and RB?
5. How are rickettsiae transmitted and by what methods are they detected?
6. How do actinomycetes differ from fungi? What is a druse?
7. How do the vegetative form and the cyst of protozoa differ?
8. Which preparations and which staining method are used in the diagnosis of malaria?

## Test questions

1. The structure that allows spirochetes to move:
   A) Flagellum
   B) Axial fibrils (axial filaments)
   C) Pili
   D) Capsule
   E) Mesosome

2. Prokaryotes without a cell wall:
   A) Chlamydiae
   B) Rickettsiae
   C) Mycoplasmas
   D) Actinomycetes
   E) Spirochetes

3. Mycoplasmas are naturally resistant to which antibiotics:
   A) Macrolides
   B) Tetracyclines
   C) Beta-lactams (penicillin)
   D) Fluoroquinolones
   E) Lincosamides

4. The infectious form of chlamydiae that lives outside the cell:
   A) Reticulate body
   B) Elementary body
   C) Cyst
   D) Spore
   E) Trophozoite

5. Used to propagate chlamydiae:
   A) Blood agar
   B) Endo medium
   C) Living cell culture
   D) Sabouraud medium
   E) Kligler medium

6. Bacteria that are obligate intracellular parasites:
   A) Mycoplasmas and spirochetes
   B) Chlamydiae and rickettsiae
   C) Actinomycetes and nocardiae
   D) Clostridia
   E) Mycobacteria

7. The causative agent and vector of epidemic typhus:
   A) *R. prowazekii* — louse
   B) *R. typhi* — tick
   C) *C. burnetii* — fly
   D) *B. recurrentis* — flea
   E) *R. prowazekii* — tick

8. The most suitable method for seeing *Treponema pallidum* alive:
   A) Gram method
   B) Dark-field microscopy
   C) Ziehl–Neelsen
   D) Neisser
   E) Ozheshko

9. A spirochete with large, irregular coils that stains blue-violet with Romanowsky–Giemsa in a blood smear:
   A) *Treponema*
   B) *Borrelia*
   C) *Leptospira*
   D) *Mycoplasma*
   E) *Chlamydia*

10. The feature that distinguishes actinomycetes from fungi:
    A) Forming mycelium
    B) Prokaryotic structure and a peptidoglycan wall
    C) Slow growth
    D) Branching
    E) Forming conidia

11. The yellow granules found in the pus in actinomycosis:
    A) Cysts
    B) Druses
    C) Volutin granules
    D) Spores
    E) Inclusions

12. The infectious form of protozoa that is resistant to the external environment:
    A) Trophozoite
    B) Cyst
    C) Gamete
    D) Merozoite
    E) Vegetative form

13. Used for identifying the species in the diagnosis of malaria:
    A) Thick smear only
    B) Thin blood smear
    C) Stool preparation
    D) Urine sediment
    E) CSF smear

14. The rule for examining a Trichomonas preparation:
    A) It is dried, stained by Gram and examined
    B) It is examined immediately, as a warm native preparation
    C) It is stored in a refrigerator overnight
    D) It is fixed first
    E) It is examined only under the electron microscope

15. The appearance typical of mycoplasma colonies on nutrient medium:
    A) “Dewdrop”
    B) “Fried egg” (depressed center)
    C) “String of pearls”
    D) “Lion’s mane”
    E) “Ant nest”

## Clinical cases

### *Case 1*

In a 25-year-old patient, a painless hard ulcer appeared on the genitalia. In a preparation made from the ulcer discharge, bright, tiny, regularly coiled motile microorganisms that flex at an angle were seen on a dark background.

**Question:** Which method was used and which causative agent should be considered?

**Answer:** Dark-field microscopy; the morphology is typical of *Treponema pallidum* — primary syphilis. Serological reactions (RPR and treponema-specific tests) are performed for confirmation.

### *Case 2*

A 19-year-old patient has had a dry cough and subfebrile fever for several weeks; the chest X-ray shows bilateral interstitial changes. Sputum culture showed no growth, and treatment with penicillin gave no result.

**Question:** Which causative agent is most likely, and why is penicillin ineffective?

**Answer:** *Mycoplasma pneumoniae* (atypical pneumonia). Mycoplasmas have no cell wall, and penicillin blocks wall synthesis — so it has no effect. Diagnosis is by PCR/serology, treatment with a macrolide or tetracycline.

### *Case 3*

A discharge taken from the cervical canal was sent for PCR, and the result was negative. On repeat testing, when a scraping (with epithelial cells) was taken, the result was positive.

**Question:** Why was the first test negative?

**Answer:** Chlamydia is an intracellular parasite; the material must contain **epithelial cells**. When only the discharge is taken, the causative agent does not get into the sample and gives a false-negative

result.

## Quick review

**Table 4.4.** *Topic 4 — key concepts*

| Concept | Essence |
|---|---|
| Structure of a spirochete | Protoplasmic cylinder + axial fibrils + outer sheath |
| Treponema / Borrelia / Leptospira | Small regular coils / few large coils / many coils + hooked ends |
| Mycoplasma | No wall, sterol-containing membrane, resistant to beta-lactams, “fried-egg” colony |
| Chlamydia | Obligate intracellular; EB (infectious) ↔ RB (multiplying) |
| Rickettsia | Obligate intracellular; via arthropods; Zdrodovsky, serology |
| Actinomycete | Prokaryotic mycelium; druses; sensitive to antibacterial antibiotics |
| Protozoa | Eukaryotic; trophozoite ↔ cyst; Romanowsky–Giemsa |
| Diagnosis of malaria | Thick smear (finding) + thin smear (species identification) |

TOPIC 5

# Bacterial physiology: chemical composition, nutrition, growth and reproduction. Nutrient media. Isolation of a pure culture — day 1

## 1. Learning objectives

In this class the student will master the following:

- To know the chemical composition of bacteria, the types of nutrition and the growth factors;
- to explain the stages of the growth curve of a bacterial population in liquid medium and their practical importance;
- to know the classification of nutrient media, the requirements for their composition and their role in diagnosis;
- to acquire the skill of day 1 of isolating a pure culture (taking the material, inoculation technique, isolation streaking);
- Independent work: to gain an understanding of prions and chromogenic nutrient media.

## 2. Chemical composition of bacteria

**Water** makes up 75–85% of the bacterial cell (the larger part of it is free water — the medium and solvent for reactions). The main components of the dry residue:

- **proteins** (50–80%) — structural proteins, enzymes, toxins, antigens;
- **nucleic acids** — DNA (nucleoid, plasmid) and RNA (r-, m-, t-RNA);
- **carbohydrates** — cell wall, capsule, reserve substances;
- **lipids** — membrane, mycolic acid in mycobacteria, lipid A in gram-negative bacteria;
- **minerals** — phosphorus, potassium, magnesium, iron, sulfur and trace elements.

## 3. Types of nutrition and growth factors

Bacteria are classified by their source of **carbon** and **energy**:

**Table 5.1.** *Types of nutrition*

| Criterion | Type | Description | Example |
|---|---|---|---|
| Carbon source | Autotroph | Synthesizes organic matter from $CO_2$ | Soil bacteria |
| | **Heterotroph** | Uses ready-made organic matter | All pathogenic bacteria |
| Energy source | Phototroph | Light | Photosynthetic bacteria |
| | **Chemotroph** | Energy of chemical reactions | Pathogenic bacteria |
| Lifestyle | Saprophyte | On dead organic matter | Putrefactive bacteria |
| | **Parasite** | In a living organism | Pathogens |

**REMEMBER**

All bacteria that cause disease in humans are **chemoorganoheterotrophs**: they obtain both carbon and energy from ready-made organic substances.

**Entry of substances into the cell** occurs by passive diffusion, facilitated diffusion, **active transport** (spending energy, against the concentration gradient) and group translocation. Active transport is the

main route.

**Growth factors** — substances that the bacterium cannot synthesize itself and must obtain ready-made: some amino acids, vitamins (B group), purine and pyrimidine bases, heme. Bacteria that need them are called **auxotrophs**, and those that do not are called **prototrophs**.

**CLINICAL SIGNIFICANCE**

*Haemophilus influenzae* needs **factor X (hemin)** and **factor V (NAD)** to grow. That is why it does not grow on ordinary blood agar, but grows well on **chocolate agar** (heated blood — the erythrocytes break down and release the factors). This requirement is taken into account directly in routine laboratory work.

## 4. Growth and reproduction

Bacteria multiply by **binary fission**. The time between two cell divisions is the **generation time**; in *E. coli* it is 20–30 minutes under favorable conditions, and in *Mycobacterium tuberculosis* up to 14–24 hours.

The **growth curve in liquid medium** consists of four phases:

1. **Lag phase (adaptation)** — the number of cells does not increase; the enzyme system adapts to the new medium.
2. **Logarithmic (exponential) phase** — the number of cells increases in geometric progression; the bacteria are most active, their morphology is typical, and they are most sensitive to antibiotics and disinfectants. Cultures for research are taken from this phase.
3. **Stationary phase** — balance between multiplication and death; the cell count is at its maximum (M-concentration). Many secondary metabolites, toxins, pigments and spores are formed in this phase.
4. **Death (decline) phase** — nutrients run out, metabolites accumulate, the number of living cells falls, and forms change (pleomorphism, involution forms).

**CAUTION**

In determining antibiotic susceptibility and testing a disinfectant, a **24-hour (young, logarithmic-phase)** culture is used. In an old culture the cells are polymorphic, and the result comes out wrong.

**Growth on solid medium** — in the form of colonies. A colony is a visible collection of cells that has arisen by multiplication from a single cell; therefore **a single colony = a source of pure culture**.

## 5. Nutrient media

**Requirements for a nutrient medium:** it must be sterile; contain the necessary nutrients and growth factors; have an optimal pH (7.2–7.6 for most pathogens); be isotonic; have moisture and transparency; have a standard composition.

**Table 5.2.** *Classification of nutrient media*

| Criterion | Type | Example | Task |
|---|---|---|---|
| Consistency | Liquid | Meat-peptone broth (MPB) | Enrichment, accumulation of pure culture |
| | Semisolid | Medium with 0.5% agar | Determining motility |
| | Solid | Meat-peptone agar (MPA) | Isolating colonies |
| Composition | Simple (universal) | MPA, MPB | General cultivation |

| Criterion | Type | Example | Task |
|---|---|---|---|
| | Complex (enriched) | Blood agar, serum agar, chocolate agar | Fastidious microbes |
| Task | Selective | Alkaline peptone water (cholera), egg-yolk salt agar | Preferentially grows the desired microbe |
| | Differential-diagnostic | Endo, Levine, Ploskirev, Kligler, Hiss | Differentiation by biochemical property |
| | Transport (preservative) | Glycerol mixture, Cary–Blair | Delivering the material |

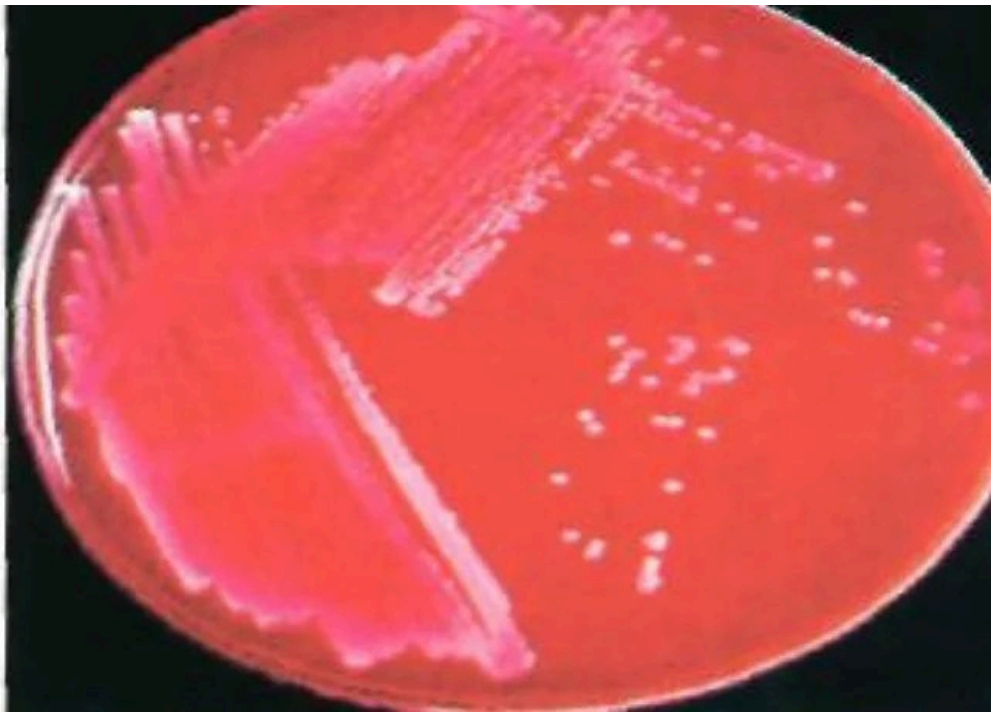


**Figure 5.1.** *Colonies grown on blood agar — the medium is enriched and allows the type of hemolysis to be assessed. Source: Muhamedov I.M. et al., 2019.*

**MEMORY AID**

Remember the types of media by function: **S**elective — **S**ingles out the one you want to grow; **D**ifferential — **D**istinguishes (tells organisms apart); **T**ransport — **T**o carry the specimen safely.

## 6. Isolation of a pure culture: general scheme and day 1

A **pure culture** is a culture consisting of bacteria belonging to a single species. A bacteriological examination is a multistep process that usually extends over three to four days:

**Table 5.3.** *Stages of pure culture isolation*

| Day | Work performed | Result |
|---|---|---|
| **Day 1** | Taking the material, microscopy, isolation streaking onto solid medium, placing in the incubator | A plate inoculated to obtain separate colonies |
| Day 2 | Studying the colonies (cultural properties), a smear from a suspicious colony, transfer to an agar slant | Pure culture accumulates |
| Day 3 | Checking the purity of the pure culture, biochemical and serological identification, setting up the antibiogram | The species is identified |

| Day | Work performed | Result |
|---|---|---|
| Day 4 | Reading the results, final report | Diagnosis + antibiotic susceptibility |

**Tasks of day 1:**

1. **Taking the material correctly** — before antibiotics are given, from the organ appropriate to the stage of the disease, into a sterile container, in sufficient quantity. Each material has its own rule (blood — at the height of fever, urine — the midstream portion, sputum — morning, deep).
2. **Preliminary microscopy** — a smear stained by the Gram method helps in choosing the next step.
3. **Choosing the medium and inoculating** — the material is inoculated onto selective and differential media suited to its composition.
4. **Isolation streaking** — the main technique for obtaining a separate colony.

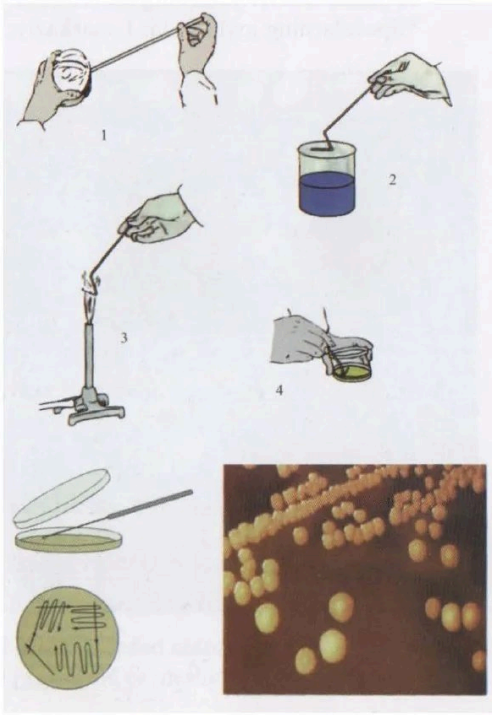


**Figure 5.2.** *The technique of taking the material and inoculating it onto a Petri dish with a bacteriological loop: flaming the loop, taking the material, streaking by sectors and, as a result, obtaining separate colonies. Source: Aliyev Sh.R. et al., 2013.*

## 7. Independent work 1: Prions

A **prion** is an infectious particle that has no nucleic acid and consists of **protein** alone. In the brains of humans and animals the normal protein **$PrP^{C}$** is normally present; the pathological form **$PrP^{Sc}$** refolds the normal protein to resemble itself (changes its conformation). As a result, insoluble aggregates accumulate in the brain and the tissue becomes porous — **spongiform encephalopathy**.

**Features:** they do not provoke an immune response (no inflammation and no fever), are extremely resistant to ordinary sterilization regimens and to irradiation, the incubation period is measured in years, the disease develops gradually and has no cure.

**Prion diseases:** Creutzfeldt–Jakob disease, kuru, Gerstmann–Sträussler–Scheinker syndrome, fatal familial insomnia; in animals — bovine spongiform encephalopathy, scrapie.

**REMEMBER**

A prion is an infectious agent **without nucleic acid**. This is an exception to the classic “central dogma” in microbiology, and for this reason it is studied separately.

## 8. Independent work 2: Chromogenic nutrient media

**Chromogenic media** are modern differential media that contain a colorless **chromogenic substrate**. If a bacterium has a particular enzyme, it breaks down the substrate and a colored product is released — the colony is stained a **characteristic color**.

**Advantages:**

- the species can be preliminarily identified **by colony color** — the answer is 18–24 hours faster;
- it becomes easier to separate species in mixed growth (for example, in a urine culture);
- the number of additional biochemical tests is reduced.

Examples used in practice: chromogenic agar for urine culture (E. coli — pink-red, enterococcus — blue-green colonies), chromogenic agar for distinguishing *Candida* species, and special chromogenic media for screening for MRSA and Salmonella.

**CLINICAL SIGNIFICANCE**

A chromogenic medium gives a quick **preliminary** answer, but it does not replace final identification: a biochemical or molecular method (in modern laboratories, mass spectrometry or PCR) is still used for confirmation.

### *Practical skill: the algorithm for streaking by sectors (quadrant method)*

1. Draw 4 sectors on the back of the plate and mark them; write the patient information and the date.
2. Flame the bacteriological loop in the alcohol burner **until red-hot** and cool it for 10–15 seconds.
3. Take the material with the loop and spread it in a dense zigzag in **sector 1**.
4. **Flame and cool the loop again**; draw it 3–4 times from the edge of sector 1 and spread into **sector 2**.
5. Continue in the same way into sectors 2→3 and 3→4. The loop is flamed before each transition — the number of cells decreases in each sector.
6. Place the plate **upside down** (so that condensation drops do not fall on the streak) in the incubator at 37 °C for 18–24 hours.

**CAUTION**

Touching the material with the loop **without cooling** it after flaming is the most common mistake: a hot loop kills the cells and the result becomes a false negative. The loop can be cooled by touching it to an empty area of the medium.

**Pasteur’s dilution method** and **Koch’s plate method** (serially diluting the material and pouring it mixed with molten agar) are other classic variants of isolation; in current practice, streaking by sectors is used more often.

## 9. Review questions

1. State the chemical composition of the bacterial cell. What is the share of water?

2. How are the types of nutrition classified? Which group do pathogenic bacteria belong to?
3. What are growth factors? What is the difference between an auxotroph and a prototroph?
4. Explain the phases of the growth curve and the practical importance of each.
5. What requirements are set for a nutrient medium? How are media classified?
6. What is the difference between a selective and a differential-diagnostic medium? Give an example.
7. List the stages of pure culture isolation. What is done on day 1?
8. Why does streaking by sectors give separate colonies?
9. What is a prion and how does it differ from viruses?
10. How does a chromogenic medium work and what advantage does it have?

## Test questions

1. The amount of water in the bacterial cell:
   A) 20–30%
   B) 40–50%
   C) 75–85%
   D) 95–99%
   E) 5–10%

2. The type of nutrition of pathogenic bacteria:
   A) Photoautotroph
   B) Chemoautotroph
   C) Chemoorganoheterotroph
   D) Photoheterotroph
   E) Lithotroph

3. A bacterium that needs a substance it cannot synthesize itself:
   A) Prototroph
   B) Auxotroph
   C) Autotroph
   D) Phototroph
   E) Saprophyte

4. The factors required for the growth of *Haemophilus influenzae*:
   A) X (hemin) and V (NAD)
   B) Glucose only
   C) Sodium chloride
   D) Teichoic acid
   E) Mycolic acid

5. The method of reproduction of bacteria:
   A) Budding
   B) Binary fission
   C) Mitosis
   D) Meiosis
   E) Reproduction by spores

6. The culture of which phase is taken to determine antibiotic susceptibility:
   A) Lag phase
   B) Logarithmic phase
   C) Stationary phase
   D) Death phase
   E) No difference

7. In which phase are toxins, pigments and spores mainly formed:
   A) Lag phase
   B) Logarithmic phase
   C) Stationary phase
   D) Death phase
   E) In none of the phases

8. The optimal pH for most pathogenic bacteria:
   A) 4,0–4,5
   B) 5,0–5,5
   C) 7,2–7,6
   D) 8,5–9,0
   E) 9,5–10,0

9. Which group does Endo medium belong to:
   A) Simple
   B) Enriched
   C) Differential-diagnostic
   D) Transport
   E) Semisolid

10. The consistency of the medium used to determine motility:
    A) Liquid
    B) Semisolid
    C) Solid
    D) Dry
    E) Gel

11. What is a colony:
    A) A mixture of bacteria of different species
    B) A collection of cells that has arisen by multiplication from a single cell
    C) Turbidity in liquid medium
    D) A collection of spores
    E) A collection of bacteriophages

12. What is done on day 1 of pure culture isolation:
    A) Biochemical identification
    B) Taking the material, microscopy and isolation streaking
    C) Giving the final report
    D) Reading the antibiogram
    E) Serological identification

13. In streaking by sectors, the loop is flamed before each transition because:
    A) The medium is sterilized
    B) The number of cells decreases in each sector and separate colonies form
    C) The loop lasts a long time
    D) The color of the colony changes
    E) The pH of the medium changes

14. The main property of a prion:
    A) Contains DNA
    B) Contains RNA
    C) Has no nucleic acid, consists of protein only
    D) Has a cell wall
    E) Is retained by a bacterial filter

15. What the color of a colony on a chromogenic medium depends on:
    A) The size of the colony
    B) The breakdown of the chromogenic substrate by a bacterial enzyme
    C) The temperature of the medium
    D) The motility of the bacterium
    E) The incubation period

## Clinical cases

### *Case 1*

Material taken from the CSF was inoculated onto blood agar, but after 24 hours there was no growth. In the smear, however, small gram-negative rods had been seen. Growth appeared when it was inoculated onto chocolate agar.

**Question:** Why was there no growth on ordinary blood agar?

**Answer:** The causative agent needs growth factors (factors X and V) — probably *Haemophilus influenzae*. These factors are present only in blood agar in which the erythrocytes have been broken down, that is, heated blood (chocolate) agar.

### *Case 2*

To determine antibiotic susceptibility, a student used a culture that had stood in the incubator for three days. The results differed sharply from previous tests.

**Question:** What is the error?

**Answer:** A three-day culture is in the death phase; the cells are polymorphic and some have died. For susceptibility testing, a young culture in the **24-hour, logarithmic phase** and a standard turbidity (for example, 0.5 McFarland) must be used.

### *Case 3*

A urine culture was inoculated onto chromogenic medium; after 18 hours pink-red and blue-green colonies grew at the same time.

**Question:** What does this result mean and what is the next step?

**Answer:** Two different causative agents have grown (judging by the colors, probably E. coli and enterococcus). The next step is to subculture each colony separately, assess the quantity (CFU/mL),

confirm the identification and set up a separate antibiogram for each.

## Quick review

**Table 5.4.** *Topic 5 — key concepts*

| Concept | Essence |
|---|---|
| Chemical composition | Water 75–85%; protein, nucleic acid, carbohydrate, lipid, minerals |
| Nutrition of pathogens | Chemoorganoheterotroph |
| Growth factors | Vitamins, amino acids, heme; auxotroph ↔ prototroph |
| Growth phases | Lag → logarithmic (sensitive, typical) → stationary (toxin, spore) → death |
| pH of the medium | 7,2–7,6 |
| Types of media | Simple, enriched, selective, differential, transport |
| Colony | From a single cell — the source of a pure culture |
| Day 1 | Material + microscopy + isolation streaking + incubator 37 °C |
| Prion | Protein without nucleic acid; spongiform encephalopathy |
| Chromogenic medium | Enzyme + chromogenic substrate → colored colony |

TOPIC 6

# Catabolic metabolism: types of respiration. Isolation of pure cultures of aerobes and anaerobes. Cultural properties — day 2

## 1. Learning objectives

In this class the student will master the following:

- To know the ways of obtaining energy and the types of respiration (aerobic, anaerobic, facultative) in bacteria;
- to divide bacteria into groups by their relation to oxygen and to explain the methods of creating anaerobic conditions;
- to acquire the skill of fully describing the cultural properties of colonies;
- to perform day 2 of pure culture isolation (studying the colonies, subculturing onto an agar slant);
- Independent work: to gain an understanding of modern molecular-genetic methods of rapid diagnosis.

## 2. Catabolism and ways of obtaining energy

**Metabolism** consists of two directions: **catabolism** (breakdown, release of energy) and **anabolism/constructive metabolism** (synthesis, expenditure of energy — topic 7). In bacteria, energy is stored as **ATP** and is produced in the **cytoplasmic membrane** (there are no mitochondria).

The main pathways of carbohydrate breakdown: **glycolysis (Embden–Meyerhof–Parnas pathway)** — the most widespread; the **pentose phosphate pathway**; the **Entner–Doudoroff pathway** (for example, in *Pseudomonas*). The product of glycolysis is pyruvate, and its further fate depends on the type of respiration.

**Table 6.1.** *Comparison of the ways of obtaining energy*

| Process | Final acceptor | ATP (from 1 glucose, approximate) | End product |
|---|---|---|---|
| Aerobic respiration | Oxygen ($O_2$) | ≈ 38 | $CO_2$ and $H_2O$ |
| Anaerobic respiration | Nitrate, sulfate, fumarate | ≈ 10–20 | $N_2$, $H_2S$, etc. |
| Fermentation | **Organic substance** (pyruvate and its derivatives) | 2 | Lactic acid, alcohol, butyric acid, gas |

**REMEMBER**

**The main feature of fermentation** is that the electron acceptor is an **organic substance** and oxygen does not take part. That is why fermentation yields little ATP, but allows life in the absence of oxygen.

## 3. Bacteria by relation to oxygen

**Table 6.2.** *Groups by type of respiration*

| Group | Relation to oxygen | Example | Where it grows (in a tube) |
|---|---|---|---|
| Obligate aerobes | Only in the presence of $O_2$ | *Mycobacterium tuberculosis*, *Pseudomonas* | Surface layer |
| Microaerophiles | Require little $O_2$ (5–10%) and $CO_2$ | *Campylobacter*, *Helicobacter* | Slightly below the surface |
| **Facultative anaerobes** | Both with and without $O_2$ | Enterobacteriaceae, staphylococci | Throughout the column |
| Aerotolerant | Does not use $O_2$, but tolerates it | Some streptococci and lactobacilli | Throughout the column, uniformly |
| Obligate anaerobes | $O_2$ is toxic | Clostridia, Bacteroides | Only at the bottom |

**Why is oxygen toxic to anaerobes?** In oxygen metabolism, toxic intermediate products form — the **superoxide anion ($O_2^-$)** and **hydrogen peroxide ($H_2O_2$)**. Aerobes have enzymes that neutralize them — **superoxide dismutase** and **catalase**; obligate anaerobes lack these enzymes or have very little, so they die in the presence of oxygen.

> **MEMORY AID**
>
> **Catalase present → tolerates oxygen.** The catalase test is also used in practice to distinguish staphylococci (catalase +) from streptococci (catalase −): if bubbles appear when hydrogen peroxide is dropped on, the catalase is positive.

**Methods of creating anaerobic conditions:**

1. **Physical:** anaerostat (replacing the air with a gas mixture), boiling the medium to drive off oxygen and sealing it with a layer of oil, deep inoculation (a tall column of medium).
2. **Chemical:** gas-generating packs, oxygen-absorbing substances (pyrogallol + alkali).
3. **Biological:** Fortner's method — an aerobe and an anaerobe are inoculated side by side on one plate; the aerobe consumes the oxygen and the anaerobe begins to grow.
4. **Special media:** Kitt–Tarozzi medium (with liver pieces and a layer of oil), thioglycolate medium.

## 4. Cultural properties: describing a colony

The appearance of a colony is a species-specific feature that allows **preliminary identification**. A colony is described by the following features:

**Table 6.3.** *Scheme for describing a colony*

| Feature | Variants |
|---|---|
| Size | Small (up to 1 mm), medium (1–3 mm), large (more than 3 mm) |
| Shape | Round, irregular, rhizoid (root-like) |
| Edge | Smooth, undulate, serrated, filamentous |
| Surface | Smooth (S form), rough (R form), mucoid (M form) |
| Profile (elevation) | Flat, convex, with a depressed center, dome-shaped |

| Feature | Variants |
|---|---|
| Color (pigment) | Colorless, white, yellow, golden, blue-green, red |
| Transparency | Transparent, translucent, opaque |
| Consistency | Soft, dry, sticky, easily picked up with a loop |
| Change in the medium | Hemolysis, change in the color of the medium, diffusion of pigment into the medium |

**S and R form colonies. S (smooth)** colonies are smooth, convex and moist; they are usually typical of encapsulated strains or those with complete LPS, which are more virulent. **R (rough)** colonies are dry with an uneven edge; in some species (tubercle bacillus, anthrax bacillus, diphtheria bacillus), by contrast, the R form is typical and is considered virulent.

**Hemolysis on blood agar** — the main feature in distinguishing streptococci:

- **alpha-hemolysis** — a green-gray zone around the colony (partial breakdown of hemoglobin);
- **beta-hemolysis** — a completely clear, colorless zone (the erythrocytes are completely lysed);
- **gamma** — no hemolysis.

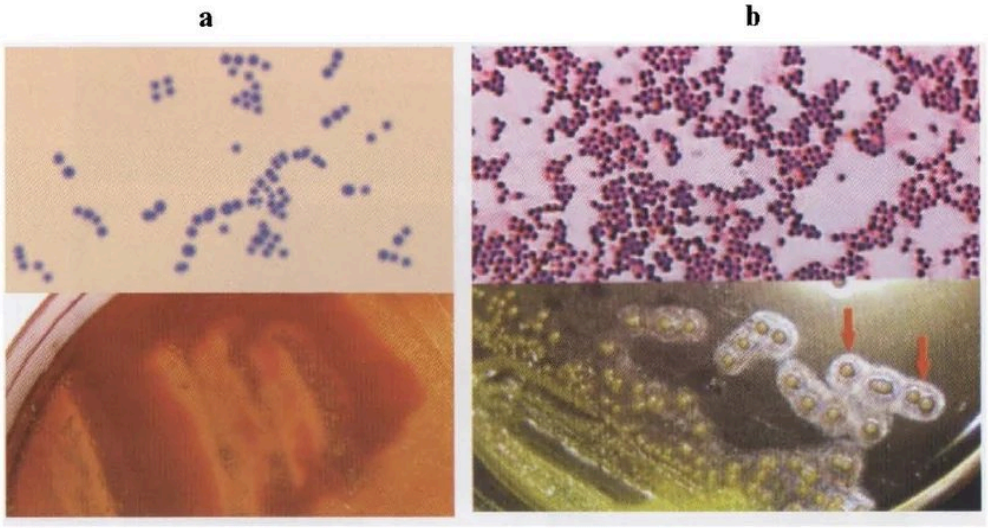



**Figure 6.1.** *Pure culture and colonies of pathogenic cocci: streptococcus (stained with methylene blue) and staphylococcus (by Gram), and their colonies on nutrient medium. Source: Aliyev Sh.R. et al., 2013.*

**Types of growth in liquid medium:** uniform turbidity (most facultative anaerobes), sediment at the bottom (streptococcus, species that settle out), a pellicle on the surface (obligate aerobes — the tubercle bacillus, *Pseudomonas aeruginosa*), pellicle + sediment.

## 5. Isolation of a pure culture: day 2

**Goal** — to choose the needed colony from the inoculated plate and accumulate a pure culture.

**Work to be done:**

1. Examine the plate: whether there is growth, the number of colony types, and record the cultural properties of each.
2. **Prepare a smear and stain it by the Gram method** from the chosen (suspicious) colony — check that the morphology and tinctorial property match the colony description.
3. Take the remainder of that colony with the loop and subculture it onto a **meat-peptone agar slant** (or a selective medium) — this is the stage of accumulating the pure culture.

4. Place in the incubator at 37 °C for 18–24 hours.

Day 2 is slightly different **for anaerobes**: inoculation is done in an anaerostat or on Kitt–Tarozzi medium; because growth is slower, the result is assessed after 48–72 hours.

**CAUTION**

When choosing a colony, **only an isolated colony** is picked. Material taken from a confluent ("fused") area gives a mixed culture, and all the following steps (biochemical, serological, antibiogram) will show wrong results.

**CLINICAL SIGNIFICANCE**

Small beta-hemolytic colonies on blood agar + gram-positive cocci in chains + material taken from the throat → suspicion of group A beta-hemolytic streptococcus; this is important in view of tonsillitis and its complications (rheumatic fever, glomerulonephritis) (topic 16).

## 6. Independent work: Modern methods of rapid diagnosis of bacteria

The classic bacteriological method is reliable, but it takes **3–4 days**. Modern methods can give an answer within hours.

**1. Polymerase chain reaction (PCR).** A specific fragment of the pathogen's DNA (or RNA — via reverse transcription) is amplified in vitro. Steps: **denaturation** (≈95 °C) → **primer binding (annealing)** (50–65 °C) → **elongation** (72 °C); the cycle is repeated 30–40 times and the target sequence multiplies exponentially. **Real-time PCR** measures the result during amplification by fluorescence and gives a quantitative assessment.

*Advantages:* very sensitive and specific; indispensable for pathogens that do not grow or grow slowly (chlamydia, mycoplasma, tuberculosis, viruses); gives a result even after antibiotics have been started.

*Limitations:* does not distinguish live from dead microbes; does not give full antibiotic susceptibility (detects only known resistance genes); sensitive to contamination.

**2. Sequencing the 16S ribosomal RNA gene.** The 16S rRNA gene is present in all bacteria; within it, regions common to all species (conserved) alternate with species-specific (variable) ones. Primers are placed on the common regions to amplify the gene, then its sequence is read and compared with a database. This method is used to identify **uncultivable or unusual** microorganisms, and also to study the composition of the microbiota.

**3. Electromigration methods. Capillary zone electrophoresis** — separates particles by charge and size in a thin capillary; **capillary isoelectric focusing** — separates proteins by their isoelectric point. They allow microbial proteins and nucleic acids to be analyzed with high accuracy.

**4. Mass spectrometry (MALDI-TOF).** The spectrum of proteins in a colony is compared with a reference database, and the species is identified **in a few minutes**. It has now become the main method of identification in many laboratories.

**REMEMBER**

Rapid methods **complement, not replace,** classic culture: a live culture is still needed for full assessment of antibiotic susceptibility and for epidemiological analysis.

### *Practical skill: describing a colony and accumulating a pure culture*

1. Take the plate out of the incubator and, **without turning it over**, hold it up to the light and examine it: determine whether there is growth and how many kinds of colonies there are.
2. Describe each type of colony according to the scheme: size, shape, edge, surface, profile, color, transparency, consistency and change in the medium (hemolysis, pigment).
3. Assess the type of hemolysis on blood agar: clear zone (beta), green zone (alpha), no zone (gamma). Look at the zone from the bottom of the plate, holding it up to the light.
4. Write the description in the notebook and mark the suspicious colony — **only an isolated** colony is chosen.
5. Prepare a smear from part of that colony and stain it by the Gram method: the morphology must match the colony description.
6. Take the rest of the colony with the loop and inoculate it in a zigzag onto a **meat-peptone agar slant** (to accumulate a pure culture).
7. Label the tube (patient, date, colony number) and place it in the incubator at 37 °C for 18–24 hours.

> **CAUTION**
> Touching a neighboring colony when picking a colony with the loop is the most common mistake: as a result a mixed culture accumulates and all the following steps give a wrong answer.

## 7. Review questions

1. Where is ATP produced in bacteria? Name the main pathways of carbohydrate breakdown.
2. How do aerobic respiration, anaerobic respiration and fermentation differ?
3. Into what groups are bacteria divided by their relation to oxygen? Give an example of each.
4. Why is oxygen toxic to obligate anaerobes?
5. What methods of creating anaerobic conditions are there?
6. By what features is a colony described? What are the S and R forms?
7. Name the types of hemolysis and explain their diagnostic significance.
8. What work is done on day 2 of pure culture isolation?
9. What are the steps and advantages of PCR? And its limitations?
10. In which cases is 16S rRNA gene sequencing used?

## Test questions

1. The final electron acceptor in fermentation:
   A) Oxygen
   B) Nitrate
   C) Sulfate
   D) Organic substance (pyruvate and its derivatives)
   E) Iron

2. The approximate amount of ATP produced from 1 molecule of glucose in aerobic respiration:
   A) 2
   B) 8
   C) 18
   D) 38
   E) 60

3. Bacteria that can grow both in the presence and in the absence of oxygen:
   A) Obligate aerobes
   B) Obligate anaerobes
   C) Facultative anaerobes
   D) Microaerophiles
   E) Aerotolerant organisms

4. The toxic effect of oxygen in obligate anaerobes is associated with:
   A) A rise in pH
   B) A lack of enzymes that neutralize superoxide and hydrogen peroxide
   C) A rise in temperature
   D) Osmotic pressure
   E) Exhaustion of nutrients

5. Bubbles appearing when hydrogen peroxide is dropped on in the catalase test mean:
   A) Catalase-negative
   B) Catalase-positive
   C) Oxidase-positive
   D) Hemolysis is present
   E) Motile

6. Fortner’s method belongs to which group of methods for creating anaerobic conditions:
   A) Physical
   B) Chemical
   C) Biological
   D) Mechanical
   E) Irradiation method

7. The medium used for growing anaerobes:
   A) Endo
   B) Kitt–Tarozzi
   C) Sabouraud
   D) Löwenstein–Jensen
   E) Alkaline peptone water

8. Formation of a completely clear zone around a colony on blood agar:
   A) Alpha-hemolysis
   B) Beta-hemolysis
   C) Gamma (no hemolysis)
   D) Pigment production
   E) Lysogeny

9. The color of the zone around the colony in alpha-hemolysis:
    A) Clear, colorless
    B) Green-gray
    C) Red
    D) Black
    E) Yellow

10. Smooth, convex, moist colonies are designated as:
    A) R form
    B) S form
    C) M form
    D) L form
    E) D form

11. Growth as a pellicle on the surface of liquid medium is more typical of:
    A) Obligate aerobes
    B) Obligate anaerobes
    C) Aerotolerant organisms
    D) Mycoplasmas
    E) Chlamydiae

12. What is done on day 2 of pure culture isolation:
    A) Taking the material and inoculating it
    B) Studying the colonies, making a smear and subculturing onto an agar slant
    C) The final report
    D) Reading the antibiogram
    E) Setting up a serological reaction

13. The first step of PCR:
    A) Elongation
    B) Primer binding
    C) Denaturation (≈95 °C)
    D) Measuring fluorescence
    E) Electrophoresis

14. 16S rRNA gene sequencing is mainly used:
    A) In determining antibiotic susceptibility
    B) In identifying uncultivable or unusual bacteria
    C) In assessing hemolysis
    D) In determining colony color
    E) In monitoring sterilization

15. What MALDI-TOF mass spectrometry is based on:
    A) The antigen–antibody reaction
    B) Comparing the spectrum of proteins in a colony with a reference database
    C) DNA amplification
    D) Enzymatic activity
    E) Hemolysis

## Clinical cases

### *Case 1*

Material taken from a deep, ulcerated wound was inoculated onto ordinary blood agar and incubated for 24 hours under aerobic conditions — there was no growth. In the smear, however, a large number of gram-positive rods had been seen.

**Question:** What mistake was made and how is it corrected?

**Answer:** The causative agent may be an obligate anaerobe. The material must be inoculated under anaerobic conditions (anaerostat, Kitt–Tarozzi medium) and observed for 48–72 hours.

### *Case 2*

Material taken from the throat was inoculated onto blood agar. After 24 hours, a completely clear zone appeared around small, transparent colonies; gram-positive cocci in chains were seen in the smear. The catalase test is negative.

**Question:** What is the preliminary conclusion?

**Answer:** Beta-hemolytic streptococcus (catalase-negative, gram-positive cocci in chains). To determine the group, bacitracin susceptibility and serological grouping (for group A) are carried out.

### *Case 3*

Material was taken after antibiotics had been started in the patient. Culture gave no growth, but PCR showed a positive result.

**Question:** How is this explained and how should the PCR result be interpreted?

**Answer:** Under the action of the antibiotic the microbes have died or cannot grow, but their DNA has been preserved. PCR does not distinguish live from dead microbes — so the result confirms the presence of the causative agent, but in assessing the effect of treatment it is interpreted together with the clinical picture.

## Quick review

**Table 6.4.** *Topic 6 — key concepts*

| Concept | Essence |
|---|---|
| Aerobic respiration / fermentation | $O_2$ acceptor, ≈38 ATP / organic acceptor, 2 ATP |
| Groups | Obligate aerobe, microaerophile, facultative anaerobe, aerotolerant, obligate anaerobe |
| $O_2$ is toxic to anaerobes | Lack of superoxide dismutase and catalase |
| Anaerobic conditions | Anaerostat, gas packs, Fortner, Kitt–Tarozzi |
| Colony description | Size, shape, edge, surface, profile, pigment, transparency, consistency |
| S / R form | Smooth (often virulent) / rough (typical in tuberculosis, anthrax, diphtheria) |
| Hemolysis | Alpha — green zone; beta — clear zone; gamma — none |
| Day 2 | Studying the colony + smear + subculture onto an agar slant |
| PCR | Denaturation → annealing → elongation; does not distinguish live from dead |
| MALDI-TOF | Identification in minutes by protein spectrum |

TOPIC 7

# Constructive metabolism: enzymes and products of vital activity. Identification of a pure culture — day 3

## 1. Learning objectives

In this class the student will master the following:

- To know the constructive metabolism of bacteria, their enzymes and their classification;
- to explain the products of bacterial vital activity (enzymes, pigments, aromatic substances, toxins, antibiotics) and their diagnostic significance;
- to master the methods of identifying a pure culture by its biochemical properties (Hiss, Kligler, oxidase, catalase, indole, urease);
- to perform day 3 of pure culture isolation and prepare for the final report;
- Independent work: to know the forms of obtaining energy in microorganisms, the types of fermentation and the pathways of carbohydrate fermentation.

## 2. Constructive metabolism (anabolism)

**Constructive metabolism** is the process of synthesizing cell components (protein, nucleic acid, polysaccharide, lipid); energy is expended. It is closely linked to catabolism: catabolism provides **ATP and intermediate metabolites** (carbon skeletons), and anabolism uses them. Reactions common to both directions are called **amphibolic pathways** (for example, the Krebs cycle).

The bacterial cell synthesizes very quickly: under favorable conditions the entire cell content doubles in 20–30 minutes. The direction and rate of synthesis are controlled by **enzymes**.

## 3. Bacterial enzymes

An **enzyme** is a biological catalyst of protein nature. Enzymes serve as the main feature in the identification of bacteria, because **each species has its own characteristic set of enzymes** and it is passed on from generation to generation.

**Table 7.1.** *Classification of bacterial enzymes*

| Criterion | Type | Description and example |
|---|---|---|
| Arrangement | Exoenzymes | Secreted outside the cell; break down large molecules (protease, lipase, amylase, hyaluronidase) |
| | Endoenzymes | Work inside the cell (respiratory and synthetic enzymes) |
| Formation | **Constitutive** | Always synthesized, independent of the substrate (glycolytic enzymes) |
| | **Inducible (adaptive)** | Synthesized only when the substrate is present (if lactose is present — beta-galactosidase) |
| Task | Constructive / catabolic | Synthesis / breakdown |
| Role in pathogenicity | **Aggression (pathogenicity) enzymes** | Hyaluronidase, plasmacoagulase, fibrinolysin, lecithinase, neuraminidase, collagenase |

**REMEMBER**

**Constitutive** enzymes — "permanent staff" (always working); **inducible** — "on call" (synthesis begins when the substrate arrives). The lac operon is the classic example of induction.

**CLINICAL SIGNIFICANCE**

Aggression enzymes are themselves pathogenicity factors: **plasmacoagulase** (*S. aureus*) forms a fibrin shell that protects against phagocytosis and is the main test for distinguishing *S. aureus* from *S. epidermidis*; **hyaluronidase** "dissolves" the intercellular space of the tissue and makes spread easier (the "spreading factor"); **lecithinase** destroys the cell membrane.

## 4. Products of bacterial vital activity

1. **Enzymes** — discussed above; the basis of identification.
2. **Pigments** — species-specific colored substances: *S. aureus* — golden-yellow (carotenoid), *Pseudomonas aeruginosa* — **pyocyanin** (blue-green, diffuses into the medium) and pyoverdine (fluorescent), *Serratia marcescens* — red prodigiosin. Functions of pigment: protection from UV rays, participation in respiration, antibiotic action.
3. **Aromatic (odorous) substances** — some species give off a characteristic smell: *P. aeruginosa* — a jasmine (hyacinth) smell, *Proteus* — a foul, putrid smell; in practice noted as an auxiliary sign.
4. **Toxins** — exotoxins (protein, species-specific, potent) and endotoxin (LPS) — in detail in topic 11.
5. **Antibiotics and bacteriocins** — substances that inhibit other microbes (topic 9).
6. **Vitamins and organic acids** — synthesized by the normal intestinal microflora (topic 10) and used in industry.

**CAUTION**

Pigment and odor are only **auxiliary** signs. They speed up identification, but a final answer is not given without biochemical and serological confirmation.

## 5. Methods of biochemical identification

For identification, the pure culture is tested for **saccharolytic** (carbohydrate breakdown), **proteolytic** (protein breakdown), **oxidation–reduction** and other enzyme activities.

### *5.1. Saccharolytic activity*

**Hiss media (the "variegated series")** — each tube contains one carbohydrate (glucose, lactose, maltose, sucrose, mannitol) + an indicator + a gas-collecting float (Durham tube). If the carbohydrate is broken down, acid forms → the indicator changes color (**A — acid**); if gas is also released, a bubble is seen in the tube (**AG — acid + gas**).

**Kligler medium (two-sugar iron medium)** — a medium that gives a lot of information in one tube. It contains glucose (little), lactose (much), an iron salt and an indicator; the tube consists of a slant (aerobic) and a butt (anaerobic) part.

**Table 7.2.** *Reading the results on Kligler medium*

| Slant | Butt | Interpretation | Example |
|---|---|---|---|
| Red (alkaline) | Yellow (acid) | Only glucose fermented | Salmonella, Shigella |

| Slant | Butt | Interpretation | Example |
|---|---|---|---|
| Yellow | Yellow | Glucose + lactose fermented | E. coli, Klebsiella |
| Red | Red | Carbohydrates not fermented | *Pseudomonas* |
| Cracks in the butt, lifting | — | Gas formed | E. coli |
| Black color in the butt | — | **$H_2S$** formed | Salmonella, Proteus |

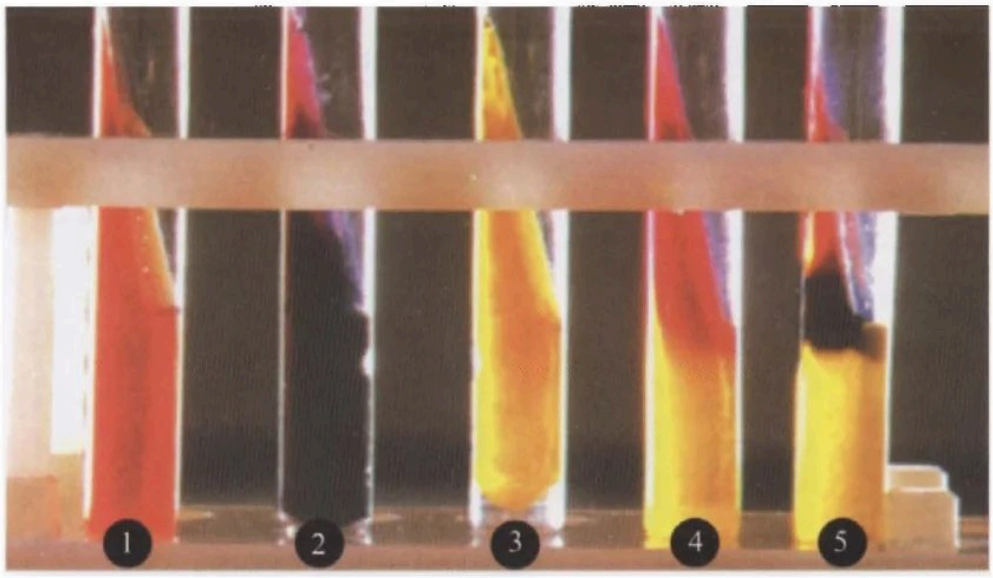



**Figure 7.1.** *Enzymatic activity of enteric bacteria on Kligler two-sugar medium: different results (acid, gas, $H_2S$). Source: Aliyev Sh.R. et al., 2013.*

**MEMORY AID**

Remember Kligler as a "**traffic light**": **yellow = acid** (sugar fermented), **red = alkaline** (not fermented), **black = $H_2S$**, **crack/bubble = gas**.

## 5.2. Proteolytic activity

The breakdown of proteins and amino acids is tested: gelatin liquefaction, milk coagulation, and production of **indole**, **hydrogen sulfide ($H_2S$)** and **ammonia**. To detect indole, Kovacs reagent is added — if a red ring appears, the result is positive (typical of E. coli).

## 5.3. Commonly used individual tests

**Table 7.3.** *Main differential tests*

| Test | What it detects | Positive result | Diagnostic significance |
|---|---|---|---|
| Catalase | The enzyme catalase | Bubbles from $H_2O_2$ | Staphylococcus (+) / streptococcus (−) |
| Oxidase | Cytochrome oxidase | The reagent paper turns blue | *Pseudomonas*, vibrio, Neisseria (+) / Enterobacteriaceae (−) |
| Plasmacoagulase | Coagulation of plasma | Plasma clots | *S. aureus* (+) / other staphylococci (−) |
| Urease | Breakdown of urea | The medium turns pink | *Proteus*, *Helicobacter* (+) |
| Indole | Indole from tryptophan | Red ring with Kovacs reagent | E. coli (+) |

| Test | What it detects | Positive result | Diagnostic significance |
|---|---|---|---|
| Lecithinase | Breakdown of egg yolk | A cloudy zone around the colony | *S. aureus*, clostridia |

## 6. Isolation of a pure culture: day 3

**Goal** — to confirm the purity of the accumulated pure culture and determine its species.

1. **Checking purity:** a smear is prepared from the growth on the slant and stained by the Gram method. All cells must have the same morphology and tinctorial property.
2. **Biochemical identification:** the Hiss series, Kligler, indole, $H_2S$, urease and other tests are set up; modern laboratories use ready-made test systems or automated analyzers.
3. **Serological identification:** a **slide agglutination** reaction with diagnostic sera (dividing enteric bacteria into serovars — topic 22).
4. **Antibiogram:** set up by disk diffusion or another method (topic 9).
5. If necessary, additional tests: typing with phages, determination of toxigenicity, biological test.

**Day 4** — all the results are put together and the final report is given: **the species of the causative agent (if necessary — the serovar) and its susceptibility to antibiotics.**

> **CAUTION**
> Biochemical results must be read **at the specified time**. Kligler is read after 18–24 hours: if read too early, the breakdown of lactose is not seen; if read too late, the acid in the slant turns alkaline and a lactose-negative result may be misinterpreted as falsely "lactose-positive."

## 7. Independent work: Forms of energy production and types of fermentation

**Forms of cell energy: ATP** (the chemical form) and the **proton gradient across the membrane (proton-motive force)**. The respiratory chain pumps protons out through the membrane, and the gradient formed is converted into ATP by **ATP synthase** (chemiosmotic theory). The proton gradient is also used directly for flagellar rotation and active transport.

**Ways of producing energy:** substrate-level phosphorylation (in fermentation, in a direct reaction), oxidative phosphorylation (in the respiratory chain), photophosphorylation (in phototrophs).

**Pathways of carbohydrate fermentation:** glycolysis (EMP), the pentose phosphate pathway, the Entner–Doudoroff pathway.

**Table 7.4.** *Types of fermentation and their significance*

| Type of fermentation | Main product | Representatives | Significance |
|---|---|---|---|
| Lactic acid (homofermentative) | Lactic acid only | *Lactobacillus*, *Streptococcus* | Normal microflora, dairy products |
| Lactic acid (heterofermentative) | Lactic acid + alcohol + $CO_2$ | *Leuconostoc*, bifidobacteria | Intestinal microbiota |
| Alcoholic | Ethanol + $CO_2$ | Yeasts | Industry, baking |
| Butyric acid | Butyric acid + gas | Clostridia | Food spoilage, gas formation |
| Formic acid (mixed) | A mixture of acids, $CO_2$, $H_2$ | Enterobacteriaceae | Gas formation in Kligler and Hiss media |
| Propionic acid | Propionic acid | Propionibacteria | Cheese ripening, skin microflora |

**REMEMBER**

In Enterobacteriaceae, **mixed (formic acid) fermentation** is exactly the reason **gas** forms in Hiss and Kligler media — this result is seen in daily identification.

## 8. Practical skill: the algorithm for reading a biochemical series

1. Take the tubes out of the incubator at the specified time (usually 18–24 hours).
2. Compare with a control (uninoculated) tube — to see the color difference clearly.
3. For each carbohydrate: write **A** if the color has changed, **AG** if there is a bubble in the Durham tube, and − if there is no change.
4. In Kligler, assess the slant and the butt separately and note $H_2S$ (black) and gas (cracks).
5. Perform the additional tests (indole, urease, oxidase, catalase).
6. Compare the set of results with an identification table or with **Bergey's Manual** and determine the species.
7. In doubtful cases, confirm by serological identification or a molecular method.

## 9. Review questions

1. What is constructive metabolism? How is it linked to catabolism?
2. How are enzymes classified? What is the difference between a constitutive and an inducible enzyme?
3. Give examples of aggression enzymes and explain their role.
4. List the products of bacterial vital activity.
5. How are Hiss media constructed and how is the result read?
6. What results can be obtained on Kligler medium? Interpret each.
7. What is the diagnostic significance of the catalase, oxidase and plasmacoagulase tests?
8. What work is done on day 3? What is stated in the final report?
9. What is the proton-motive force and where is it used?
10. Name the main types of fermentation and their practical significance.

## Test questions

1. Enzymes synthesized only when the substrate is present:
   A) Constitutive
   B) Inducible (adaptive)
   C) Endoenzymes
   D) Exoenzymes
   E) Oxidoreductases
2. The main enzyme test for distinguishing *S. aureus* from other staphylococci:
   A) Catalase
   B) Oxidase
   C) Plasmacoagulase
   D) Urease
   E) Indole

3. The aggression enzyme called the “spreading factor”:
   A) Plasmacoagulase
   B) Hyaluronidase
   C) Catalase
   D) Beta-lactamase
   E) Peroxidase

4. The blue-green pigment of *Pseudomonas aeruginosa* that diffuses into the medium:
   A) Prodigiosin
   B) Pyocyanin
   C) Carotenoid
   D) Melanin
   E) Flavin

5. If only a color change is seen in a Hiss medium, this means:
   A) The carbohydrate has not been broken down
   B) Acid has formed (A)
   C) Acid and gas have formed (AG)
   D) $H_2S$ has formed
   E) Indole has formed

6. If on Kligler medium the slant is red and the butt is yellow:
   A) Both glucose and lactose have been broken down
   B) Only glucose fermented
   C) No carbohydrate has been broken down
   D) Only lactose has been broken down
   E) The result is wrong

7. What blackening of the butt of Kligler medium indicates:
   A) Gas formation
   B) $H_2S$ formation
   C) Indole formation
   D) Urease activity
   E) Hemolysis

8. Bacteria with a positive oxidase test:
   A) Enterobacteriaceae
   B) *Pseudomonas*, vibrios, Neisseria
   C) Staphylococci
   D) Streptococci
   E) Clostridia

9. A bacterium with a positive urease test:
   A) E. coli
   B) *Proteus*
   C) Shigella
   D) Streptococcus
   E) Diphtheria bacillus

10. The reagent used to detect indole:
    A) Lugol
    B) Kovacs reagent
    C) Hydrogen peroxide
    D) Tetramethylphenylenediamine
    E) Fuchsin

11. The work that is not done on day 3 of pure culture isolation:
    A) Checking purity (smear)
    B) Biochemical identification
    C) Setting up the antibiogram
    D) Taking the material from the patient
    E) Serological identification

12. The enzyme that produces ATP from the proton gradient in the respiratory chain:
    A) ATP synthase
    B) Catalase
    C) Transpeptidase
    D) Polymerase
    E) Ligase

13. The type of fermentation characteristic of Enterobacteriaceae:
    A) Alcoholic
    B) Propionic acid
    C) Mixed (formic acid)
    D) Homofermentative lactic acid
    E) Methane-producing

14. The way energy is produced in fermentation:
    A) Oxidative phosphorylation
    B) Substrate-level phosphorylation
    C) Photophosphorylation
    D) Chemosynthesis
    E) Beta-oxidation

15. The reference with which biochemical results are compared to identify the species:
    A) Pharmacopoeia
    B) Bergey’s Manual
    C) International Classification of Diseases
    D) Sanitary norms
    E) State register

## Clinical cases

### *Case 1*

A pure culture isolated from stool on Kligler medium: the slant is red, the butt is yellow, there is gas, and black color appeared in the butt. It is motile, oxidase-negative and does not break down lactose.

**Question:** Which group should be considered and what is the next step?

**Answer:** A lactose-negative, $H_2S$-positive, motile gram-negative rod — **Salmonella** is highly likely. The next step is serological identification with diagnostic agglutinating sera (by O and H antigens).

### *Case 2*

Gram-positive cocci isolated from pus are catalase-positive and plasmacoagulase-positive.

**Question:** Which causative agent is this and what do these tests prove?

**Answer:** Catalase-positive — staphylococcus (not streptococcus); plasmacoagulase-positive — *Staphylococcus aureus*. Coagulase is the main pathogenicity and differential feature of this species.

### *Case 3*

A student read the Hiss series after 48 hours and wrote “lactose-positive,” but on serological testing the culture turned out to be Shigella.

**Question:** What is the error related to?

**Answer:** Reading the result later than the specified time. Some shigellae can break down lactose **late** (after 48 hours); the standard reading time is 18–24 hours. A result read late misleads identification.

## Quick review

**Table 7.5.** *Topic 7 — key concepts*

| Concept | Essence |
|---|---|
| Enzymes | Exo/endo; constitutive/inducible; aggression enzymes |
| Aggression enzymes | Plasmacoagulase, hyaluronidase, lecithinase, fibrinolysin |
| Pigments | *S. aureus* — golden; *P. aeruginosa* — pyocyanin; *Serratia* — prodigiosin |
| Hiss series | A — acid; AG — acid + gas |
| Kligler | Color of slant/butt, gas (cracks), $H_2S$ (black) |
| Catalase / oxidase | Staphylo(+)/strepto(−); *Pseudomonas*(+)/Enterobacteriaceae(−) |
| Coagulase | The main test that distinguishes *S. aureus* |
| Day 3 | Purity + biochemistry + serology + antibiogram |
| Forms of energy | ATP and the proton gradient (proton-motive force) |
| Types of fermentation | Lactic acid, alcoholic, butyric acid, mixed, propionic acid |

TOPIC 8

# Distribution of microbes in the environment. The microbiota of water, air and soil. Basics of sanitary microbiology

## 1. Learning objectives

In this class the student will master the following:

- To know the composition of the microbiota of water, air, soil and dwellings and its effect on human health;
- to explain the aim of sanitary microbiology, the concept of sanitary indicator microorganisms and the requirements for them;
- to master the methods of determining the total microbial count (TMC) and the coli index;
- Independent work: to know the principles of sanitary-bacteriological assessment of milk and meat products.

## 2. General concept of the ecology of microorganisms

Microorganisms are distributed in all layers of the biosphere. They play a decisive role in **biogeochemical cycles** (the cycles of nitrogen, carbon, sulfur and phosphorus), break down organic residues and maintain soil fertility. From the medical point of view, however, the external environment can be a **source of infection and a factor of transmission**.

Relationships of microorganisms in the external environment: **symbiosis** (mutualism), **commensalism**, **neutralism**, **antagonism** (one species inhibits another — the natural basis of antibiotics and bacteriocins), **parasitism**.

## 3. Soil microbiota

Soil is the environment richest in microorganisms (billions of cells in 1 g of fertile soil). They are mainly **saprophytes**: bacteria, actinomycetes, fungi, algae, protozoa. The most microbes are at a depth of 10–20 cm; at the surface they decrease under the action of UV rays and drying.

**Function:** mineralization of organic residues, nitrogen fixation (*Azotobacter*, nodule bacteria), nitrification and denitrification, cellulose breakdown.

**Medical significance.** Most pathogenic microorganisms die quickly in soil, but **spore forms** persist for a long time; for this reason soil is a natural reservoir for the causative agents of tetanus, gas gangrene and anthrax. The causative agents of intestinal infections enter the soil with feces and can persist for several days to weeks.

**Sanitary indicators:** the total number of microbes, the **coli titer** and **perfringens titer** (the smallest amount of soil in which *E. coli* and *C. perfringens* are found), the number of thermophilic bacteria. A small coli titer is a sign of fresh fecal contamination, a small perfringens titer — of old contamination.

## 4. Water microbiota

The composition of microbes in water depends on the source: **artesian waters** are almost sterile; **open water bodies** (river, lake) are contaminated from soil, precipitation and wastewater. Water has a

capacity for self-purification (sunlight, sedimentation, microbial antagonism, grazing by protozoa), but it is limited.

**Waterborne infections:** cholera, typhoid and paratyphoid fevers, dysentery, escherichiosis, leptospirosis, tularemia, hepatitis A and E, poliomyelitis, rotavirus gastroenteritis, giardiasis and amebiasis.

**Table 8.1.** *Main microbiological indicators of drinking water*

| Indicator | Meaning |
|---|---|
| TMC (total microbial count) | The number of mesophilic aerobic and facultative anaerobic microbes in 1 mL of water (CFU/mL) |
| **Coli index** | The number of E. coli (enteric group bacteria) in 1 **liter** of water |
| **Coli titer** | The smallest volume of water (mL) in which a single E. coli is found |
| Thermotolerant coliform bacteria | An indicator of fresh fecal contamination |
| Coli phage | An indicator of viral contamination |

> **REMEMBER**
> **The coli index and the coli titer are inversely related**: the larger the index, the smaller the titer, that is, the more polluted the water. Coli titer (mL) ≈ 1000 / coli index.

## 5. Air microbiota

Air is not a favorable environment for microbes: there are no nutrients, it dries out, and sunlight acts on it. Therefore the microbes in the air are temporary and float together with **dust particles and droplets**.

**Two kinds of aerosol:**

- **droplet (large-droplet) phase** — formed by coughing and sneezing, settles quickly; ensures transmission at close range;
- **droplet-nuclei and dust phase** — fine, stays suspended for a long time; resistant pathogens (tubercle bacillus, streptococcus, fungal spores) spread by this route.

**Airborne infections:** influenza and other acute respiratory viral infections, COVID-19, measles, chickenpox, whooping cough, diphtheria, meningococcal infection, tuberculosis, streptococcal tonsillitis.

**Sanitary indicators:** per 1 m³ of air, the **total microbial count**, and the number of **hemolytic streptococci** and **Staphylococcus aureus** (an indicator of contamination from the human respiratory tract). In enclosed rooms the indicators are standardized according to the summer–winter season and the density of people; the requirements are strictest for operating rooms and children’s institutions.

**Methods of taking an air sample:** sedimentation (Koch’s method — leaving an open Petri dish for a certain time; simplified, approximate) and **aspiration** (the Krotov apparatus and similar devices — blowing a certain volume of air onto the surface of the medium; quantitative and accurate).

## 6. Basics of sanitary microbiology

**Sanitary microbiology** is the branch that assesses the microbiological state of environmental objects (water, air, soil, food, objects, hands) and their danger to human health.

**Two main directions:**

1. Determining the **total microbial count** — the overall level of contamination of the object.
2. Determining **sanitary indicator microorganisms** — **indirect** evidence of contamination with human or animal excretions.

**Requirements for a sanitary indicator microorganism:**

- it must live permanently in the human (animal) body and be excreted with discharges;
- it must not multiply in the external environment, but survive for no shorter a time than pathogens;
- the method of detection must be simple and reliable;
- there must be no natural “native” representative of it in the external environment.

**Table 8.2.** *Main sanitary indicator microorganisms*

| Object | Indicator microorganism | What it indicates |
|---|---|---|
| Water | Enteric group bacteria (coliforms), *E. coli* | Fecal contamination |
| Water | Coli phages | Possible contamination with enteroviruses |
| Soil | *E. coli* (coli titer) | Fresh fecal contamination |
| Soil | *C. perfringens* (perfringens titer) | Old fecal contamination |
| Air | Hemolytic streptococcus, *S. aureus* | Contamination from the respiratory tract |
| Objects, hands | E. coli | Violation of the sanitary regime |

**CLINICAL SIGNIFICANCE**
In sanitary examination, looking for the pathogen itself takes a long time and is often ineffective (pathogens are present in small numbers). That is why an **indirect indicator** — coliforms — is used: where they are found, the conclusion is drawn that intestinal pathogens may also be present.

## 7. Independent work: Sanitary-bacteriological assessment of milk and meat products

**Milk.** After milking, milk contains natural microflora; the main cause of spoilage is the multiplication of lactic acid bacteria (souring). Sanitary indicators: **TMC (CFU/mL)**, coliforms (coli titer), somatic cell count, absence of pathogens (Salmonella, *S. aureus*, Listeria). The **reductase test** is a method for rapidly assessing the number of microbes in milk by the time it takes methylene blue or resazurin to decolorize: the faster the color disappears, the more microbes there are.

Infections that can be transmitted through milk: brucellosis, tuberculosis, salmonellosis, listeriosis, staphylococcal poisoning, campylobacteriosis. **Pasteurization** (for example, 72 °C for 15 s or 63 °C for 30 minutes) destroys vegetative pathogens.

**Meat.** The muscle tissue of a healthy animal is practically sterile; contamination occurs during slaughter, processing and storage. Indicators: TMC per 1 g, coliforms, sulfite-reducing clostridia, absence of Salmonella and Listeria in 25 g. In assessing freshness, organoleptic and chemical tests (ammonia, hydrogen sulfide reaction) are also used.

**REMEMBER**
The microbiological safety of food consists of three indicators: **total contamination (TMC)** + **sanitary indicator microbes (coliforms)** + **absence of pathogens in the specified mass**.

## 8. Practical skill: microbiological examination of water

### *7.1. Determining the total microbial count (pour plate method)*

1. The water sample is collected in a sterile container and examined within 2 hours (or within 6 hours if kept cooled).
2. 1 mL of water is placed in a sterile Petri dish.
3. 10–12 mL of melted meat-peptone agar cooled to 45 °C is poured over it and mixed by rotating the dish.
4. After the agar sets, the dish is inverted and incubated at 37 °C for 24 hours.
5. The colonies that have grown are counted — the result is expressed in **CFU/mL** (colony-forming units).

### *7.2. Membrane filter method (determining coliforms)*

1. A certain volume of water (usually 100–333 mL) is passed through a **membrane filter** — the bacteria are retained on the surface of the filter.
2. The filter is removed with sterile forceps and placed on the surface of **Endo medium**.
3. It is incubated at 37 °C for 24 hours.
4. The **red colonies with a metallic sheen** that have grown on the filter are counted as lactose-positive coliforms; they are confirmed with a smear and an oxidase test.
5. The coli index is calculated from the result.

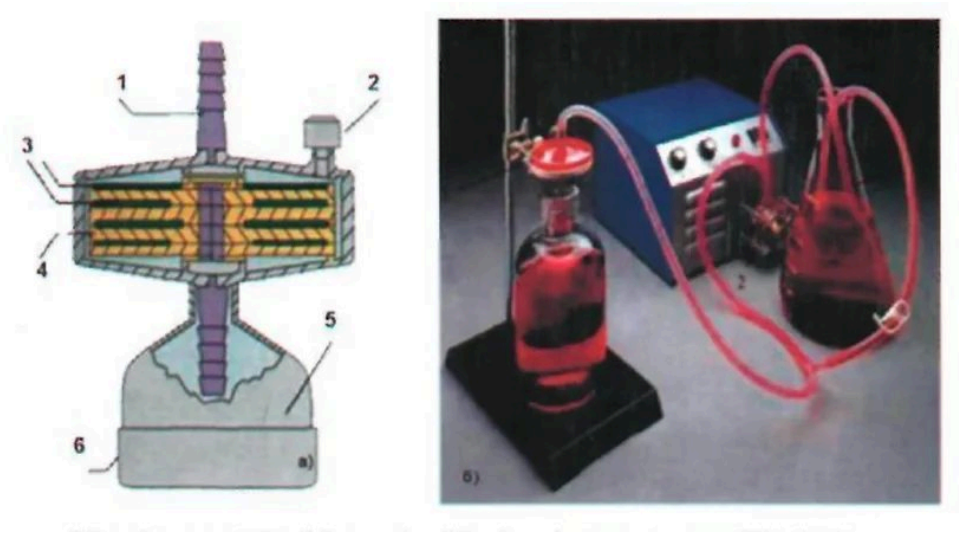



**Figure 8.1.** *The membrane filter apparatus used in the microbiological examination of water and how it is used. Source: Muhamedov I.M. et al., 2019.*

> **CAUTION**
> Pouring agar that is **hotter than 45 °C** kills the microbes, and the TMC comes out falsely low. Placing the plate upside down prevents condensation drops from “spreading” the colonies.

### *7.3. Titration (fermentation) method*

Different volumes of water are inoculated into lactose medium; from the tubes with gas formation, subcultures are made onto Endo medium, and the coli index is calculated from the confirmed results using a special table (most probable number).

## 9. Review questions

1. What types of relationships exist in the ecology of microorganisms?
2. What is the composition and medical significance of the soil microbiota?
3. What infections are transmitted through water? What is the self-purification of water based on?
4. What are the coli index and the coli titer, and how are they related to each other?
5. State the phases of aerosol in the air and their epidemiological significance.
6. What requirements are set for a sanitary indicator microorganism?
7. Which indicators distinguish fresh from old contamination in soil?
8. Describe the method of determining the total microbial count step by step.
9. What is the advantage of the membrane filter method?
10. Which indicators are considered in the sanitary-bacteriological assessment of milk and meat?

## Test questions

1. The pathogenic forms that persist longest in soil:
   A) Vegetative cocci
   B) Spore-forming bacteria
   C) Mycoplasmas
   D) Chlamydiae
   E) Spirochetes

2. The coli titer is:
   A) The number of E. coli in 1 liter of water
   B) The smallest volume of water in which a single E. coli is found
   C) The number of microbes in 1 mL of water
   D) The diameter of colonies
   E) The pH of the water

3. When the coli index increases, the coli titer:
   A) Increases
   B) Decreases
   C) Does not change
   D) Becomes zero
   E) Doubles

4. The microorganism that indicates old fecal contamination in soil:
   A) *E. coli*
   B) *C. perfringens*
   C) *S. aureus*
   D) *P. aeruginosa*
   E) *B. subtilis*

5. The indicator microorganisms determined in assessing the sanitary state of air:
   A) E. coli and enterococcus
   B) Hemolytic streptococcus and *Staphylococcus aureus*
   C) Clostridia
   D) Lactobacilli
   E) Bifidobacteria

6. The method used for quantitative sampling of air:
   A) Koch (sedimentation) method
   B) Aspiration method (Krotov apparatus)
   C) Membrane filter
   D) Titration method
   E) Reductase test

7. At what temperature is the agar poured in the microbiological examination of water:
   A) 100 °C
   B) 80 °C
   C) Cooled to 45 °C
   D) 20 °C
   E) 4 °C

8. In the membrane filter method, the filter is placed on the surface of which medium:
   A) Blood agar
   B) Endo medium
   C) Sabouraud
   D) Kligler
   E) Kitt–Tarozzi

9. The appearance typical of coliform colonies on Endo medium:
   A) Colorless, transparent
   B) Red, with a metallic sheen
   C) With a black center
   D) Blue-green
   E) Yellow

10. What CFU/mL stands for:
    A) Colony-forming unit (per 1 mL)
    B) Disease-causing unit
    C) Chemical temperature unit
    D) Coli titer unit
    E) Microbe volume unit

11. Not a requirement for a sanitary indicator microorganism:
    A) Living permanently in the human body
    B) Not multiplying in the external environment
    C) Persisting in the external environment for less time than pathogens
    D) Having a simple method of detection
    E) Being excreted with discharges

12. The rapid method for assessing the number of microbes in milk:
    A) Reductase test
    B) Oxidase test
    C) Coagulase test
    D) Kligler
    E) Ozheshko

13. What pasteurization destroys:
    A) All microbes, including spores
    B) Vegetative pathogenic microbes
    C) Only viruses
    D) Only fungi
    E) Only spores

14. An infection that can be transmitted through milk:
    A) Brucellosis
    B) Tetanus
    C) Malaria
    D) Epidemic typhus
    E) Rabies

15. An indirect indicator of viral contamination in water:
    A) Coli phages
    B) Lactobacilli
    C) Thermophilic bacteria
    D) Fungi
    E) Protozoa

## Clinical cases

### *Case 1*

A well-water sample was examined: the coli index was sharply elevated and thermotolerant coliforms were found. Cases of acute intestinal infection have increased in the village.

**Question:** What does the result mean and what measures are taken?

**Answer:** The water is contaminated with fresh feces; intestinal pathogens are likely to be present. Use of the water is stopped, the source of contamination is identified, the well is cleaned and disinfected, and use is permitted after repeat testing.

### *Case 2*

An operating-room air sample was examined: the total microbial count was above the norm and hemolytic streptococcus was found.

**Question:** What does this indicate?

**Answer:** The room air is contaminated with the microflora of the human respiratory tract (staff, ventilation, violation of the sanitary regime). The air exchange, disinfection and UV irradiation regimens are reviewed, and the staff are examined.

### *Case 3*

In the laboratory, agar was poured while still hot when determining the TMC of a water sample. The result was very low, but many coliforms were found by the membrane filter method.

**Question:** Why do the results contradict each other?

**Answer:** Hot agar killed the microbes, so the TMC came out falsely low. The agar should have been cooled to 45 °C. The result of the membrane filter method is reliable.

## Quick review

**Table 8.3.** *Topic 8 — key concepts*

| Concept | Essence |
|---|---|
| Soil | Spore-forming pathogens persist for a long time; coli titer (fresh), perfringens titer (old contamination) |
| Water | TMC, coli index, coli titer, coliphages; cholera, typhoid, hepatitis A |
| Air | Droplet and droplet-nuclei/dust phases; streptococcus and *S. aureus* as indicators |
| Sanitary indicator microbe | Permanent in humans, does not multiply in the environment, easy to detect |
| Determining TMC | 1 mL + 45 °C agar → CFU/mL |
| Membrane filter | Filter → Endo → red colonies with a metallic sheen |
| Coli index ↔ coli titer | Inversely related |
| Milk | TMC, coliforms, reductase test; pasteurization |
| Meat | TMC, coliforms, clostridia, absence of Salmonella/Listeria |

TOPIC 9

# Factors affecting microorganisms. Sterilization, disinfection, asepsis and antisepsis. Antibiotics and bacteriocins

## 1. Learning objectives

In this class the student will master the following:

- To know the effect of physical, chemical and biological factors on microorganisms;
- to distinguish the concepts of sterilization, disinfection, asepsis and antisepsis and to describe their methods;
- to explain the classification, mechanism of action and antimicrobial spectrum of antibiotics;
- to master the methods of determining susceptibility to antibiotics (disk diffusion, MIC) and the interpretation of the result;
- Independent work: to know the mechanisms by which resistance to antibiotics develops and how to give a bacteriological assessment of sterilization.

## 2. Physical factors

**Table 9.1.** *Effect of physical factors on microorganisms*

| Factor | Effect | Practical use |
|---|---|---|
| High temperature | Protein denaturation, loss of enzyme activity | Sterilization, pasteurization, boiling |
| Low temperature | Metabolism slows; most microbes are not killed | Preserving cultures, lyophilization |
| Drying | Dehydration, metabolism stops | Lyophilization (vaccines, culture preservation) |
| UV rays | Formation of thymine dimers in DNA | Disinfecting air and surfaces (bactericidal lamp) |
| Ionizing radiation | DNA breaks, free radicals | Industrial sterilization of single-use medical products |
| Ultrasound | Cell disruption (cavitation) | Obtaining cell lysate, cleaning |
| Osmotic pressure | Plasmolysis or plasmoptysis | Preserving (salt, sugar) |
| Filtration | Mechanical retention of microbes | Sterilizing heat-labile solutions (serum, antibiotics) |

**REMEMBER**

**Low temperature does not kill, it preserves.** That is why freezing is not a method of disinfection; on the contrary, it is used to store cultures and samples.

## 3. Sterilization and disinfection

**Sterilization** is the complete elimination of **all** microorganisms, including spores, from an object.

**Disinfection** is the elimination of pathogenic and opportunistic microbes (mainly vegetative forms) to reduce the risk of transmission; spores may survive.

**Table 9.2.** *Main sterilization methods*

| Method | Regimen | Used for |
|---|---|---|
| **Autoclave (steam under pressure)** | 121 °C / 15–20 minutes or 132 °C / shorter | Nutrient media, used materials, fabrics, rubber |
| **Dry-air oven** | 160–180 °C / 60–150 minutes | Glassware, metal instruments |
| Boiling | 100 °C | Disinfection level (unreliable against spores) |
| Tyndallization (fractional) | 56–100 °C, repeated over several days | Heat-labile media |
| Filtration | Membrane filters (0.22 µm) | Serum, antibiotic and vitamin solutions |
| Irradiation | Gamma rays, UV | Single-use items; surfaces |
| Gas method | Ethylene oxide, etc. | Optics, plastic instruments |

**Asepsis** is a system of measures to **prevent** microbes from entering a wound, tissue or an object that must be sterile (sterilized instruments, gloves, mask, laminar flow cabinet, preparation of the operative field).

**Antisepsis** is the elimination or reduction of microbes **already present** in a wound or living tissue (treatment of skin and mucous membranes).

**Groups of disinfectant and antiseptic substances:** halogens (chlorine and iodine preparations), alcohols (70% ethanol is the most effective), oxidizing agents (hydrogen peroxide, potassium permanganate), aldehydes (formaldehyde, glutaraldehyde), phenols, salts of heavy metals, surfactants (chlorhexidine, quaternary ammonium compounds), dyes (brilliant green).

> **CAUTION**
> **95% alcohol disinfects less effectively than 70% alcohol**: high-concentration alcohol immediately coagulates the protein on the cell surface, forms a crust and cannot penetrate inside.

## 4. Antibiotics

**Antibiotics** are substances produced by microorganisms, plants or animals, or obtained synthetically or semisynthetically, that stop the growth of other microorganisms or kill them.

**Classification:**

- **By origin:** natural (penicillin, streptomycin), semisynthetic (ampicillin, amoxicillin), synthetic (sulfonamides, fluoroquinolones).
- **By the nature of action: bactericidal** (kill — beta-lactams, aminoglycosides, fluoroquinolones, vancomycin) and **bacteriostatic** (stop multiplication — tetracyclines, macrolides, chloramphenicol (levomycetin), sulfonamides).
- **By spectrum of action:** narrow-spectrum (mainly Gram-positive or mainly Gram-negative) and broad-spectrum.

**Table 9.3.** *Groups of antibiotics by mechanism of action*

| Target | Mechanism | Groups |
|---|---|---|
| **Cell wall** | Stopping the synthesis of peptidoglycan (transpeptidase) | Beta-lactams (penicillins, cephalosporins, carbapenems), glycopeptides (vancomycin), bacitracin |

| Target | Mechanism | Groups |
|---|---|---|
| **Cytoplasmic membrane** | Disrupting permeability | Polymyxins, polyene antimycotics (amphotericin B, nystatin) |
| **Protein synthesis (30S)** | Binding to the 30S subunit | Aminoglycosides, tetracyclines |
| **Protein synthesis (50S)** | Binding to the 50S subunit | Macrolides, lincosamides, chloramphenicol (levomycetin) |
| **Nucleic acids** | Inhibition of DNA gyrase, RNA polymerase | Fluoroquinolones, rifampicin |
| **Folate metabolism** | Stopping the steps of folate synthesis | Sulfonamides, trimethoprim |

**MEMORY AID**

Remember the targets of protein synthesis inhibitors: **“30 — AT, 50 — CLM”**: **A**minoglycoside and **T**etracycline — **30S**; **C**hloramphenicol, **L**incosamide, **M**acrolide — **50S**.

**CLINICAL SIGNIFICANCE**

The selective action (selective toxicity) of an antibiotic is based on the difference between the bacterial and the human cell: humans have no peptidoglycan wall (which is why beta-lactams are relatively safe) and the ribosome is 80S (not 70S). For this reason, the groups acting on the wall are usually considered the least toxic.

## 5. Determining susceptibility to antibiotics

### *5.1. Disk diffusion method (Kirby–Bauer)*

1. A suspension of standard turbidity (0.5 McFarland) is prepared from a pure culture.
2. The suspension is spread evenly (in three directions) over the surface of **Mueller–Hinton agar** with a swab.
3. **Disks** impregnated with antibiotics are placed with forceps at specified distances.
4. It is incubated at 35–37 °C for 16–18 hours.
5. The **diameter of the zone of growth inhibition** around the disk is measured with a ruler and interpreted using a standard table: **S — susceptible**, **I — intermediate (susceptible at an increased dose)**, **R — resistant**.

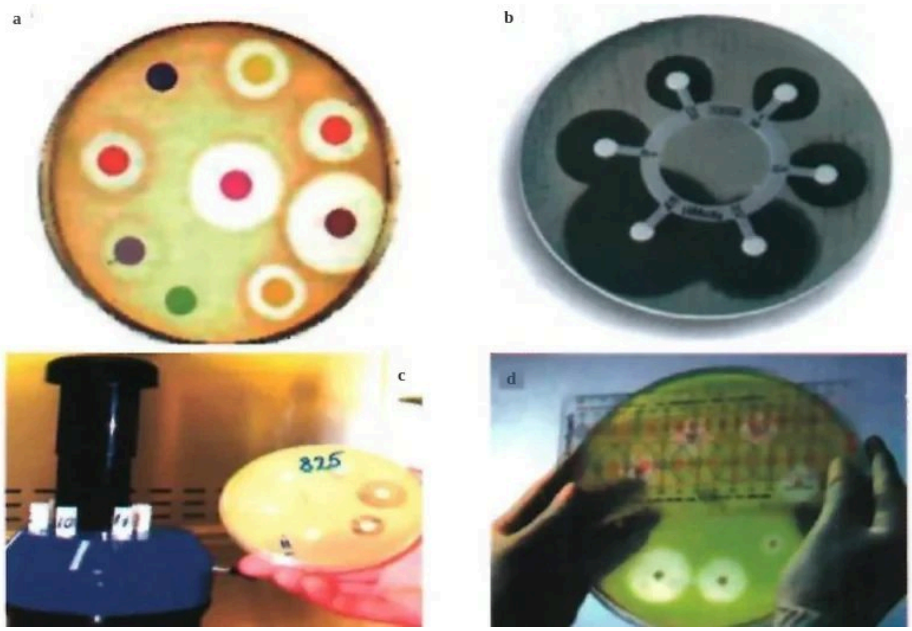



**Figure 9.1.** *Determining susceptibility to antibiotics: zones of growth inhibition around the disks in the disk diffusion method and reading of the result. Source: Muhamedov I.M. et al., 2019.*

> **CAUTION**
> The diameter of the zone by itself does not mean “good or bad”: every antibiotic and every species has its **own cutoff value**. That is why the result is interpreted only according to the current standard table.

### *5.2. Dilution method — MIC and MBC*

**MIC (minimum inhibitory concentration)** is the lowest concentration of an antibiotic that stops visible growth. **MBC (minimum bactericidal concentration)** is the lowest concentration that kills a specified proportion of the microbes. MIC is determined by serial dilution in liquid or solid medium; in practice the E-test (gradient strip) is also used.

**Clinical significance:** MIC helps in choosing the drug dose and route of administration; in severe infections (sepsis, endocarditis, meningitis) a quantitative MIC value is preferable to a qualitative “S/I/R” answer.

## 6. Bacteriocins

**Bacteriocins** are protein substances produced by bacteria that act against **closely related species**. They are named after the producing species: colicins (*E. coli*), staphylococcins, pyocins. They are encoded on plasmids.

**Significance:** a factor of competition (antagonism) between microbial populations; **bacteriocin typing** is a method of distinguishing strains in epidemiological investigation; a natural preservative in the food industry (nisin); being studied as new antibacterial agents.

## 7. Independent work: Mechanisms of resistance to antibiotics

**Types of resistance:**

- **Natural (innate)** — arises from the structure of the species and is present in all its representatives (mycoplasmas — to beta-lactams; the outer membrane of gram-negative bacteria — to many drugs).

- **Acquired** — arises as a result of mutation or gene transfer; this is the type that causes clinical problems.

**Genetic basis of acquired resistance:** chromosomal mutations and **horizontal gene transfer** — conjugation (R plasmids), transduction, transformation, transposons and integrons (topic 11).

**Table 9.4.** *Biochemical mechanisms of resistance*

| Mechanism | Meaning | Example |
|---|---|---|
| Enzymatic degradation | An enzyme that destroys the antibiotic | Beta-lactamases, extended-spectrum beta-lactamases, carbapenemases |
| Altering the target | The antibiotic cannot bind | An altered penicillin-binding protein in MRSA; ribosome methylation |
| Reduced permeability | The antibiotic cannot enter the cell | Loss of porin channels |
| Active efflux | A pump pushes the antibiotic out | Resistance to tetracyclines and fluoroquinolones |
| Switching the metabolic pathway | Synthesis via another pathway | Resistance to sulfonamides |

**Factors that lead to the spread of resistance:** unjustified prescription of antibiotics (in viral infections), not completing the course of treatment, wrong choice of dose, mass use in livestock farming, spread of resistant strains inside hospitals.

**Prevention:** prescribing antibiotics **based on an antibiogram**, completing the course, infection control and hand hygiene, antibiotic stewardship programs, monitoring of resistance.

**Bacteriological assessment of sterilization and disinfection.** The quality of sterilization is monitored with **biological indicators** (an ampoule or strip containing spores of a resistant test microorganism): after the cycle is finished, the indicator is cultured and there must be no growth. Physical (temperature, pressure, time record) and chemical (indicator strips) monitoring are also used in addition. The quality of disinfection is assessed by culturing wash samples taken from treated surfaces — no sanitary indicator microbes should be found.

> **REMEMBER**
> Only **biological monitoring** fully confirms the reliability of sterilization: physical and chemical indicators show that the regimen was carried out, but they do not directly prove the result.

### *Practical skill: setting up and reading the disk diffusion method*

1. From a 24-hour pure culture, prepare a suspension in sterile saline and adjust its turbidity to the **0.5 McFarland** standard.
2. Dip a sterile swab into the suspension and press out the excess liquid against the wall of the tube.
3. Spread the swab evenly over the surface of **Mueller–Hinton agar** in **three directions** (rotating the dish 60° each time); also cover the edges.
4. Wait 3–5 minutes until the surface is dry.
5. Place the antibiotic disks with sterile forceps at least 24 mm apart and press them lightly so that they stick to the surface.
6. Invert the dish and incubate at 35–37 °C for 16–18 hours.
7. Measure the diameter of the zone of growth inhibition from the bottom of the dish with a ruler in **mm**; take the whole diameter of the zone.

8. Compare the result with the current standard table, mark it as **S / I / R** and write it in the report.

> **CAUTION**
> If the suspension is denser than the standard, the zone becomes smaller (false resistance); if it is thinner, the zone becomes larger (false susceptibility). That is why adjusting the turbidity is the main condition for a reliable result.

## 8. Review questions

1. List the effects of physical factors on microbes. What is the mechanism of action of UV rays?
2. How do sterilization and disinfection differ?
3. State the regimens and uses of the autoclave and the dry-air oven.
4. What is the difference between asepsis and antisepsis?
5. Why is 70% alcohol more effective than 95% alcohol?
6. Into what groups are antibiotics divided by mechanism of action?
7. What is the difference between bactericidal and bacteriostatic action?
8. Describe the disk diffusion method step by step. How is the result interpreted?
9. What are MIC and MBC? When are they needed?
10. State the biochemical mechanisms of resistance to antibiotics.

## Test questions

1. The standard regimen of sterilization in an autoclave:
   A) 100 °C, 30 minutes
   B) 121 °C, 15–20 minutes
   C) 160 °C, 60 minutes
   D) 56 °C, 1 hour
   E) 80 °C, 10 minutes

2. Glass and metal instruments are sterilized by which method:
   A) Autoclave
   B) Dry-air oven (160–180 °C)
   C) Boiling
   D) Filtration
   E) Pasteurization

3. Heat-labile serum and antibiotic solutions are sterilized:
   A) In an autoclave
   B) In a dry-air oven
   C) By membrane filtration
   D) By boiling
   E) By UV light

4. The mechanism of action of UV rays on DNA:
   A) Formation of thymine dimers
   B) Protein denaturation
   C) Membrane rupture
   D) Change in pH
   E) Osmotic shock

5. How low temperature acts on microbes:
   A) Kills them immediately
   B) Slows metabolism, most are preserved
   C) Activates spores
   D) Causes mutation
   E) Has no effect

6. Asepsis is:
   A) Eliminating microbes in a wound
   B) A system of measures to prevent microbes from entering
   C) Eliminating all microbes
   D) Treatment with antibiotics
   E) Vaccination

7. The target of penicillin’s action:
   A) 30S ribosome
   B) 50S ribosome
   C) Cell wall (transpeptidase)
   D) DNA gyrase
   E) Cytoplasmic membrane

8. Antibiotics acting on the 30S ribosomal subunit:
   A) Macrolides and lincosamides
   B) Aminoglycosides and tetracyclines
   C) Beta-lactams
   D) Fluoroquinolones
   E) Polymyxins

9. The group that inhibits DNA gyrase:
   A) Fluoroquinolones
   B) Tetracyclines
   C) Cephalosporins
   D) Vancomycin
   E) Nystatin

10. The group with bacteriostatic action:
    A) Beta-lactams
    B) Aminoglycosides
    C) Tetracyclines
    D) Vancomycin
    E) Fluoroquinolones

11. The medium used in the disk diffusion method:
    A) Endo
    B) Mueller–Hinton agar
    C) Sabouraud
    D) Kligler
    E) Kitt–Tarozzi

12. MIC is:
    A) The lowest concentration that kills microbes
    B) The lowest concentration that stops visible growth
    C) The zone diameter
    D) The maximum value of the dose
    E) The amount of drug in the blood serum

13. The main mechanism of resistance in MRSA:
    A) Production of beta-lactamase
    B) Alteration of the target (penicillin-binding protein)
    C) Efflux pump
    D) Loss of porins
    E) Switching the metabolic pathway

14. The main property of bacteriocins:
    A) Broad-spectrum action
    B) A protein-based action against closely related species
    C) Antiviral action
    D) Antifungal action
    E) Action only on spores

15. The type of monitoring that fully confirms the quality of sterilization:
    A) Physical (temperature record)
    B) Chemical (indicator strip)
    C) Biological (spores of a test microorganism)
    D) Visual
    E) Time count

## Clinical cases

### *Case 1*

A patient has a purulent infection; the *S. aureus* isolated was found to be oxacillin-resistant in the antibiogram.

**Question:** What does this result mean and what is important in treatment?

**Answer:** A methicillin (oxacillin)-resistant staphylococcus — MRSA. The resistance is linked to an altered penicillin-binding protein, so **all beta-lactams** are considered ineffective. Treatment is carried out according to the antibiogram with drugs of another group; the patient is placed under infection control measures.

### *Case 2*

In a laboratory, after the autoclave cycle was finished, the color of the chemical indicator changed, but growth was detected when the biological indicator was cultured.

**Question:** What is the conclusion?

**Answer:** Sterilization was in fact not achieved. The chemical indicator only shows that the temperature reached the set value; the exposure time or the penetration of steam into the material may have been insufficient. The autoclave is sent for technical inspection and the material is re-sterilized.

### *Case 3*

For an acute respiratory viral infection, a doctor prescribed a broad-spectrum antibiotic to a patient; when the patient felt better, he stopped the drug on day 3.

**Question:** What two mistakes are there here, and what is the consequence?

**Answer:** First, an antibiotic in a viral infection is unjustified (it does not act on viruses). Second, the course was not completed. Both mistakes lead to the selection of resistant strains and to disruption of the normal microflora (dysbacteriosis).

## Quick review

**Table 9.5.** *Topic 9 — key concepts*

| Concept | Essence |
|---|---|
| Sterilization / disinfection | Everything (spores too) / mainly vegetative pathogens |
| Autoclave / dry oven | 121 °C 15–20 min / 160–180 °C |
| Asepsis / antisepsis | Preventing entry / eliminating what is already present |
| Alcohol | 70% — the most effective |
| Action on the wall | Beta-lactams, vancomycin, bacitracin |
| 30S / 50S | Aminoglycoside, tetracycline / macrolide, lincosamide, chloramphenicol |
| Nucleic acids | Fluoroquinolone (DNA gyrase), rifampicin (RNA polymerase) |
| Disk diffusion | Mueller–Hinton + 0.5 McFarland → zone → S/I/R |
| MIC / MBC | The lowest concentration that stops growth / kills |
| Mechanisms of resistance | Enzyme, target alteration, permeability, efflux, pathway switching |
| Bacteriocin | A protein against related species; for typing |

TOPIC 10

# The normal microflora of the human body. Dysbacteriosis. Probiotics

## 1. Learning objectives

In this class the student will master the following:

- To know the concept of the normal microflora (microbiota), its formation and its composition by organ;
- to explain the functions of the normal microflora for the body;
- to master the causes, degrees and laboratory diagnostic methods of dysbacteriosis (dysbiosis);
- Independent work: to have an understanding of probiotics, prebiotics and synbiotics.

## 2. General concept of the normal microflora

**The normal microflora (microbiota)** is the totality of microorganisms that permanently inhabit the organs and tissues of a healthy body that are in contact with the external environment. The number of microbial cells in the human body is comparable to the number of body cells; most of them are located in the large intestine.

**Two parts:**

- **Resident (permanent, obligate) microflora** — of stable composition, characteristic of the organ (bifidobacteria, lactobacilli, bacteroides, *E. coli*);
- **Transient microflora** — species that come from the external environment and do not persist for long; among them there are also opportunistic pathogens.

**Sterile sites (microbe-free in the norm):** blood, cerebrospinal fluid, internal organs, synovial fluid, the bladder and the upper parts of the urinary tract, the uterine cavity, the alveolar part of the lung.

> **REMEMBER**
> If a microorganism is isolated from a site that should normally be **sterile** (blood, cerebrospinal fluid), this is almost always a sign of **pathology** — even if the isolated species is “a member of the normal microflora.”

## 3. Microflora by organ

**Table 10.1.** *Main locations of the normal microflora*

| Organ / region | Main representatives | Note |
|---|---|---|
| Skin | *S. epidermidis*, micrococci, corynebacteria, *Propionibacterium acnes* | *S. aureus* carriage — often found in the nasal cavity |
| Oral cavity | Viridans streptococci, Veillonella, actinomycetes, lactobacilli | Dental plaque (biofilm) — plays a role in caries and periodontitis |
| Upper respiratory tract | Streptococci, Neisseria, corynebacteria | Carriage of pneumococcus and meningococcus is possible |
| Stomach | Very few (acid); *Helicobacter pylori* may be present | Acid is a natural barrier |
| Small intestine | Few in the upper part, increasing downward | Lactobacilli, enterococci |

| Organ / region | Main representatives | Note |
|---|---|---|
| **Large intestine** | **Bifidobacteria and bacteroides — the main mass**; lactobacilli, *E. coli*, enterococci, clostridia | Anaerobes outnumber aerobes by about 100–1000 times |
| Vagina | In the reproductive years — **lactobacilli (Döderlein bacilli)** | They produce lactic acid and keep the pH at 3.8–4.5 |
| The outer part of the urethra | Few cocci, corynebacteria | The upper parts are sterile |

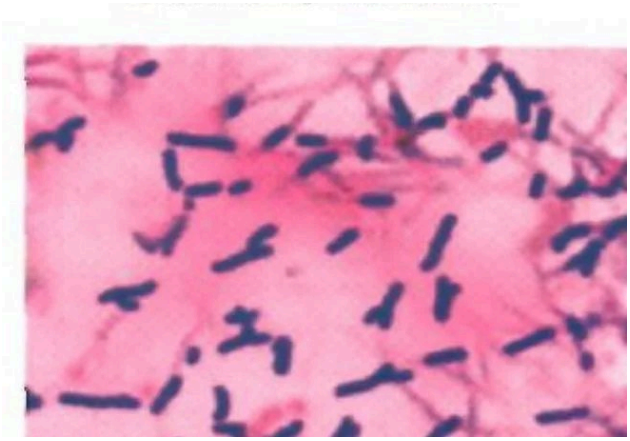


**Figure 10.1.** *Bifidobacteria — the main representatives of the normal microflora of the large intestine (Muhamedov I.M. et al., 2019)*



**Figure 10.2.** *Lactobacilli — the protective representatives of the intestinal and vaginal microflora (Muhamedov I.M. et al., 2019)*

**MEMORY AID**

In the large intestine **anaerobes predominate**: **BI-BA** — **Bi**fidobacteria and **Ba**cteroides. Although *E. coli* is mentioned often, it makes up **less than 1%** of the total mass.

## 4. Functions of the normal microflora

1. **Colonization resistance (protection)** — occupies the space and the nutrients, preventing pathogens from settling; produces bacteriocins, organic acids and hydrogen peroxide.
2. **Shaping immunity** — stimulates the gut-associated lymphoid tissue (GALT), supports the production of secretory IgA, provides “training” of the immune system.
3. **Metabolic and digestive function** — ferments undigested carbohydrates, forms short-chain fatty acids (an energy source for the intestinal epithelium), takes part in the metabolism of bile acids and cholesterol.
4. **Vitamin synthesis** — vitamin K, B vitamins ($B_{12}$, folate, biotin).
5. **Detoxification** — breaks down certain toxic substances and speeds up their excretion.
6. **Morphokinetic effect** — influences the normal development of the intestinal mucosa and peristalsis.

**CLINICAL SIGNIFICANCE**

When colonization resistance is disrupted (especially after a broad-spectrum antibiotic), the vacated place is taken by resistant or opportunistic species. Antibiotic-associated diarrhea and pseudomembranous colitis caused by *Clostridioides difficile* are the clinical expression of this condition.

## 5. Dysbacteriosis (dysbiosis)

**Dysbacteriosis** is a persistent disturbance in the qualitative and quantitative composition of the normal microflora: the resident representatives decrease, and opportunistic pathogens and unusual species increase. It is not an independent nosological entity but a **syndrome** — the cause must always be sought.

**Causes:** antibiotics and other drugs (chemotherapy, immunosuppressants), acute intestinal infections, chronic gastrointestinal diseases, improper nutrition, operations, immunodeficiency, stress and irradiation; in a newborn — disruption of the breastfeeding regimen.

**Table 10.2.** *Conventional degrees of intestinal dysbacteriosis*

| Degree | Laboratory picture | Clinical picture |
|---|---|---|
| I (latent) | Bifido-/lactobacilli slightly decreased, the amount of *E. coli* changed | Often asymptomatic |
| II | Anaerobes markedly decreased, opportunistic pathogens increased, lactose-negative and hemolytic *E. coli* have appeared | Flatulence, unstable stool |
| III | Opportunistic pathogens (Proteus, Klebsiella, staphylococcus, fungi) predominate | Persistent diarrhea, pain, decreased appetite |
| IV | Profound disturbance, risk of overgrowth and spread of opportunistic pathogens | Severe clinical picture, possibility of generalization |

**Laboratory diagnosis:**

1. **Classic bacteriological method** — serial dilution of stool (from $10^{-1}$ to $10^{-9}$) and culture on selective and differential media; for each group (bifidobacteria, lactobacilli, *E. coli* and its variants, enterococci, staphylococci, Proteus, Klebsiella, fungi) the **CFU/g** is calculated and compared with the norm.
2. **Gas-liquid chromatography** — assessing the state of the anaerobic microbiota from the spectrum of short-chain fatty acids in stool.
3. **Molecular methods** — PCR, sequencing of the 16S rRNA gene; they also cover species that cannot be cultured and show the composition of the microbiota more fully.
4. **Additional** — examination of biopsy material from the intestinal mucosa (mucosa-associated microflora).

> **CAUTION**
> A stool culture reflects only part of the **luminal** microflora of the intestine, and not all anaerobes can be cultured. That is why the result of a “dysbacteriosis analysis” is interpreted together with the clinical picture; prescribing treatment on the basis of the analysis alone is not correct.

## 6. Independent work: Probiotics, prebiotics and synbiotics

**Probiotics** are **live microorganisms** that, when taken in adequate amounts, have a beneficial effect on health. By composition:

- **monocomponent** — a single strain (bifidumbacterin, lactobacterin, colibacterin);
- **polycomponent (symbiotic)** — several strains (Bifiform and similar preparations);
- **self-eliminating antagonists** — species that do not multiply in the intestine but displace pathogens (preparations based on *Bacillus*, *Saccharomyces boulardii*);
- **sorbed** — strains immobilized on a sorbent.

**Mechanism of action:** restoring colonization resistance, producing organic acids and bacteriocins, strengthening the barrier function of the epithelium, stimulating local immunity (secretory IgA).

**Prebiotics** are substances that are not digestible but stimulate the growth of the beneficial microflora (lactulose, inulin, oligofructose).

**Synbiotics** are combined preparations of a probiotic and a prebiotic.

**Use:** prevention of antibiotic-associated diarrhea, recovery after acute intestinal infections, supporting the formation of the microbiota in newborns.

> **REMEMBER**
> The effect of a probiotic is **strain-specific**: a benefit proven for one strain is not automatically transferred to the whole species or to another preparation. In patients with severely reduced immunity, preparations of live microorganisms are used with caution.

## 7. Practical skill: algorithm for examining stool for dysbacteriosis

1. Stool is collected into a **sterile container**, before antibiotics are taken (or 2–3 days after stopping them); the sample is delivered to the laboratory within 2 hours.
2. 1 g of stool is mixed in 9 mL of sterile solution — a $10^{-1}$ dilution; serial dilutions are made in the same way up to $10^{-9}$.
3. From each dilution, the appropriate media are inoculated: Blaurock (bifidobacteria), MRS (lactobacilli), Endo/Levine (E. coli and enterobacteria), egg-yolk salt agar (staphylococci), enterococcus agar, Sabouraud (fungi), blood agar (hemolytic variants).
4. For anaerobes, culture is done under anaerobic conditions for 48–72 hours; aerobes for 24–48 hours, at 37 °C.
5. The colonies are counted and multiplied by the dilution factor — the result is given in **CFU/g**.
6. Identification is done (Gram stain, biochemical tests); hemolytic and lactose-negative variants are recorded separately.
7. The results are compared with age norms and a conclusion is given on the degree of dysbacteriosis.

## 8. Review questions

1. What is the normal microflora? What is the difference between resident and transient microflora?
2. Which sites of the body are sterile in the norm?
3. Who are the main representatives of the microflora of the large intestine? What is the share of *E. coli*?
4. Explain the protective mechanism of the vaginal microflora.
5. List the functions of the normal microflora.
6. What is colonization resistance and what happens when it is disrupted?
7. State the causes and degrees of dysbacteriosis.
8. How is the laboratory diagnosis of dysbacteriosis carried out? What are the limitations of the method?
9. What is the difference between a probiotic, a prebiotic and a synbiotic?
10. Explain the mechanism of action of probiotics.

## Test questions

1. The site that should be sterile in the norm:
   A) Large intestine
   B) Cerebrospinal fluid
   C) Skin
   D) Oral cavity
   E) Vagina

2. What makes up the main mass of the microflora of the large intestine:
   A) E. coli
   B) Bifidobacteria and bacteroides
   C) Staphylococci
   D) Fungi
   E) Proteus

3. The main representative of the vaginal microflora of a woman of reproductive age:
   A) Bifidobacteria
   B) Lactobacilli (Döderlein bacilli)
   C) Enterococci
   D) Corynebacteria
   E) Clostridia

4. The vaginal pH is kept at 3.8–4.5 owing to:
   A) Hydrogen sulfide
   B) Lactic acid
   C) Ammonia
   D) Urea
   E) Carbon dioxide

5. Colonization resistance is:
   A) Resistance to antibiotics
   B) The ability of the normal microflora to prevent pathogens from settling
   C) Phagocytosis
   D) Antibody formation
   E) Complement activity

6. Vitamins synthesized by the normal microflora:
   A) Vitamin K and B vitamins
   B) A and D
   C) C and E
   D) Only D
   E) Only A

7. The microbiota product that is the main energy source for the intestinal epithelium:
   A) Short-chain fatty acids
   B) Ammonia
   C) Indole
   D) Hydrogen sulfide
   E) Methane

8. The usual causative agent of pseudomembranous colitis that develops after a broad-spectrum antibiotic:
   A) *E. coli*
   B) *Clostridioides difficile*
   C) *S. epidermidis*
   D) *Lactobacillus*
   E) *Bifidobacterium*

9. Dysbacteriosis is:
   A) An independent infectious disease
   B) A persistent disturbance of the composition of the microflora (a syndrome)
   C) An allergic reaction
   D) A type of immunodeficiency
   E) An intoxication syndrome

10. How the result is expressed in the bacteriological examination for dysbacteriosis:
    A) CFU/g
    B) Titer
    C) Diameter (mm)
    D) Percentage
    E) mg/L

11. The medium used to culture bifidobacteria:
    A) Endo
    B) Blaurock
    C) Sabouraud
    D) Kligler
    E) Mueller–Hinton

12. The main limitation of a stool culture:
    A) It is too expensive
    B) It reflects only part of the luminal microflora; not all anaerobes can be cultured
    C) It detects only viruses
    D) The result is ready in 1 hour
    E) It is used only in children

13. A probiotic is:
    A) An indigestible nutrient for the beneficial microflora
    B) Live microorganisms that provide a benefit when given in adequate amounts
    C) An antibiotic
    D) A vaccine
    E) An immunoglobulin preparation

14. An example of a prebiotic:
    A) Lactulose
    B) Bifidumbacterin
    C) Lactobacterin
    D) Colibacterin
    E) Bacteriophage

15. The correct statement about the effect of a probiotic:
    A) All strains act in the same way
    B) The effect is strain-specific
    C) It works only in children
    D) It is never used together with an antibiotic
    E) It does not affect immunity

## Clinical cases

### *Case 1*

After a patient took a broad-spectrum antibiotic for 10 days, frequent loose stools and abdominal pain appeared. In the stool culture, bifidobacteria were sharply decreased, while hemolytic *E. coli* and fungi had increased.

**Question:** How do you assess the condition and what must be taken into account?

**Answer:** Antibiotic-associated dysbacteriosis. Colonization resistance is disrupted. In severe cases, colitis associated with *C. difficile* must be excluded (special testing). Treatment consists of eliminating the cause and, if necessary, agents that support the microbiota; the prescription is determined by the physician.

### *Case 2*

*Staphylococcus epidermidis* was isolated from a blood culture. The patient has a central venous catheter and a raised temperature.

**Question:** Is it correct to assess this result as “skin microflora, insignificant”?

**Answer:** No. Blood is sterile in the norm. Although *S. epidermidis* is a member of the normal skin microflora, when a catheter is present it can form a biofilm and cause a true catheter-related infection. It is assessed together with a repeat culture and clinical data.

### *Case 3*

On examination of a woman’s vaginal discharge, lactobacilli are almost absent, the pH is above 5.5, and various opportunistic anaerobes have increased.

**Question:** What does this picture mean?

**Answer:** Disruption of the vaginal microbiota (the picture of bacterial vaginosis). The decrease in lactobacilli stops the production of lactic acid, the pH rises and anaerobes multiply. The diagnosis is made on the basis of the clinical picture, pH, a smear (clue cells) and other criteria.

## Quick review

**Table 10.3.** *Topic 10 — key concepts*

| Concept | Essence |
|---|---|
| Resident / transient microflora | Stable, characteristic of the organ / from the external environment, temporary |
| Sterile sites | Blood, cerebrospinal fluid, internal organs, synovial fluid |
| Large intestine | Bifidobacteria + bacteroides predominate; *E. coli* — less than 1% |
| Vagina | Lactobacilli → lactic acid → pH 3.8–4.5 |

| Concept | Essence |
|---|---|
| Functions | Colonization resistance, immunity, metabolism, vitamin synthesis, detoxification |
| Dysbacteriosis | A syndrome, 4 degrees; the cause must be sought |
| Diagnosis | Serial dilution culture → CFU/g; GLC; PCR/16S rRNA |
| Probiotic / prebiotic / synbiotic | Live microbe / nutrient substrate / both together |

TOPIC 11

# The doctrine of infection. Pathogenicity and virulence. Genetics and variability of microorganisms

## 1. Learning objectives

In this class the student will master the following:

- To distinguish the concepts of infection, infectious process and infectious disease;
- to know the factors of pathogenicity and virulence and the types of toxins;
- to explain the stages, forms and routes of transmission of the infectious process;
- to know systematically the main methods of diagnosing infectious diseases;
- to master the genetics of bacteria, the types of variability and the ways genetic material is transferred;
- Independent work: to know the significance of plasmids, transposons, IS elements and temperate phages in medicine.

## 2. Infection and the infectious process

**Infection (infectious process)** is the totality of interactions between a microorganism and a macroorganism under certain environmental conditions. **Infectious disease** is the form of the infectious process that is manifested by clinical signs; that is, not every infection ends in disease.

**Table 11.1.** *Forms of the infectious process*

| Form | Description |
|---|---|
| Manifest (clinically expressed) | Acute, chronic, recurrent; typical or atypical course |
| Hidden (inapparent, subclinical) | No clinical signs, but an immunological shift is present |
| Microbe carriage | Healthy, convalescent (after recovery), acute or chronic |
| Persistence | Long-term retention of the pathogen in the body |
| Latent infection | The pathogen “sleeps” and is activated when conditions change (herpes, tuberculosis) |
| Slow infection | Long incubation, gradual progression (prion diseases) |

**By origin:** exogenous (from outside) and **endogenous** (from the body’s own opportunistic microflora, when immunity falls). By the number of pathogens: monoinfection and **mixed infection**. By repeat infection: **reinfection** (infection again with the same species after recovery), **superinfection** (infection again without having recovered), **relapse** (recurrence due to the pathogen that remained in the body).

**Periods of the disease:** incubation → prodromal (initial, vague signs) → peak (typical clinical picture) → recovery (convalescence). For each period, **the appropriate material and method** are chosen — this is the main rule of laboratory diagnosis.

## 3. Pathogenicity and virulence

**Pathogenicity** is the ability of a species to cause disease; it is a **species trait**, genetically fixed, a qualitative concept.

**Virulence** is the **strain-specific quantitative** expression of pathogenicity ("the power to cause disease"); the units of measurement are DLM (minimum lethal dose), $LD_{50}$ (the dose that kills 50% of animals) and $ID_{50}$ (infectious dose).

**Table 11.2.** *Factors of pathogenicity (virulence)*

| Group | Factors | Task |
|---|---|---|
| **Adhesion** | Pili, adhesins, teichoic acids | Attachment to the epithelium — the first stage of infection |
| **Colonization** | Enzymes, biofilm formation | Multiplication and settling |
| **Invasion** | Hyaluronidase, neuraminidase, collagenase, invasins | Penetration into tissue and spread |
| **Aggression (protection)** | Capsule, plasmacoagulase, catalase, protein A, IgA protease | Protection from phagocytosis and immune factors |
| **Toxigenicity** | Exotoxins, endotoxin | Tissue damage and general intoxication |

### *3.1. Exotoxin and endotoxin*

**Table 11.3.** *Comparison of exotoxin and endotoxin*

| Feature | Exotoxin | Endotoxin (LPS) |
|---|---|---|
| Chemical nature | Protein | Lipopolysaccharide (the active part is lipid A) |
| Source | Secreted by Gram-positive and Gram-negative bacteria | Only the Gram-negative wall; released when the cell breaks down |
| Potency | Very strong | Relatively weak |
| Specificity of action | High (the target organ/tissue is clear) | Low, the same type of response |
| Relation to heat | Most are unstable (thermolabile) | Resistant (thermostable) |
| Under the action of formalin | Turns into a **toxoid** | Does not |
| Immunogenicity | Strong, antitoxic immunity forms | Weak |
| Clinical expression | A species-specific syndrome (tetanus, diphtheria, botulism) | Fever, drop in blood pressure, DIC, septic shock |

**Types of exotoxins by mechanism of action:** cytotoxins (stop protein synthesis — diphtheria, Shigella), membrane-damaging toxins (hemolysins, lecithinase), functional blockers (neurotoxins — tetanus, botulism; enterotoxins — cholera), exfoliatins and superantigens (toxic shock syndrome toxin).

> **REMEMBER**
> **Toxoid (anatoxin)** is an exotoxin that has lost its toxicity under the action of formalin and heat but has retained its immunogenicity. It is the basis of the vaccines against tetanus and diphtheria.

## 4. Mechanisms and routes of transmission

**Table 11.4.** *Mechanism of transmission, routes and examples*

| Mechanism | Routes | Diseases |
|---|---|---|
| Fecal-oral | Water, food, contact-household | Cholera, typhoid, dysentery, hepatitis A, poliomyelitis |
| Airborne | Airborne droplet, airborne dust | Influenza, diphtheria, whooping cough, tuberculosis, meningococcal infection |
| Contact | Direct, indirect (objects), sexual | Syphilis, gonorrhea, scabies, skin infections |
| Vector-borne | Through bites of arthropods | Malaria, typhus, Lyme disease, plague |
| Vertical | Transplacental, birth canal, milk | Toxoplasmosis, rubella, HIV, hepatitis B |
| Artificial | Parenteral, medical manipulations | Hepatitis B and C, HIV |

## 5. Methods of diagnosing infectious diseases

**Table 11.5.** *Main diagnostic methods*

| Method | Meaning | When used | Limitations |
|---|---|---|---|
| **Microscopic (bacterioscopic)** | A stained smear from the material | When a rapid answer is needed; if the pathogen has typical morphology | Low sensitivity, often presumptive |
| **Bacteriological (culture)** | Isolation of a pure culture, identification | The “gold standard”; essential for an antibiogram | 3–4 days; ineffective after antibiotics are started |
| **Serological** | Detecting antibody (or antigen) in the blood serum | From week 2 of the illness; retrospective diagnosis | Negative in the early period; paired sera are needed |
| **Molecular (PCR)** | Detecting the nucleic acid of the pathogen | Non-growing and slow-growing pathogens; rapid answer | Does not distinguish live from dead |
| **Allergological** | Skin allergy tests | Tuberculosis, brucellosis, tularemia | Shows exposure to infection, not disease |
| **Biological test** | Infection of a laboratory animal | Determining toxigenicity, isolation | Time, ethical and technical limitations |

**CAUTION**

**Paired sera rule:** in serological diagnosis, serum is taken at the beginning of the illness and after 10–14 days. For diagnosis, a **rise in titer of at least 4-fold** is significant; a positive result in a single serum can often also indicate a past infection or vaccination.

## 6. Genetics of bacteria

**Genetic apparatus:** a circular **chromosome** (nucleoid) and extrachromosomal elements — **plasmids**, **transposons**, **IS elements**, and the genome of a temperate phage (prophage).

**Genotype** is the totality of genes; **phenotype** is the expression of the genotype under particular conditions.

**Table 11.6.** *Types of variability*

| Type | Mechanism | Heritable? | Example |
|---|---|---|---|
| **Modification (phenotypic)** | The genotype does not change, the expression changes | Absent | Formation of a capsule or pigment depending on the composition of the medium |
| **Mutational** | Change in the DNA sequence | Yes | Antibiotic resistance mutation |
| **Recombinational** | Acquiring a gene from another cell | Yes | Transfer of an R plasmid |

**Mutations:** spontaneous (natural, low frequency) and induced (under the action of mutagens — UV light, chemicals); point (a single nucleotide) and genomic (large rearrangements). Under conditions of **selection** (for example, antibiotic pressure), mutations come to predominate in the population.

### *6.1. Ways of transfer of genetic material (recombination)*

1. **Transformation** — a bacterium takes up **free DNA** from the environment. For this the cell must be in a special state (competence). Historical significance: the experiments of Griffith (1928) and Avery proved that DNA is the hereditary material.
2. **Transduction** — the transfer of genes by a **bacteriophage**. In **general (generalized)** transduction any piece of the chromosome is transferred; in **specialized** transduction, particular genes next to the prophage are transferred.
3. **Conjugation** — two cells in direct contact transfer DNA (often a plasmid) through a **sex (F) pilus**. The donor is an $F^+$ or Hfr cell, the recipient is $F^-$.

**MEMORY AID**
Remember the three routes by their source: **Trans-FOR-mation** — **free DNA** (from the environment); **Trans-DUC-tion** — a **phage** carries it; **Conjugation** — **contact** (through a pilus).

**CLINICAL SIGNIFICANCE**
Horizontal gene transfer is the main reason for the **rapid spread of antibiotic resistance among species**. That is why a resistant strain that has arisen in one ward can pass the resistance gene to other species and become a problem throughout the hospital.

## 7. Independent work: Plasmids, transposons, IS elements and temperate phages

**Plasmids** are extrachromosomal, independently replicating circular DNA. Properties: they are not essential for the life of the cell but give it additional capabilities; they can pass to another cell by conjugation; they can also be lost (elimination).

**Table 11.7.** *Plasmids of significance in medicine*

| Plasmid | Encoded property | Significance |
|---|---|---|
| **R plasmid** | Resistance to antibiotics (often to several) | Spread of multidrug resistance |
| **F plasmid** | Sex pilus, donor ability | Conjugation |
| **Ent, Hly plasmids** | Enterotoxin, hemolysin | Virulence |
| Col plasmids | Colicins (bacteriocins) | Antagonism, typing |

| Plasmid | Encoded property | Significance |
|---|---|---|
| Biodegradation plasmids | Enzymes that break down substances | Ecological significance |

**IS elements (insertion sequences)** are the simplest transposable elements, containing only the gene required for their own transposition. They move from place to place in the genome and can inactivate genes or, conversely, activate them.

**Transposons**, unlike IS elements, carry **additional genes** besides the transposition genes (often an antibiotic resistance gene or a toxin gene). They move between the chromosome and plasmids and speed up the spread of resistance genes. **Integrons** are systems that "collect" resistance gene cassettes and play an important role in the development of multidrug resistance.

**Temperate phages and lysogeny.** A temperate phage does not lyse the bacterium immediately but integrates its genome into the chromosome and is passed from generation to generation as a **prophage** (a **lysogenic culture**). If the prophage genes give the bacterium a new property, this is called **phage conversion**.

**The most important clinical examples:** toxin production by the diphtheria agent is controlled by a prophage gene — **only lysogenic strains are toxigenic**; the toxin genes of the cholera vibrio and of some *Shigella*/*E. coli* strains are also phage-associated.

> **REMEMBER**
> The toxigenicity of the diphtheria bacillus is a classic example of **phage conversion**. That is why, in the diagnosis of diphtheria, it is essential to **determine the toxigenicity** of the isolated culture separately (topic 18).

### *Practical skill: choosing the method and material according to the stage of the disease*

1. **Determine the stage of the disease:** incubation, prodromal, peak or recovery. At each stage the pathogen is in different places in the body.
2. **In the early stage**, look for the pathogen itself: take the material **before antibiotics** and choose a bacteriological or molecular method.
3. **From week 2**, assess the body's response: for the serological method take **paired sera** (at the beginning of the illness and after 10–14 days).
4. **Choose the material:** according to where the pathogen is multiplying — blood (bacteremia), stool (intestine), sputum (respiratory tract), cerebrospinal fluid (CNS), urine, wound discharge.
5. **Collect and transport the sample correctly:** a sterile container, a transport medium if necessary; for fragile pathogens (meningococcus, gonococcus) — warm and fast.
6. **Fill in the referral:** clinical diagnosis, date of onset of the illness, whether antibiotics have been taken, vaccination history.
7. **Interpret the result:** not a single positive serological result but the **dynamics of the titer**; and with PCR, take into account that live and dead microbes are not distinguished.

> **CAUTION**
> There is no single universal method: the method is matched to **the stage of the disease**. Material taken at the wrong time gives a false-negative result even in the best laboratory.

## 8. Review questions

1. How do infection, infectious process and infectious disease differ?

2. List the forms of the infectious process. What is the difference between reinfection, superinfection and relapse?
3. How do pathogenicity and virulence differ? How is virulence measured?
4. Divide the factors of pathogenicity into groups and explain them.
5. Compare exotoxin and endotoxin. How is a toxoid obtained?
6. State the mechanisms and routes of transmission with examples.
7. State the strengths and weaknesses of each diagnostic method. What is the paired sera rule?
8. State the types of variability. Why is modification variability not inherited?
9. Explain the difference between transformation, transduction and conjugation.
10. What is the difference between a plasmid, a transposon and an IS element? What are lysogeny and phage conversion?

## Test questions

1. Pathogenicity is:
   A) A quantitative trait of a strain
   B) The ability of a species to cause disease (a species trait)
   C) The amount of toxin
   D) Resistance to antibiotics
   E) The growth rate

2. The unit of measurement of virulence:
   A) CFU/mL
   B) $LD_{50}$
   C) MIC
   D) Titer
   E) pH

3. The chemical nature of endotoxin:
   A) Protein
   B) Lipopolysaccharide (lipid A)
   C) Polysaccharide
   D) Nucleic acid
   E) Lipid

4. It turns into a toxoid under the action of formalin:
   A) Endotoxin
   B) Exotoxin
   C) LPS
   D) Peptidoglycan
   E) Teichoic acid

5. A clinical manifestation characteristic of endotoxin:
   A) Neuromuscular paralysis
   B) Fever, drop in blood pressure, septic shock
   C) Selective inhibition of protein synthesis
   D) Exfoliation
   E) Selective action on the intestinal epithelium

6. Repeat infection with the same pathogen before recovery:
   A) Relapse
   B) Reinfection
   C) Superinfection
   D) Persistence
   E) Latent infection

7. An example of the vector-borne mechanism of transmission:
   A) Cholera
   B) Malaria
   C) Influenza
   D) Tetanus
   E) Hepatitis B

8. The sign of diagnostic significance in serological diagnosis:
   A) Finding antibody in a single serum
   B) A rise in titer of at least 4-fold in paired sera
   C) Complete absence of antibodies
   D) Leukocytosis
   E) Raised ESR

9. A method that gives a result even after antibiotics are started:
   A) Bacteriological culture
   B) PCR
   C) Biological test
   D) Allergological test
   E) Microscopy

10. Only the expression of a trait changes, with no change in the genotype:
    A) Mutational variability
    B) Modification variability
    C) Recombination
    D) Transduction
    E) Conjugation

11. A bacterium taking up free DNA from the environment:
    A) Transformation
    B) Transduction
    C) Conjugation
    D) Lysogeny
    E) Translation

12. The transfer of genes by a bacteriophage:
    A) Transformation
    B) Transduction
    C) Conjugation
    D) Mutation
    E) Modification

13. The plasmid that encodes resistance to antibiotics:
    A) F plasmid
    B) R plasmid
    C) Col plasmid
    D) Ent plasmid
    E) Hly plasmid

14. Unlike an IS element, a transposon:
    A) Does not transpose
    B) Carries additional genes (for example, a resistance gene)
    C) Occurs only in viruses
    D) Is protein in nature
    E) Cannot be located in the chromosome

15. What the toxigenicity of the diphtheria bacillus depends on:
    A) A plasmid
    B) Prophage genes (phage conversion)
    C) The capsule
    D) The flagellum
    E) Volutin granules

## Clinical cases

### *Case 1*

A patient has fever, a sharp drop in blood pressure and signs of DIC. A gram-negative rod grew in the blood culture. After antibiotics were started, the condition at first worsened.

**Question:** What is this picture related to?

**Answer:** The effect of **endotoxin (LPS)** in the wall of gram-negative bacteria — a picture of septic shock. When bacteria break down under the action of the antibiotic, a large amount of endotoxin is released, and the condition may temporarily worsen. Treatment is carried out in intensive care.

### *Case 2*

A culture resembling the diphtheria bacillus, isolated from the throat, matched morphologically and biochemically, but the toxigenicity test was negative.

**Question:** How should the result be interpreted?

**Answer:** The culture may be a **nontoxigenic** strain: toxin production depends on prophage genes (phage conversion). A nontoxigenic strain does not produce the typical clinical picture of diphtheria. That is why the diagnosis is made together with determining toxigenicity.

### *Case 3*

In an intensive care unit, various gram-negative bacteria isolated from several patients proved resistant to the same group of antibiotics.

**Question:** How is such a picture explained?

**Answer:** The resistance genes may have spread among species by **horizontal gene transfer** (R plasmids, transposons, integrons). Measures: infection control, hand hygiene, rational use of antibiotics and monitoring of resistance.

## Quick review

**Table 11.8.** *Topic 11 — key concepts*

| Concept | Essence |
|---|---|
| Infection ↔ disease | Not every infection ends in disease |
| Pathogenicity / virulence | Species trait (qualitative) / strain trait (quantitative, $LD_{50}$) |
| Factors of pathogenicity | Adhesion, colonization, invasion, aggression, toxigenicity |
| Exotoxin | Protein, strong, specific, gives a toxoid |
| Endotoxin | LPS, thermostable, fever and shock |
| Transmission | Fecal-oral, airborne, contact, vector-borne, vertical, artificial |
| Diagnosis | Microscopic, bacteriological, serological, PCR, allergological, biological |
| Paired sera | A rise in titer of ≥4-fold is diagnostic |
| Recombination | Transformation (free DNA), transduction (phage), conjugation (pilus) |
| Plasmids | R (resistance), F (conjugation), Col, Ent/Hly |
| Phage conversion | A prophage gives a new property — diphtheria toxin |

TOPIC 12

# General virology: the structure, reproduction and classification of viruses. Bacteriophages. Isolation, indication and identification of viruses

## 1. Learning objectives

In this class the student will master the following:

- To know the structure, chemical composition and principles of classification of viruses;
- to explain the stages of virus reproduction and the types of virus–cell relationships;
- to master the methods of culturing viruses (cell culture, chick embryo, laboratory animals), indication and identification;
- to know the structure, life cycle and practical uses of bacteriophages;
- Independent work: the use of bacteriophages in treatment, identification and prophylaxis.

## 2. The structure and chemical composition of viruses

**Viruses** are acellular forms of life; they are **obligate intracellular parasites**. Their main distinguishing features:

- they contain only **one** type of nucleic acid — **DNA or RNA** (not both together);
- they have no protein-synthesizing system (ribosomes) or energy-producing system of their own — they use those of the host cell;
- they multiply not by binary fission but by the **disjunctive (separated)** method: the components are synthesized separately and then assembled;
- they pass through bacterial filters;
- they are not susceptible to antibiotics.

**Virion** is the mature, extracellular virus particle. Structure:

- **Nucleoid (genome)** — DNA or RNA; single-stranded or double-stranded, linear or circular, whole or segmented.
- **Capsid** — the protein shell, built of **capsomers**; it protects the genome and takes part in attachment to the cell and in antigenic properties.
- **Supercapsid (outer envelope)** — present in some viruses; it is formed from the host cell membrane and equipped with viral glycoproteins (spikes).

**Types of symmetry: helical** (the nucleocapsid is coiled in a spiral — influenza, rabies), **cubic (icosahedral)** (adenovirus, poliovirus, herpes), **complex (mixed)** (smallpox virus, most bacteriophages).

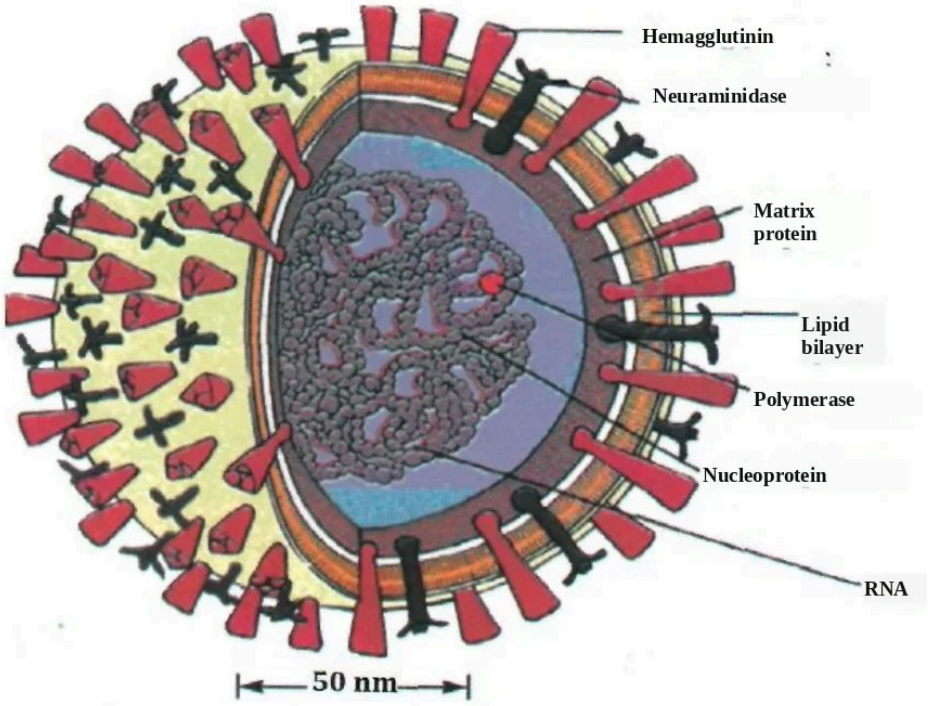



**Figure 12.1.** *The structure of the influenza virus: supercapsid, hemagglutinin and neuraminidase spikes, matrix protein, nucleoprotein with segmented RNA. Source: Muhamedov I.M. et al., 2019.*

**REMEMBER**

**Enveloped (supercapsid)** viruses are unstable in the external environment: they are sensitive to lipid solvents, alcohol and soap. **Non-enveloped** viruses are much more resistant and adapted to transmission by the fecal-oral route (enteroviruses, hepatitis A, rotavirus).

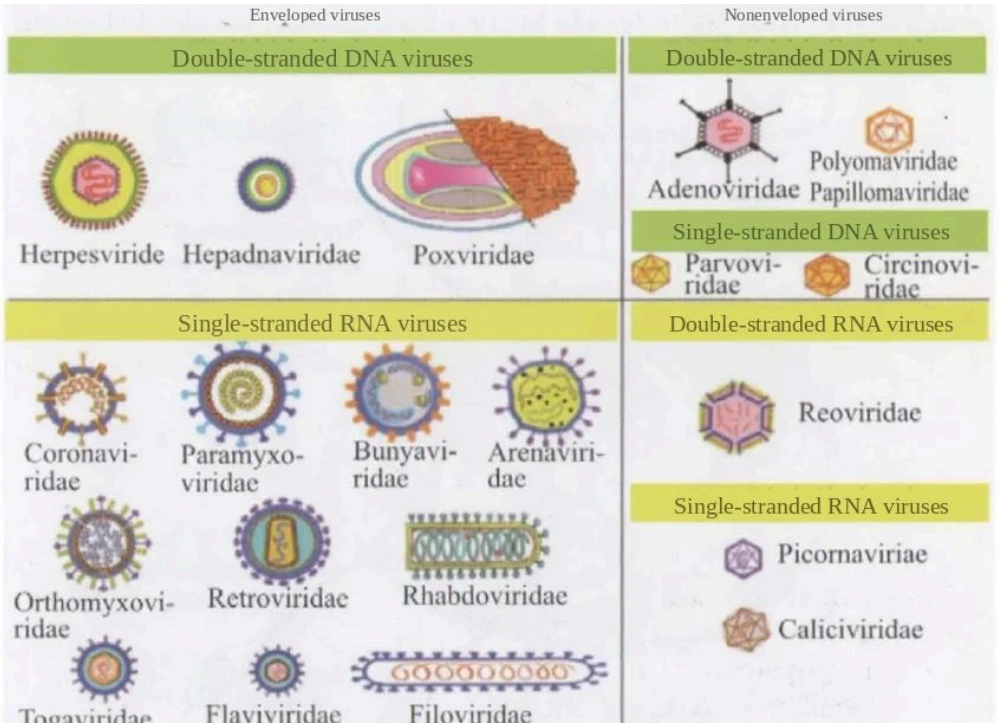



**Figure 12.2.** *The main virus families that cause disease in humans: classification into DNA and RNA viruses and enveloped and non-enveloped groups. Source: Aliyev Sh.R. et al., 2013.*

**Principles of classification:** type of nucleic acid (DNA/RNA), strandedness (single/double), type of symmetry, presence of an envelope, virion size, site of replication (nucleus or cytoplasm), antigenic

properties.

## 3. The interaction of a virus with the cell

**Three kinds of outcome:**

1. **Productive (lytic)** — the full cycle is completed, new virions form, and the cell often dies.
2. **Abortive** — the cycle stalls (the cell is unsuitable or the virus is defective); no new virions form.
3. **Integrative (virogeny)** — the viral genome integrates into the cell chromosome and replicates with it (retroviruses, hepatitis B, some oncoviruses).

### *Stages of the productive cycle*

1. **Adsorption** — binding of the viral spikes to **specific receptors** on the cell surface. This very stage determines the tissue and species specificity (tropism) of the virus.
2. **Penetration (entry)** — endocytosis through the receptor, or fusion of the envelope with the cell membrane.
3. **Uncoating (“undressing”)** — the genome is freed from the envelope and capsid and exposed.
4. **Eclipse (hidden) period and synthesis** — the viral genome and proteins are synthesized in the host cell’s systems; during this time no whole virion can be found in the cell.
5. **Assembly (morphogenesis)** — the nucleocapsid is assembled from the components.
6. **Release** — by cell lysis (non-enveloped viruses) or by **budding** (enveloped viruses take their envelope from the membrane).

> **MEMORY AID**
> Remember the stages in order: **A–P–U–S–A–R**: **A**dsorption → **P**enetration → **U**ncoating → **S**ynthesis → **A**ssembly → **R**elease.

## 4. Methods of culturing viruses

Viruses do not grow in artificial nutrient media, so **living systems** are needed:

**Table 12.1.** *Systems for culturing viruses*

| System | Use | Methods of indication (detecting presence) |
|---|---|---|
| **Cell culture** | The main modern method | Cytopathic effect (CPE), plaques, hemadsorption, color test, immunofluorescence |
| **Chick embryo** | Influenza, smallpox, some other viruses | Embryo death, plaques on the chorioallantoic membrane, hemagglutination test |
| **Laboratory animals** | Rabies, some neurotropic viruses | Signs of disease, paralysis, pathomorphological changes |

**Types of cell cultures:** primary (directly from tissue, limited lifespan), semi-continuous (diploid strains) and **continuous (permanent) lines** (unlimited growth — HeLa, Vero and others).

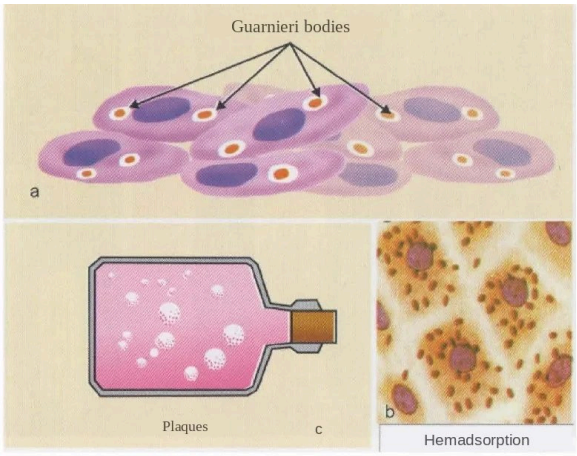



**Figure 12.3.** *Cytopathic effect of viruses on cell culture: a) perinuclear inclusions; b) hemadsorption reaction; c) virus colonies (plaques). Source: Aliyev Sh.R. et al., 2013.*

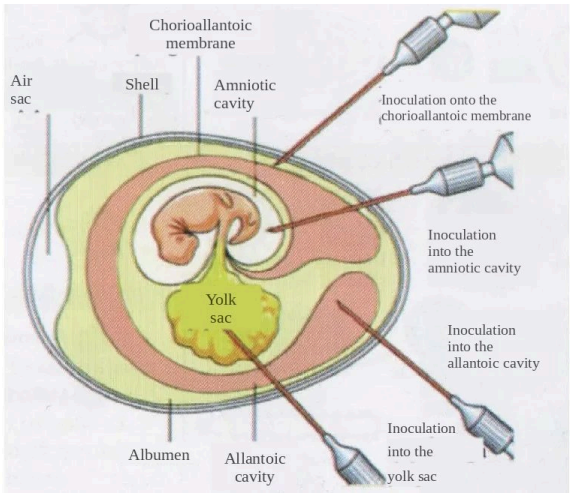



**Figure 12.4.** *The structure of a chick embryo (8–10 days old) and the routes of infecting viruses: onto the chorioallantoic membrane, into the amniotic and allantoic cavities, into the yolk sac. Source: Aliyev Sh.R. et al., 2013.*

**Methods of detecting the presence of a virus (indication):**

- **Cytopathic effect (CPE)** — rounding of cells, formation of syncytia (multinucleated cells), appearance of inclusions, breakdown of the cell layer;
- **Intracellular inclusions** — characteristic of certain viruses (**Babes–Negri** bodies in rabies, **Guarnieri** bodies in smallpox);
- **Hemadsorption** — adhesion of erythrocytes to the surface of infected cells (influenza, parainfluenza);
- **Hemagglutination (HA)** — fluid containing the virus causes erythrocytes to agglutinate;
- **Plaque method** — counting virus colonies under an agar overlay (quantitative assessment).

**Identification** — determining the type of virus — is done with **specific immune sera** in serological reactions: the **hemagglutination inhibition test (HI)**, the **neutralization test (NT)**, the complement fixation test (CFT), ELISA, immunofluorescence, as well as PCR and sequencing.

> **CLINICAL SIGNIFICANCE**
> The basis of modern virological diagnosis is **PCR** (and real-time PCR): it is fast and sensitive, detects the virus without culturing it and quantitatively assesses the viral load. The classic virological method (isolation) has remained mainly for scientific purposes and for selecting vaccine strains.

## 5. Bacteriophages

**Bacteriophage (phage)** — a virus that infects bacteria. It was discovered in 1915–1917 by F. Twort and F. d'Hérelle.

**Structure** (in the example of T-even phages): an icosahedral **head** (DNA inside), a **tube (tail)** and a contractile sheath, a **baseplate** and **tail fibers** (for adsorption).

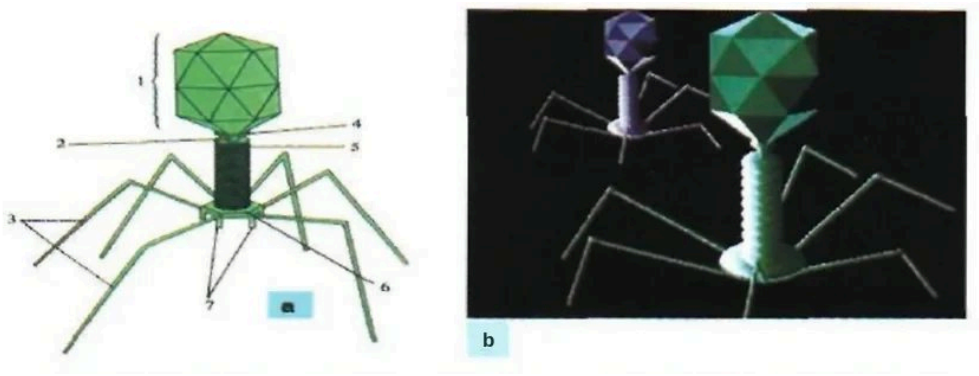



**Figure 12.5.** *The structure of the T4 coliphage: head, tube, baseplate and tail fibers. Source: Muhamedov I.M. et al., 2019.*

**The life cycle of a phage runs in two ways:**

| Feature | Lytic cycle (virulent phage) | Lysogenic cycle (temperate phage) |
|---|---|---|
| Result | Bacterial lysis, release of new phages | The phage DNA integrates into the chromosome (prophage) |
| Bacterium | Dies | Lives, passes the prophage to its progeny at division |
| Culture | Zone of lysis (a sterile spot) | **Lysogenic culture** |
| Additional effect | — | **Phage conversion** — a new property (diphtheria toxin) |

**The stages of the phage cycle** are like those of the viral cycle: adsorption (through the tail fibers, onto a species-specific receptor) → injection of the nucleic acid into the cell → synthesis → assembly → lysis of the bacterium.

**Specificity of the phage:** polyvalent (for several genera), monovalent (for a single species), type-specific (for particular variants within a species) phages.

## 6. Independent work: Uses of bacteriophages in practice

**1. Identification and epidemiological typing (phage typing).** A set of type-specific phages is applied to the isolated culture; from which phages produce lysis, the **phagovar** (phage type) is determined. This helps to distinguish strains within the same species and to find the source of infection (for example, in identifying the focus of a hospital staphylococcal infection).

**2. Phage diagnosis.** Isolating **the phage itself** from the material being examined also indirectly shows the presence of the pathogen (for example, detection of coliphages in water).

**3. Phage therapy.** Preparations of live virulent phages are used to treat infections (staphylococcal, streptococcal, dysentery, typhoid fever, Proteus, *Pseudomonas aeruginosa* phages, polyvalent pyobacteriophage). Advantages: narrow and precise action (does not disturb the normal microflora), and it may also act on antibiotic-resistant strains. Limitations: narrow spectrum (the phage must match the strain), development of phage resistance in bacteria, problems of standardization. The order of prescribing and using the preparation is determined by the physician.

**4. Phage prophylaxis.** Used on epidemic indications (for contacts in a focus).

**5. Genetic engineering and research.** Phages are a means of transferring genetic material (transduction), vectors, and also a classic model of molecular biology.

**REMEMBER**

The main difference between phage therapy and an antibiotic: an antibiotic is a **chemical substance with broad action**; a phage is a **live, highly selective** biological agent that lyses only the bacteria it matches.

## 7. Practical skill: the general algorithm of a virological examination

1. **Collecting material** — according to the stage of the disease and the tropism of the virus: nasopharyngeal wash/swab for respiratory viruses, stool for enteroviruses, vesicle fluid for herpes, cerebrospinal fluid for neuroinfections, blood serum for serology.
2. **Transport** — on ice (2–8 °C), in a special transport medium, fast; many viruses are unstable in the external environment.
3. **Rapid diagnosis** — detecting **viral antigen** (ELISA, immunofluorescence, immunochromatography) or **nucleic acid** (PCR) in the material.
4. **Virological method** — inoculation into cell culture/chick embryo → indication (CPE, HA, hemadsorption) → identification with specific sera (HI, NT).
5. **Serological method** — a ≥4-fold rise in antibody titer in **paired sera**; IgM — recent infection, IgG — past infection or a sign of immunity.
6. **Interpreting the result** — together with clinical and epidemiological data.

**CAUTION**

In virology, **the time of collecting the material** is decisive: virus shedding is often highest in the first days of the illness. In material taken late, even PCR may be negative, so it is supplemented with serology.

## 8. Review questions

1. By which features do viruses differ from cellular organisms?
2. State the structure of the virion and the types of symmetry.
3. How does the resistance of enveloped and non-enveloped viruses to the external environment differ? How does this affect the route of transmission?
4. Explain the three types of virus–cell relationship.
5. Describe the stages of the productive cycle in order. What is the eclipse period?
6. State the systems for culturing viruses and their methods of indication.
7. What do cytopathic effect, hemadsorption and hemagglutination mean?
8. How are viruses identified?
9. What is the structure of a bacteriophage, and what is the difference between the lytic and lysogenic cycles?
10. What are phage therapy, phage typing and phage prophylaxis?

## Test questions

1. A feature characteristic of viruses:
   A) They contain both nucleic acids
   B) Only one type of nucleic acid (DNA or RNA)
   C) They have ribosomes
   D) They multiply by binary fission
   E) They are susceptible to antibiotics

2. The capsid is:
   A) Nucleic acid
   B) A protein shell built of capsomers
   C) An outer envelope formed from the cell membrane
   D) An enzyme
   E) A receptor

3. Where the supercapsid comes from:
   A) Entirely from viral protein
   B) From the host cell membrane
   C) From nucleic acid
   D) From a ribosome
   E) From capsomers

4. A property characteristic of non-enveloped viruses:
   A) Resistant to the external environment, often transmitted by the fecal-oral route
   B) Very sensitive to alcohol
   C) Transmitted only through blood
   D) Do not multiply inside the cell
   E) Without a capsid

5. At which stage the tropism of a virus for cells is determined:
   A) Adsorption (binding to the receptor)
   B) Assembly
   C) Release
   D) Uncoating
   E) Eclipse

6. In the eclipse period:
   A) Virions leave the cell
   B) No whole virion can be found in the cell; the components are being synthesized
   C) The virus is adsorbed
   D) The cell undergoes lysis
   E) The capsid is assembled

7. Integration of the viral genome into the cell chromosome:
   A) Productive infection
   B) Abortive infection
   C) Integrative infection (virogeny)
   D) Lysis
   E) Hemadsorption

8. The intracellular inclusions found in rabies:
   A) Guarnieri bodies
   B) Babes–Negri bodies
   C) Volutin granules
   D) Druses
   E) Clue cells

9. Adhesion of erythrocytes to the surface of an infected cell:
    A) Hemagglutination
    B) Hemadsorption
    C) Hemolysis
    D) Precipitation
    E) Flocculation

10. A reaction used to identify viruses:
    A) Hemagglutination inhibition test (HI)
    B) Catalase test
    C) Oxidase test
    D) Ozheshko method
    E) Coagulase test

11. The state of a temperate phage integrated into the bacterial chromosome:
    A) Virion
    B) Prophage
    C) Capsomer
    D) Plasmid
    E) Transposon

12. The appearance of a new property in a lysogenic culture due to phage genes:
    A) Transduction
    B) Phage conversion
    C) Conjugation
    D) Transformation
    E) Modification

13. What phage typing is used for:
    A) Determining susceptibility to antibiotics
    B) Distinguishing strains and finding the source of infection
    C) Assessing hemolysis
    D) Measuring the viral load
    E) Monitoring sterilization

14. The main advantage of phage therapy over an antibiotic:
    A) Broad-spectrum action
    B) Narrow, selective action — disturbs the normal microflora little
    C) Low cost
    D) It acts only on viruses
    E) It acts on spores

15. The system classically used to culture influenza virus:
    A) Blood agar
    B) Chick embryo
    C) Sabouraud medium
    D) Kligler medium
    E) Semi-solid agar

## Clinical cases

### *Case 1*

After 48 hours, cells in a cell culture infected with the material rounded up and the cell layer began to break up; when erythrocytes were added, they stuck to the cell surface.

**Question:** What phenomena were observed and what do they mean?

**Answer:** Cytopathic effect (CPE) and hemadsorption indicate that a virus is present in the material. To determine its type, HI or the neutralization test is set up with specific immune sera.

### *Case 2*

In a nasopharyngeal swab taken on day 8 of a respiratory infection, PCR was negative, but the clinical picture fits a viral infection.

**Question:** How should the result be interpreted and what is the next step?

**Answer:** Virus shedding is usually highest in the first days; in material taken late the result may be negative. The next step is serological examination with **paired sera** (a ≥4-fold rise in titer or detection of IgM).

### *Case 3*

*S. aureus* was isolated from several patients in a hospital ward. All the strains proved to belong to the same phagovar.

**Question:** What does this conclusion mean epidemiologically?

**Answer:** The strains have most likely spread from a single source (a common source or a carrier). Phage typing is used precisely to establish such an epidemiological link; the measures are to find the source and to apply infection control.

## Quick review

**Table 12.2.** *Topic 12 — key concepts*

| Concept | Essence |
|---|---|
| Virus | One nucleic acid, no ribosomes, disjunctive reproduction |
| Parts of the virion | Genome + capsid (+ supercapsid) |
| Symmetry | Helical, cubic (icosahedral), complex |
| Enveloped / non-enveloped | Unstable, sensitive to alcohol / resistant, fecal-oral |
| Cycle | Adsorption → penetration → uncoating → synthesis → assembly → release |
| Types of relationship | Productive, abortive, integrative |
| Culturing | Cell culture, chick embryo, animal |
| Indication | CPE, inclusions, hemadsorption, HA, plaques |
| Identification | HI, NT, CFT, ELISA, PCR |
| Phage cycle | Lytic (lysis) / lysogenic (prophage, phage conversion) |
| Phages in practice | Phage typing, phage diagnosis, phage therapy, phage prophylaxis |

TOPIC 13

# Immunity. The structure and functions of the immune system. Innate immunity: complement, phagocytes. Antigen-presenting cells

## 1. Learning objectives

In this class the student will master the following:

- To know the concept and types of immunity and the structure of the immune system;
- to explain the factors of innate (nonspecific) immunity — barriers, phagocytosis, complement, interferons, NK cells;
- to master the stages of phagocytosis and the pathways of complement activation;
- to know the antigen-presenting cells and their role in linking innate and adaptive immunity;
- Independent work: to have an understanding of the theories of immunity and of the major histocompatibility complex class I (HLA-A, -B, -C).

## 2. General concept of immunity

**Immunity** is the body's way of defending itself against genetically foreign substances (microorganisms, altered own cells, foreign proteins) and of maintaining the constancy of the internal environment.

**Table 13.1.** *Types of immunity*

| Criterion | Type | Description |
|---|---|---|
| Origin | **Innate (nonspecific, natural)** | Hereditary, present from birth; acts quickly, leaves no memory |
| | **Acquired (adaptive, specific)** | Forms during life; slow, but specific and leaves **memory** |
| Forms of acquired immunity | Natural active | After the disease |
| | Natural passive | From mother to child (IgG, IgA via the placenta and milk) |
| | Artificial active | After a vaccine |
| | Artificial passive | When serum or immunoglobulin is administered |
| Direction | Antimicrobial, antitoxic, antiviral, transplantation, antitumor | By target |

**Table 13.2.** *Comparison of innate and adaptive immunity*

| Feature | Innate | Adaptive |
|---|---|---|
| Speed of response | Minutes–hours | 4–7 days or more |
| Specificity | General (recognizes structures characteristic of groups of microbes) | High (for a specific antigen) |

| Feature | Innate | Adaptive |
|---|---|---|
| Memory | Absent | Present (the secondary response is rapid and strong) |
| Main participants | Barriers, phagocytes, NK cells, complement, interferons | T and B lymphocytes, antibodies |
| Recognized structures | Pathogen-associated molecular patterns (PAMPs) — through Toll-like receptors | Specific antigenic determinants (epitopes) |

**REMEMBER**
Innate immunity is the **first line of defense** — it limits the infection and buys time until the adaptive response forms; in addition, through antigen presentation it **triggers** the adaptive response.

## 3. Structure of the immune system

**Central (primary) organs** — lymphocytes mature here:

- **Bone marrow** — the source of all blood cells; **B lymphocytes** mature here.
- **Thymus** — **T lymphocytes** mature here; positive and negative selection take place here (cells directed against the body's own tissue are eliminated).

**Peripheral (secondary) organs** — the encounter with antigen and the immune response take place here: **lymph nodes**, **spleen**, mucosa-associated lymphoid tissue (MALT; in the intestine — **GALT**, Peyer's patches, tonsils, appendix), and free lymphocytes in the blood and lymph.

**Immunocompetent cells:** T lymphocytes, B lymphocytes, NK cells, macrophages, dendritic cells, neutrophils, eosinophils, basophils and mast cells.

## 4. Factors of innate immunity

### *4.1. Barriers and humoral factors*

- **Mechanical barriers:** intact skin, mucous membranes, movement of the ciliated epithelium, the cough and sneeze reflex, the flow of urine.
- **Chemical factors:** fatty acids and sweat acids of the skin, gastric hydrochloric acid, vaginal lactic acid, bile acids.
- **Biological barrier:** the normal microflora (colonization resistance — topic 10).
- **Humoral factors: lysozyme** (breaks down peptidoglycan; abundant in tears, saliva, mucus), **complement**, **interferons**, **C-reactive protein** and other acute-phase proteins, properdin, beta-lysins, lactoferrin.

### *4.2. Phagocytosis*

**Phagocytosis** is the process by which a cell engulfs and digests particles; it was discovered by I.I. Mechnikov.

**Phagocytes:**

- **Microphages** — neutrophils (abundant in the blood, arrive quickly, short-lived);
- **Macrophages** — arise from monocytes; located in tissues (Kupffer cells in the liver, alveolar macrophages in the lung, osteoclasts in bone tissue, microglia in the brain).

**Stages of phagocytosis:**

1. **Chemotaxis** — the movement of a phagocyte toward the object along chemical signals (bacterial products, complement C5a, cytokines).
2. **Adhesion (attachment)** — sticking to the object; **opsonins** (IgG antibodies, complement C3b) sharply speed up this stage.
3. **Endocytosis (engulfment)** — enclosure with pseudopodia, formation of a **phagosome**.
4. **Formation of the phagolysosome** — fusion with a lysosome.
5. **Digestion (killing)** — oxygen-dependent mechanisms (the oxidative burst: superoxide, hydrogen peroxide, hypochlorite) and oxygen-independent mechanisms (lysozyme, defensins, lactoferrin, proteases).

**By outcome:**

- **Complete phagocytosis** — the microbe is digested;
- **Incomplete phagocytosis** — the microbe stays alive inside the phagocyte and even multiplies.

**CLINICAL SIGNIFICANCE**

**Incomplete phagocytosis** is seen with *Mycobacterium tuberculosis*, *Brucella*, gonococci and *Leishmania*. This explains the long persistence of the pathogen in the body, the chronic course and the relapses of the disease.

**MEMORY AID**

The stages of phagocytosis: **C–A–E–P–D**: **C**hemotaxis → **A**dhesion → **E**ndocytosis → **P**hagolysosome → **D**igestion.

### *4.3. The complement system*

**Complement** is a system of more than 30 proteins in the blood serum that are activated in sequence (a cascade). Activation proceeds by **three pathways**:

**Table 13.3.** *Pathways of complement activation*

| Pathway | What starts it | Role in the immune response |
|---|---|---|
| **Classical** | Antigen–antibody complex (IgM, IgG) → C1 | Linked to adaptive immunity |
| **Alternative** | Microbial surface (LPS, polysaccharides), without antibody | Innate immunity, the fastest |
| **Lectin** | Binding of mannose-binding lectin to microbial carbohydrates | Innate immunity |

All three pathways converge at the **cleavage of C3** and finally form the **membrane attack complex (MAC, C5b–C9)** — it opens a pore in the cell membrane and causes lysis.

**Main effects of complement:**

- **Lysis** (through the MAC);
- **Opsonization** (C3b — enhances phagocytosis);
- **Chemotaxis and inflammation** (C3a, C5a — anaphylatoxins; release of histamine from mast cells, recruitment of leukocytes);
- **Clearance of immune complexes**.

**REMEMBER**

**C3 — the central component**: all three pathways converge here. **C3b — opsonin**, **C3a and C5a — anaphylatoxin and chemoattractant**, **C5b–C9 — MAC (lysis)**.

### 4.4. Interferons and NK cells

**Interferons (IFN)** are the main innate factor of antiviral defense; they are protein cytokines. Types: **IFN-α** (leukocyte) and **IFN-β** (fibroblast) — mainly antiviral; **IFN-γ** (immune) — mainly regulates immunity and activates macrophages.

**Mechanism of action:** interferon does not kill the virus directly; acting on **neighboring cells**, it switches on enzymes in them that stop the synthesis of viral nucleic acid and proteins — the cell enters a "resistant" state against the virus.

**NK (natural killer) cells** destroy virus-infected cells and tumor cells in an antigen-independent manner. They detect **a decrease in MHC class I molecules** on the surface of the target cell ("missing-self recognition"): many viruses and tumors try to hide MHC I, and NK cells kill exactly such cells. They release perforin and granzymes and trigger apoptosis.

## 5. Antigen-presenting cells (APCs) and MHC

**Antigen-presenting cells: dendritic cells** (the most effective, "professional"), **macrophages**, **B lymphocytes**.

**Function:** to engulf an antigen, break it into fragments (peptides) and show it to a T lymphocyte on the cell surface together with an **MHC (HLA in humans)** molecule. A T lymphocyte recognizes an antigen **only together with MHC** — this is called **MHC-restricted recognition**.

**Table 13.4.** *MHC class I and class II*

| Feature | MHC I (HLA-A, -B, -C) | MHC II (HLA-DR, -DQ, -DP) |
|---|---|---|
| In which cells | In almost **all nucleated** cells | Only in APCs (dendritic cells, macrophages, B lymphocytes) |
| Which peptide it presents | Synthesized **inside** the cell (viral, tumor proteins) | Taken up from outside (bacterial, foreign protein) |
| Which T cell recognizes it | **$CD8^+$** (cytotoxic T lymphocyte) | **$CD4^+$** (T helper) |
| Result | The infected cell is destroyed | The immune response is regulated, B cells receive help |

> **MEMORY AID**
> The **"8 × 1 = 8, 4 × 2 = 8"** rule: **CD8 — MHC I**; **CD4 — MHC II** (the product is always 8).

**The dendritic cell takes a central place in starting the immune response:** it engulfs the antigen in the tissue → migrates to the lymph node → there it shows the antigen to a "naive" T lymphocyte → the adaptive response begins. That is why APCs are the **bridge linking innate and adaptive immunity**.

## 6. Independent work: Theories of immunity and class I HLA molecules

**Development of the theories:**

1. **Phagocytic (cellular) theory** — I.I. Mechnikov (1883): the basis of defense is phagocytes.
2. **Humoral theory** — P. Ehrlich: defense is due to antibodies in the blood; the "side-chain" theory was the first attempt to explain antibody formation.

3. **Instructive (template) theories** — assumed that the antigen acts as a "template" in antibody synthesis; later rejected.
4. **Clonal selection theory** — F. Burnet (1957): every lymphocyte has **in advance** a receptor matching one specific antigen; the antigen **selects** the matching clone and expands it. This is the foundation of modern immunology and explains both immunological memory and tolerance to self (clones that respond to the body's own tissue are eliminated during maturation).

The theories of Mechnikov and Ehrlich do not exclude each other — it was later confirmed that they are the two arms of immunity (cellular and humoral); both scientists shared the Nobel Prize in 1908.

**Major histocompatibility complex class I (HLA-A, -B, -C).** These molecules are encoded on chromosome 6 and are extremely **polymorphic** (almost every person has a unique set). Structure: a heavy chain ($\alpha_1$, $\alpha_2$, $\alpha_3$ domains) + **$\beta_2$-microglobulin**; the peptide-binding groove lies between the $\alpha_1$ and $\alpha_2$ domains and holds a peptide of 8–10 amino acids.

**Functions:**

- presenting peptides synthesized inside the cell (viral, tumor-specific) to **$CD8^+$ cytotoxic T lymphocytes**;
- serving as a "self-recognition" signal for NK cells (if MHC I decreases, the NK cell kills the cell);
- determining tissue compatibility in **transplantation** — donor and recipient are matched by HLA;
- some HLA alleles are linked to susceptibility to particular diseases (for example, HLA-B27 and certain rheumatologic diseases).

> **CLINICAL SIGNIFICANCE**
> A virus multiplies inside the cell, so the main defense against it is the **MHC I + $CD8^+$ T lymphocyte** pathway. For this very reason, antibodies alone are not enough in intracellular infections.

## 7. Practical skill: assessing phagocytic activity

**Procedure for setting up the phagocytic reaction:**

1. A suspension of leukocytes is prepared from the patient's blood.
2. A certain amount of test culture (usually killed staphylococci or standard particles) is added to it.
3. The mixture is incubated at 37 °C for 30 minutes.
4. A smear is prepared and stained by the Romanowsky–Giemsa method.
5. 100 neutrophils are viewed under the microscope and counted:

- **phagocytic activity (percentage)** — the percentage of cells that have engulfed microbes;

- **phagocytic number (index)** — the average number of microbes per phagocyte;

- **completion index** — the proportion of microbes digested (also assessed by repeat culture).

**NBT test (nitroblue tetrazolium test)** — assesses the oxygen-dependent killing ability of phagocytes: in an activated neutrophil the colorless dye is converted into blue formazan granules. The test is used to detect defects of the phagocytic system (for example, chronic granulomatous disease).

## 8. Review questions

1. What is immunity? State its types and the forms of acquired immunity.

2. Compare innate and adaptive immunity.
3. State the central and peripheral organs of the immune system; what is the function of each?
4. List the barrier and humoral factors of innate immunity. How does lysozyme act?
5. Explain the stages of phagocytosis. What are opsonins?
6. What is the difference between complete and incomplete phagocytosis? Give examples.
7. State the pathways of complement activation and its main effects.
8. What is the mechanism of action of interferon? How do NK cells recognize their target?
9. Which are the antigen-presenting cells? What is the difference between MHC I and MHC II?
10. What is the essence of the clonal selection theory?

## Test questions

1. In which organ T lymphocytes mature:
   A) Bone marrow
   B) Thymus
   C) Spleen
   D) Lymph node
   E) Liver

2. Belongs to the peripheral immune organs:
   A) Thymus
   B) Bone marrow
   C) Spleen
   D) Kidney
   E) Lung

3. The target of lysozyme action:
   A) Viral nucleic acid
   B) Peptidoglycan
   C) Lipid A
   D) Ribosome
   E) Membrane sterols

4. The first stage of phagocytosis:
   A) Adhesion
   B) Chemotaxis
   C) Endocytosis
   D) Formation of the phagolysosome
   E) Digestion

5. Opsonins include:
   A) IgG and C3b
   B) Lysozyme and interferon
   C) Perforin and granzyme
   D) Histamine
   E) Properdin

6. Incomplete phagocytosis is characteristic of:
   A) Pneumococcus
   B) *Mycobacterium tuberculosis*
   C) *E. coli*
   D) *Klebsiella*
   E) *Vibrio cholerae*

7. The classical pathway of complement begins with:
   A) The microbial surface, without antibody
   B) An antigen–antibody complex (C1)
   C) Mannose-binding lectin
   D) Interferon
   E) Perforin

8. The central component of the complement system:
   A) C1
   B) C3
   C) C5
   D) C9
   E) C4

9. The composition of the membrane attack complex:
   A) C1–C4
   B) C3a and C5a
   C) C5b–C9
   D) C2 and C4
   E) C3b

10. Acts as an anaphylatoxin and chemoattractant:
    A) C3b
    B) C3a and C5a
    C) C5b–C9
    D) C1q
    E) Properdin

11. The mechanism of action of interferon:
    A) Kills the virus directly
    B) Puts neighboring cells into a state of resistance to the virus
    C) Forms antibody
    D) Activates complement
    E) Stops phagocytosis

12. How NK cells recognize a target cell:
    A) By a decrease in MHC I molecules
    B) They do not recognize without antibody
    C) Only by MHC II
    D) Only with complement
    E) Through interferon

13. In which cells MHC II molecules are found:
    A) In all nucleated cells
    B) Only in antigen-presenting cells
    C) Only in erythrocytes
    D) Only in neutrophils
    E) Only in NK cells

14. The T lymphocyte that recognizes a peptide on an MHC I molecule:
    A) $CD4^+$ T helper
    B) $CD8^+$ cytotoxic T lymphocyte
    C) B lymphocyte
    D) NK cell
    E) Macrophage

15. The author of the clonal selection theory:
    A) I.I. Mechnikov
    B) P. Ehrlich
    C) F. Burnet
    D) L. Pasteur
    E) R. Koch

## Clinical cases

### *Case 1*

A child has recurrent purulent infections. When the NBT test was done, formazan granules did not form in the neutrophils.

**Question:** Which stage is impaired?

**Answer:** The **digestion (killing)** stage of phagocytosis — the oxygen-dependent mechanism is not working. This picture is characteristic of a defect of the phagocytic system (a condition of the chronic granulomatous disease type); the engulfment stage may be preserved.

### *Case 2*

A patient was found to have repeated episodes of meningococcal infection. Examination found a deficiency of the terminal complement components (C5–C9).

**Question:** Why do exactly these infections recur?

**Answer:** The terminal components form the **MAC**; the MAC is decisive in lysing *Neisseria* (meningococcus, gonococcus). Their deficiency creates susceptibility to exactly this group of infections.

### *Case 3*

A virus-infected cell reduced the MHC I molecules on its surface and in this way “hid” from $CD8^+$ T lymphocytes.

**Question:** Which other defense mechanism is triggered in the body?

**Answer: NK cells.** By the principle of “missing-self recognition,” they detect a cell with decreased MHC I and trigger apoptosis through perforin and granzymes.

## Quick review

**Table 13.5.** *Topic 13 — key concepts*

| Concept | Essence |
|---|---|
| Innate immunity | Fast, no memory, general; barrier, phagocyte, complement, IFN, NK |
| Central organs | Bone marrow (B), thymus (T) |
| Peripheral organs | Lymph nodes, spleen, MALT/GALT |
| Lysozyme | Breaks down peptidoglycan |
| Phagocytosis | Chemotaxis → adhesion → endocytosis → phagolysosome → digestion |
| Opsonins | IgG, C3b |
| Incomplete phagocytosis | Tuberculosis, *Brucella*, gonococcus, *Leishmania* |
| Complement pathways | Classical (Ag–Ab), alternative, lectin → C3 → MAC |
| C3b / C3a, C5a / C5b–C9 | Opsonin / anaphylatoxin / MAC |
| Interferon | Makes the neighboring cell resistant |
| NK | Kills a cell with decreased MHC I |
| MHC I / II | All nucleated cells, CD8 / APCs only, CD4 |

TOPIC 14

# Antigens and antibodies. Serological (immunological) reactions. Diagnosticums and diagnostic sera

## 1. Learning objectives

In this class the student will master the following:

- To know the concept of antigen, its properties and the types of bacterial antigens;
- to explain the structure, classes and functions of immunoglobulins;
- to master the principle, components and purpose of setting up serological reactions;
- to acquire the skill of setting up the main reactions (agglutination, precipitation, CFT, ELISA, immunoblot, immunofluorescence) and interpreting the result;
- Independent work: to know monoclonal antibodies, how they are obtained and their use in practice.

## 2. Antigens

**Antigen (Ag)** — a genetically foreign substance that, on entering the body, can provoke an immune response (antibody formation and/or a cellular response) and can bind **specifically** with the products of that response.

**Main properties:**

- **Antigenicity** — the ability to bind with the products of the immune response;
- **Immunogenicity** — the ability to provoke an immune response (it is high if the molecular mass is large and the chemical structure is complex);
- **Specificity** — determined by the **epitope (antigenic determinant)**, the small part of the antigen molecule that binds directly to the antibody.

**Types:**

- **Complete antigens** — both immunogenic and antigenic (proteins);
- **Haptens** — small molecules; they do not provoke an immune response by themselves but do so when joined to a carrier protein; they bind to antibody independently (many drugs can cause allergy as haptens).

**Table 14.1.** *The main antigens of bacteria*

| Antigen | Location/nature | Property | Significance |
|---|---|---|---|
| **O antigen** | Cell wall LPS (somatic) | Thermostable | Determines the serological group/serovar |
| **H antigen** | Flagellum (flagellin protein) | Thermolabile | Serovar, in motile species |
| **K antigen** | Capsule/surface polysaccharide | “Masks” the O antigen | Virulence, vaccine component |
| **Vi antigen** | A type of K antigen (typhoid fever) | Linked to virulence | In detecting carriage |
| Exotoxins and enzymes | Protein | Strong immunogen | Antitoxic immunity, toxoid |

**Immunologically important concepts: superantigens** (activate a large proportion of T lymphocytes at once — toxic shock syndrome); **cross-reactive (shared) antigens** — similarity between microbial and human tissue antigens (group A streptococcus and heart tissue — important in the pathogenesis of rheumatic fever); **autoantigens** and **alloantigens** (blood groups, HLA).

## 3. Antibodies (immunoglobulins)

**Antibody** — a protein (immunoglobulin, Ig) produced by B lymphocytes and plasma cells in response to an antigen, which binds specifically with the antigen.

**Structure:** two **heavy (H)** and two **light (L)** chains; the **Fab fragment** binds antigen (variable — V domains), the **Fc fragment** carries out the effector function (activating complement, binding to phagocytes, crossing the placenta). On cleavage with enzymes, papain gives 2 Fab + 1 Fc.

**Table 14.2.** *Classes of immunoglobulins*

| Class | Structure | Share in blood serum | Main properties |
|---|---|---|---|
| **IgG** | Monomer | ≈ 75–80% | The main serum antibody; **crosses the placenta**; opsonization, activates complement (classical pathway); a sign of the **secondary response** |
| **IgM** | **Pentamer** | ≈ 5–10% | Appears **first in the primary response**; the strongest activator of complement; does not cross the placenta; a sign of new infection |
| **IgA** | Monomer (in serum), **dimer + secretory component** (in mucus) | ≈ 10–15% | Protection of mucous membranes, saliva, tears, breast milk |
| **IgE** | Monomer | Very little | Binds to mast cells; **immediate-type allergy**, defense against helminths |
| **IgD** | Monomer | Little | Receptor on the surface of B lymphocytes |

**REMEMBER**

**IgM — "Most recent"** (first, a new infection); **IgG — "Gestation"** (crosses the placenta, protects the newborn) and long-term immunity; **IgA — "At the surfaces"** (mucous membranes); **IgE — "Eczema/allergy"**.

**Dynamics of the immune response. Primary response:** latent period 3–5 days, first **IgM**, then IgG; the titer is low and falls quickly. **Secondary response (due to memory):** the latent period is short (1–2 days), a large amount of **IgG** forms at once, and the titer is high and persists for a long time. For this reason **revaccination** is done in vaccination.

## 4. Serological reactions: the general principle

**Serological reactions** are reactions based on the specific binding between antigen and antibody *in vitro*.

**They are set up for two purposes:**

1. **Detecting an unknown antigen with a known antibody** (identifying the pathogen) — for this a **diagnostic serum** is needed;

2. **Detecting unknown antibodies with a known antigen** (in the patient's serum) — for this a **diagnosticum** is needed.

**Reagents:**

- **Diagnostic serum** — serum containing known antibodies, obtained by immunizing animals with a known antigen (agglutinating, precipitating, luminescent, enzyme-conjugated and others).
- **Diagnosticum** — a preparation of a known antigen (a suspension of killed microbes, an erythrocyte diagnosticum, an antigen extract).

**General conditions of the reaction:** the antigen and antibody must match, an electrolyte (physiological saline) must be present, the pH and temperature must be optimal (often 37 °C), and the ratio of the components must be correct.

> **CAUTION**
> **Prozone phenomenon:** if there is too much antibody (the serum is only slightly diluted), the reaction may not be visible. That is why the serum is set up in **serial dilutions** and the result is assessed by titer.

## 5. The main serological reactions

### *5.1. Agglutination reaction (AR)*

**Principle:** a corpuscular (large, particulate) antigen — bacterial cells, erythrocytes — sticks together under the action of antibody and forms a visible granular precipitate.

**Variants:**

- **On a glass slide (preliminary, rapid)** — mixing the isolated culture with diagnostic serum → granularity appears; for identification;
- **Extended (in a tube)** — serially diluting the patient's serum and adding a diagnosticum → the **titer** is determined (Widal reaction — in typhoid fever; Wright reaction — in brucellosis);
- **Indirect (passive) hemagglutination (PHA)** — the antigen is "loaded" onto the surface of erythrocytes; high sensitivity;
- **Coagglutination** — antibodies are attached to staphylococcal protein A;
- **Latex agglutination** — antibodies are attached to latex particles (rapid tests).

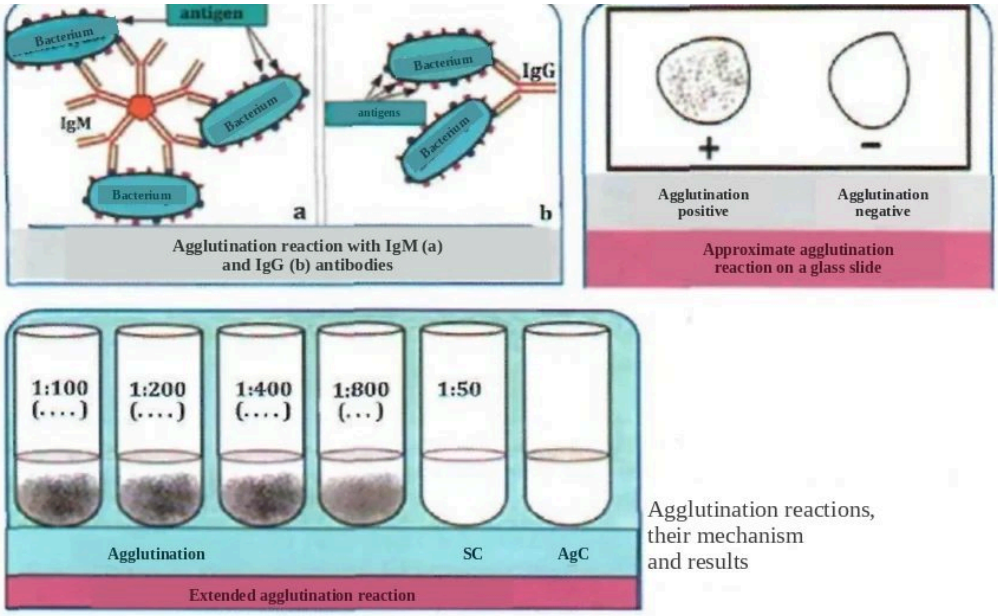



**Figure 14.1.** *Agglutination reactions: the mechanism, the titration series in a tube and the preliminary reaction on a glass slide; positive and negative result. Source: Muhamedov I.M. et al., 2019.*

## *5.2. Precipitation reaction (PR)*

**Principle:** a **soluble (molecular)** antigen combines with antibody and forms visible turbidity or a precipitate. The result is seen best in the **equivalence zone**, where the ratio of antigen and antibody is optimal.

**Variants:** ring precipitation (a white ring at the boundary of two liquids in a tube — the Ascoli reaction), **double immunodiffusion (Ouchterlony)** in gel, radial immunodiffusion (Mancini — measuring the amount of immunoglobulin), immunoelectrophoresis, **flocculation** (measuring toxoid activity).

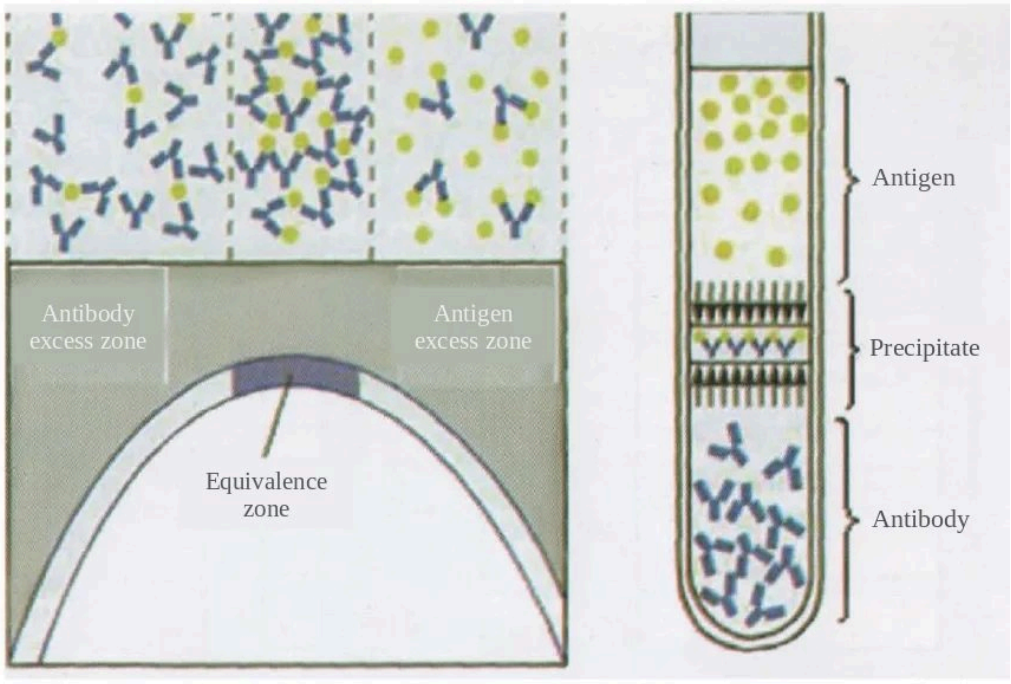



**Figure 14.2.** *The mechanism of the precipitation reaction: the zones of antibody excess and antigen excess, the equivalence zone and ring precipitation. Source: Aliyev Sh.R. et al., 2013.*

## 5.3. Complement fixation test (CFT)

**It consists of two systems:**

1. **Test system** — patient serum (antibody) + known antigen + complement;
2. **Indicator (hemolytic) system** — sheep erythrocytes + hemolytic serum.

**Result:** if an antigen–antibody complex forms in the test system, the complement is **bound** there and does not reach the indicator system → **there is no hemolysis** → the reaction is **positive**. If no complex forms, the complement stays free and lyses the erythrocytes → **there is hemolysis** → the reaction is **negative**.

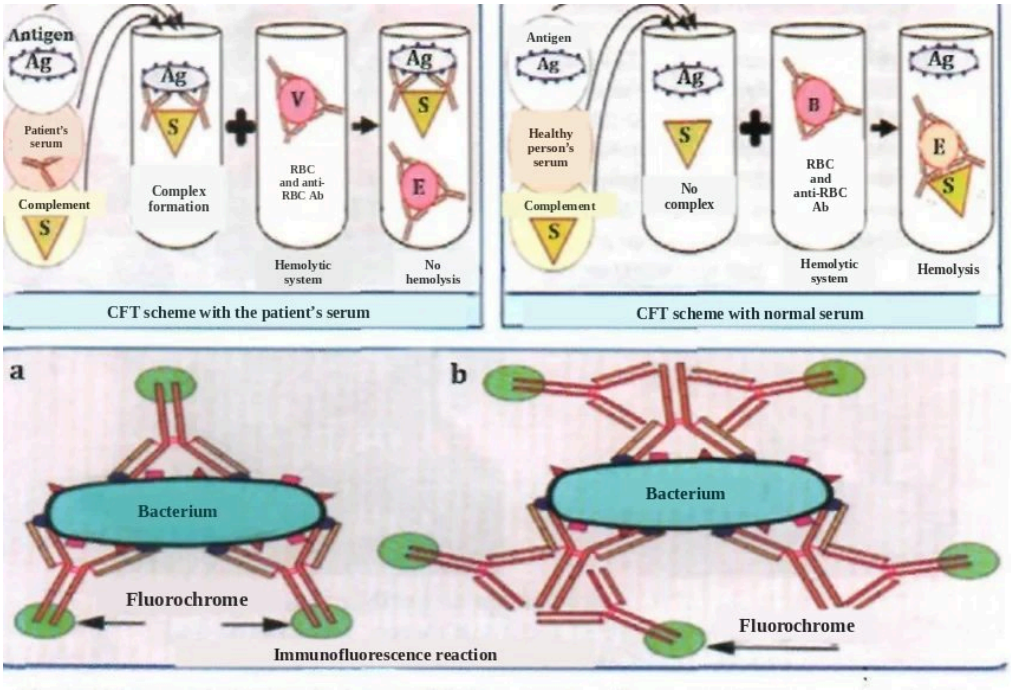



**Figure 14.3.** *Scheme of setting up the complement fixation test (with the serum of a sick and a healthy person) and the principle of the immunofluorescence reaction. Source: Muhamedov I.M. et al., 2019.*

> **MEMORY AID**
> Remember the result of CFT: **"No hemolysis — reaction present"** (positive); **"Hemolysis — no reaction"** (negative). This is where students make the most mistakes.

## 5.4. Reactions based on labeled antibodies

**Immunofluorescence reaction (IF, Coons method)** — the antibody is labeled with a fluorochrome (FITC); in a fluorescence microscope it glows bright green. In the **direct** method the labeled antibody is applied directly to the antigen; in the **indirect** method an ordinary antibody is added first, then a labeled antibody against it (more sensitive).

**Enzyme-linked immunosorbent assay (ELISA)** — the antibody or antigen is labeled with an **enzyme**; when substrate is added a color appears, and the color intensity is measured in a photometer. Direct, indirect and "sandwich" variants exist. It is currently the most widespread serological method.

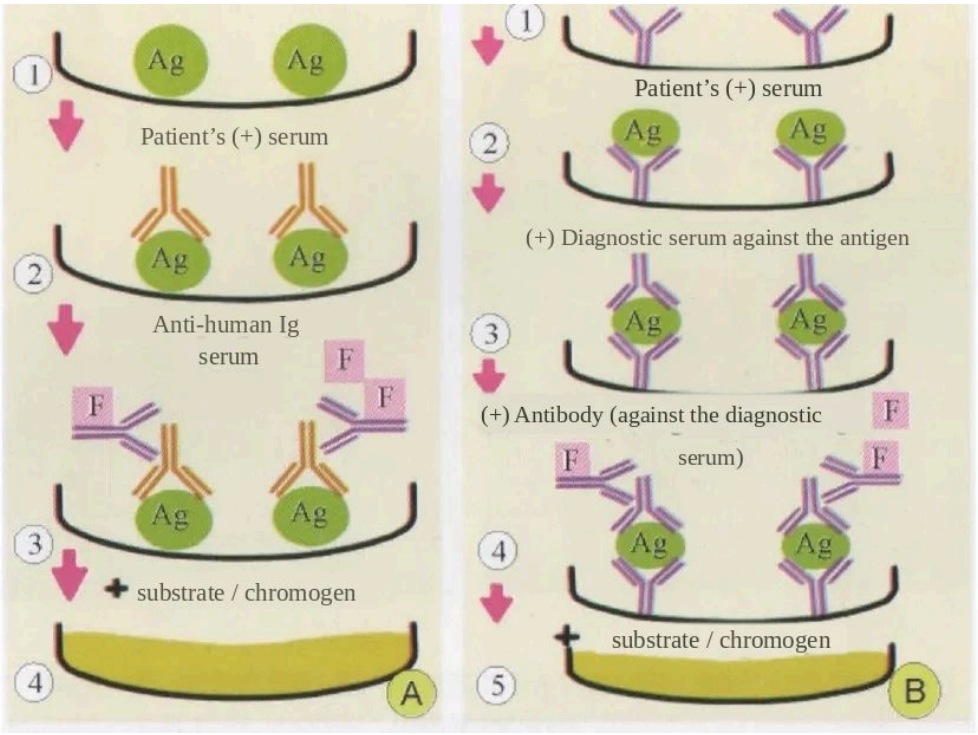



**Figure 14.4.** *The schematic mechanism of the enzyme-linked immunosorbent assay (ELISA): a) direct, b) indirect method; the steps of antigen, patient serum, enzyme-conjugated antibody and substrate–chromogen. Source: Aliyev Sh.R. et al., 2013.*

**Radioimmunoassay (RIA)** — labeling with an isotope; very sensitive, but replaced by ELISA in practice because of the radiation hazard.

**Immunoblot (Western blot)** — antigens are separated by electrophoresis, transferred to a membrane and treated with the patient's serum; as a result the **antibody against each antigen appears as a separate band**. That is why it is used as a **confirmatory (verification)** test (for example, in HIV diagnosis — topic 29).

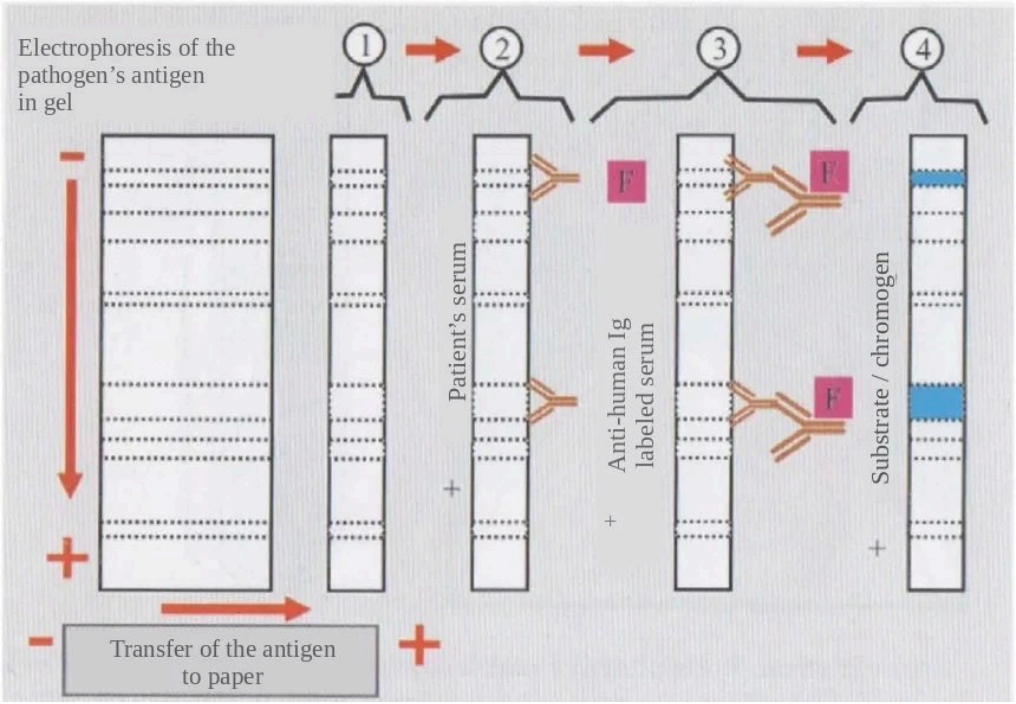



**Figure 14.5.** *The scheme of performing the immunoblot (Western blot) reaction: electrophoresis of the antigens in gel, transfer to a membrane and detection of the antibodies. Source: Aliyev Sh.R. et al., 2013.*

**Neutralization test (NT)** — the antibody removes the biological activity of a toxin or virus; the result is assessed in a living system (animal, cell culture).

**Table 14.3.** *Comparison of the reactions*

| Reaction | Type of antigen | Appearance of the result | Main application |
|---|---|---|---|
| Agglutination | Corpuscular | Granular precipitate | Identification, Widal, Wright |
| Precipitation | Soluble | Turbidity, ring, line in gel | Ascoli, Mancini, toxin activity |
| CFT | Any | Presence/absence of hemolysis | Serodiagnosis |
| IF | Any | Green glow (luminescence) | Rapid detection |
| ELISA | Any | Color change (optical density) | Screening, antibody/antigen |
| Immunoblot | Protein spectrum | Bands | Confirmatory test |

## 6. Independent work: Monoclonal antibodies

**Monoclonal antibodies (mAbs)** are completely identical antibodies obtained from a single clone of B lymphocytes and directed against **a single epitope**. An ordinary immune serum, by contrast, is **polyclonal** — the product of many clones, a mixture of antibodies against different epitopes.

**Production — hybridoma technology (Köhler and Milstein, 1975):**

1. A mouse is immunized with the required antigen.
2. **B lymphocytes** are taken from the spleen (they produce the required antibody but do not live long).
3. They are fused with **myeloma cells** (myeloma cells divide without limit but do not produce the required antibody).
4. In a special selective medium only the fused cells — **hybridomas** — survive.
5. The hybridomas are separated into individual clones, the clone that produces the required antibody is selected and multiplied without limit.

As a result a cell line that is **“immortal + produces the required antibody”** is obtained.

**Advantages:** absolute specificity, standardization (the same from batch to batch), and the possibility of obtaining unlimited amounts.

**Uses:**

- **In diagnosis** — ELISA, immunofluorescence, immunochromatographic rapid tests, detecting CD markers by flow cytometry (topic 15), determining blood groups, measuring hormones and tumor markers;
- **In treatment** — targeted therapy (directed at a chosen target): in oncology, autoimmune diseases and some infections; preventing rejection in transplantation;
- **In research** — detecting, purifying and studying proteins.

**Types:** mouse, chimeric, humanized and fully human antibodies — progressively “humanized” to reduce the response to the foreign protein (indicated by the corresponding suffixes in their names).

> **REMEMBER**
> **Polyclonal** serum — many kinds of antibodies to many epitopes (the product of natural immunization). **Monoclonal** — one kind of antibody to one epitope (the product of a hybridoma). The accuracy of standard diagnostic test systems is ensured precisely by monoclonality.

## 7. Practical skill: setting up the extended agglutination reaction

1. Physiological saline is poured into a row of tubes and the patient's serum is serially **diluted two-fold** (1:50, 1:100, 1:200, 1:400, 1:800...).
2. The same volume of **diagnosticum** is added to each tube.
3. Control tubes are set up: a **serum control** (serum + physiological saline, without diagnosticum) and an **antigen control** (diagnosticum + physiological saline).
4. They are left in a thermostat at 37 °C for 2 hours, then at room temperature for 18–20 hours.
5. The result is assessed: clearing of the liquid with a granular sediment at the bottom is positive; depending on the amount of sediment it is graded from "++++" to "+".
6. The **titer** is the highest dilution that gives a positive result. There must be no reaction in the control tubes.

> **CAUTION**
> The result can be read **only if the controls are correct**. If there is spontaneous sediment in the antigen control (the diagnosticum is of poor quality), the whole reaction is considered unreliable.

## 8. Review questions

1. What is an antigen? What determines antigenicity, immunogenicity and specificity?
2. What is a hapten and in what cases does it provoke an immune response?
3. State the main antigens of bacteria and their significance.
4. Explain the structure of an immunoglobulin. What is the function of the Fab and Fc fragments?
5. State the classes of Ig and their main properties.
6. How do the primary and secondary immune responses differ?
7. What are a diagnosticum and a diagnostic serum? For what purpose are they used?
8. How do the agglutination and precipitation reactions differ?
9. Explain the principle of CFT and the interpretation of its result.
10. How do ELISA, IF and immunoblot work? Why is immunoblot considered a confirmatory test?
11. How are monoclonal antibodies obtained and where are they used?

## Test questions

1. The part of the antigen molecule that binds directly to the antibody:
   A) Fab fragment
   B) Epitope (antigenic determinant)
   C) Fc fragment
   D) Hapten
   E) Paratope

2. A substance that does not provoke an immune response itself but does when combined with a carrier protein:
   A) A complete antigen
   B) Hapten
   C) A superantigen
   D) An adjuvant
   E) A toxoid

3. A thermostable somatic antigen:
   A) H-antigen
   B) O-antigen
   C) K-antigen
   D) Vi-antigen
   E) A phage receptor

4. The immunoglobulin class that crosses the placenta:
   A) IgM
   B) IgG
   C) IgA
   D) IgE
   E) IgD

5. The antibody that appears first in the primary immune response:
   A) IgG
   B) IgM
   C) IgA
   D) IgE
   E) IgD

6. The class that provides protection of the mucous membranes:
   A) IgG
   B) Secretory IgA
   C) IgM
   D) IgE
   E) IgD

7. The immunoglobulin involved in immediate-type allergic reactions:
   A) IgA
   B) IgE
   C) IgM
   D) IgG
   E) IgD

8. The reaction becoming invisible when antibody is in excess:
   A) Prozone phenomenon
   B) Equivalence zone
   C) Flocculation
   D) Opsonization
   E) Hemadsorption

9. A reaction set up with a corpuscular antigen:
   A) Agglutination
   B) Precipitation
   C) Flocculation
   D) Immunodiffusion
   E) Radial immunodiffusion

10. The disease in which the Widal reaction is used:
    A) Brucellosis
    B) Typhoid fever
    C) Syphilis
    D) Tuberculosis
    E) Cholera

11. The appearance of a positive CFT result:
    A) Hemolysis is present
    B) No hemolysis
    C) Granular precipitate
    D) Ring formation
    E) Change of color

12. The main purpose for which immunoblot is used:
    A) Screening
    B) A confirmatory (verification) test
    C) Determining susceptibility to antibiotics
    D) Counting colonies
    E) Assessing hemolysis

13. What is used as the label in ELISA:
    A) A fluorochrome
    B) An enzyme
    C) A radioactive isotope
    D) Latex
    E) Complement

14. The cell from which monoclonal antibodies are obtained:
    A) A plasma cell
    B) A hybridoma (B lymphocyte + myeloma cell)
    C) Macrophage
    D) NK cell
    E) A dendritic cell

15. The main difference of monoclonal antibodies from polyclonal serum:
    A) Low cost
    B) Directed against a single epitope, completely identical
    C) They work only in animals
    D) They do not activate complement
    E) Heat resistance

## Clinical cases

### *Case 1*

**IgM** against a known pathogen was detected in the patient's serum, while IgG is at a low titer.

**Question:** What does this result mean?

**Answer:** IgM is the class that appears first in the primary response; this is characteristic of a **new (acute) infection**. For confirmation, a rise in IgG titer is looked for in paired sera after 10–14 days.

### *Case 2*

In an extended agglutination reaction, the result was negative in the first two tubes (the least diluted) and positive in the following ones.

**Question:** What is this phenomenon called and what should be done?

**Answer: Prozone phenomenon** — because antibody is in excess, the reaction is not visible in the first tubes. The result is read by titer, from the diluted tubes; that is why the reaction is always set up with serial dilution.

### *Case 3*

A screening ELISA test was positive. The doctor did not make a final diagnosis right away and sent the patient for confirmatory testing.

**Question:** Why, and which method is used?

**Answer:** Screening tests are tuned for high sensitivity, so a **false-positive** result is possible. For confirmation a method with high specificity — **immunoblot** (or a molecular method) — is used.

## Quick review

**Table 14.4.** *Topic 14 — key concepts*

| Concept | Essence |
|---|---|
| Epitope | The part of the antigen that binds to the antibody |
| Hapten | Not immunogenic by itself, immunogenic with a carrier |
| Antigens of bacteria | O (somatic, thermostable), H (flagellar), K/Vi (capsule) |
| Ig structure | 2H + 2L; Fab — binding, Fc — effector |
| IgG / IgM / IgA / IgE | Placenta, secondary / first, pentamer / mucosa / allergy |
| Diagnosticum / diagnostic serum | Known antigen / known antibody |
| Prozone | Excess antibody — the reaction is not visible |
| Agglutination / precipitation | Corpuscular / soluble antigen |
| CFT | No hemolysis — positive; hemolysis — negative |
| IF / ELISA / immunoblot | Fluorochrome / enzyme / confirmatory bands |
| Monoclonal antibody | Hybridoma; a single epitope; diagnosis and targeted therapy |

TOPIC 15

# Adaptive immunity. T and B lymphocytes, their subpopulations. CD markers. Cytokines, interferons. Immunoprophylaxis

## 1. Learning objectives

In this class the student will master the following:

- To know the formation of adaptive immunity and the mechanisms of the cellular and humoral response;
- to explain the subpopulations, CD markers and functions of T and B lymphocytes;
- to master the laboratory assessment of lymphocytes (E-RFC, flow cytometry);
- Independent work: to know the basics of cytokines and interferons, immunodeficiency states, immunoprophylaxis and immunotherapy.

## 2. General scheme of adaptive immunity

Adaptive immunity forms after an encounter with antigen, is **specific** and leaves **immunological memory**. It appears in two forms:

- **Humoral response** — B lymphocytes → plasma cells → **antibodies**; effective against extracellular bacteria and toxins;
- **Cellular response** — T lymphocytes; directed against **intracellular** pathogens (viruses, *Mycobacterium tuberculosis*), tumor cells and the transplant.

**Three stages of the immune response:** recognition and presentation of the antigen (APC) → activation, proliferation and differentiation of lymphocytes → the effector stage (antibody production or destruction of the target cell) and the formation of **memory cells**.

## 3. T lymphocytes and their subpopulations

T lymphocytes arise in the bone marrow and mature in the **thymus**. They recognize antigen **only together with MHC** (through the T cell receptor — TCR). The common marker is **CD3** (the complex linked to the TCR).

**Table 15.1.** *The main subpopulations of T lymphocytes*

| Subpopulation | Marker | Which MHC it works with | Main function |
|---|---|---|---|
| **T helpers (Th)** | $CD3^+$ **$CD4^+$** | MHC II | Regulating the immune response, secreting cytokines, helping B cells |
| **Cytotoxic T lymphocytes (CTL)** | $CD3^+$ **$CD8^+$** | MHC I | Destroying virus-infected and tumor cells (perforin, granzyme, Fas pathway) |
| **Regulatory T cells (Treg)** | $CD4^+$ $CD25^+$ $FoxP3^+$ | — | Limiting the immune response, maintaining self-tolerance |
| **Memory T cells** | $CD45RO^+$ | — | Rapid response on a repeat encounter |

**Directions (polarization) of T helpers:**

- **Th1** — secretes IFN-γ and IL-2; **activates macrophages**, strengthens the response against intracellular pathogens (tuberculosis, leishmaniasis);

- **Th2** — IL-4, IL-5, IL-13; **helps B cells**, strengthens the IgE and eosinophil response (helminths, allergy);
- **Th17** — IL-17; recruits neutrophils, protects the mucous membranes against extracellular bacteria and fungi;
- **Tfh (follicular helpers)** — in the lymph node they help B cells in class switching of antibodies.

**REMEMBER**

**Th1 — against intracellular pathogens (activates macrophages), Th2 — the antibody and allergy pathway.** These two directions inhibit each other; their balance determines the course of the infection.

## 4. B lymphocytes and the humoral response

In humans B lymphocytes mature in the **bone marrow**. Markers: **CD19, CD20, CD21**; on the surface there is **membrane-bound immunoglobulin (IgM and IgD)** — the B cell receptor.

**Stages of the humoral response:**

1. The receptor on the surface of the B cell recognizes the antigen directly and engulfs it;
2. It shows a piece of the antigen with **MHC II** to a T helper (Tfh);
3. With the help of the T helper (cytokines + CD40L), the B cell is activated and multiplies;
4. Some become **plasma cells** (the antibody “factory”), others remain as **memory B cells**;
5. In the germinal centers of the lymph node, **class switching** (IgM → IgG, IgA, IgE) and **affinity maturation** take place.

**T-dependent and T-independent antigens.** Protein antigens require the help of a T helper and give a full response (IgG, memory). Polysaccharide antigens (for example, the pneumococcal capsule) provoke a T-independent response: mainly IgM, with weak memory.

**CLINICAL SIGNIFICANCE**

Polysaccharide vaccines do not work well in young children — their T-independent response is not sufficiently developed. That is why **conjugate vaccines** were created: the polysaccharide is attached to a carrier protein, and the response becomes T-dependent, strong and with memory.

## 5. CD markers and methods of assessing lymphocytes

**CD (Cluster of Differentiation)** — differentiation markers on the cell surface; detected with the help of monoclonal antibodies.

**Table 15.2.** *The main CD markers*

| Marker | In which cell | Note |
|---|---|---|
| CD3 | All T lymphocytes | The TCR complex |
| CD4 | T helpers | The main target of HIV |
| CD8 | Cytotoxic T lymphocytes | Works with MHC I |
| CD19, CD20 | B lymphocytes | Target of B cell therapy |
| CD16, CD56 | NK cells | Natural killers |
| CD25 | Activated T cells, Treg | The α chain of the IL-2 receptor |

**Methods of assessment:**

1. **Flow cytometry** — the modern standard. The cells are stained with monoclonal antibodies labeled with different fluorochromes, passed one by one under a laser beam, and the set of markers is recorded for each cell. As a result the percentage and absolute number of the subpopulations are determined, including the **CD4/CD8 ratio** (normally about 1.5–2.5).
2. **E-RFC method (rosette formation reaction)** — the classical method: T lymphocytes spontaneously form "rosettes" with sheep erythrocytes; the cells that have formed rosettes are counted. For B lymphocytes, EAC rosettes (with complement and antibody) are used.
3. **Separation of lymphocytes** — from peripheral blood by the **density gradient** (Ficoll–Verografin) method: after centrifugation the lymphocytes collect in the interphase layer and the erythrocytes settle to the bottom.

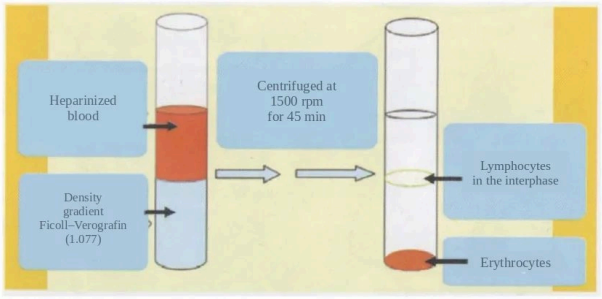



**Figure 15.1.** *Separation of lymphocytes from peripheral blood by the density gradient (Ficoll–Verografin) method: centrifugation and the layer of lymphocytes in the interphase. Source: Aliyev Sh.R. et al., 2013.*

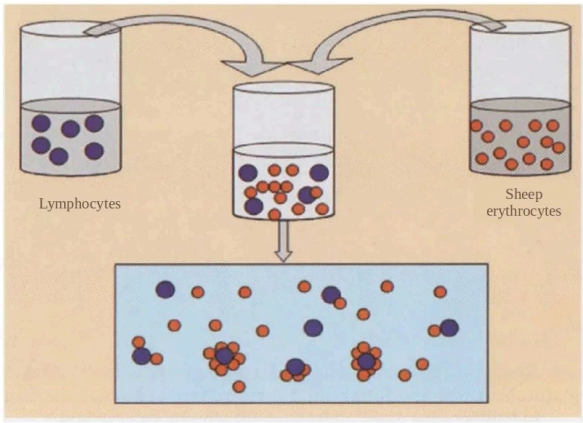



**Figure 15.2.** *Determining the number of T lymphocytes — the scheme of the E-RFC (rosette formation) method. Source: Aliyev Sh.R. et al., 2013.*

**CAUTION**

**The CD4⁺ cell count** is the main indicator for assessing the stage of disease and for starting treatment in HIV infection (topic 29). That is why flow cytometry is the main method of clinical immunology.

## 6. Independent work 1: Cytokines and interferons

**Cytokines** are protein–peptide molecules produced by immune and other cells that provide cell-to-cell communication. Properties: they act in very small amounts, work through receptors, are **pleiotropic** (one cytokine has many effects) and **redundant** (different cytokines have similar effects), and act mainly locally.

**Table 15.3.** *The main groups of cytokines*

| Group | Representatives | Main effect |
|---|---|---|
| Interleukins | IL-1 … IL-38 | Cell-to-cell communication; IL-1, IL-6, TNF-α — inflammation and fever; IL-2 — T cell proliferation; IL-10 — anti-inflammatory |
| Interferons | IFN-α, -β, -γ | Antiviral (α, β), regulation of immunity and activation of macrophages (γ) |
| Tumor necrosis factors | TNF-α, TNF-β | Inflammation, apoptosis, cachexia |
| Chemokines | IL-8 and others | Recruiting leukocytes to the focus of inflammation |
| Colony-stimulating factors | G-CSF, GM-CSF | Stimulating the formation of blood cells |

| Group | Representatives | Main effect |
|---|---|---|
| Growth factors | TGF-β | Repair, suppression of the immune response |

**Interferons:** IFN-α (leukocytes), IFN-β (fibroblasts) — **type I**, direct antiviral defense; IFN-γ — **type II**, produced by T and NK cells, activates macrophages and increases MHC expression. Production: natural (from donor leukocytes) and **recombinant** (by genetic engineering — the main method now). Use: certain chronic viral hepatitides, some oncological and neurological diseases — only on a physician's prescription.

> **REMEMBER**
> **IFN-α/β — "antiviral," IFN-γ — "immune regulator."** IFN-γ is mainly linked to the Th1 pathway and strengthens the macrophage's ability to kill intracellular microbes.

## 7. Independent work 2: Immunodeficiency states

**Primary (congenital) immunodeficiency** — the result of a genetic defect; it usually appears in childhood with recurrent and unusual infections. Depending on the impaired link:

- **B-system (humoral)** defect — recurrent purulent infections with encapsulated bacteria;
- **T-system (cellular)** defect — viral, fungal and intracellular infections;
- **combined** defects — the most severe course;
- **phagocytosis** defects — recurrent abscesses (the NBT test in topic 13);
- **complement** defects — susceptibility to *Neisseria* infections.

**Secondary (acquired) immunodeficiency** — much more common; causes: viral infections (the most important — HIV), protein-energy malnutrition, severe chronic diseases, tumors and chemotherapy, irradiation, long-term glucocorticoid and immunosuppressive treatment, extensive surgical interventions, old age, chronic stress.

**Laboratory assessment:** complete blood count and leukocyte differential, immunoglobulin class levels, lymphocyte subpopulations (flow cytometry), phagocytic activity and the NBT test, complement components, and genetic testing when necessary.

## 8. Independent work 3: Immunoprophylaxis and immunotherapy

**Immunoprophylaxis** — preventing infections with the help of vaccines and immune preparations.

**Table 15.4.** *The main types of vaccines*

| Type | Meaning | Examples |
|---|---|---|
| **Live attenuated** | A weakened but live pathogen; strong and long-lasting immunity | Measles, rubella, mumps, chickenpox, tuberculosis (BCG), oral poliomyelitis |
| **Killed (inactivated)** | A killed pathogen; safer, revaccination is needed | Pertussis (whole-cell), inactivated poliomyelitis, rabies, hepatitis A |
| **Chemical (subunit)** | Isolated parts of the antigen | Pneumococcal, meningococcal and influenza subunit vaccines |
| **Toxoid** | A neutralized (detoxified) exotoxin | Diphtheria and tetanus toxoids |
| **Recombinant** | An antigen obtained by genetic engineering | Hepatitis B, papillomavirus |

| Type | Meaning | Examples |
|---|---|---|
| **Conjugate** | Polysaccharide + carrier protein | *H. influenzae* type b and pneumococcal conjugate vaccines |

**Immunotherapy and passive immunization:** sera and immunoglobulins provide **ready-made antibodies** — protection begins immediately but is short-lived (2–4 weeks). **Antitoxic sera** (tetanus, diphtheria, botulism) neutralize the toxin; that is why they are given as early as possible. When foreign (horse) serum is given, special rules are followed because of the risk of serum sickness and anaphylaxis.

**Table 15.5.** *Active and passive immunization*

| Feature | Active (vaccine) | Passive (serum, Ig) |
|---|---|---|
| Onset of protection | 2–4 weeks | Immediately |
| Duration | Long (years), memory is present | Short (weeks), no memory |
| When | Prophylaxis | Emergency prophylaxis and treatment |

**CLINICAL SIGNIFICANCE**

Tetanus prophylaxis in a contaminated wound is determined according to the **vaccination history**: for a vaccinated person, usually an additional dose of toxoid; for an unvaccinated person or one with an unknown history, active and passive immunization are used together (the prescription is by the physician, according to the current protocol).

## 9. Practical skill: a stepwise approach to assessing immune status

1. **History and clinical picture:** frequently recurring infections, unusual pathogens, severe course, poor response to antibiotics, family history.
2. **First-stage tests:** complete blood count (absolute numbers of lymphocytes and neutrophils), immunoglobulin classes (IgG, IgM, IgA), the NBT test or phagocytic indices.
3. **Second-stage tests:** lymphocyte subpopulations (CD3, CD4, CD8, CD19, CD16/56) and the **CD4/CD8 ratio**, complement components, the response of lymphocytes to mitogens.
4. **Special investigations:** HIV test, genetic analysis, antibody response to specific antigens (assessing the response to vaccination).
5. **Interpretation:** the results are compared with **age-appropriate norms** and assessed together with the clinical picture.

## 10. Review questions

1. How do the humoral and cellular forms of adaptive immunity differ?
2. State the subpopulations and markers of T lymphocytes.
3. What are the functions of the Th1, Th2 and Th17 pathways?
4. Explain the stages of the humoral response. What is class switching?
5. What is the difference between T-dependent and T-independent antigens? Why was the conjugate vaccine created?
6. What are CD markers? List the main ones.
7. How do flow cytometry and the E-RFC method work?
8. State the main groups of cytokines and their effects.

9. What is the difference between IFN-α/β and IFN-γ?
10. State the causes of primary and secondary immunodeficiency.
11. Explain the types of vaccines and the difference between active and passive immunization.

## Test questions

1. The common marker of all T lymphocytes:
   A) CD19
   B) CD3
   C) CD20
   D) CD16
   E) CD56
2. The marker of T helpers and the MHC class they work with:
   A) CD8, MHC I
   B) CD4, MHC II
   C) CD19, MHC I
   D) CD3, MHC II
   E) CD56, MHC I
3. The cell that directly destroys a virus-infected cell:
   A) T helper
   B) Cytotoxic T lymphocyte ($CD8^+$)
   C) A plasma cell
   D) B lymphocyte
   E) Treg
4. The pathway that activates macrophages and strengthens the response against intracellular infections:
   A) Th1
   B) Th2
   C) Th17
   D) Treg
   E) Tfh
5. The pathway that strengthens the IgE and eosinophil response:
   A) Th1
   B) Th2
   C) Th17
   D) Treg
   E) CTL
6. The marker of B lymphocytes:
   A) CD3
   B) CD19, CD20
   C) CD4
   D) CD8
   E) CD16

7. The mature cell that produces antibody:
   A) Memory B cell
   B) A plasma cell
   C) T helper
   D) Macrophage
   E) NK cell

8. A feature of the response to a polysaccharide antigen:
   A) Strong IgG and memory
   B) Mainly IgM, weak memory
   C) Only a cellular response
   D) No response at all
   E) Only IgE

9. Why the conjugate vaccine was created:
   A) To be cheaper
   B) To make the response to the polysaccharide T-dependent and with memory
   C) To extend the shelf life
   D) To be given without injection
   E) To strengthen the action of antibiotics

10. The modern standard method of determining lymphocyte subpopulations:
    A) E-RFC
    B) Flow cytometry
    C) Agglutination
    D) CFT
    E) Precipitation

11. Normally the CD4/CD8 ratio is approximately:
    A) 0,2–0,5
    B) 1,5–2,5
    C) 5–7
    D) 10–12
    E) 0,8–1,0

12. The interferon that activates macrophages and increases MHC expression:
    A) IFN-α
    B) IFN-β
    C) IFN-γ
    D) IL-4
    E) IL-10

13. An example of a live attenuated vaccine:
    A) Tetanus toxoid
    B) Measles vaccine
    C) Hepatitis B recombinant vaccine
    D) Inactivated poliomyelitis vaccine
    E) Diphtheria toxoid

14. An example of a toxoid vaccine:
    A) BCG
    B) Diphtheria and tetanus toxoids
    C) Influenza subunit vaccine
    D) Varicella vaccine
    E) Rotavirus vaccine

15. A feature of passive immunization:
    A) Protection begins after 2–4 weeks
    B) Protection begins immediately but is short-lived
    C) It leaves immunological memory
    D) It is done only with a vaccine
    E) It lasts for years

## Clinical cases

### *Case 1*

In a patient, the $CD4^+$ cell count on flow cytometry has fallen sharply and the CD4/CD8 ratio has dropped to 0.4. Infections caused by unusual pathogens have been found in the patient.

**Question:** What condition does this picture suggest?

**Answer:** Profound deficiency of cellular immunity; selective loss of $CD4^+$ cells is very characteristic of HIV infection. For confirmation, HIV-specific testing (screening ELISA + a confirmatory method, viral load) is done.

### *Case 2*

A 2-year-old child has recurrent pneumococcal otitis and pneumonias. Immunoglobulin levels are low and the response to polysaccharide vaccine is weak.

**Question:** Which link is impaired and which vaccine is preferable in this situation?

**Answer:** The **humoral (B-system)** link is impaired. Polysaccharide vaccines are ineffective at this age and in this condition; **conjugate** vaccines are preferable because they give a T-dependent response. The treatment plan is determined by the physician.

### *Case 3*

A patient not vaccinated against tetanus came with a deep wound contaminated with soil.

**Question:** Which of active and passive immunization is needed, and why?

**Answer:** Both: **passive** (ready-made antibodies) gives immediate protection, and **active** (toxoid) builds long-term immunity. The exact scheme is determined by the physician according to the current protocol and the vaccination history.

## Quick review

**Table 15.6.** *Topic 15 — key concepts*

| Concept | Essence |
|---|---|
| Humoral / cellular response | B → antibody / T → intracellular and antitumor |
| T subpopulations | CD4 helper (MHC II), CD8 CTL (MHC I), Treg, memory |

| Concept | Essence |
|---|---|
| Th1 / Th2 / Th17 | Macrophage activation / IgE and eosinophils / neutrophils and mucosal defense |
| B cell markers | CD19, CD20, CD21 |
| Class switching | IgM → IgG/IgA/IgE, with the help of a T helper |
| T-independent antigen | Polysaccharide → IgM, no memory → conjugate vaccine |
| Methods of assessment | Flow cytometry, E-RFC, Ficoll gradient |
| CD4/CD8 | Normally ≈1.5–2.5 |
| Cytokines | IL, IFN, TNF, chemokines, CSF, TGF-β |
| IFN-α/β / IFN-γ | Antiviral / immune-regulating |
| Types of vaccines | Live, killed, chemical, toxoid, recombinant, conjugate |
| Active / passive | Slow but long-lasting / immediate but short-lived |

# PART II

## *SYSTEMATIC MICROBIOLOGY AND VIROLOGY*

TOPIC 16

# Causative agents of purulent inflammatory diseases: staphylococci, streptococci, *Pseudomonas aeruginosa*

## 1. Learning objectives

In this class the student will master the following:

- To know the morphological, cultural and biochemical properties of staphylococci, streptococci and *Pseudomonas aeruginosa*;
- to explain their pathogenicity factors and the diseases they cause;
- to master the collection of material in purulent infections, the algorithm of bacteriological diagnosis and the differential tests;
- Independent work: to know the significance and laboratory diagnosis of opportunistic anaerobic cocci (*Peptococcus*, *Peptostreptococcus*, *Veillonella*).

## 2. Staphylococci (*Staphylococcus*)

**Morphology:** gram-positive cocci (0.5–1.5 μm); because division occurs in different planes, they form **grape-like clusters**; nonmotile, do not form spores, may have a microcapsule.

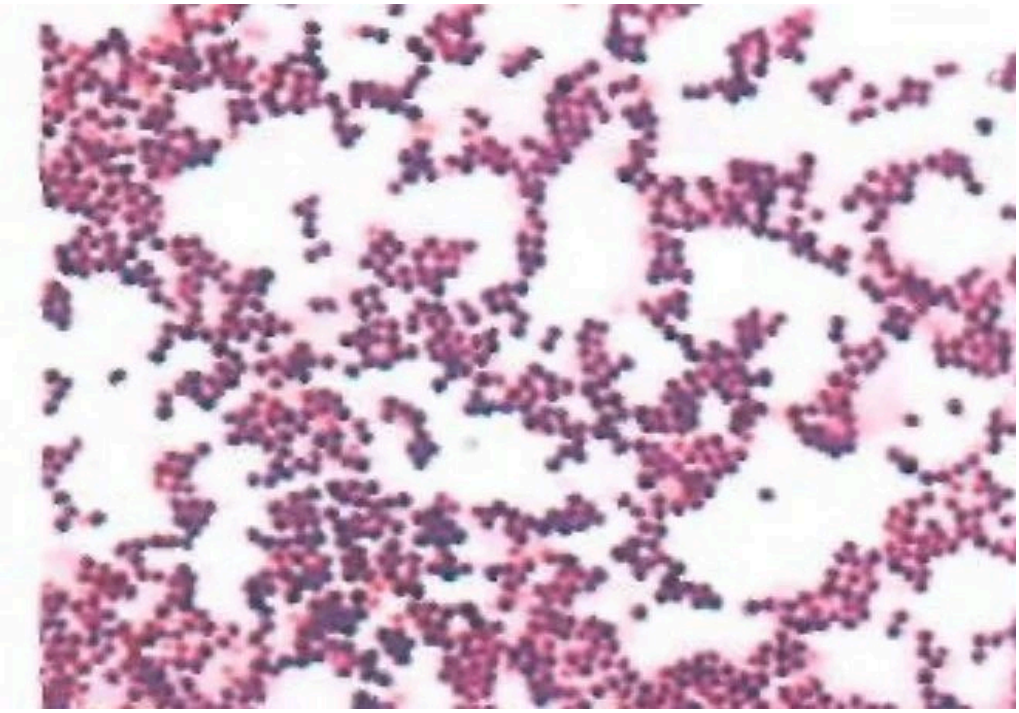


**Figure 16.1.** *Staphylococci: gram-positive cocci forming typical grape-like clusters (pure culture, Gram stain). Source: Muhamedov I.M. et al., 2019.*

**Cultural properties:** facultative anaerobe; grows on ordinary media, **resistant to a high concentration of NaCl (7.5–10%)** — this property was the basis for creating a selective medium (egg-yolk salt agar). Colonies are round, smooth, convex; *S. aureus* forms a **golden-yellow pigment**. On blood agar it often produces beta-hemolysis.

**The main species and how to distinguish them:**

**Table 16.1.** *Clinically significant staphylococci*

| Species | Coagulase | Pigment | Main significance |
|---|---|---|---|
| ***S. aureus*** | + | Golden-yellow | Purulent infections, food poisoning, toxic shock, osteomyelitis, sepsis |
| *S. epidermidis* | − | White | Prosthesis- and catheter-related infections (biofilm) |
| *S. saprophyticus* | − | White/yellow | Urinary tract infection in young women |

**Pathogenicity factors:**

- **Enzymes: plasmacoagulase** (a fibrin envelope — protection from phagocytosis, the main marker of *S. aureus*), hyaluronidase, fibrinolysin, lecithinase, DNase, beta-lactamase;
- **Toxins:** hemolysins (α, β, γ, δ), **leukocidin**, **exfoliative toxin** (scalded skin syndrome), **toxic shock syndrome toxin (TSST-1)** — a superantigen, **enterotoxins** (A–E; thermostable — food poisoning, topic 23);
- **Surface structures: protein A** (binds to the Fc part of IgG and disrupts opsonization), teichoic acids, microcapsule.

**Diseases:** skin and soft tissue infections (furuncle, carbuncle, abscess, phlegmon, mastitis), pneumonia, osteomyelitis, endocarditis, meningitis, sepsis, food poisoning, toxic shock syndrome, hospital-acquired infections.

> **CLINICAL SIGNIFICANCE**
> **MRSA** (methicillin-resistant *S. aureus*) is an important problem in hospital and community settings. The resistance is linked to an altered penicillin-binding protein, so **all beta-lactams** are considered ineffective (topic 9). Treatment is chosen on the basis of the antibiogram.

## 3. Streptococci (*Streptococcus*)

**Morphology:** gram-positive cocci, dividing in one plane to form **chains**; nonmotile, nonspore-forming; some species are encapsulated.

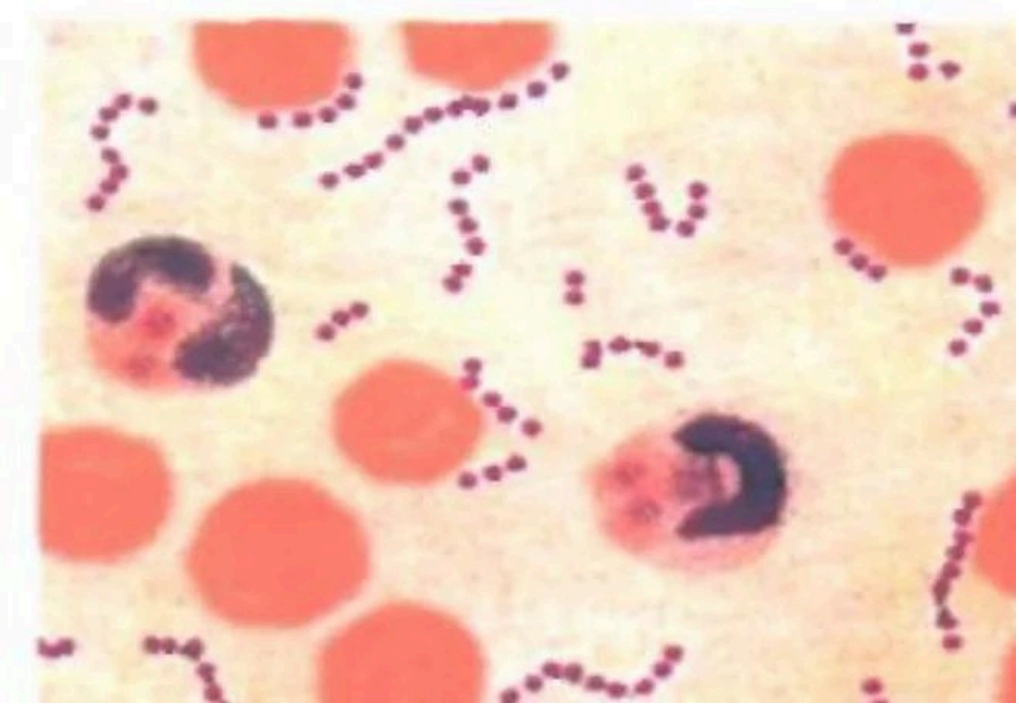


**Figure 16.2.** *Streptococci: gram-positive cocci forming chains in pus. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties:** fastidious — requires medium with added blood or serum; small, transparent colonies; **catalase-negative** (the main difference from staphylococci).

**Classification:**

1. **By type of hemolysis:** beta-hemolytic (a completely clear zone), alpha-hemolytic (a green zone — the viridans group, pneumococcus), gamma (no hemolysis).
2. **Serological groups by Lancefield** (by the cell wall polysaccharide): A, B, C, D and others.

**Table 16.2.** *Clinically significant streptococci*

| Species/group | Hemolysis | Differential test | Diseases |
|---|---|---|---|
| ***S. pyogenes*** (group A) | Beta | **Sensitive to bacitracin** | Tonsillitis, scarlet fever, erysipelas, impetigo, phlegmon, sepsis; complications — **rheumatic fever, acute glomerulonephritis** |
| *S. agalactiae* (group B) | Beta | **CAMP test positive** | Sepsis and meningitis in newborns |
| *S. pneumoniae* | Alpha | **Sensitive to optochin**, dissolves in bile | Pneumonia, otitis, sinusitis, meningitis (topic 18) |
| Viridans group | Alpha | Resistant to optochin | Oral microflora; **subacute endocarditis**, caries |
| Enterococci (group D) | Gamma/alpha | Grows in 6.5% NaCl, positive on bile-esculin medium | Urinary tract and abdominal cavity infections; multidrug resistance |

**Pathogenicity factors of group A streptococcus: M protein** (the main virulence factor — antiphagocytic, type-specific), capsule (hyaluronic acid), **streptolysin O and S**, **erythrogenic (pyrogenic) toxin** (the rash in scarlet fever), streptokinase, hyaluronidase, DNase.

**CLINICAL SIGNIFICANCE**

Group A streptococcal antigens resemble antigens of heart, joint and kidney tissue (**cross-reactive antigens**). That is why **rheumatic fever** or **acute glomerulonephritis** may develop 2–4 weeks after tonsillitis. For this very reason, complete treatment of streptococcal tonsillitis is important.

**MEMORY AID**

Remember the complications of streptococcus: **"High ASO — had tonsillitis."** The antistreptolysin O (ASO) titer shows a past group A streptococcal infection and helps in the diagnosis of rheumatic fever.

## 4. *Pseudomonas aeruginosa*

**Morphology:** gram-negative, straight or slightly curved rod; **monotrichous** (a single polar flagellum) — motile; nonspore-forming, with a slime layer.

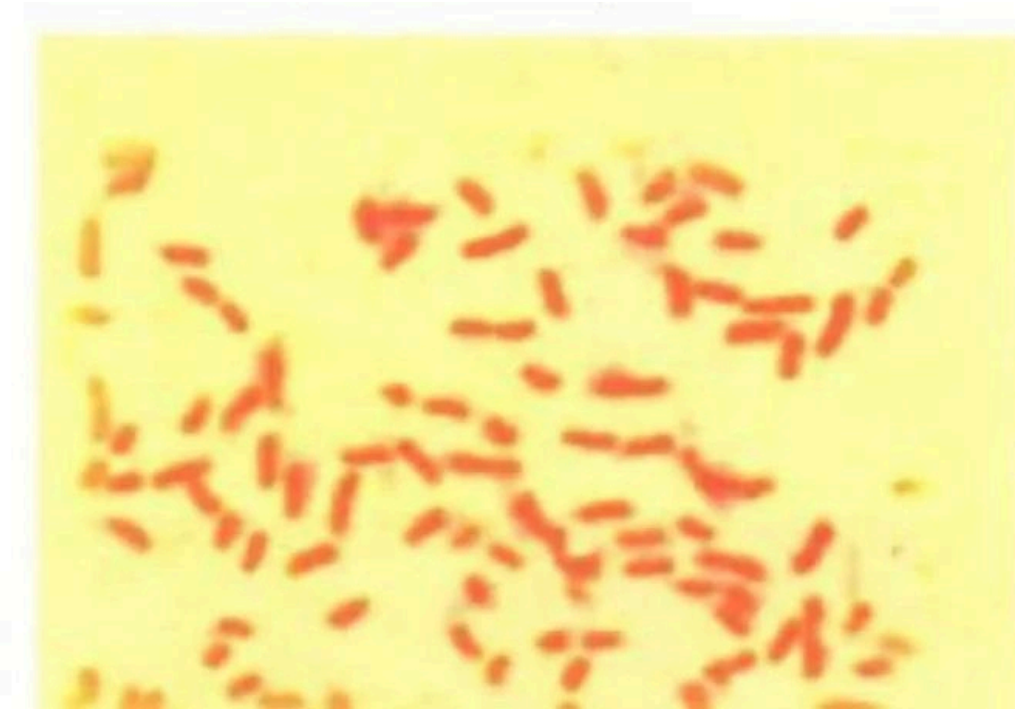


**Figure 16.3.** *Pseudomonas aeruginosa — gram-negative motile rods. Source: Muhamedov I.M. et al., 2019.*

**Cultural and biochemical properties: obligate aerobe**; grows on ordinary media, not demanding of nutrients; **oxidase-positive** (the main difference from Enterobacteriaceae); does not break down lactose. **Pyocyanin** (blue-green) and pyoverdine pigments stain the medium; it has a characteristic **hyacinth-like odor**.

**Pathogenicity factors:** exotoxin A (stops protein synthesis), elastase, proteases, phospholipase C, pyocyanin, LPS, **biofilm formation**.

**Clinical significance:** a typical **opportunistic and hospital-acquired** pathogen. It causes infection in burn patients, in patients on artificial ventilation, in patients with catheters and in the immunocompromised: suppuration of wounds (blue-green pus), pneumonia, urinary tract infection, otitis, eye infections, sepsis.

**CAUTION**

*P. aeruginosa* is resistant to many disinfectants and persists in wet places (sinks, respiratory equipment, solutions). That is why infection control in hospitals is directed at the regimen related to exactly these wet environments and instruments.

## 5. Laboratory diagnosis

**Material:** pus, wound discharge, sputum, blood (in sepsis), urine, cerebrospinal fluid, puncture fluid; in hospital surveillance — a nasal swab (staphylococcal carriage).

**Bacterioscopic method** — a Gram-stained smear: gram-positive clustered cocci (staphylococcus), chained cocci (streptococcus), gram-negative rods (*Pseudomonas*). It gives a preliminary direction.

**Bacteriological method (main):**

1. **Day 1:** the material is inoculated onto blood agar and a selective medium (egg-yolk salt agar for staphylococcus).
2. **Day 2:** the colonies are studied (hemolysis, pigment, odor), a smear is made, and a pure culture is collected.
3. **Day 3:** differential tests and antibiogram.
4. **Day 4:** final report.

**Table 16.3.** *The main differential tests*

| Test | Result and interpretation |
|---|---|
| **Catalase** | Staphylococcus (+) ↔ streptococcus (−) |
| **Plasmacoagulase** | *S. aureus* (+) ↔ other staphylococci (−) |
| **Lecithinase** | A cloudy zone around the colony in *S. aureus* |
| **Sensitivity to bacitracin** | Group A streptococcus is sensitive |
| **CAMP test** | Group B streptococcus is positive |
| **Sensitivity to optochin** | Pneumococcus is sensitive ↔ viridans is resistant |
| **Oxidase** | *P. aeruginosa* (+) ↔ Enterobacteriaceae (−) |
| Pigment and odor | Pyocyanin, hyacinth-like odor — *P. aeruginosa* |

**Serological method:** determining the ASO titer (assessing a past group A streptococcal infection and its complications).

**Additional:** phage typing (epidemiological investigation), detecting toxin genes by PCR, MALDI-TOF identification.

## 6. Independent work: Opportunistic anaerobic cocci

**Table 16.4.** *Brief characteristics of anaerobic cocci*

| Representative | Gram | Normal location | Clinical significance |
|---|---|---|---|
| *Peptococcus* | Gram-positive coccus | Skin, mouth, intestine | Abscesses, mixed purulent infections |
| *Peptostreptococcus* | Gram-positive coccus (chain) | Mouth, intestine, vagina | Gynecologic infections, aspiration pneumonia, abscesses |

| Representative | Gram | Normal location | Clinical significance |
|---|---|---|---|
| *Veillonella* | **Gram-negative** coccus | Oral cavity, intestine | Oral cavity infections, mixed anaerobic infections |

**General features:** members of the normal microflora; against a background of injury, circulatory disturbance, surgery or reduced immunity they cause an **endogenous** infection. They usually occur as part of a **mixed (polymicrobial)** infection.

**Clinical signs:** a **foul smell** of the pus, gas in the tissue, necrosis, poor response to the usual antibiotics — grounds for suspecting a mixed anaerobic infection.

**Special features of laboratory diagnosis:**

1. **Collecting material** — not with a swab but as a **puncture sample or tissue**, taken with limited contact with air;
2. **Transport** — in a special anaerobic transport medium, fast;
3. **Culture** — under anaerobic conditions (anaerostat, gas packs), on enriched blood agar, for 48–72 hours;
4. **Identification** — Gram stain, oxygen tolerance, biochemical tests, gas-liquid chromatography (fatty acid spectrum), MALDI-TOF, PCR.

> **REMEMBER**
> The most frequent reason for "missing" an anaerobic infection is **incorrect collection and late delivery of the material**. In a sample transported under aerobic conditions the anaerobes die, and the result comes out falsely negative.

## 7. Practical skill: collecting material from a purulent wound

1. Clean the skin around the wound with an antiseptic (the antiseptic must not get into the sample).
2. Remove the dried surface layer or crust — the material is taken from the **deep, active** part.
3. The preferred method is taking a **puncture sample with a syringe** (especially in a closed abscess and when anaerobes are suspected); a swab is a less informative option.
4. Label the sample: patient, site, time of collection, whether antibiotics have been taken.
5. Deliver it to the laboratory in a transport medium as fast as possible (within 1–2 hours); a special medium is essential for anaerobic examination.
6. In the laboratory: Gram smear + blood agar (aerobic) + anaerobic culture + selective medium.

## 8. Review questions

1. State the morphological and cultural properties of staphylococci. What is the selective medium based on?
2. How do you distinguish *S. aureus* from other staphylococci?
3. Explain the pathogenicity factors of staphylococcus by group. What is the role of protein A?
4. How are streptococci classified? What are the Lancefield groups?
5. What are the pathogenicity factors and the complications of group A streptococcus?
6. What is ASO and when is it determined?
7. State the characteristic features of *P. aeruginosa* (pigment, odor, oxidase).
8. Describe the algorithm of bacteriological diagnosis in purulent infections.

9. In which cases do anaerobic cocci cause disease? What are the clinical signs?
10. What is special about collecting and transporting material in anaerobic infection?

## Test questions

1. The arrangement of staphylococci:
   A) Chain
   B) Grape-like cluster
   C) Pairs
   D) Packets (sarcina)
   E) Single

2. What the selective medium for staphylococci is based on:
   A) Addition of blood
   B) Resistance to a high concentration of NaCl
   C) Low pH
   D) Anaerobic conditions
   E) Addition of an antibiotic

3. The main differential enzyme of *S. aureus*:
   A) Catalase
   B) Plasmacoagulase
   C) Oxidase
   D) Urease
   E) Beta-galactosidase

4. The staphylococcal factor that binds to the Fc part of IgG and disrupts opsonization:
   A) M protein
   B) Protein A
   C) Teichoic acid
   D) Leukocidin
   E) Exfoliative toxin

5. How the toxic shock syndrome toxin (TSST-1) acts:
   A) It activates T cells en masse as a superantigen
   B) It stops protein synthesis
   C) It perforates the membrane
   D) It stops the release of acetylcholine
   E) It activates adenylate cyclase

6. The main test that distinguishes streptococci from staphylococci:
   A) Coagulase
   B) Catalase (streptococcus is negative)
   C) Oxidase
   D) Indole
   E) Urease

7. The main virulence factor of group A streptococcus:
   A) Capsule
   B) M protein
   C) Streptokinase
   D) DNase
   E) Erythrogenic toxin

8. A complication that may develop 2–4 weeks after tonsillitis:
   A) Botulism
   B) Rheumatic fever and acute glomerulonephritis
   C) Tetanus
   D) Diphtheria
   E) Cholera

9. The susceptibility test used to identify group A streptococcus:
   A) Optochin
   B) Bacitracin
   C) Novobiocin
   D) Penicillin
   E) Furazolidone

10. The test used to identify group B streptococcus:
    A) CAMP test
    B) Catalase
    C) Coagulase
    D) Indole
    E) Oxidase

11. A cultural property of *P. aeruginosa*:
    A) Anaerobe, lactose-positive
    B) Obligate aerobe, oxidase-positive, pyocyanin pigment
    C) Microaerophile, urease-positive
    D) Facultative anaerobe, $H_2S$-positive
    E) Grows only on blood agar

12. The test that distinguishes Enterobacteriaceae from *Pseudomonas*:
    A) Catalase
    B) Oxidase
    C) Coagulase
    D) Hemolysis
    E) Motility

13. A pathogen common in burn patients and in patients on artificial ventilation:
    A) *S. pyogenes*
    B) *P. aeruginosa*
    C) *S. saprophyticus*
    D) *Veillonella*
    E) *S. agalactiae*

14. A gram-negative anaerobic coccus:
    A) *Peptococcus*
    B) *Peptostreptococcus*
    C) *Veillonella*
    D) *Staphylococcus*
    E) *Enterococcus*

15. The material when an anaerobic infection is suspected:
    A) Taken with an ordinary swab and kept at room temperature
    B) A puncture sample/tissue, delivered quickly in anaerobic transport medium
    C) Frozen
    D) Preserved in formalin
    E) Dried

## Clinical cases

### *Case 1*

A pus smear from a furuncle showed gram-positive clustered cocci. The culture gave golden-yellow colonies on blood agar, with beta-hemolysis; it was catalase-positive and coagulase-positive.

**Question:** Which is the pathogen, and which tests were decisive?

**Answer:** *Staphylococcus aureus*. Catalase positivity distinguishes a staphylococcus from a streptococcus, and coagulase positivity distinguishes *S. aureus* from other staphylococci.

### *Case 2*

An 8-year-old child developed joint pain and a heart murmur 3 weeks after tonsillitis. The ASO titer is high.

**Question:** What is this condition linked to?

**Answer:** A complication after group A beta-hemolytic streptococcal infection — acute rheumatic fever. The cause is the similarity between streptococcal antigens and human tissue antigens (cross-reactive antigens). ASO confirms a past streptococcal infection.

### *Case 3*

A burn patient in intensive care developed blue-green pus and a characteristic smell in the wound. The culture grew an oxidase-positive, lactose-negative gram-negative rod, and the medium turned blue-green.

**Question:** Which is the pathogen and what measures are important?

**Answer:** *Pseudomonas aeruginosa*. Measures — treatment according to the antibiogram, local treatment of the wound, control of wet equipment and solutions, and strict observance of infection control rules.

## Quick review

**Table 16.5.** *Topic 16 — key concepts*

| Concept | Essence |
|---|---|
| Staphylococcus | Gram+, clusters, catalase +, resistant to 7.5–10% NaCl |

| Concept | Essence |
|---|---|
| *S. aureus* | Coagulase +, golden pigment, beta-hemolysis |
| Factors of *S. aureus* | Coagulase, protein A, leukocidin, TSST-1, exfoliatin, enterotoxin |
| MRSA | Altered penicillin-binding protein — all beta-lactams are ineffective |
| Streptococcus | Gram+, chains, catalase − |
| Group A (*S. pyogenes*) | M protein, sensitive to bacitracin; rheumatic fever, glomerulonephritis; ASO |
| Group B | CAMP +, sepsis/meningitis in newborns |
| *P. aeruginosa* | Gram−, aerobe, oxidase +, pyocyanin, hyacinth-like odor, biofilm |
| Anaerobic cocci | *Peptococcus*, *Peptostreptococcus* (Gram+), *Veillonella* (Gram−); mixed infection |
| Material | A puncture sample is preferred; for anaerobes — a special transport medium |

TOPIC 17

# Wound infections: causative agents of tetanus and gas gangrene. Non-clostridial anaerobic infections

## 1. Learning objectives

In this class the student will master the following:

- To know the morphological, cultural and toxigenic properties of the causative agents of tetanus and gas gangrene;
- to explain the main links in their pathogenesis and their clinical expression;
- to master the algorithm of laboratory diagnosis in anaerobic wound infections and the principles of specific prophylaxis and treatment;
- Independent work: to know the non-clostridial anaerobic pathogens (*Bacteroides, Prevotella, Fusobacterium*) and their diagnosis.

## 2. General description: clostridia

**Clostridia** (*Clostridium*) are gram-positive, **spore-forming**, **obligate anaerobic** rods. The position of the spore and the shape of the cell are important signs in preliminary identification (topic 3). The natural habitat is soil and the intestines of humans and animals.

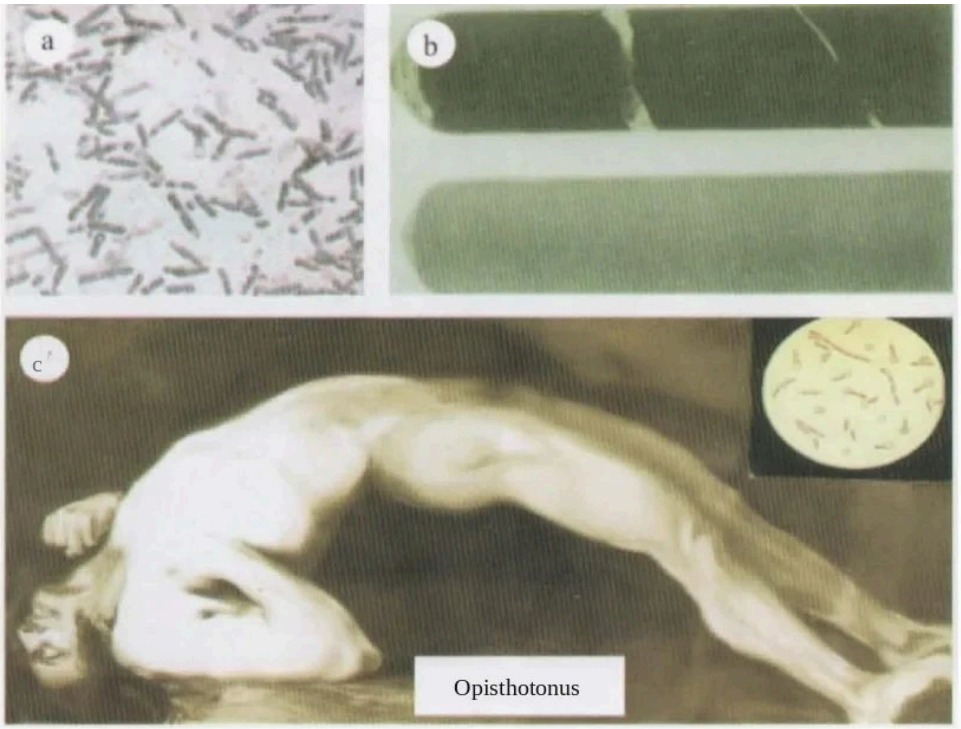



**Figure 17.1.** *Causative agents of anaerobic wound diseases: a) pure culture of Cl. perfringens (Gram stain); b) its growth in a special medium; c) the classic clinical picture of tetanus. Source: Aliyev Sh.R. et al., 2013.*

## 3. The causative agent of tetanus — *Clostridium tetani*

**Morphology:** gram-positive, thin, long rod; **peritrichous** (motile); forms a **round, terminally located spore** — the cell takes the shape of a **"drumstick"** or a "matchstick."

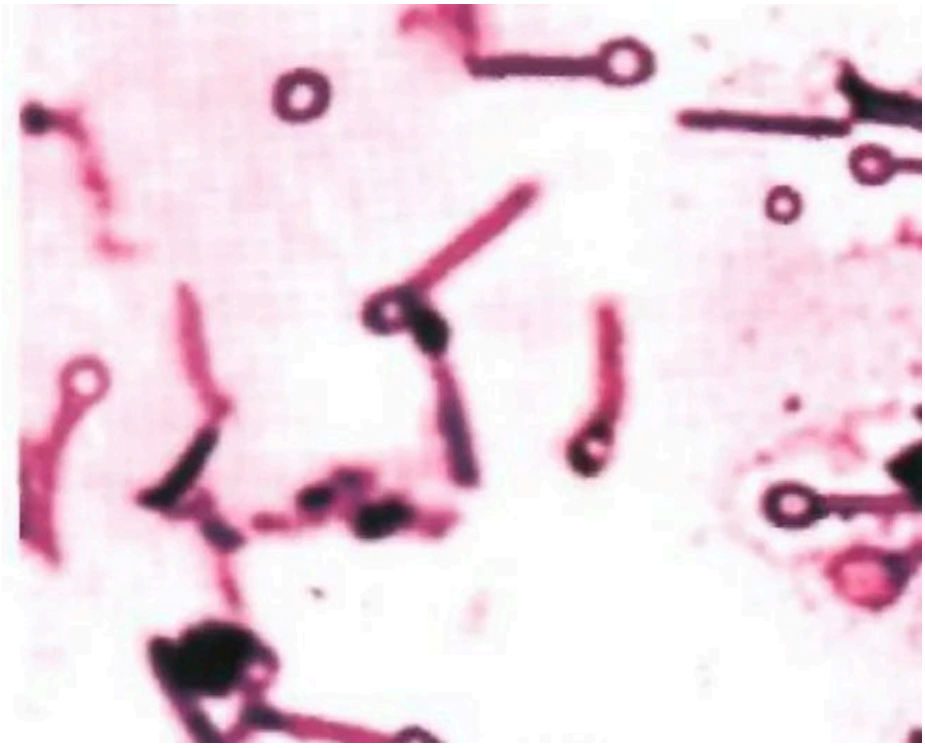


**Figure 17.2.** *The causative agent of tetanus: gram-positive rods with terminal spores — the "drumstick" appearance. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties:** obligate anaerobe; in Kitt–Tarozzi medium it grows, making the medium turbid and giving a characteristic smell; on blood agar it gives thin, "creeping" growth and hemolysis.

**Pathogenicity factors — two exotoxins:**

- **Tetanospasmin (neurotoxin)** — the main toxin; it reaches the central nervous system along the nerve fibers and through the blood, and **stops the release of inhibitory mediators (glycine and GABA)**. As a result the inhibition reaching the muscles is lost and a constant **tonic contraction (spasm)** develops;
- **Tetanolysin** — lyses erythrocytes and contributes to local tissue damage.

**The main condition of pathogenesis:** the pathogen stays **local** in the wound and does not spread into the blood — the clinical picture is determined only by the **toxin**. That is why tetanus is a disease of **toxinemia**.

**Clinical picture:** contraction of the masseter muscles (**trismus**), drawing of the facial muscles ("risus sardonicus"), difficulty swallowing, then generalized tonic convulsions and arching of the trunk (**opisthotonus**); consciousness is preserved.

**REMEMBER**

In tetanus consciousness **is not impaired** — the patient feels everything. This is because the toxin acts selectively on inhibitory neurons.

**CLINICAL SIGNIFICANCE**

In tetanus **no immunity develops**: the amount of toxin that causes disease is too small for an immune response. That is why **vaccination** is done even after recovery from the disease.

## 4. Causative agents of gas gangrene

**The main pathogen — *Clostridium perfringens***; in addition *C. novyi*, *C. septicum* and *C. histolyticum* take part. It often occurs as a **mixed (several species)** infection.

***C. perfringens* features:** large gram-positive rod, **nonmotile** (rare among clostridia), **encapsulated**; in the body it often does not form spores. On blood agar it gives **double-zone hemolysis**, and in milk medium **"stormy fermentation"** (violent gas formation).

**Pathogenicity factors:** numerous toxins and enzymes — **alpha-toxin (lecithinase)** (damages the membrane, the main factor), collagenase, hyaluronidase, proteases, DNase, neuraminidase; **enterotoxin** (in some strains — food poisoning).

**Pathogenesis:** when an oxygen-free environment forms in the tissue (a crushed wound with impaired circulation), clostridia multiply; enzymes dissolve tissue, **gas** forms from the fermentation of carbohydrates (crepitus under the skin), necrosis spreads, and severe intoxication develops.

> **MEMORY AID**
> The triad of gas gangrene: **gas** (crepitus) + **necrosis** + **severe intoxication**. In tetanus, by contrast, **convulsions**, with consciousness intact.

## 5. Laboratory diagnosis

**In tetanus the diagnosis is made mainly clinically** — waiting for the laboratory result delays treatment. Laboratory examination has an auxiliary role: wound discharge, necrotic tissue and operative material are examined; detection is possible by bacterioscopy, anaerobic culture and the **toxin neutralization test**.

**Laboratory diagnosis in gas gangrene:**

1. **Material:** a piece of tissue taken from the wound, discharge from the edge of the necrosis, puncture fluid, blood (when generalized).
2. **Bacterioscopy** — Gram stain: large gram-positive rods, capsule; gives a rapid direction.
3. **Anaerobic culture** — Kitt–Tarozzi medium, Wilson–Blair medium (black colonies), blood agar; in an anaerostat for 24–48 hours.
4. **Identification** — morphology, lecithinase activity (a zone on egg-yolk agar), the reaction in milk medium, biochemical tests, determining the toxin type with the neutralization test.
5. **Modern methods** — detecting toxin genes by PCR, MALDI-TOF.

**Table 17.1.** *Comparison of tetanus and gas gangrene*

| Feature | Tetanus | Gas gangrene |
|---|---|---|
| Causative agent | *C. tetani* | *C. perfringens* and others (mixed) |
| Spore | Terminal ("drumstick") | Central/subterminal; often absent in the body |
| Motility | Motile (peritrichous) | *C. perfringens* is nonmotile |
| Main factor | Tetanospasmin (neurotoxin) | Alpha-toxin (lecithinase) and enzymes |
| Spread | Stays local, the toxin spreads | Spreads in the tissue |
| Clinical picture | Trismus, convulsions, opisthotonus; consciousness intact | Gas, necrosis, intoxication |

| Feature | Tetanus | Gas gangrene |
|---|---|---|
| Diagnosis | Mainly clinical | Clinical + bacteriological |

## 6. Principles of specific prophylaxis and treatment

**Prevention of tetanus:**

- **Planned (active) — tetanus toxoid** (as part of combination vaccines), according to the national immunization schedule, later with booster vaccinations;
- **Emergency (when a wound is received)** — depending on the vaccination history, toxoid and/or ready-made antibodies (passive immunization) are given; the exact scheme is determined by a **physician** on the basis of the current protocol;
- **Primary surgical debridement of the wound** — removal of necrotic tissue and foreign bodies is the main part of prophylaxis.

**In gas gangrene:** the main measure is **surgery** (wide surgical debridement, necrectomy); antibacterial therapy and intensive care; oxygen-based methods; antitoxic serum is used in some protocols. All prescriptions are made by the physician.

> **CAUTION**
> In anaerobic wound infections **the time factor** is decisive: the earlier the surgical treatment is done, the better the outcome. Treatment cannot be delayed to wait for the laboratory answer.

## 7. Independent work: Non-clostridial anaerobic infections

Non-spore-forming (non-clostridial) anaerobes are members of the normal microflora and a frequent cause of endogenous infections.

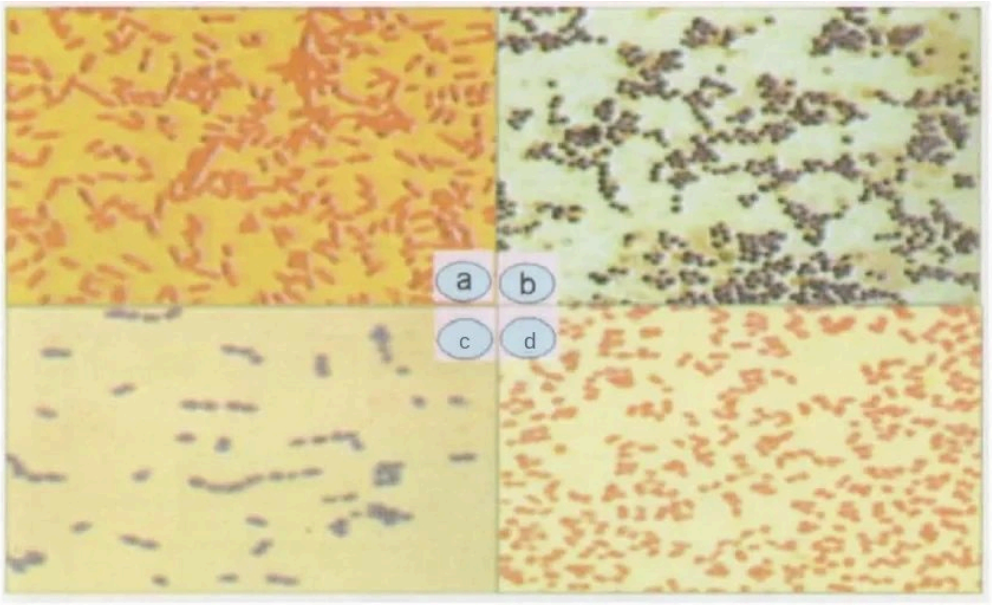



**Figure 17.3.** *Morphotinctorial properties of the causative agents of non-spore-forming anaerobic purulent-inflammatory processes: a) Bacteroides fragilis; b) Peptococcus niger; c) Peptostreptococcus; d) Veillonella parvula. Source: Aliyev Sh.R. et al., 2013.*

**Table 17.2.** *The main non-clostridial anaerobes*

| Group | Gram | Normal location | Clinical significance |
|---|---|---|---|
| ***Bacteroides fragilis*** group | Gram-negative rod | Large intestine | Intra-abdominal abscesses, peritonitis, sepsis; produces beta-lactamases |
| ***Prevotella*** | Gram-negative rod | Oral cavity, urogenital tract | Odontogenic infections, gynecologic infections |
| ***Fusobacterium*** | Gram-negative, **spindle-shaped** rod | Oral cavity | Necrotizing tonsillitis (Vincent), lung abscesses, periodontitis |
| *Peptostreptococcus*, *Veillonella* | Cocci | Mouth, intestine, vagina | Mixed purulent infections (topic 16) |

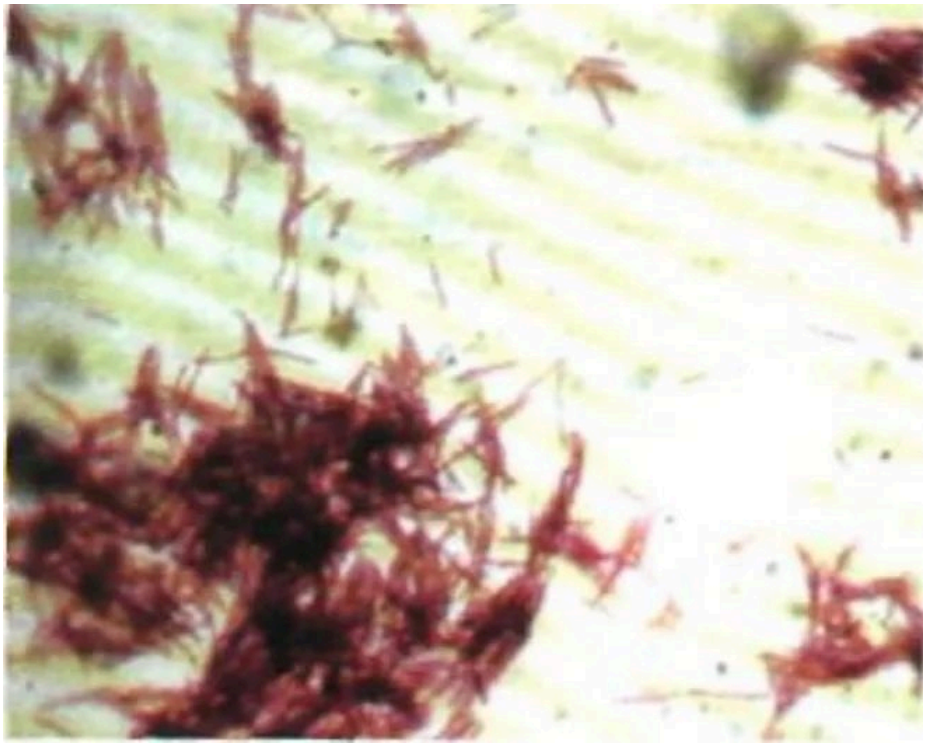


**Figure 17.4.** *Fusobacteria — an anaerobic representative of the oral microflora (spindle-shaped rods). Source: Muhamedov I.M. et al., 2019.*

**General features of pathogenesis:** endogenous origin; a fall in the oxygen level in the tissue (ischemia, necrosis, foreign body, surgery) is the decisive condition; it often occurs **together with aerobes (synergism)**.

**Signs that raise suspicion:** a foul smell of the pus, gas formation, necrosis, no growth on ordinary culture even though the smear shows many microbes, poor response to commonly used antibiotics.

**Diagnosis:** properly collected material (puncture sample/tissue) + anaerobic transport + anaerobic culture; in identification — Gram stain, relation to oxygen, biochemical tests, gas-liquid chromatography, MALDI-TOF, PCR.

> **REMEMBER**
>
> **Vincent's angina** (necrotizing ulcerative tonsillitis) — the combined action of oral anaerobes (fusobacteria) and spirochetes; the diagnosis is made by seeing both morphotypes together in a smear.

## 8. Practical skill: collecting and culturing material for an anaerobic infection

1. The skin around the wound is cleaned with an antiseptic; wait until the antiseptic dries.

2. The material is taken as a **puncture sample with a syringe** or as a **piece of tissue** from the edge of the necrosis; the syringe must be free of air.
3. The sample is placed in **anaerobic transport medium**; it is not cooled and is delivered as fast as possible.
4. In the laboratory the following are inoculated at the same time: anaerobic blood agar, Kitt–Tarozzi medium, Wilson–Blair medium and an **aerobic control** plate (to distinguish whether the culture is aerobic or anaerobic).
5. In an anaerostat at 37 °C for 48–72 hours; the aerobic plate — 24 hours.
6. Smears (Gram) and differential tests are made from the colonies that grew; a culture that did not grow on the aerobic plate but grew anaerobically is judged to be an **obligate anaerobe**.

## 9. Review questions

1. State the general properties of clostridia.
2. What is the morphological sign of *C. tetani*? Where is the spore located?
3. Explain the mechanism of action of tetanospasmin. Why is consciousness not impaired?
4. Why does immunity not develop in tetanus?
5. How does *C. perfringens* differ from other clostridia?
6. How is gas formed in the pathogenesis of gas gangrene?
7. How does the diagnosis of tetanus differ from that of gas gangrene?
8. What do the active and passive parts of tetanus prophylaxis consist of?
9. Which are the non-clostridial anaerobes, and when do they cause disease?
10. State the rules for collecting material in anaerobic infection.

## Test questions

1. The position of the spore of *C. tetani*:
   A) Central
   B) Subterminal
   C) Terminal (“drumstick”)
   D) No spore
   E) Exosporial

2. The main mechanism of action of tetanus toxin:
   A) It stops protein synthesis
   B) It stops the release of inhibitory mediators (glycine, GABA)
   C) It stops the release of acetylcholine
   D) It activates adenylate cyclase
   E) It perforates the membrane

3. The state of consciousness in tetanus:
   A) Lost
   B) Preserved
   C) Lost only at the beginning
   D) Passes into coma
   E) Delirium is seen

4. Tonic contraction of the masseter muscles:
   A) Opisthotonus
   B) Trismus
   C) Crepitus
   D) Ptosis
   E) Diplopia

5. Immunity after the disease in tetanus:
   A) Persists for life
   B) Does not develop, so vaccination is needed
   C) Lasts 10 years
   D) Is only passive
   E) Develops only in children

6. The active means of tetanus prophylaxis:
   A) Antitoxic serum
   B) Tetanus toxoid
   C) An antibiotic
   D) Bacteriophage
   E) Immunoglobulin

7. The main causative agent of gas gangrene:
   A) *C. tetani*
   B) *C. perfringens*
   C) *C. botulinum*
   D) *C. difficile*
   E) *B. fragilis*

8. The feature that distinguishes *C. perfringens* from other clostridia:
   A) Nonmotility and encapsulation
   B) A terminal spore
   C) Gram-negativity
   D) Aerobic nature
   E) Absence of spores

9. The main toxin of *C. perfringens*:
   A) Tetanospasmin
   B) Alpha-toxin (lecithinase)
   C) Botulinum toxin
   D) Erythrogenic toxin
   E) Diphtheria toxin

10. The physical sign of gas accumulation under the skin in gas gangrene:
    A) Fluctuation
    B) Crepitus
    C) Zone of fluctuation
    D) Percussion dullness
    E) Murmur

11. The characteristic phenomenon that *C. perfringens* produces in milk medium:
    A) Slow clotting
    B) “Stormy fermentation”
    C) No clotting
    D) Darkening of color
    E) Formation of a film

12. The most informative material in anaerobic infections:
    A) A swab from the surface
    B) A puncture sample or a piece of tissue
    C) Dried crust
    D) Urine
    E) Hair

13. A frequent causative agent of intra-abdominal anaerobic infections:
    A) *Bacteroides fragilis*
    B) *Fusobacterium*
    C) *Prevotella*
    D) *Veillonella*
    E) *Peptococcus*

14. A spindle-shaped gram-negative anaerobic rod:
    A) *Bacteroides*
    B) *Fusobacterium*
    C) *Prevotella*
    D) *Clostridium*
    E) *Veillonella*

15. The causative agents of Vincent’s angina:
    A) Fusobacteria and spirochetes together
    B) Only streptococcus
    C) Only staphylococcus
    D) Diphtheria bacillus
    E) Pneumococcus

## Clinical cases

### *Case 1*

A patient who received a deep puncture wound to the foot during field work developed, after 8 days, difficulty opening the mouth, drawing of the facial muscles, and then generalized convulsions. Consciousness is intact.

**Question:** What is the probable diagnosis and how is it made?

**Answer:** Tetanus. The diagnosis is **clinical** — typical trismus, risus sardonicus, convulsions and preserved consciousness. Without waiting for laboratory confirmation, urgent measures are taken in the manner prescribed by the physician; the wound is debrided surgically.

### *Case 2*

A patient with a crush wound developed, on day 2, edema around the wound, crepitus under the skin, a foul smell and rapidly spreading necrosis. The smear shows large gram-positive rods.

**Question:** What should be considered and what is the first measure?

**Answer:** Gas gangrene (*C. perfringens* and others). The first and decisive measure is **urgent wide surgical debridement**; laboratory examination is carried out in parallel.

### *Case 3*

An abscess developed after abdominal surgery. There was no growth on the usual aerobic culture, but the pus smelled foul and the smear showed many gram-negative rods.

**Question:** Why was the culture negative and how is it corrected?

**Answer:** The pathogen is an obligate anaerobe (probably *Bacteroides fragilis*). The material should have been collected in anaerobic transport medium and cultured under anaerobic conditions for 48–72 hours. The repeat examination is done in this way.

## Quick review

**Table 17.3.** *Topic 17 — key concepts*

| Concept | Essence |
|---|---|
| Clostridia | Gram+, spore-forming, obligate anaerobe |
| *C. tetani* | Terminal spore, motile; tetanospasmin |
| Pathogenesis of tetanus | Local infection + toxinemia; consciousness intact |
| Clinical picture of tetanus | Trismus, risus sardonicus, opisthotonus |
| Prevention of tetanus | Toxoid (active) + passive immunization; surgical debridement |
| *C. perfringens* | Nonmotile, encapsulated, alpha-toxin (lecithinase) |
| Gas gangrene | Gas + necrosis + intoxication; the main treatment is surgery |
| Non-clostridial anaerobes | *Bacteroides*, *Prevotella*, *Fusobacterium* |
| Signs that raise suspicion | Foul smell, gas, necrosis, no growth on aerobic culture |
| Material | Puncture sample/tissue + anaerobic transport + anaerobic culture |

TOPIC 18

# Airborne infections: diphtheria, pertussis, meningococci. Pneumococcus and *Haemophilus influenzae*

## 1. Learning objectives

In this class the student will master the following:

- To know the biological properties of the causative agents of diphtheria, pertussis and meningococcal infection;
- to explain their pathogenicity factors and the pathogenesis of the diseases they cause;
- to master the algorithm of laboratory diagnosis of each infection (material, medium, identification, determining toxigenicity);
- Independent work: to know the description, diseases and diagnosis of pneumococcus and *Haemophilus influenzae*.

## 2. The causative agent of diphtheria — *Corynebacterium diphtheriae*

**Morphology:** gram-positive, straight or slightly curved rod; the ends are thickened owing to **volutin granules** (resembling a matchstick); the cells lie at an angle to each other, forming the shape of **the Latin letters V, X, Y** (**pleomorphism and pleomorphic arrangement**). Nonmotile, nonspore-forming, noncapsulated.

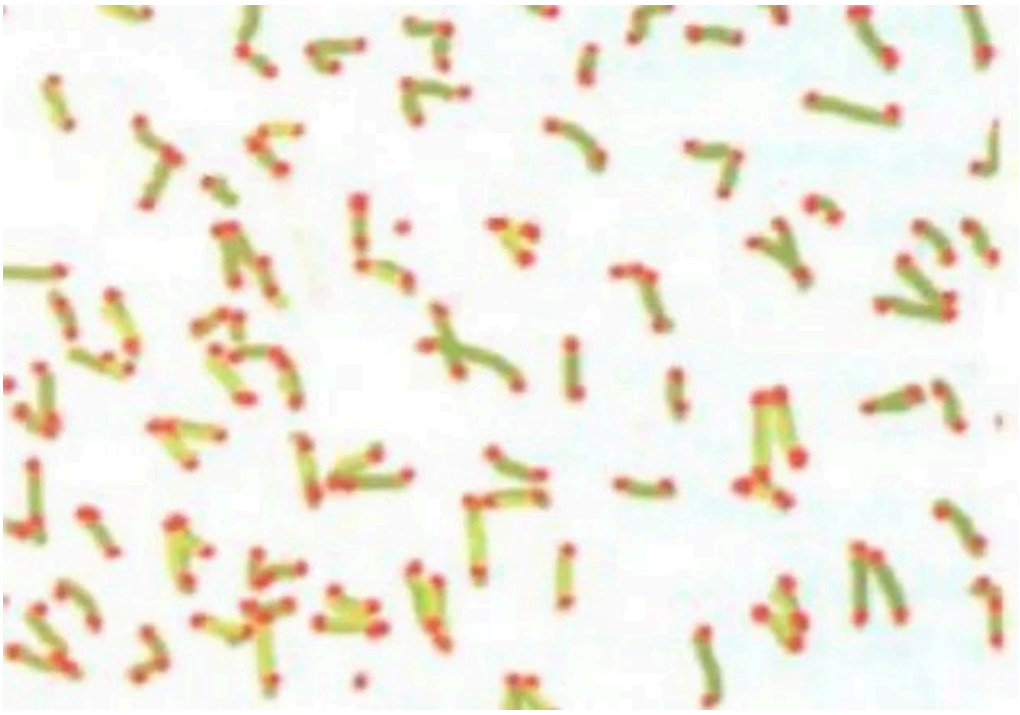


**Figure 18.1.** *The diphtheria bacillus stained by the Neisser method: dark volutin granules at the ends of the cells and angular arrangement. Source: Muhamedov I.M. et al., 2019.*

**Staining:** Gram (positive), **Neisser** (the cell is yellow-brown, the volutin granules dark blue-black), **Löffler's blue**.

**Cultural properties:** facultative anaerobe; fastidious — requires **serum medium** (Löffler's coagulated serum) and **tellurite-containing medium** (selective; the tellurite is reduced, making the colony **black/gray**).

**Biovars:** *gravis* (large, rough, “resembling a daisy”), *mitis* (small, smooth, black, shiny), *intermedius*.

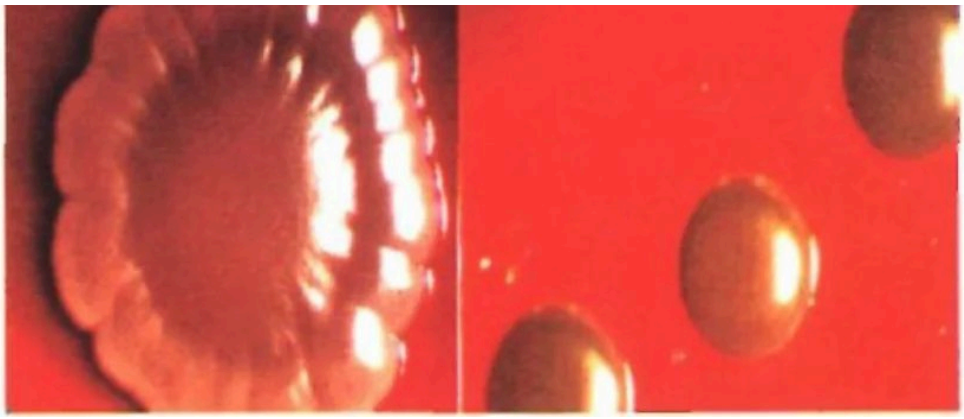


**Figure 18.2.** *Colonies of the gravis (left) and mitis (right) biovars of C. diphtheriae. Source: Muhamedov I.M. et al., 2019.*

**Pathogenicity factor — diphtheria exotoxin.** The toxin is encoded by **prophage genes**: only **lysogenic (toxigenic)** strains cause disease (phage conversion — topic 11). The toxin’s mechanism of action is blocking **elongation factor EF-2** in the cell by ADP-ribosylation → **stopping of protein synthesis** → cell necrosis.

**Pathogenesis:** the pathogen stays **local** at the portal of entry (throat, nose, rarely the eye, skin, wound); the toxin is absorbed into the blood and damages **the heart muscle, nervous system and kidneys**. The local change is fibrinous inflammation, with formation of a **dense gray-white membrane**; when the membrane is removed, the mucosa bleeds.

> **CLINICAL SIGNIFICANCE**
> The main danger of diphtheria is **asphyxia** (the laryngeal membrane blocks the airway — “true croup”) and the late complications: **myocarditis**, polyneuritis, nephrosis. That is why antitoxic serum is given on clinical suspicion, without waiting for the laboratory answer.

**Laboratory diagnosis:**

1. **Material:** a swab from the throat and nose taken **at the edge of the membrane**; before antibiotics and throat rinses are used, on an empty stomach.
2. **Bacterioscopy** — Neisser method: rods with volutin granules arranged at angles (**presumptive result**).
3. **Bacteriological (main):** inoculation onto tellurite-containing and serum media → black colonies → pure culture → biochemical identification (cystinase-positive — **Pizu test**; urease-negative — differs from *C. ulcerans*).
4. **Determining toxigenicity (the decisive step): precipitation in gel (Elek method)** — precipitation lines between a paper strip impregnated with antitoxin and the culture; also detecting the **tox gene** by PCR, and the biological test.
5. **Serological:** assessing the level of antitoxic immunity (PHA).

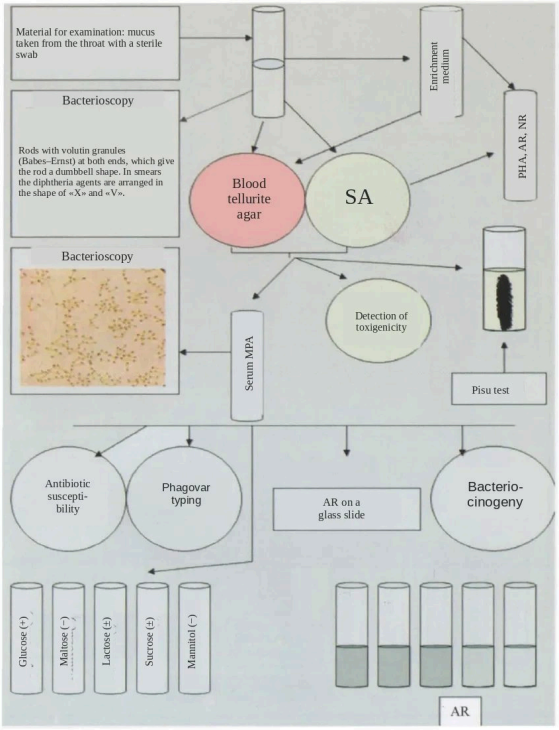



**Figure 18.3.** *Scheme of the bacteriological examination of the diphtheria causative agent: collecting material, inoculation onto tellurite and serum media, identification, determining toxigenicity, susceptibility to antibiotics. Source: Aliyev Sh.R. et al., 2013.*

> **REMEMBER**
> In the diagnosis of diphtheria, **determining toxigenicity is essential**: nontoxigenic corynebacteria can occur in the throat as carriage and do not cause disease.

**Prophylaxis:** planned vaccination with **toxoid** (as part of combination vaccines, according to the national schedule). In treatment, **antitoxic serum** and antibiotics are used (as prescribed by a physician).

## 3. The causative agent of pertussis (whooping cough) — *Bordetella pertussis*

**Morphology:** very small, ovoid gram-negative rod (coccobacillus); nonmotile, encapsulated.

**Cultural properties: obligate aerobe**, extremely fastidious. The classic medium is **Bordet–Gengou** (potato-glycerin blood agar) and **charcoal-casein agar**; growth is slow (3–5 days), the colonies are small and shiny — resembling a **"mercury droplet"** or a pearl.

**Pathogenicity factors: pertussis toxin** (disrupts the adenylate cyclase system), filamentous hemagglutinin and pertactin (adhesion), **tracheal cytotoxin** (damages the ciliated epithelium), adenylate cyclase–hemolysin, LPS.

**Pathogenesis:** the pathogen **adheres to the respiratory epithelium** and does not enter the blood; the movement of the cilia is disrupted, mucus is not cleared, and toxins constantly stimulate the cough center. As a result, **spasmodic (paroxysmal) coughing fits** — a series of cough bursts and a deep whooping inspiration (reprises), often ending in vomiting.

**Stages of the disease:** catarrhal (resembling an acute respiratory viral infection — **the most contagious stage**), spasmodic (typical fits), recovery.

**Laboratory diagnosis:**

1. **Material:** a swab taken from the nasopharynx **from the posterior wall** (special swab), the **“cough plate”** method (an open dish is held in front of the patient’s mouth while he coughs).
2. **Bacteriological** — immediate inoculation onto special medium; the pathogen is very unstable in the external environment, so **inoculation is done at the patient’s bedside**.
3. **PCR** — currently the most sensitive and practical method, especially in the catarrhal stage.
4. **Serological** — detecting antibodies by ELISA; significant in the late stage.

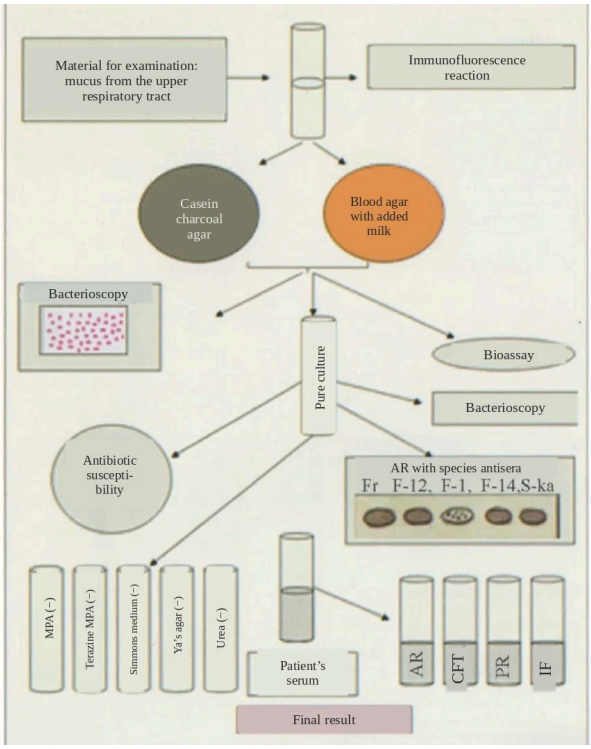



**Figure 18.4.** *Scheme of the microbiological examination of the pertussis causative agent: collecting material, inoculation onto special media, identification and serological examination. Source: Aliyev Sh.R. et al., 2013.*

> **CAUTION**
> In pertussis the effectiveness of the bacteriological method falls sharply depending on the stage of the disease: the best result is obtained **in the catarrhal stage and at the beginning of the spasmodic stage**; later only PCR and serology give information.

**Prophylaxis:** vaccine (whole-cell or acellular) — within the complex immunization schedule.

## 4. The meningococcus — *Neisseria meningitidis*

**Morphology:** gram-negative **diplococcus** with flattened facing sides — resembling a **“coffee bean”** or a kidney bean; in cerebrospinal fluid and blood smears it is often located **inside leukocytes** (intracellularly). Encapsulated, nonmotile.

**Cultural properties:** fastidious aerobe; requires **serum agar** and chocolate agar; grows well in an atmosphere with **5–10% $CO_2$**; **oxidase-positive**, breaks down glucose and maltose (differs from the gonococcus, which breaks down only glucose).

**Pathogenicity factors:** the **capsule** (antiphagocytic; determines the serological groups — A, B, C, W, Y and others), pili (adhesion), **LPS (endotoxin)** — the main cause of severe shock, IgA protease.

**Clinical forms:** inflammation of the nasopharynx (nasopharyngitis) and **healthy carriage** (the most widespread form), **generalized forms** — meningococcemia (heterogeneous rash, shock) and

**purulent meningitis**.

**Laboratory diagnosis:**

1. **Material:** cerebrospinal fluid (lumbar puncture), blood, nasopharyngeal swab, scraping from a rash element.
2. **Transporting the material — warm**, without cooling, immediately: the meningococcus is extremely sensitive to low temperature and drying.
3. **Bacterioscopy** — Gram smear from the CSF sediment: intracellular gram-negative diplococci in leukocytes → a rapid presumptive answer.
4. **Bacteriological** — serum/chocolate agar, $CO_2$ conditions; identification — oxidase, carbohydrates (glucose + maltose), serological grouping.
5. **Rapid methods** — detecting the capsular antigen in CSF by **latex agglutination**, and **PCR**.

**MEMORY AID**
Distinguish the Neisseria by carbohydrate: **MeninGoMaltose** — meningococcus: **G**lucose + **M**altose; gonococcus — only **G**lucose.

**CLINICAL SIGNIFICANCE**
In meningococcal infection, collecting and culturing the CSF must not be delayed; at the same time, in severe cases antibacterial treatment is started without waiting for the laboratory answer. After antibiotics are started the culture may be negative — in which case **PCR and antigen tests** are valuable.

**Prophylaxis:** conjugate and polysaccharide vaccines (according to serological groups), preventive measures for contacts in a focus — as prescribed by a physician.

## 5. Independent work: Pneumococcus and *Haemophilus influenzae*

### *5.1. Streptococcus pneumoniae (pneumococcus)*

**Morphology:** gram-positive **lancet-shaped diplococcus** with the ends pointing in opposite directions; **encapsulated** (clearly visible in pathological material).

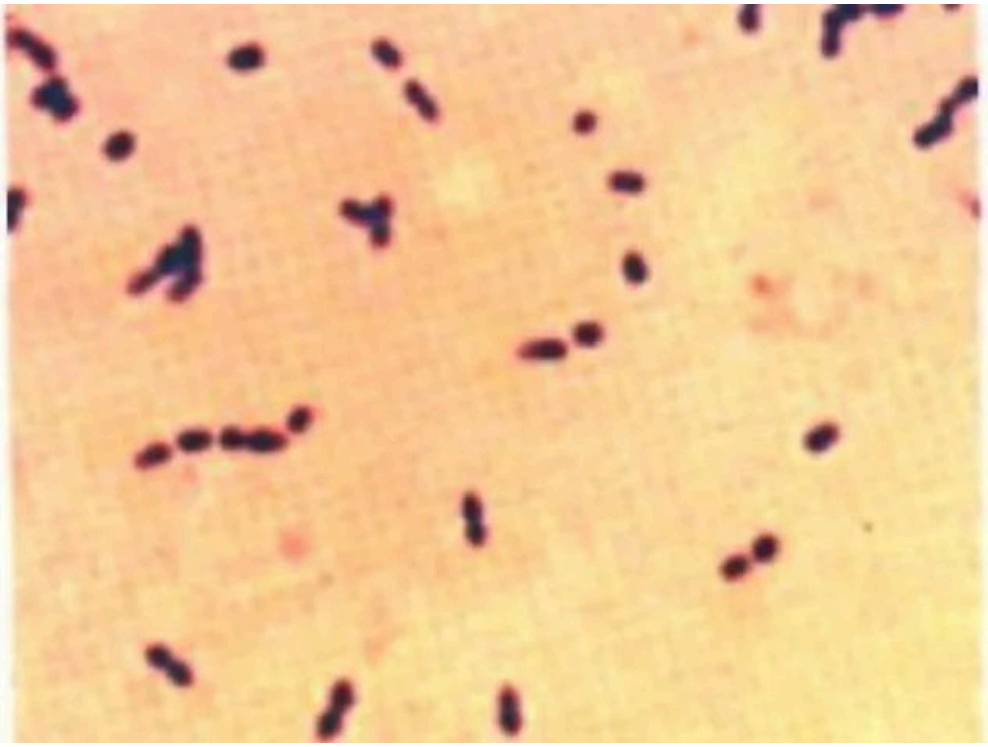


**Figure 18.5.** *Pneumococci: lancet-shaped gram-positive diplococci. Source: Muhamedov I.M. et al., 2019.*

**Cultural and differential properties: alpha-hemolysis** on blood agar (green zone); **sensitive to optochin**, **soluble in bile acids** (this distinguishes it from the viridans streptococci); the **"capsular swelling" reaction** (Quellung; with a special serum) is used to detect the capsule.

**Pathogenicity factor:** the main one is the **polysaccharide capsule** (more than 90 serotypes); pneumolysin, IgA protease, teichoic acids.

**Diseases:** acute pneumonia (classic **lobar pneumonia**), otitis media, sinusitis, meningitis, sepsis.

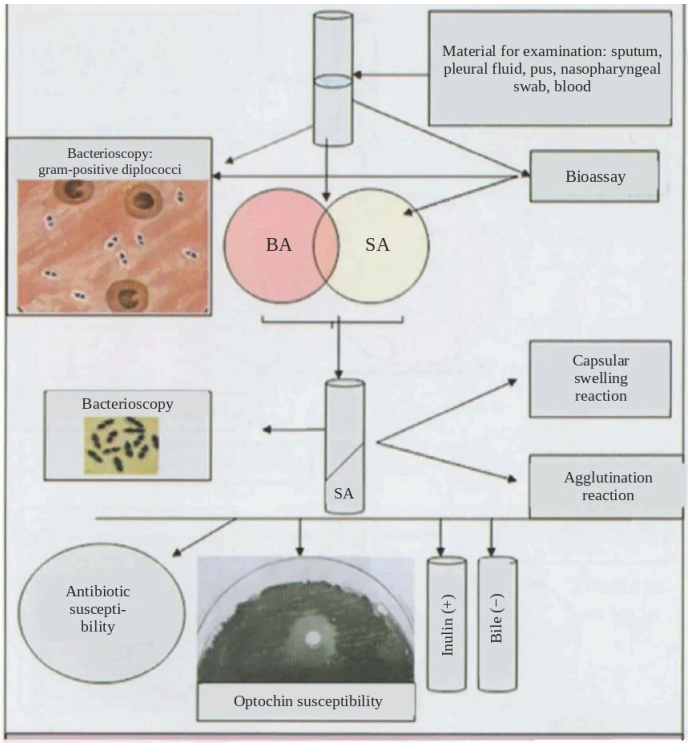



**Figure 18.6.** *Scheme of the microbiological examination of diseases caused by pneumococcus: material, bacterioscopy, culture, capsule reaction, sensitivity to optochin and bile acid. Source: Aliyev Sh.R. et al., 2013.*

**Diagnosis:** sputum, cerebrospinal fluid, blood → bacterioscopy (encapsulated diplococci) → blood agar → optochin and bile tests → serotyping; antigen tests and PCR in cerebrospinal fluid.

**Prophylaxis:** polysaccharide and **conjugate** vaccines (for the reason explained in topic 15, the conjugate form is preferable in children).

## *5.2. Haemophilus influenzae*

**Morphology:** very small gram-negative coccobacillus, pleomorphic; type b is encapsulated.

**Cultural property:** requires **factor X (hemin)** and **factor V (NAD)** — therefore it does not grow on ordinary blood agar but grows on **chocolate agar** (topic 5). When it grows next to *S. aureus*, the **"satellite" phenomenon** is seen — the staphylococcus lyses erythrocytes and releases factor V.

**Diseases:** type b (Hib) — meningitis, epiglottitis, pneumonia and sepsis in children; nonencapsulated strains — otitis, sinusitis, exacerbations of bronchitis.

**Diagnosis:** cerebrospinal fluid, blood, sputum → chocolate agar + $CO_2$ → determining the requirement for factors X and V (disk method) → serotyping; PCR and antigen tests.

**Prophylaxis:** the **Hib conjugate vaccine** has sharply reduced the frequency of Hib meningitis in children.

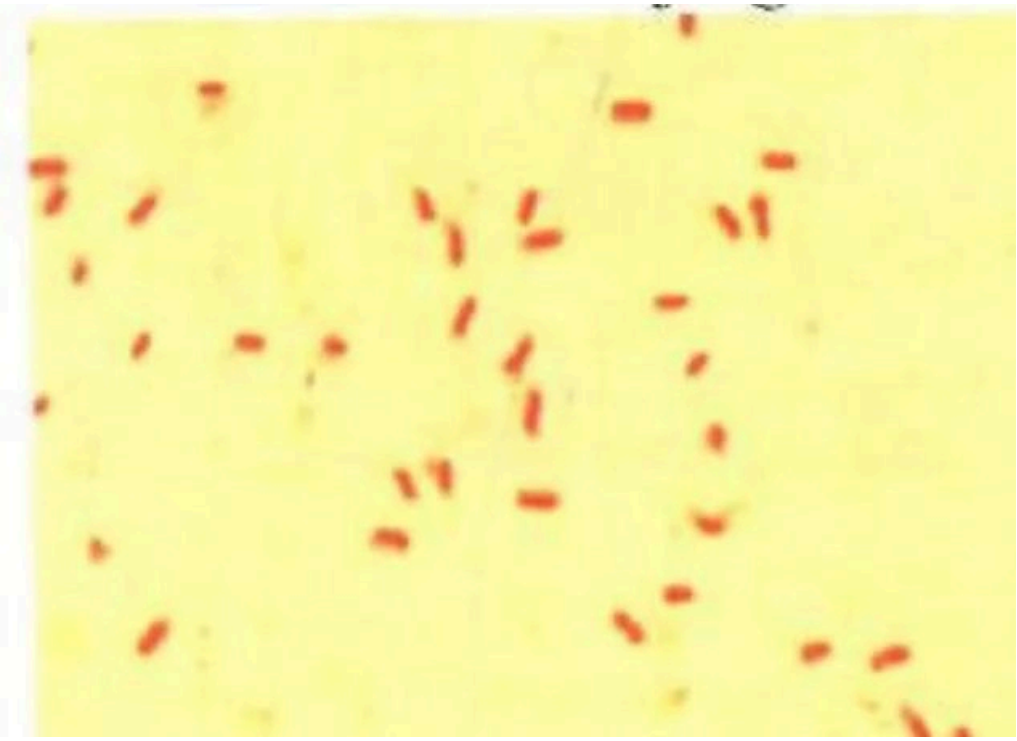


**Figure 18.7.** *Legionellae — gram-negative rods transmitted by the airborne route (mainly through water aerosol). Source: Muhamedov I.M. et al., 2019.*

## 6. Practical skill: taking a throat and nose swab

1. The patient is examined on an empty stomach, before throat rinses and antibiotics.
2. The tongue is pressed with a spatula; with a light, the **membrane or the border of the inflamed area** is identified.
3. With a sterile swab a sample is taken **from the edge of the membrane** (the border between healthy and altered tissue); the swab must not touch the tongue, lips or teeth.
4. Material is taken from the nose with a separate swab.
5. The swabs are placed in transport medium; when diphtheria is suspected, inoculation is done **as soon as possible** (within 2–3 hours), in pertussis — **at the patient's bedside**.
6. The referral states the clinical diagnosis, vaccination history and whether antibiotics were taken.

## 7. Review questions

1. State the morphological and tinctorial features of the diphtheria bacillus.
2. How is diphtheria toxin encoded and how does it act?
3. Why is determining toxigenicity essential in the diagnosis of diphtheria? What methods are there?
4. What are the principles of prevention and treatment of diphtheria?
5. State the cultural requirements and pathogenicity factors of *B. pertussis*.
6. By what mechanism does spasmodic cough arise in pertussis?
7. How does the choice of method in the diagnosis of pertussis depend on the stage of the disease?
8. What is the morphology of the meningococcus and the rule for transporting the material?
9. Which test distinguishes the meningococcus from the gonococcus?
10. How do you distinguish the pneumococcus from the viridans streptococci?

11. Why does *H. influenzae* not grow on ordinary blood agar?

## Test questions

1. The granules at the ends of the diphtheria bacillus cells:
   A) Spore
   B) Volutin (metachromatic) granules
   C) Capsule
   D) Inclusion bodies
   E) Mesosome

2. The method for detecting volutin granules:
   A) Ozheshko
   B) Neisser
   C) Burri–Hiss
   D) Ziehl–Neelsen
   E) Gram

3. Diphtheria toxin is encoded:
   A) On a plasmid
   B) In prophage genes
   C) Permanently on the chromosome
   D) On a transposon
   E) In the capsule

4. The mechanism of action of diphtheria toxin:
   A) It blocks EF-2 and stops protein synthesis
   B) It stops the release of GABA
   C) It activates adenylate cyclase
   D) It perforates the membrane
   E) It breaks down DNA

5. The substance in the selective medium for the diphtheria causative agent:
   A) NaCl
   B) Potassium tellurite
   C) Bile acid
   D) Optochin
   E) Bacitracin

6. The method of determining toxigenicity in diphtheria:
   A) Precipitation in gel (Elek method)
   B) Catalase test
   C) Coagulase test
   D) Ozheshko
   E) CAMP test

7. The classic nutrient medium for *B. pertussis*:
   A) Endo
   B) Bordet–Gengou
   C) Sabouraud
   D) Kligler
   E) Löwenstein–Jensen

8. The location of the pathogen in pertussis:
   A) It multiplies in the blood
   B) It adheres to the respiratory epithelium and does not enter the blood
   C) It multiplies in the intestine
   D) In nerve cells
   E) In the liver

9. The most contagious stage of pertussis:
   A) Incubation
   B) Catarrhal
   C) Spasmodic
   D) Recovery
   E) The stage of complications

10. The morphological appearance of the meningococcus:
    A) A gram-positive chain-forming coccus
    B) A gram-negative diplococcus (“coffee bean”), often inside a leukocyte
    C) Gram-negative rod
    D) A gram-positive rod
    E) Spirochete

11. The rule for transporting meningococcal material:
    A) Cooled on ice
    B) Warm, immediately
    C) In formalin
    D) Dried
    E) In alcohol

12. The feature that distinguishes the meningococcus from the gonococcus:
    A) Oxidase positivity
    B) Breakdown of maltose
    C) Gram-negativity
    D) Diplococcal shape
    E) Its capsule

13. The test that distinguishes the pneumococcus from the viridans streptococci:
    A) Catalase
    B) Sensitivity to optochin and solubility in bile acid
    C) Coagulase
    D) Oxidase
    E) Urease

14. The main pathogenicity factor of the pneumococcus:
    A) M protein
    B) The polysaccharide capsule
    C) Erythrogenic toxin
    D) Lecithinase
    E) Coagulase

15. Factors required for the growth of *H. influenzae*:
    A) X (hemin) and V (NAD)
    B) NaCl and glycerin
    C) Tellurite
    D) Optochin
    E) Mycolic acid

## Clinical cases

### *Case 1*

An unvaccinated 6-year-old child has a dense gray-white membrane in the throat that bleeds when removed; the neck is swollen and the condition is severe.

**Question:** What are the probable diagnosis, the material and the decisive laboratory step?

**Answer:** Diphtheria. The material is a throat and nose swab from the border of the membrane. The decisive step is determining the **toxigenicity** of the culture (the Elek method, or the tox gene by PCR). Antitoxic serum is given by a physician on clinical suspicion, without waiting for the laboratory answer.

### *Case 2*

A 3-month-old infant has had a cough for 2 weeks; in the last few days paroxysmal coughing, a whooping inspiration and vomiting have appeared. The bacteriological culture is negative.

**Question:** Why might the culture be negative, and which method is preferable?

**Answer:** In pertussis, the chance of isolating the pathogen falls sharply in the spasmodic stage; in addition, *B. pertussis* is unstable in the external environment. The preferable method is **PCR**, and in the late stage, serology.

### *Case 3*

A Gram smear of the cerebrospinal fluid sediment showed gram-negative diplococci inside leukocytes.

**Question:** What is the preliminary conclusion and the next step?

**Answer:** Suspected meningococcal purulent meningitis. The next step is inoculation onto serum/chocolate agar under $CO_2$ conditions, oxidase and carbohydrate tests (glucose + maltose), and serological grouping; in parallel, latex agglutination and PCR. Treatment is not delayed.

## Quick review

**Table 18.1.** *Topic 18 — key concepts*

| Concept | Essence |
|---|---|
| *C. diphtheriae* | Volutin granules, V/X arrangement; Neisser stain; tellurite medium |
| Diphtheria toxin | Prophage genes; EF-2 block → protein synthesis stops |
| Toxigenicity test | Elek (precipitation in gel), PCR (tox gene) |
| Prevention of diphtheria | Toxoid; treatment — antitoxic serum |
| *B. pertussis* | Bordet–Gengou, “mercury droplet”; adheres to the epithelium |
| Stages of pertussis | Catarrhal (most contagious) → spasmodic → recovery |

| Concept | Essence |
|---|---|
| Diagnosis of pertussis | Early — culture; later PCR and serology |
| Meningococcus | Gram-negative diplococcus, intracellular; warm transport; glucose + maltose |
| Pneumococcus | Lancet-shaped diplococcus, capsule; optochin +, soluble in bile |
| *H. influenzae* | Factors X and V; chocolate agar; satellite phenomenon; Hib vaccine |

TOPIC 19

# Airborne infections: the causative agents of tuberculosis and leprosy. Actinomycosis and atypical mycobacteria

## 1. Learning objectives

In this class the student will master the following:

- To know the general properties of mycobacteria and the reason for acid-fastness;
- to explain the biological properties of the tuberculosis causative agent, its pathogenesis and the course of the disease;
- to master the methods of laboratory diagnosis of tuberculosis (bacterioscopy, enrichment, culture, rapid molecular methods, skin test);
- to know the distinctive features of the leprosy causative agent;
- Independent work: to know the significance of the causative agents of actinomycosis and of atypical mycobacteria.

## 2. General characteristics of mycobacteria

**Mycobacteria** (*Mycobacterium*) are slender, slightly curved rods; nonmotile, nonspore-forming, noncapsulated. Their main distinguishing feature is the special composition of the cell wall: it contains very large amounts of **mycolic acids**, waxes and lipids (40–60% of the wall is lipid).

**Consequences:**

- **Acid-fastness (acid resistance)** — after staining they are not decolorized by acid and alcohol; for this reason the Gram method is unsuitable and the **Ziehl–Neelsen** method is used;
- resistance to environmental factors, to many disinfectants and to drying;
- **slow growth** (generation time 14–24 hours) — in culture the result is obtained in **3–8 weeks**;
- the ability to survive inside the phagocyte (**incomplete phagocytosis**).

> **REMEMBER**
> **Ziehl–Neelsen result:** the acid-fast rods are **red (carbol fuchsin)**; the background and other cells are **blue (methylene blue)**.

## 3. The causative agent of tuberculosis — *Mycobacterium tuberculosis*

**Historical note:** discovered in 1882 by **Robert Koch** (hence it is also called “Koch’s bacillus”).

**Morphology:** a slender, straight or slightly curved rod; in smears it lies singly, in pairs or in groups, sometimes **branching**. Under unfavorable conditions (antibacterial treatment, immune pressure) it can change into **L-forms**, filterable and granular forms — which matters in relapse of the disease.

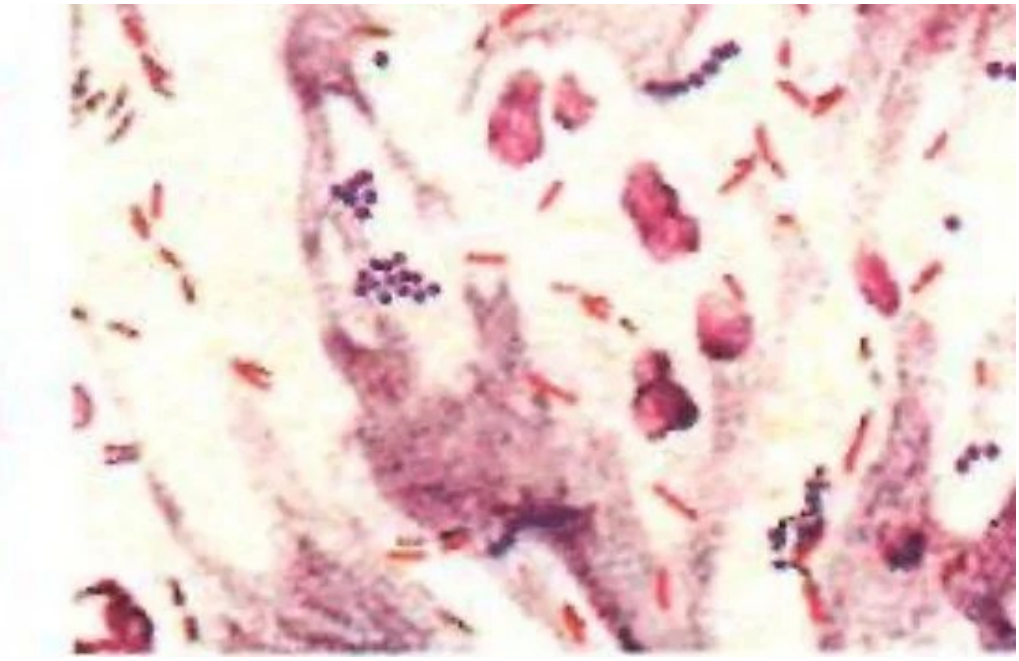


**Figure 19.1.** *M. tuberculosis in a sputum smear: red rods stained by the Ziehl–Neelsen method on a blue background. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties: obligate aerobe**; grown on **egg-based Löwenstein–Jensen** medium with added glycerin, or on Finn-2 medium. Growth takes **3–8 weeks**; the colonies are dry, rough, **"cauliflower"** or "bread-crumb" in appearance, cream-colored (**R-form — a sign of virulence**). In liquid medium, owing to **cord factor**, it grows in "rope"-like (ribbon-like) strands.

**Pathogenicity factors:**

- **Cord factor** (trehalose-6,6-dimycolate) — the main marker of virulence; it stops phagosome–lysosome fusion and damages macrophages;
- **Sulfatides** — block the formation of the phagolysosome;
- **Mycolic acids and lipids** — resistance and granuloma formation;
- **Tuberculin (a protein antigen)** — causes the immune-allergic response.

> **CLINICAL SIGNIFICANCE**
>
> In tuberculosis there is **no exotoxin**. The picture of the disease is determined mainly by the **delayed-type (type IV) hypersensitivity immune response** and the formation of granulomas. That is why tuberculosis runs as an "immunological disease": tissue damage is largely the result of the body's own response.

**Pathogenesis:** mycobacteria that enter by the airborne route are engulfed by alveolar macrophages, but because of **incomplete phagocytosis** they survive and multiply in them. As a result a **specific granuloma** forms (caseous necrosis in the center, surrounded by epithelioid cells and Pirogov–Langhans giant cells, and lymphocytes). In most people the process is contained at this stage — **latent infection**; when immunity falls (HIV, diabetes mellitus, malnutrition, immunosuppressive therapy, old age) **reactivation** occurs.

**Clinical forms:** pulmonary tuberculosis (the most frequent) and extrapulmonary forms — lymph nodes, bone and joint, kidney, intestine, meningitis, disseminated (miliary) tuberculosis.

## 4. Laboratory diagnosis of tuberculosis

**Material:** sputum (morning, 2–3 times, with a deep cough), bronchial washings, gastric washings (in children), urine, cerebrospinal fluid, puncture fluid, tissue biopsy.

### 4.1. Bacterioscopic method

- Staining by the **Ziehl–Neelsen** method — rapid and inexpensive; its limitation is low sensitivity (about $10^4$–$10^5$ cells per 1 ml of material are needed);
- **Fluorescence microscopy** (auramine–rhodamine) — more sensitive and convenient for viewing many samples;
- **Enrichment methods** — **flotation** (shaking with xylene/benzine and making a smear from the layer that rises to the surface) and **homogenization/sedimentation** — raise the sensitivity.

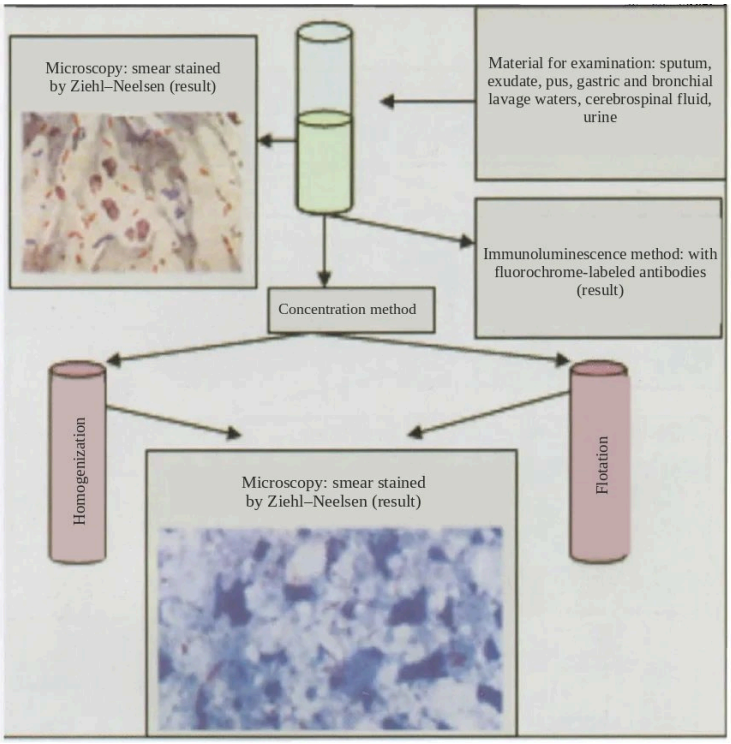



**Figure 19.2.** *Scheme of the bacterioscopic examination in tuberculosis: material, Ziehl–Neelsen staining, the fluorescence method, enrichment (flotation and homogenization). Source: Aliyev Sh.R. et al., 2013.*

### 4.2. Bacteriological (cultural) method — the gold standard

1. The material is subjected to **decontamination** with alkali or acid (to remove the accompanying flora; mycobacteria survive because they are resistant), neutralized and centrifuged.
2. The sediment is inoculated onto **Löwenstein–Jensen** (or Finn-2) medium; modern laboratories use automated **liquid medium systems** (the result comes much faster).
3. At 37 °C for 3–8 weeks, observed weekly.
4. Identification: growth rate, colony appearance, **cord factor**, the **niacin test** (positive for *M. tuberculosis*), nitrate reductase, pigment formation; modern methods — molecular identification and MALDI-TOF.
5. **Antibiotic susceptibility** — by the cultural or molecular method (resistance genes).

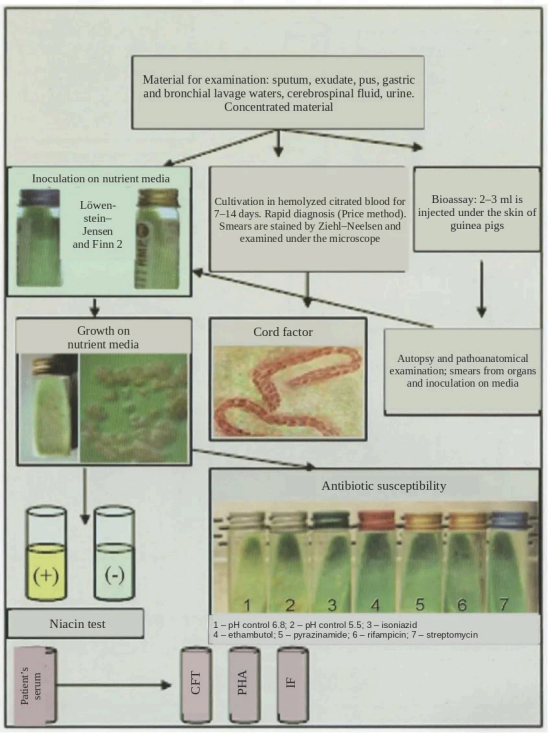



**Figure 19.3.** *Scheme of the bacteriological examination in tuberculosis: decontamination, inoculation onto Löwenstein–Jensen and Finn-2 media, cord factor, niacin test, antibiotic susceptibility. Source: Aliyev Sh.R. et al., 2013.*

### *4.3. Molecular and immunological methods*

- **Rapid molecular tests (PCR-based)** — detect the pathogen and, at the same time, the **genes for resistance to the main drugs** within a few hours; at present one of the first-line methods;
- **Tuberculin test (Mantoux)** — tuberculin is injected into the skin and the papule is measured after 72 hours; it shows **infection (sensitization)**, not disease. It can also be positive after BCG vaccination;
- **Blood tests based on IFN-γ release (IGRA)** — independent of BCG, used to detect latent infection;
- **Biological test** — now rarely used.

> **CAUTION**
> A positive Mantoux test does not mean **"tuberculosis disease"**: it shows that the body has met mycobacterial antigens. The diagnosis is made from a combination of clinical findings, X-ray and microbiological results.

**Prophylaxis: BCG** — a live attenuated vaccine; it is important for protection against severe disseminated forms. Additional measures: early detection, isolation of the source, examination of contacts, social and hygienic measures.

## 5. The causative agent of leprosy — *Mycobacterium leprae*

**Features:** an acid-fast rod; **it does not grow on artificial nutrient media at all** — this is its main laboratory feature. In experiments it is propagated in armadillos and in the mouse footpad. It multiplies very slowly (generation time 12–14 days), and the incubation period is several years.

**Morphological feature:** in tissue and mucosal scrapings it lies **inside cells, in orderly bundles resembling "cigarettes in a pack"** (globi).

**Clinical forms: lepromatous** (weak immune response, many organisms — more contagious) and **tuberculoid** (strong cell-mediated immunity, few organisms), as well as intermediate forms. The skin, peripheral nerves and mucous membranes are mainly affected; nerve damage leads to loss of sensation and deformities.

**Diagnosis:** nasal mucosal scrapings, material from skin lesions, biopsy → **Ziehl–Neelsen** → seeing orderly intracellular bundles; the **lepromin test** (to assess the type of immune response, not for diagnosis); PCR.

## 6. Independent work: Actinomycosis and atypical mycobacteria

### *6.1. Actinomycosis*

**Causative agent:** *Actinomyces israelii* and other actinomycetes — gram-positive, branching filamentous **bacteria** (topic 4), members of the normal flora of the oral cavity, intestine and urogenital tract.

**Pathogenesis:** an **endogenous** disease — when the mucosa is damaged (tooth extraction, jaw injury, appendicitis, an intrauterine device) the organism enters the tissue; under anaerobic conditions chronic **granulomatous** inflammation, a dense infiltrate and **fistulas** develop.

**Forms:** orofacial (the most frequent), thoracic, abdominal, urogenital, and central nervous system forms.

**Diagnosis:** fistula discharge or tissue → looking for visible yellow **druses** → microscopy (mycelium in the center of the druse, a radiating structure at the edge) → **anaerobic culture** (grows slowly, 7–14 days) → identification. Because actinomycetes are bacteria, they **respond to antibacterial treatment**.

> **MEMORY AID**
> The triad of actinomycosis: **chronic dense infiltrate + fistulas + yellow druses in the pus**.

### *6.2. Atypical (nontuberculous) mycobacteria*

**General name:** NTM — mycobacteria that do not belong to the tuberculosis group. Their natural habitat is **water, soil, water pipes and moist environments**; they are hardly ever transmitted from person to person (the main epidemiological difference from tuberculosis).

**Classic groups by Runyon:** photochromogens (produce pigment in light — *M. kansasii*), scotochromogens (pigmented even in the dark — *M. scrofulaceum*), nonphotochromogens (*M. avium* complex), and **rapid growers** (*M. fortuitum*, *M. chelonae* — grow within up to 7 days).

**Clinical significance:** disseminated infection in the immunocompromised (especially in HIV infection — *M. avium* complex); pulmonary forms on the background of chronic lung disease; cervical lymphadenitis in children; skin and soft-tissue infections after surgical and cosmetic procedures (rapid-growing species).

**Diagnostic particularity:** finding an acid-fast rod does not in itself mean tuberculosis — **the species must be identified** (molecular methods, MALDI-TOF). Atypical mycobacteria are naturally resistant to most antituberculosis drugs, so the treatment regimen is entirely different.

> **REMEMBER**
> When an acid-fast rod is found in sputum, **two questions** are asked: (1) is it *M. tuberculosis* or an atypical mycobacterium? (2) is it drug-resistant? Modern molecular tests answer both questions quickly.

## 7. Practical skill: collecting sputum for tuberculosis testing

1. The patient is told: the material needed is **not saliva but sputum brought up by a deep cough**.
2. The sample is collected **in the morning, on an empty stomach**, after rinsing the mouth with water.
3. The patient breathes in deeply, coughs forcefully and spits the sputum into a **wide-mouthed sterile container**; the volume is usually 3–5 ml.
4. The sample is delivered to the laboratory in a tightly closed container, **upright**, cooled (2–8 °C).
5. Usually **2–3** samples are collected (on different days or according to a set scheme) — this raises sensitivity.
6. In the laboratory: a smear (Ziehl–Neelsen or fluorescence), decontamination and culture, a rapid molecular test.
7. Safety: the material can generate aerosols — the work is done **in a special cabinet**, with protective equipment.

## 8. Review questions

1. What is special about the mycobacterial cell wall? What is acid-fastness related to?
2. What does the result of the Ziehl–Neelsen method look like?
3. State the cultural properties and growth time of *M. tuberculosis*.
4. What is cord factor and how does it act?
5. What is the role of the granuloma and of delayed-type hypersensitivity in the pathogenesis of tuberculosis?
6. How can the bacterioscopic diagnosis of tuberculosis be made more sensitive?
7. Why is decontamination needed in the cultural method?
8. What does the Mantoux test show, and what does it not show?
9. What is the main laboratory feature of *M. leprae*?
10. How do atypical mycobacteria differ from the tuberculosis bacillus?

## Test questions

1. The substance that provides acid-fastness in mycobacteria:
   A) Teichoic acid
   B) Mycolic acid
   C) Lipopolysaccharide
   D) Dipicolinic acid
   E) Peptidoglycan

2. The color of acid-fast rods in the Ziehl–Neelsen method:
   A) Blue
   B) Red
   C) Green
   D) Yellow
   E) Purple

3. The medium used to grow *M. tuberculosis*:
   A) Endo
   B) Löwenstein–Jensen
   C) Sabouraud
   D) Kligler
   E) Bordet–Gengou

4. The growth time of the tuberculosis causative agent:
   A) 18–24 hours
   B) 48 hours
   C) 3–8 weeks
   D) 2 hours
   E) 6 months

5. The factor considered the main marker of virulence:
   A) Cord factor
   B) Capsule
   C) Exotoxin
   D) Flagellum
   E) Plasmacoagulase

6. The main immunological mechanism in the pathogenesis of tuberculosis:
   A) Immediate-type allergy (type I)
   B) Delayed-type hypersensitivity (type IV) and granuloma
   C) Immune complexes (type III)
   D) Cytotoxic (type II)
   E) No immune response takes part

7. The exotoxin in tuberculosis:
   A) Present, neurotoxic
   B) Absent
   C) Present, enterotoxic
   D) Present, cytotoxic
   E) Present, erythrogenic

8. The method of increasing the sensitivity of bacterioscopy:
   A) Flotation (enrichment)
   B) Gram method
   C) Ozheshko
   D) Burri–Hiss
   E) CAMP test

9. Why the material is decontaminated before cultural examination:
   A) To kill mycobacteria
   B) To remove the nonresistant accompanying flora
   C) To raise the pH
   D) To change the color
   E) To activate spores

10. The biochemical test characteristic of *M. tuberculosis*:
    A) Niacin test positive
    B) Oxidase-positive
    C) Coagulase positive
    D) Urease positive
    E) Indole positive

11. What a positive Mantoux test means:
    A) Tuberculosis is certain
    B) Contact with mycobacterial antigens (sensitization)
    C) Absence of immunity
    D) Resistance to antibiotics
    E) Ineffectiveness of the vaccine

12. The live attenuated vaccine used against tuberculosis:
    A) BCG
    B) A toxoid
    C) Recombinant vaccine
    D) Polysaccharide vaccine
    E) Inactivated vaccine

13. The main laboratory feature of *M. leprae*:
    A) It grows quickly
    B) It does not grow on artificial nutrient media
    C) Gram-negative
    D) It forms spores
    E) Motile

14. The diagnostic structure found in pus in actinomycosis:
    A) Druses
    B) Globi
    C) Volutin granules
    D) Clue cells
    E) Negri bodies

15. The main epidemiological difference of atypical mycobacteria from the tuberculosis bacillus:
    A) Almost never transmitted from person to person; the source is water and soil
    B) Transmitted only by air
    C) Occurs only in children
    D) Not acid-fast
    E) Never causes disease

## Clinical cases

### *Case 1*

In the sputum of a patient with a cough lasting 3 weeks, evening low-grade fever and weight loss, red rods were found by the Ziehl–Neelsen method.

**Question:** What are the preliminary conclusion and the next steps?

**Answer:** Acid-fast rods have been found — tuberculosis is suspected. Next steps: **identify the species** (a rapid molecular test, culture), assess **drug resistance**, a chest X-ray and examination by a

phthisiologist. An acid-fast rod alone does not prove that it is *M. tuberculosis*.

### *Case 2*

A patient with HIV infection has a disseminated infection; in culture, acid-fast rods grew in 7 days, and the niacin test is negative.

**Question:** What should be considered?

**Answer:** Atypical (nontuberculous) mycobacteria — rapid growth and niacin negativity are not characteristic of *M. tuberculosis*. The species is identified by a molecular method; the treatment regimen differs from that for tuberculosis.

### *Case 3*

A healthy child vaccinated with BCG had a positive Mantoux test; there are no changes in the clinical picture or on X-ray.

**Question:** How should the result be interpreted?

**Answer:** A positive Mantoux test can also be due to sensitization after BCG; it does not establish a diagnosis of disease. Where necessary the child is assessed with **IGRA** (a blood test independent of BCG) and clinical and radiological follow-up.

## Quick review

**Table 19.1.** *Topic 19 — key concepts*

| Concept | Essence |
|---|---|
| The mycobacterial wall | Mycolic acid, wax → acid-fastness, slow growth |
| Ziehl–Neelsen | Red rod, blue background |
| Medium | Löwenstein–Jensen, Finn-2; 3–8 weeks; R-form colony |
| Cord factor | Marker of virulence; blocks the phagolysosome |
| Pathogenesis | Incomplete phagocytosis → granuloma → latency → reactivation |
| Bacterioscopy | Ziehl–Neelsen, fluorescence, enrichment by flotation |
| Cultural method | Decontamination → inoculation → niacin test → susceptibility |
| Rapid methods | PCR-based tests: species + resistance genes |
| Mantoux / IGRA | Sensitization / independent of BCG |
| *M. leprae* | Does not grow on artificial media; orderly bundles inside cells |
| Actinomycosis | Endogenous, granuloma, fistula, yellow druses |
| Atypical mycobacteria | Water/soil, not transmitted from person to person, different treatment |

TOPIC 20

# Bacterial intestinal infections: enterobacteria, coli infections, yersiniosis. Campylobacter and Helicobacter

## 1. Learning objectives

In this class the student will master the following:

- To know the general properties, taxonomy and antigenic structure of the family Enterobacteriaceae;
- to understand the role of *Escherichia coli* in the normal flora and to distinguish its diarrheagenic (pathogenic) variants;
- to master the algorithm of microbiological diagnosis of coli infections and yersiniosis;
- Independent work: to know the significance and laboratory diagnosis of Campylobacter and Helicobacter.

## 2. The family Enterobacteriaceae (*Enterobacteriaceae*)

**General properties:**

- gram-negative, nonspore-forming rods; most are **peritrichous** (motile), some are nonmotile (Shigella, Klebsiella);
- **facultative anaerobes**; grow on ordinary media;
- **oxidase negative** (the main difference from Pseudomonas), **catalase positive**;
- **they ferment glucose with production of acid (often gas)**; they reduce nitrate to nitrite;
- on differential media they are separated by their **fermentation of lactose**.

> **REMEMBER**
> Two main “filter” tests for enterobacteria: **oxidase negative** (membership of the family) and **lactose** (differentiation within the family). On Endo/Levine medium **lactose-positive organisms give colored** colonies and **lactose-negative organisms give colorless** colonies.

**Antigenic structure: O** (somatic, LPS, heat-stable), **H** (flagellar, heat-labile), **K** (capsular/surface; in typhoid fever, **Vi**). Serological identification is based on exactly these antigens.

**Table 20.1.** *Main differential media*

| Medium | Principle | Result |
|---|---|---|
| **Endo** | Lactose + fuchsin-sulfite | Lactose + → red, with a metallic sheen; lactose − → colorless |
| **Levine** | Lactose + eosin–methylene blue | Lactose + → dark purple-black; lactose − → colorless |
| **Ploskirev** | Selective (bile salts) + lactose | Selective for pathogens |
| **Kligler** | Glucose/lactose + iron salt | Interpreted by sugar, gas and $H_2S$ (topic 7) |
| Bismuth sulfite agar | Selective for Salmonella | Black colonies, metallic sheen |

## 3. *Escherichia coli* and coli infections

**Morphology and properties:** gram-negative, peritrichous rod; **lactose-positive** (red colony with a metallic sheen on Endo), indole positive, $H_2S$ negative, oxidase negative.

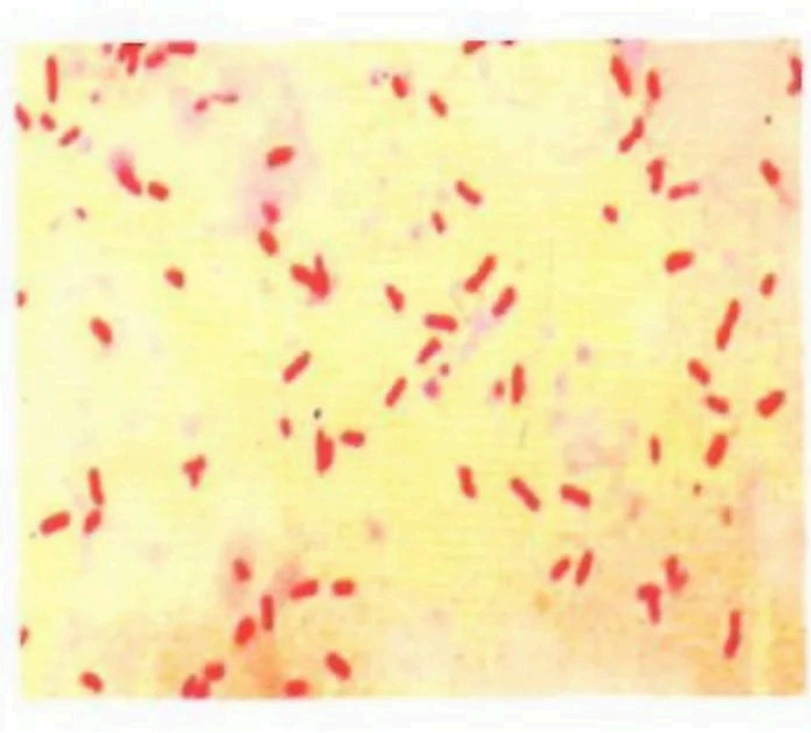


**Figure 20.1.** *Escherichia coli — gram-negative rods (smear of a pure culture). Source: Muhamedov I.M. et al., 2019.*

**Two sides:**

1. **A member of the normal flora** — lives permanently in the large intestine, synthesizes vitamin K, and takes part in colonization resistance (topic 10). It is also a **sanitary indicator microorganism** of fecal contamination of water and objects (topic 8).
2. **Pathogenic (diarrheagenic) variants** — strains that have acquired special virulence genes; they differ from the normal flora in their **serovars and virulence factors**.

**Table 20.2.** *Diarrheagenic variants of E. coli*

| Variant | In brief | Mechanism | Clinical picture |
|---|---|---|---|
| **ETEC** (enterotoxigenic) | “Traveler’s diarrhea” | Heat-stable and heat-labile enterotoxins | Watery diarrhea |
| **EPEC** (enteropathogenic) | Coli infection of newborns | Adheres to the epithelium and destroys the microvilli | Prolonged diarrhea in children |
| **EIEC** (enteroinvasive) | Dysentery-like | Enters epithelial cells and multiplies | Blood and mucus in the stool, fever |
| **EHEC** (enterohemorrhagic, e.g., O157:H7) | Hemorrhagic colitis | **Shiga-like (Vero) toxin** | Bloody diarrhea; risk of **hemolytic–uremic syndrome** |
| EAEC (enteroaggregative) | Aggregative | Adhesion like “stacked bricks” | Chronic diarrhea |

**Extraintestinal coli infections:** urinary tract infection (the most frequent causative agent), cholecystitis, peritonitis, meningitis and sepsis in newborns, wound infections.

**CLINICAL SIGNIFICANCE**

In bloody diarrhea associated with EHEC, **caution is exercised in prescribing antibiotics**: according to some data they can increase toxin release and the risk of hemolytic–uremic syndrome. The treatment approach is decided by the physician.

**Laboratory diagnosis of coli infections:**

1. **Material:** stool (in diarrhea), urine (in UTI), blood, cerebrospinal fluid.
2. **Inoculation:** Endo/Levine/Ploskirev; urine culture — **quantitative** (CFU/ml).
3. **Identification:** lactose, indole, Kligler, oxidase; then **serological identification** — agglutination with diagnostic O sera.
4. **Confirming pathogenicity:** detecting virulence genes (toxin, adhesin) by **PCR** — the most reliable way to distinguish a diarrheagenic variant from ordinary *E. coli*.

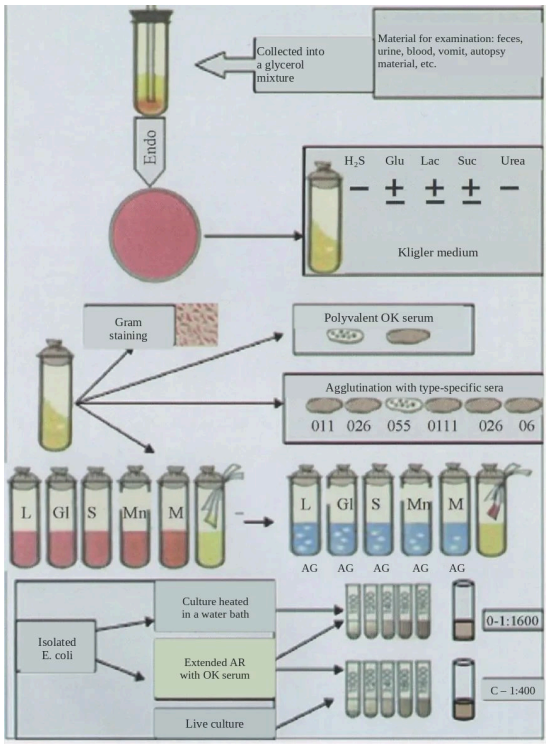



**Figure 20.2.** *Scheme of the microbiological examination in escherichiosis: material, Endo and Kligler media, biochemical properties, serological identification with diagnostic sera. Source: Aliyev Sh.R. et al., 2013.*

**CAUTION**

Growth of *E. coli* from stool is not in itself a diagnosis — it is a member of the normal flora. The diarrheagenic variant must be confirmed by **serological and molecular** methods.

## 4. Klebsiella

***Klebsiella pneumoniae*** — a gram-negative, **nonmotile**, large **encapsulated** rod. The colonies are mucoid, **sticky, stretching into a string behind the loop** (M-form); lactose-positive.

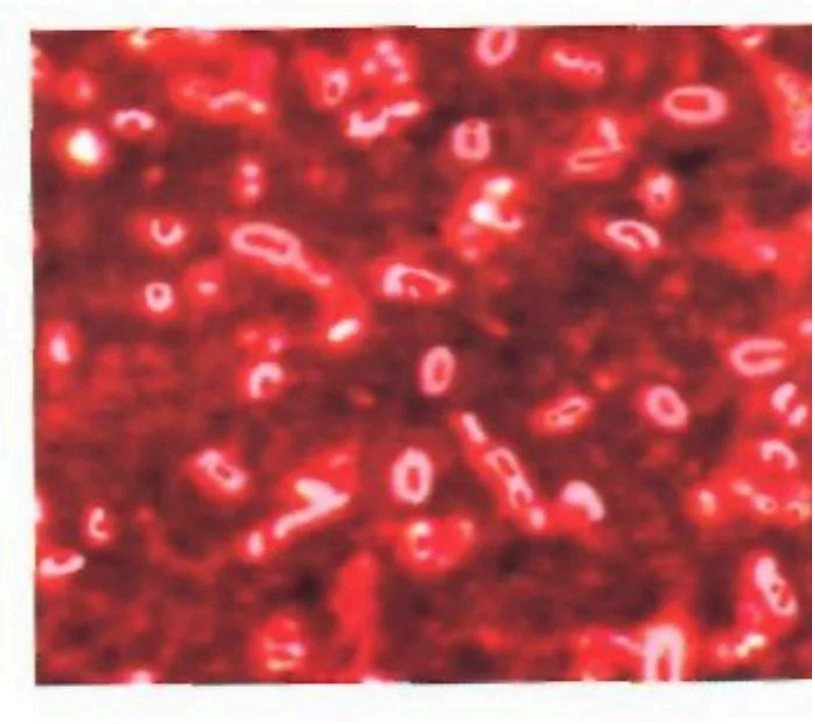


**Figure 20.3.** *Klebsiella pneumoniae — large encapsulated gram-negative rods. Source: Muhamedov I.M. et al., 2019.*

**Clinical significance:** pneumonia (classically severe, destructive), urinary tract infection, sepsis, infections of newborns; **an important causative agent of hospital-acquired infections**, notorious for multidrug resistance (including carbapenemases).

## 5. Yersiniosis

**Causative agents:** *Yersinia enterocolitica* (yersiniosis) and *Y. pseudotuberculosis* (pseudotuberculosis). (*Y. pestis* is covered in topic 24.)

**Features:** gram-negative rods; **psychrophilic — multiply at low temperature (4–8 °C)**, so they accumulate in refrigerated products and vegetables ("refrigerator infection"). Motile at 25–28 °C, nonmotile at 37 °C; urease positive.

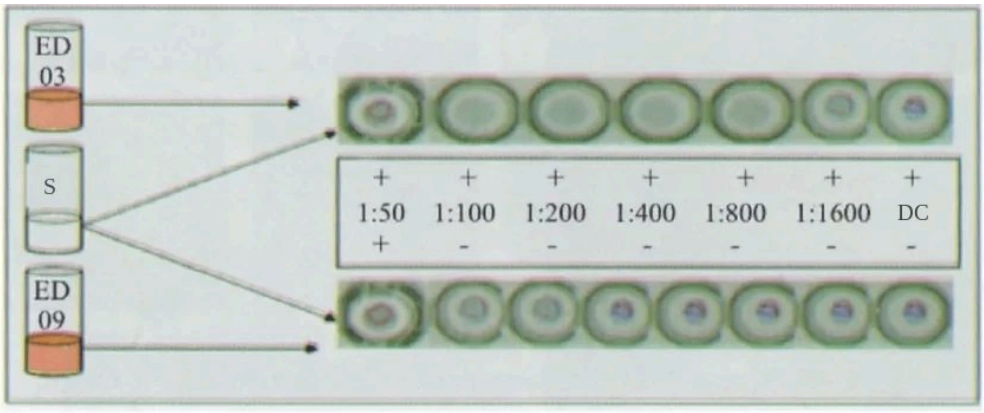



**Figure 20.4.** *Results of the indirect hemagglutination reaction (IHA) of Y. enterocolitica with an erythrocyte diagnostic reagent. Source: Aliyev Sh.R. et al., 2013.*

**Clinical picture:** gastroenteritis, mesenteric lymphadenitis (**a picture resembling acute appendicitis** — the "pseudoappendicular form"), terminal ileitis; a generalized form; complications — reactive arthritis, erythema nodosum.

**Diagnosis:**

1. **Material:** stool, blood, surgical material (lymph nodes, appendix).
2. **“Cold enrichment”** — storing the material in phosphate buffer **at 4 °C for 1–3 weeks**; Yersinia multiply while the accompanying flora is suppressed. It is then inoculated onto selective medium (CIN agar, Endo).
3. **Incubation** is carried out at 25–28 °C (not 37 °C).
4. **Identification:** biochemical tests (urease +), testing motility at two temperatures, serological identification.
5. **Serological method:** dynamics of the titer in paired sera by IHA/ELISA.

**MEMORY AID**
To remember Yersinia: it **“loves the cold”** — it multiplies in the refrigerator, is enriched in the cold, is inoculated at 25–28 °C and is motile at 25 °C.

## 6. Independent work: Campylobacter and Helicobacter

### *6.1. Campylobacter jejuni*

**Features:** gram-negative, **“gull-wing”**, S-shaped or “swallow-wing” curved rod; polar flagellum, highly motile; **microaerophilic** (5–10% $O_2$ + 10% $CO_2$) and **thermophilic** (grows well at 42 °C); oxidase positive.

**Epidemiology:** the source is poultry and other animals; it is transmitted through undercooked poultry, unpasteurized milk and contaminated water.

**Clinical picture:** acute gastroenteritis (often bloody diarrhea, severe abdominal pain, fever) — one of the main causes of bacterial diarrhea worldwide; a complication is **Guillain–Barré syndrome** (through cross-reactive antigens).

**Diagnosis:** stool → **selective medium** (with added antibiotics) → **microaerophilic conditions, 42 °C** → oxidase and motility; in a wet mount the characteristic “darting” motion; modern methods — antigen tests and PCR.

### *6.2. Helicobacter pylori*

**Features:** gram-negative, **spiral/S-shaped** curved rod; a tuft of flagella at one end (highly motile); **microaerophilic**; its most important feature is **strong urease activity**.

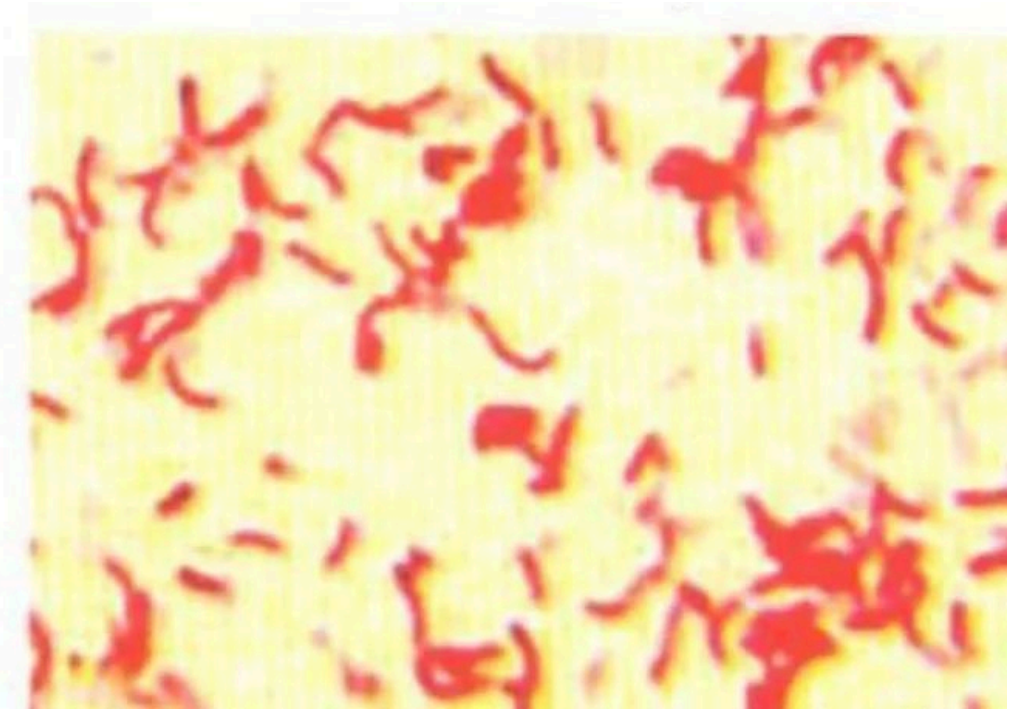


**Figure 20.5.** *Helicobacter pylori — spiral-shaped gram-negative bacteria in the gastric mucosa. Source: Muhamedov I.M. et al., 2019.*

**Pathogenesis:** urease splits urea to form **ammonia** — it neutralizes the acid around the bacterium and allows it to survive in the acidic gastric environment; motility and the spiral shape ensure penetration into the mucous layer; adhesins and the **CagA** and **VacA** factors damage the epithelium.

**Clinical significance:** chronic gastritis, gastric and duodenal ulcer disease; in the long term, it is linked to the risk of gastric cancer and MALT lymphoma.

**Diagnostic methods:**

**Table 20.3.** *Methods of detecting H. pylori*

| Method | Type | Note |
|---|---|---|
| **Urea breath test (C-urea)** | Noninvasive | Shows active infection; convenient for monitoring the effect of treatment |
| **Stool antigen test** | Noninvasive | Active infection; screening and monitoring |
| **Rapid urease test (on a biopsy)** | Invasive | During endoscopy, a quick answer |
| Histological examination of the biopsy | Invasive | Also assesses the state of the mucosa |
| Culture of the biopsy | Invasive | To determine antibiotic susceptibility |
| Serology (IgG) | Noninvasive | Also shows past infection — unsuitable for monitoring treatment |

**CAUTION**

Before breath and stool tests, proton pump inhibitors, antibiotics and bismuth preparations are stopped for a set period — otherwise the result will be a **false negative**. Serology stays positive for a long time even after treatment, so it is not suitable for monitoring eradication.

## 7. Practical skill: examining stool for intestinal infection

1. Stool is collected **before antibiotics**, if possible in the first days of the illness, into a sterile container; if there are pathological admixtures (mucus, blood), the sample is taken from exactly that part.
2. The sample is delivered within 2 hours; if delayed, **transport medium** (Cary–Blair) is used.
3. Inoculation: Endo/Levine (general), Ploskirev and bismuth sulfite agar (Salmonella/Shigella), selective media (for Campylobacter — 42 °C under microaerophilic conditions).
4. After 18–24 hours the colonies are assessed: lactose-positive (colored) and lactose-negative (colorless) colonies are separated.
5. A pure culture is picked from suspicious colonies → Kligler, the biochemical series → **serological identification** (diagnostic sera).
6. Where necessary, PCR (diarrheagenic *E. coli*, Campylobacter, viral diarrheas) and an antibiogram.
7. The final answer states the species of the causative agent, its serovar (if any) and its antibiotic susceptibility.

## 8. Review questions

1. State the general properties of the family Enterobacteriaceae. Which test shows membership of the family?
2. How is the result read on Endo and Levine media?
3. What is the antigenic structure of enterobacteria?
4. What is the role of *E. coli* in the normal flora and in sanitary microbiology?
5. State the diarrheagenic variants of *E. coli* and their mechanisms.
6. Why is finding *E. coli* in stool not sufficient for a diagnosis?
7. What are the morphological and cultural features of Klebsiella?
8. How is the psychrophilia of Yersinia used in diagnosis?
9. What are the growth conditions for Campylobacter?
10. How does *H. pylori* survive in an acidic environment? List the diagnostic methods.

## Test questions

1. The test that distinguishes Enterobacteriaceae from *Pseudomonas*:
   A) Catalase
   B) Oxidase (negative in enterobacteria)
   C) Hemolysis
   D) Motility
   E) Indole

2. A lactose-positive colony on Endo medium:
   A) Colorless
   B) Red, with a metallic sheen
   C) With a black center
   D) Blue-green
   E) Yellow

3. How the flagellar antigen is designated:
   A) O
   B) H
   C) K
   D) Vi
   E) M

4. “Traveler’s diarrhea” is most often associated with which variant:
   A) ETEC
   B) EPEC
   C) EIEC
   D) EHEC
   E) EAEC

5. The variant associated with the risk of hemolytic–uremic syndrome:
   A) ETEC
   B) EPEC
   C) EIEC
   D) EHEC (O157:H7)
   E) EAEC

6. The toxin produced by EHEC:
   A) Neurotoxin
   B) Shiga-like (Vero) toxin
   C) Erythrogenic toxin
   D) Exfoliative toxin
   E) Lecithinase

7. The most frequent causative agent of urinary tract infection:
   A) *E. coli*
   B) *P. aeruginosa*
   C) *S. aureus*
   D) *K. pneumoniae*
   E) *Proteus*

8. The morphological feature of *Klebsiella pneumoniae*:
   A) Motile, noncapsulated
   B) Nonmotile, large encapsulated
   C) Spore-forming
   D) Gram-positive
   E) Spiral-shaped

9. The distinctive feature of Yersinia:
   A) It grows only at 42 °C
   B) It multiplies at low temperature (4–8 °C)
   C) Obligate anaerobe
   D) It forms spores
   E) Oxidase-positive

10. The enrichment method used in the diagnosis of yersiniosis:
    A) Warm enrichment
    B) “Cold enrichment” (at 4 °C)
    C) Flotation
    D) Membrane filter
    E) Centrifugation

11. The “pseudoappendicular” form of yersiniosis is associated with what:
    A) Mesenteric lymphadenitis and terminal ileitis
    B) Meningitis
    C) Pneumonia
    D) Osteomyelitis
    E) Endocarditis

12. The growth conditions of *Campylobacter jejuni*:
    A) Anaerobic, 37 °C
    B) Microaerophilic, 42 °C
    C) Aerobic, 22 °C
    D) Anaerobic, 42 °C
    E) Aerobic, 4 °C

13. What can develop as a complication of campylobacteriosis:
    A) Guillain–Barré syndrome
    B) Tetanus
    C) Botulism
    D) Scarlet fever
    E) Diphtheritic croup

14. The enzyme that ensures the survival of *H. pylori* in an acidic environment:
    A) Catalase
    B) Urease
    C) Oxidase
    D) Coagulase
    E) Lecithinase

15. The method unsuitable for monitoring *H. pylori* eradication:
    A) Urea breath test
    B) Stool antigen test
    C) Serology (IgG)
    D) Culture of the biopsy
    E) Histological examination

## Clinical cases

### *Case 1*

A child has bloody diarrhea; on stool culture lactose-positive colonies grew on Endo medium, biochemically *E. coli*.

**Question:** Does this result prove a diarrheagenic strain? What should be done?

**Answer:** No — *E. coli* is a member of the normal flora. To prove it, **serological identification** (O157 and others) and **detection of virulence genes by PCR** are needed. Excluding EHEC is especially

important in bloody diarrhea.

### *Case 2*

A patient operated on with suspected acute appendicitis has an almost unchanged appendix and enlarged mesenteric lymph nodes. In recent days the patient ate a vegetable salad that had been stored in the refrigerator for a long time.

**Question:** Which pathogen should be considered, and what is special about the examination?

**Answer:** Yersiniosis (*Y. enterocolitica*) — the “pseudoappendicular” form. The examination uses **cold enrichment**, inoculation is done at 25–28 °C; the dynamics of the titer in paired sera are assessed serologically.

### *Case 3*

A patient with chronic gastritis was prescribed a urea breath test; the patient had been taking a proton pump inhibitor up to the day of the test. The result was negative, but the symptoms persist.

**Question:** How should the result be assessed?

**Answer:** A **false negative** is highly likely: PPIs and antibiotics should have been stopped for a set period before the test. The test is repeated after the drugs are stopped, or another method is chosen (stool antigen, biopsy).

## Quick review

**Table 20.4.** *Topic 20 — key concepts*

| Concept | Essence |
|---|---|
| Enterobacteriaceae | Gram−, oxidase −, glucose +, differentiated by lactose |
| Antigens | O (somatic), H (flagellar), K/Vi (capsular) |
| Endo/Levine | Lactose + → colored; lactose − → colorless |
| *E. coli* | Normal flora + sanitary indicator + diarrheagenic variants |
| ETEC/EPEC/EIEC/EHEC | Enterotoxin / newborn / invasive / Vero toxin, HUS |
| Klebsiella | Nonmotile, encapsulated, mucoid colony, hospital-acquired infection |
| Yersinia | Psychrophilic; cold enrichment; 25–28 °C; pseudoappendicular form |
| Campylobacter | Microaerophilic, 42 °C, curved motile rod; Guillain–Barré |
| *H. pylori* | Urease → ammonia; gastritis, ulcer; breath/stool tests |
| Serology (*H. pylori*) | Not suitable for monitoring eradication |

TOPIC 21

# Causative agents of shigellosis and cholera. Amebic dysentery

## 1. Learning objectives

In this class the student will master the following:

- To know the classification and biological properties of Shigella and the pathogenesis of dysentery;
- to explain the properties of *Vibrio cholerae*, its toxin and the pathogenesis of cholera;
- to master the algorithm of microbiological diagnosis of shigellosis and cholera;
- Independent work: to know the causative agent of amebic dysentery and its laboratory diagnosis.

## 2. Shigella and bacterial dysentery (shigellosis)

**Morphology and properties:** gram-negative rods; **nonmotile** (no flagella — a distinguishing feature among enterobacteria), nonspore-forming, noncapsulated. **Lactose-negative** (colorless colonies on Endo medium), **$H_2S$ negative**, usually **produce no gas**, lysine decarboxylase negative.



**Figure 21.1.** *Shigella — nonmotile gram-negative rods. Source: Muhamedov I.M. et al., 2019.*

**Table 21.1.** *Classification of Shigella*

| Group | Species | Property |
|---|---|---|
| A | *Sh. dysenteriae* | Produces **Shiga toxin** (serovar 1); the most severe course |
| B | *Sh. flexneri* | Common in developing countries; tends to become chronic |
| C | *Sh. boydii* | Relatively rare |
| D | *Sh. sonnei* | Common in developed regions; milder course; **ferments lactose late** |

**Pathogenicity factors:**

- **Invasiveness** — the main mechanism: Shigella **penetrate** the epithelium of the large intestine, multiply inside the cell and spread to neighboring cells → erosion and ulcers form;
- **Shiga toxin** (in *Sh. dysenteriae* serovar 1) — stops protein synthesis (cytotoxic), has neurotoxic and enterotoxic effects; it can be linked to hemolytic–uremic syndrome;
- **Endotoxin (LPS)** — general intoxication.

**Pathogenesis and clinical picture:** the infectious dose is very low (a few dozen cells) — so it spreads easily by the **contact-household** route. The process is mainly in the **distal large intestine**; the clinical picture: frequent, scanty, **stool mixed with mucus and blood**, **tenesmus** (a painful false urge to defecate), spasmodic abdominal pain, fever.

> **REMEMBER**
> Dysentery is an **invasive** diarrhea: the stool is scanty but **mucous and bloody**, with tenesmus. Cholera is a **secretory (toxin-mediated)** diarrhea: the stool is very copious and watery, with little pain or fever.

**Laboratory diagnosis:**

1. **Material:** stool (the mucous-bloody part), rectal swab; the sample is delivered **as quickly as possible** or in **Cary–Blair** transport medium (Shigella are unstable in the external environment).
2. **Inoculation:** Endo, Ploskirev (selective), Levine → **colorless (lactose-negative) colonies**.
3. **Identification:** Kligler (slant red, butt yellow, **no gas, no $H_2S$**), nonmotility (in semisolid agar), the biochemical series.
4. **Serological identification:** agglutination with diagnostic sera specific to group and type.
5. **Additional:** antibiogram; PCR (invasiveness and toxin genes); the colonoscopic picture — clinically supportive.

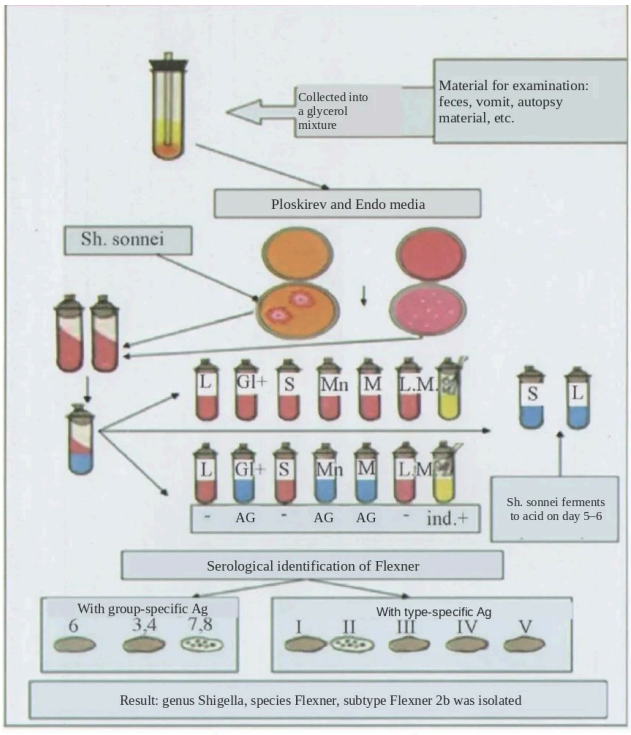



**Figure 21.2.** *Scheme of the microbiological examination in dysentery: material, Ploskirev and Endo media, biochemical properties and serological identification. Source: Aliyev Sh.R. et al., 2013.*

**CAUTION**

*Sh. sonnei* can ferment lactose **late (after 48 hours)**. If the result is read late, it may be wrongly assessed as "lactose-positive" and the organism may fail to be identified as Shigella (recall the situational case in topic 7).

## 3. The causative agent of cholera — *Vibrio cholerae*

**Morphology:** gram-negative, **comma-shaped** curved rod (a vibrio); **monotrichous** — highly motile thanks to a single polar flagellum (directed movement like a "shoal of fish"); nonspore-forming, noncapsulated.

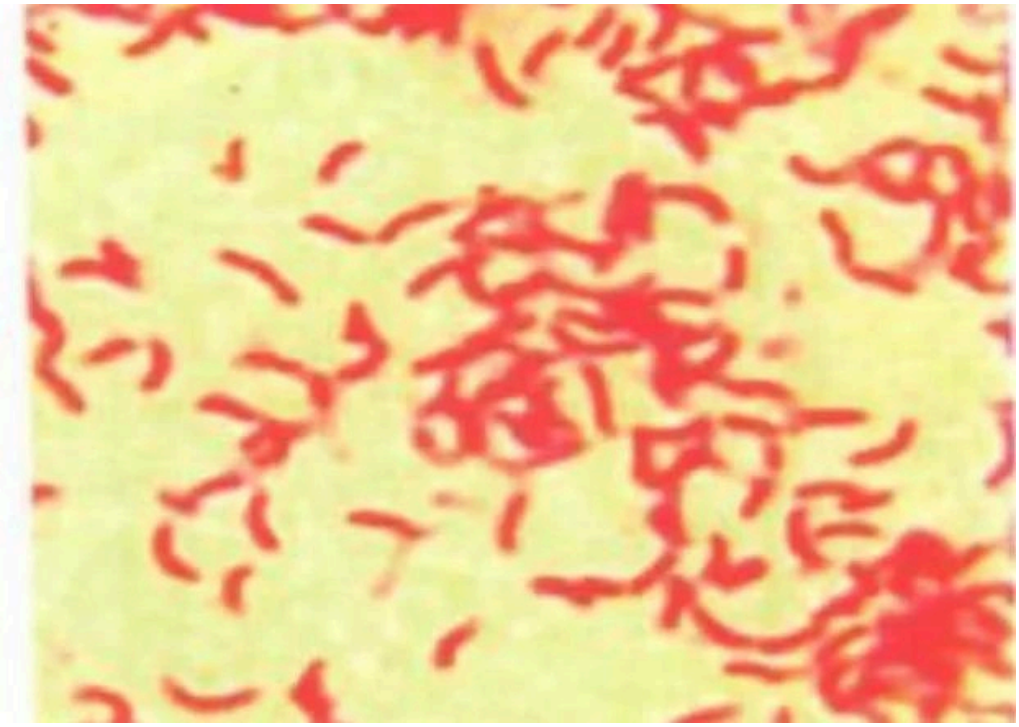


**Figure 21.3.** *Cholera vibrios — comma-shaped motile gram-negative bacteria. Source: Muhamedov I.M. et al., 2019.*

**Cultural and biochemical properties:**

- **alkaliphilic (alkali-loving)** — grows well at pH 8.5–9.0; therefore **alkaline peptone water** is used as the enrichment medium: the vibrio forms a **thin pellicle** on the surface and outgrows the accompanying flora;
- the selective solid medium is **TCBS agar** (yellow colonies) or alkaline agar (transparent, bluish colonies);
- **oxidase positive** (unlike enterobacteria!), ferments sucrose and mannose, does not ferment arabinose.

**Serological groups:** O1 (classical and **El Tor** biovars; Ogawa and Inaba serovars) and **O139** — of epidemic significance. Other (non-O1) vibrios usually cause mild gastroenteritis.

**The main pathogenicity factor — cholera toxin (choleragen):** it consists of A and B subunits; the B subunit binds to a receptor on the intestinal epithelium, and the A subunit enters the cell and **permanently activates adenylate cyclase → cAMP** rises sharply → **water and electrolytes** ($Na^+$, $K^+$, $Cl^-$, $HCO_3^-$) leave the cell into the intestinal lumen.

**Pathogenesis:** the vibrio **does not enter the epithelium**, stays in the intestinal lumen and produces toxin. Therefore:

- the stool looks like **“rice water”** — watery, odorless, with whitish flecks; no blood or mucus;
- inflammation and high fever are usually absent;
- the main danger is **severe dehydration**, loss of electrolytes, metabolic acidosis and hypovolemic shock.

**CLINICAL SIGNIFICANCE**

The basis of treatment in cholera is **rehydration** (replacing water and electrolytes): oral rehydration solutions and, in severe cases, intravenous infusion solutions. Antibiotics play a supporting role (they shorten the course of the disease and the shedding of the vibrio). It is carried out as prescribed by a physician.

**MEMORY AID**

Cholera toxin: **“cAMP goes up — water pours out.”** That is why the stool is watery, resembling “rice water.”

**Laboratory diagnosis (rapid and staged):**

1. **Material:** stool, vomit; in fatal cases — a piece of intestine; water and environmental samples (in epidemiological investigation). The requirements of the **especially dangerous infection** regime are observed.
2. **Rapid methods:** seeing **motility** in a wet mount (dark-field/phase-contrast) and the **cessation of motility** with specific serum; immunofluorescence, latex agglutination, PCR.
3. **Enrichment: alkaline peptone water**, 37 °C, 6–8 hours → subculture from the pellicle on the surface.
4. **Inoculation:** TCBS or alkaline agar.
5. **Identification:** oxidase, biochemical properties, agglutination with O1/O139 sera; additional tests to determine the biovar (classical/El Tor) (phage susceptibility, hemagglutination, polymyxin susceptibility).

**CAUTION**

Cholera is an **especially dangerous (quarantine) infection**: when it is suspected, the material is collected and transported in a special manner, and the health system is notified immediately. Laboratory work is carried out under the prescribed biosafety regime.

## 4. Independent work: Amebic dysentery (amebiasis)

**Causative agent:** *Entamoeba histolytica* — a protozoan (class Sarcodina).

**Life cycle and forms:**

- **Cyst** — resistant in the external environment, the **infective** form (4 nuclei); it is swallowed with water and food;
- **Trophozoite** — the vegetative form: *forma minuta* (in the intestinal lumen, harmless) and **forma magna (the tissue form)** — the form that has ingested erythrocytes and invades tissue.

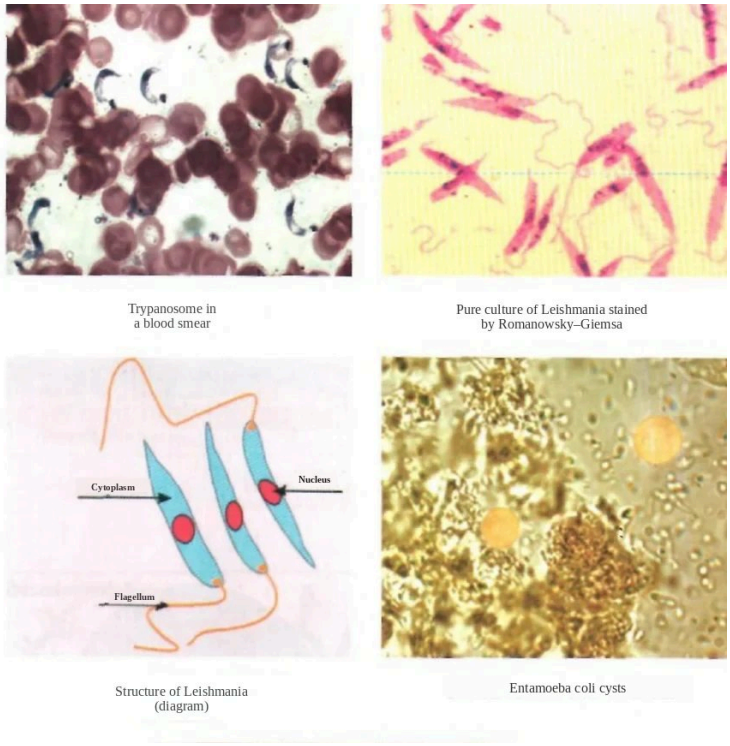



**Figure 21.4.** *Protozoa: microscopic appearance of Entamoeba cysts and other parasites. Source: Muhamedov I.M. et al., 2019.*

**Pathogenesis and clinical picture:** the tissue form penetrates the mucosa of the large intestine and forms **"flask-shaped" ulcers**; through the bloodstream it can reach the liver and cause a **liver abscess**. Clinically: a gradual onset, **blood and mucus in the stool** resembling "raspberry jelly," abdominal pain; fever is often mild.

**Table 21.2.** *Comparison of bacterial and amebic dysentery*

| Feature | Bacterial (shigellosis) | Amebic |
|---|---|---|
| Onset | Acute | Gradual |
| Fever | Often high | Often low or absent |
| Stool | Scanty, mucus and blood, tenesmus | Resembling "raspberry jelly" |
| Complications | Dehydration, HUS (Shiga toxin) | **Liver abscess** |
| Diagnosis | Culture + serological identification | Microscopic examination of stool, antigen, PCR |

**Laboratory diagnosis:**

1. **Wet mount** — from **warm, fresh** stool; seeing motile **trophozoites with ingested erythrocytes** (the main sign of the tissue form);
2. **Staining with Lugol's solution** — to see the cysts and the number of nuclei in them;
3. **Antigen detection (ELISA)** and **PCR** — distinguish *E. histolytica* from the morphologically identical but nonpathogenic *E. dispar*;
4. **Serology** — important in tissue forms, especially in **liver abscess**;
5. In liver abscess — imaging methods (ultrasound, CT) and examination of the aspirate.

**REMEMBER**

Finding an amebic **cyst** in stool may also mean carriage. The sign that proves disease is a **trophozoite with ingested erythrocytes** (the tissue form) or a positive antigen/PCR result.

## 5. Practical skill: choosing a method when examining stool for intestinal infections

1. **Mucous-bloody stool + tenesmus** → inoculation for shigellosis (Ploskirev, Endo) + excluding invasive variants of *E. coli*; in parallel — a wet mount (to exclude amebae).
2. **Watery, odorless, copious stool (rice water) → suspect cholera**: enrichment in alkaline peptone water, TCBS, rapid methods; the especially dangerous infection regime.
3. **Gradual onset, no fever, "raspberry jelly"** → examination for amebiasis: warm wet mount, antigen test, PCR.
4. **Bloody diarrhea + impaired kidney indices in a child** → excluding EHEC and Shiga toxin.
5. In all cases: the sample is taken **before antibiotics**, from the part with pathological admixtures, and delivered quickly; if delay is likely, transport medium is used.

## 6. Review questions

1. State the distinguishing features of Shigella among enterobacteria.
2. What are the groups of Shigella and their clinical significance?
3. What is the main mechanism of the pathogenesis of shigellosis?
4. How does Shiga toxin act?
5. How are the character of the stool and tenesmus in dysentery explained?
6. State the morphological and cultural properties of *Vibrio cholerae*.
7. Why is alkaline peptone water used?
8. Explain step by step the mechanism of action of cholera toxin.
9. What is the basis of treatment in cholera?
10. Which forms are distinguished in amebiasis? What proves the disease?

## Test questions

1. The distinguishing feature of Shigella among enterobacteria:
   A) Nonmotility
   B) Oxidase positivity
   C) Spore formation
   D) Gram-positivity
   E) Its capsule

2. The Shigella colony on Endo medium:
   A) Red, with a metallic sheen
   B) Colorless (lactose-negative)
   C) Black
   D) Yellow
   E) Blue-green

3. The Shigella that produces Shiga toxin:
   A) *Sh. sonnei*
   B) *Sh. dysenteriae* (serovar 1)
   C) *Sh. boydii*
   D) *Sh. flexneri*
   E) All are the same

4. The main mechanism of the pathogenesis of shigellosis:
   A) Release of water through an enterotoxin
   B) Multiplication after entering the epithelium (invasion)
   C) Spread through the blood
   D) Damage to the nervous system
   E) Breakdown of erythrocytes

5. The clinical sign characteristic of dysentery:
   A) Copious, watery, odorless stool
   B) Scanty, mucous-bloody stool and tenesmus
   C) Severe constipation
   D) Jaundice
   E) Paralysis

6. The type of flagellation of *Vibrio cholerae*:
   A) Peritrichous
   B) Monotrichous
   C) Lophotrichous
   D) Amphitrichous
   E) Atrichous

7. The medium used to enrich *Vibrio cholerae*:
   A) Alkaline peptone water
   B) Kitt–Tarozzi
   C) Sabouraud
   D) Löwenstein–Jensen
   E) Bordet–Gengou

8. The test that distinguishes *Vibrio cholerae* from enterobacteria:
   A) Catalase
   B) Oxidase (positive in the vibrio)
   C) Coagulase
   D) Indole
   E) Urease

9. The mechanism of action of cholera toxin:
   A) It stops protein synthesis
   B) It activates adenylate cyclase and raises cAMP
   C) It stops the release of GABA
   D) It perforates the membrane
   E) It breaks down DNA

10. The characteristic appearance of the stool in cholera:
    A) “Raspberry jelly”
    B) “Rice water”
    C) Black, tarry
    D) Pale
    E) Green, mucous

11. The basis of treatment in cholera:
    A) Rehydration (restoring water–electrolyte balance)
    B) Antitoxic serum
    C) Antibiotics alone
    D) Vaccination
    E) Surgery

12. The serogroups of *Vibrio cholerae* of epidemic significance:
    A) O1 and O139
    B) O157
    C) A and B
    D) I and II
    E) K and H

13. The infective form in amebiasis:
    A) Trophozoite
    B) Cyst
    C) Merozoite
    D) Gamete
    E) Sporozoite

14. The main complication of amebic dysentery:
    A) Hemolytic–uremic syndrome
    B) Liver abscess
    C) Tetanus
    D) Myocarditis
    E) Guillain–Barré syndrome

15. The microscopic sign that proves disease in amebiasis:
    A) Finding cysts
    B) Trophozoites with ingested erythrocytes (the tissue form)
    C) Leukocytes
    D) Fungal elements
    E) Crystals

## Clinical cases

### *Case 1*

A patient passes scanty stool mixed with mucus and blood 15–20 times a day, with severe tenesmus and a temperature of 38.5 °C. On stool culture colorless colonies grew on Endo medium; the culture is nonmotile, $H_2S$ negative and produces no gas.

**Question:** What are the probable causative agent and the method of final confirmation?

**Answer:** Shigella — the features (nonmotility, lactose-negative, no $H_2S$ or gas) fit. The final confirmation is **agglutination with group- and type-specific diagnostic sera** (serological identification), followed by an antibiogram.

### *Case 2*

In the hot season, several patients have very copious watery, odorless stool many times a day; there is almost no abdominal pain or fever, and rapid dehydration is developing.

**Question:** What should be suspected, and what are the first measure and the laboratory tactics?

**Answer:** Suspected cholera. The first measure is to start **rehydration** immediately. Laboratory: the material is collected under the especially dangerous infection regime, enriched in alkaline peptone water and inoculated onto TCBS; rapid methods in parallel (cessation of motility, PCR). The health system is notified.

### *Case 3*

A patient has had diarrhea that gradually worsened over several weeks, with mucus and blood in the stool; no fever. The stool culture is negative for pathogenic bacteria. A space-occupying lesion was found in the liver.

**Question:** What should be considered, and what examination is needed?

**Answer:** Amebiasis and its complication — a liver abscess. Examination: looking for **trophozoites with ingested erythrocytes** in a warm wet mount, a stool antigen test and PCR, serology (important in tissue forms), imaging methods.

## Quick review

**Table 21.3.** *Topic 21 — key concepts*

| Concept | Essence |
|---|---|
| Shigella | Gram−, **nonmotile**, lactose −, $H_2S$ −, no gas |
| Groups | A *dysenteriae* (Shiga toxin), B *flexneri*, C *boydii*, D *sonnei* |
| Shigellosis | Invasive diarrhea: scanty, mucous-bloody stool, tenesmus |
| Vibrio | Comma-shaped, monotrichous, **oxidase +**, alkaliphilic |
| Enrichment | Alkaline peptone water → pellicle on the surface |
| Cholera toxin | cAMP ↑ → water and electrolytes leave the cell |
| Cholera stool | “Rice water,” watery, odorless |
| Treatment of cholera | Rehydration is the mainstay; antibiotics are supportive |
| Serogroups | O1 (classical, El Tor), O139 |
| Amebiasis | Cyst (infective) ↔ trophozoite; the tissue form has ingested erythrocytes |
| Complication of amebiasis | Liver abscess |

TOPIC 22

# Causative agents of typhoid fever and paratyphoid A and B. Opportunistic enterobacteria and viral diarrheas

## 1. Learning objectives

In this class the student will master the following:

- To know the biological properties and antigenic structure of the causative agents of typhoid fever and paratyphoid;
- to explain the stages of the pathogenesis of typhoid fever and the corresponding examination methods;
- to master the concepts of hemoculture, coproculture, urinoculture and biliculture and the Widal reaction;
- to know the principle of detecting bacterial carriage;
- Independent work: to have an understanding of opportunistic intestinal bacteria (Citrobacter, Klebsiella, Enterobacter) and viral diarrheas (rotavirus, adenovirus).

## 2. The causative agents of typhoid fever and paratyphoid

**Causative agents:** *Salmonella* Typhi (typhoid fever), *S.* Paratyphi A and *S.* Paratyphi B.

**Morphology and properties:** gram-negative, **peritrichous** (motile) rods; nonspore-forming, noncapsulated (the Vi antigen is present). **Lactose-negative**, they ferment glucose; *S.* Typhi **produces no gas**, whereas the paratyphoids **produce gas**; *S.* Typhi produces little $H_2S$, paratyphoid B a lot.

**Antigenic structure (based on the Kauffmann–White scheme):**

- **O antigen** — somatic, heat-stable; determines the group (*S.* Typhi — group D, Paratyphi A — A, Paratyphi B — B);
- **H antigen** — flagellar, heat-labile;
- **Vi antigen** — a surface (capsule-type) antigen; associated with **virulence** and “masks” the O antigen; especially important **in detecting carriage**.

> **REMEMBER**
> The Vi antigen is the **“virulence” antigen**: it protects the live bacterium from agglutination with O serum. Therefore, if there is no O agglutination in serological identification, the culture is heated (the Vi antigen is destroyed) and the reaction is repeated.

**Pathogenesis — the stages (this determines diagnostic tactics!):**

1. **Entry and multiplication in the intestine** (incubation period, 7–14 days): the bacteria enter the mucosa of the small intestine and multiply in Peyer’s patches and the mesenteric lymph nodes.
2. **Bacteremia** (week 1): the bacteria pass into the blood — the clinical picture begins (fever, headache, intoxication, relative bradycardia, a roseolar rash on the abdomen).
3. **Parenchymatous dissemination** (week 2): they settle in the liver, spleen, bone marrow and gallbladder.
4. **Excretion stage** (weeks 2–3): they are excreted from the gallbladder with bile into the intestine and through the kidney into the urine; **necrosis and ulcers** form in Peyer’s patches owing to sensitization — complications (intestinal bleeding, perforation) occur precisely in this period.

5. **Recovery or transition to carriage:** the pathogen can persist for a long time in the gallbladder (especially in gallstone disease), and **chronic carriage** may develop.

> **CLINICAL SIGNIFICANCE**
> In typhoid fever the rule **"stage — material"** is basic: **week 1 — blood** (hemoculture), **from week 2 — stool and urine** (coproculture, urinoculture), **from week 2 — serum** (Widal/PHA), **after recovery — bile** (biliculture, detecting carriage).

## 3. Laboratory diagnosis

**Table 22.1.** *Examination in typhoid fever according to the stage of the disease*

| Period | Material | Method | Note |
|---|---|---|---|
| Week 1 (onset of fever) | **Blood** | **Hemoculture** — Rappoport or bile-broth medium | The most reliable; blood is inoculated into the medium at a 1:10 ratio |
| Weeks 2–3 | **Stool, urine** | Coproculture, urinoculture | Effective in the excretion stage |
| From week 2 | **Blood serum** | **Widal reaction**, PHA (O, H and Vi diagnosticums) | Dynamics of the titer in paired sera |
| Recovery, carriage | **Bile (duodenal juice)**, stool, urine | **Biliculture**, detecting Vi antibodies | Detecting chronic carriage |

**Hemoculture technique:** 5–10 ml of blood is drawn from a vein and inoculated into a **10-fold larger** volume of liquid medium (Rappoport — a medium containing bile, glucose and an indicator). Dilution reduces the bactericidal action of the blood serum. After 18–24 hours at 37 °C, it is assessed by color change and gas, then subcultured onto Endo/Ploskirev or bismuth sulfite agar.

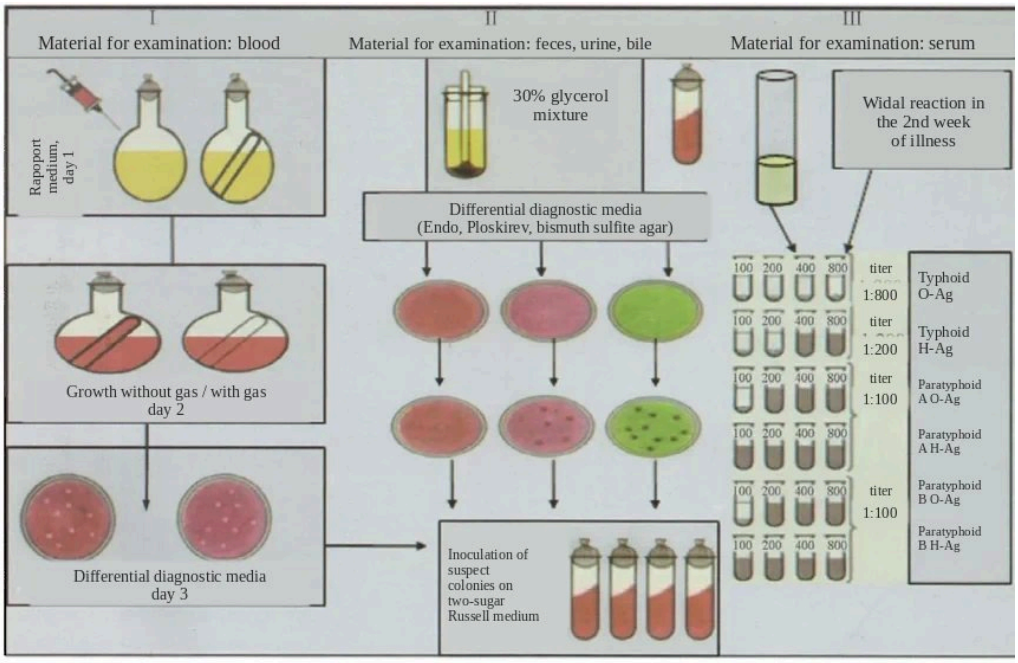
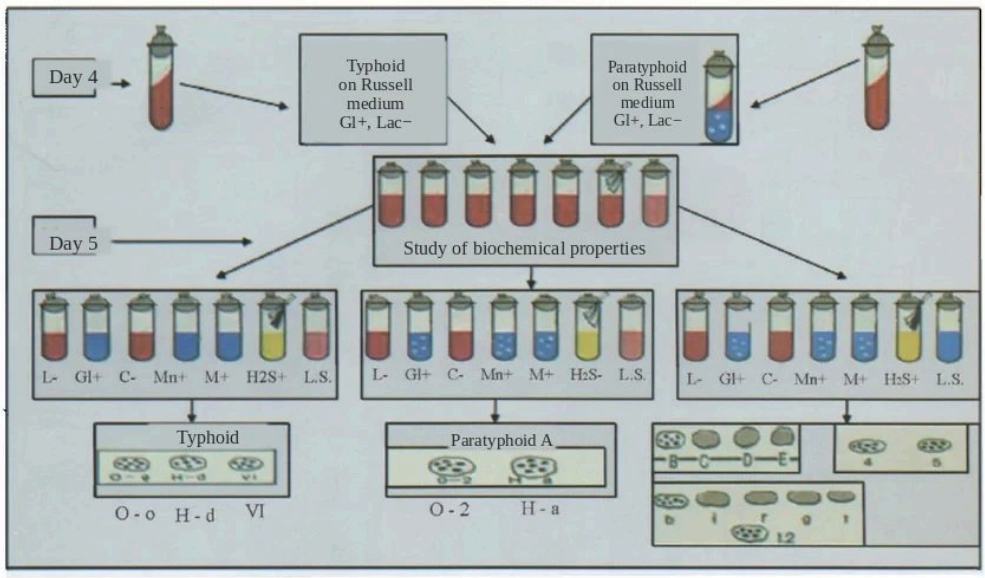



**Figure 22.1.** *Scheme of the bacteriological examination in typhoid fever and paratyphoid: hemoculture (Rappoport medium), inoculation onto differential-diagnostic media, biochemical properties, and serological examination by the Widal reaction. Source: Aliyev Sh.R. et al., 2013.*

**Identification:** lactose-negative colonies → Kligler (slant red, butt yellow; in *S.* Typhi no gas, little $H_2S$; on bismuth sulfite agar a **black colony**) → the biochemical series → agglutination with **O, H and Vi sera**.




Scheme 13. Methods of bacteriological examination for the agents of typhoid and paratyphoid A and B in different periods of the disease. I – first-period test (hemoculture); II – second-period test (coproculture); III – third-period test (Widal reaction)

**Figure 22.2.** *Scheme of the bacteriological examination of the causative agents of typhoid fever and paratyphoid A and B: I — hemoculture; II — coproculture; III — serological examination (Widal reaction). Source: Aliyev Sh.R. et al., 2013.*

**The Widal reaction** — an extended agglutination test: the patient's serum is serially diluted and set up with O, H and Vi diagnosticums.

**Interpretation:**

- **O antibodies** appear early and fall quickly → they indicate an **active (acute)** process;
- **H antibodies** appear later and persist for a long time → they may be positive after past disease or after vaccination;
- **Vi antibodies** → an indicator of **chronic carriage**.

> **CAUTION**
> A single positive Widal reaction is not enough for a final diagnosis: it can be positive in a vaccinated person or one who has had the disease before. The sign that has diagnostic value is **at least a 4-fold rise in titer in paired sera** (topic 11).

**Rapid and modern methods:** PCR (blood, stool), antigen tests; MALDI-TOF in identification.

**Prophylaxis:** sanitary and hygienic measures (water supply, sewage, food control), detecting and monitoring carriers, vaccination on indication (Vi-polysaccharide and live vaccines).

## 4. Detecting bacterial carriage

**Types of carriage:** acute (up to 3 months), **chronic** (over 3 months), transitory.

**Detection scheme:**

1. Repeated **coproculture** and **urinoculture** (by a set scheme, several times);
2. **Biliculture** — inoculation of bile portions obtained by duodenal intubation (the most reliable, because the main habitat of the pathogen is the gallbladder);
3. Detecting **Vi antibodies** (PHA) — as a screening test;
4. PCR where necessary.

**CLINICAL SIGNIFICANCE**

Chronic carriers are one of the main sources of the disease; therefore people who work in food, water supply, and children's and medical institutions are examined by a set procedure.

## 5. Independent work 1: Opportunistic intestinal bacteria

**Representatives:** *Citrobacter, Klebsiella, Enterobacter, Serratia, Proteus, Morganella, Providencia*.

**General features:** members of the normal flora or of the external environment; they cause infection **endogenously** or in hospital settings; often **multidrug-resistant**.

**Table 22.2.** *Opportunistic enterobacteria*

| Genus | Characteristic feature | Clinical significance |
|---|---|---|
| *Citrobacter* | Uses citrate, $H_2S$ + | Diarrhea in children, UTI, hospital-acquired infections |
| *Klebsiella* | Nonmotile, encapsulated, mucoid colony | Pneumonia, UTI, sepsis, neonatal infections |
| *Enterobacter* | Motile, citrate + | Hospital-acquired infections, sepsis |
| *Proteus* | **Urease positive**, spreads over the agar by **"swarming" (H growth)**, putrid odor | UTI (alkalinizes the urine, contributing to **stone formation**), wound infections |

**Rule of interpretation in diarrhea:** these bacteria can be present in stool even in health. To judge them as the causative agent: other pathogens must have been excluded, the **quantity must be high** (massive growth), it must fit the clinical picture, and it must persist on follow-up.

## 6. Independent work 2: Viral diarrheas

### *6.1. Rotavirus*

**Features:** the family **Reoviridae**; double-stranded, **segmented RNA**; nonenveloped, resistant; on electron microscopy it has a **"wheel"** appearance (Latin *rota* — wheel).

**Epidemiology and clinical picture:** in children (especially 6 months to 2 years) **the most frequent cause of acute gastroenteritis**; fecal-oral route; seasonality — the cold period. Clinically: vomiting, **watery diarrhea**, fever; the main danger is **dehydration**.

**Diagnosis: antigen detection (ELISA, immunochromatographic rapid tests)** in stool — the main method in practice; PCR; electron microscopy (research).

**Prophylaxis: live oral rotavirus vaccine** — in infancy; it significantly reduces severe forms and hospitalizations.

### *6.2. Adenovirus (enteric types)*

**Features: Adenoviridae**; double-stranded DNA, nonenveloped, icosahedral capsid, with projections (fibers).

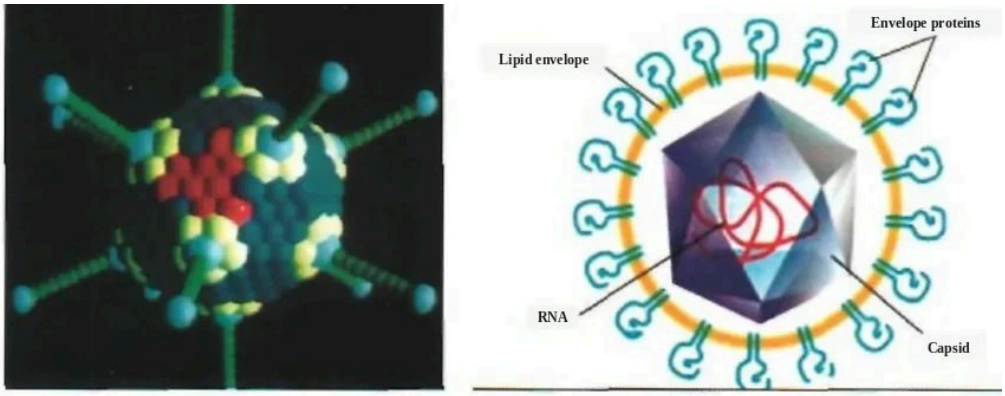



**Figure 22.3.** *The structure of adenovirus and hepatitis C virus (schematic image). Source: Muhamedov I.M. et al., 2019.*

**Clinical picture:** the enteric types (40, 41) cause longer-lasting diarrhea in children; other types — respiratory infection, conjunctivitis, pharyngoconjunctival fever.

**Diagnosis:** stool antigen test, PCR.

**REMEMBER**

In acute watery diarrhea in children, **viruses are considered first** (especially rotavirus); in mucous-bloody diarrhea — invasive bacteria (Shigella, invasive *E. coli*, Campylobacter). The basis of treatment is **rehydration**.

## 7. Practical skill: collecting a hemoculture

1. Blood is taken **during a rise in fever** and **before antibiotics**.
2. The skin is treated according to aseptic rules (wait until the antiseptic has dried) — this prevents **contamination**.
3. 5–10 ml of blood (an age-appropriate volume in children) is drawn from a vein with a sterile syringe.
4. The blood is immediately inoculated into a **10-fold larger** volume of medium (Rappoport or bile broth; in modern settings, automated bottles).
5. The bottle is placed at 37 °C; it is observed for signs of growth (turbidity, color change, gas), or the automated system gives a signal.
6. From a positive bottle a smear (Gram) and a subculture onto solid medium are made; then identification and an antibiogram.
7. For reliability, usually **several samples from different sites** are taken.

**CAUTION**

Inoculating blood into a small volume of medium or failing to treat the skin are the two most frequent mistakes: the first leads to a false-negative, the second to a false-positive (skin flora) result.

## 8. Review questions

1. State the morphological and biochemical properties of the causative agent of typhoid fever.
2. What is the significance of the Vi antigen?
3. State the stages of the pathogenesis of typhoid fever and the material suitable for each.
4. Why is hemoculture inoculated at a 1:10 ratio?

5. In which period are coproculture and urinoculture effective?
6. What do the O, H and Vi antibodies mean in the Widal reaction?
7. Why is a single positive Widal reaction not sufficient for a diagnosis?
8. Why is biliculture considered the most reliable for detecting carriage?
9. What are the criteria for judging opportunistic enterobacteria as the causative agent?
10. What are the clinical picture, diagnosis and prophylaxis of rotavirus infection?

## Test questions

1. The type of flagellation of the typhoid causative agent:
   A) Monotrichous
   B) Peritrichous
   C) Lophotrichous
   D) Amphitrichous
   E) Atrichous

2. The antigen associated with virulence that masks the O antigen:
   A) O
   B) H
   C) Vi-antigen
   D) K1
   E) M

3. The most reliable examination in week 1 of typhoid fever:
   A) Coproculture
   B) Hemoculture
   C) Urinoculture
   D) Biliculture
   E) Widal reaction

4. The ratio in which blood is inoculated into the medium in hemoculture:
   A) 1:1
   B) 1:2
   C) 1:10
   D) 1:100
   E) 1:1000

5. The reason blood is diluted when inoculated:
   A) To save medium
   B) To reduce the bactericidal action of the blood serum
   C) To change the color
   D) To raise the pH
   E) To increase gas formation

6. The medium used for hemoculture:
   A) Rappoport (bile broth)
   B) Sabouraud
   C) Löwenstein–Jensen
   D) Bordet–Gengou
   E) TCBS

7. Salmonella colonies on bismuth sulfite agar:
    A) Red
    B) Black, with a metallic sheen
    C) Colorless
    D) Yellow
    E) Blue

8. The antibodies indicating an active (acute) process in the Widal reaction:
    A) O antibodies
    B) H antibodies
    C) Vi antibodies
    D) IgE
    E) IgD

9. The antibodies indicating chronic carriage:
    A) O antibodies
    B) H antibodies
    C) Vi antibodies
    D) IgM
    E) Complement

10. The most reliable material for detecting carriage:
    A) Blood
    B) Bile (duodenal juice)
    C) Cerebrospinal fluid
    D) Sputum
    E) Throat swab

11. The period of highest risk of intestinal bleeding and perforation in typhoid fever:
    A) Incubation period
    B) Week 1
    C) Weeks 2–3
    D) Recovery period
    E) Carrier period

12. The characteristic features of *Proteus*:
    A) Urease positive, spreads over the agar by “swarming”
    B) Nonmotile, encapsulated
    C) Oxidase-positive
    D) Lactose positive, indole negative
    E) Gram-positive

13. The most frequent viral cause of acute gastroenteritis in children:
    A) Adenovirus
    B) Rotavirus
    C) Herpesvirus
    D) Rhinovirus
    E) Coronavirus

14. The main method in practice in the diagnosis of rotavirus infection:
    A) Antigen detection in stool (ELISA/rapid test)
    B) Blood culture
    C) Skin test
    D) Coagulase test
    E) Oxidase test

15. The particularity of rotavirus prophylaxis:
    A) Toxoid is given
    B) A live oral vaccine is used
    C) Antitoxic serum is given
    D) It is prevented with antibiotics alone
    E) No vaccine exists

## Clinical cases

### *Case 1*

A patient has had a stepwise rising fever since day 6, with headache and intoxication; single roseolas on the abdomen. The stool culture was negative.

**Question:** What error might there be in the examination, and what is the correct tactic?

**Answer:** In week 1 of the disease the pathogen is mainly **in the blood**; excretion into the stool begins later. The correct tactic is **hemoculture** (before antibiotics), later coproculture/urinoculture, and from week 2 serology (paired sera).

### *Case 2*

A woman who works in a canteen has no complaints, but a preventive examination detected **Vi antibodies** by PHA.

**Question:** What does this mean, and what is the next step?

**Answer:** Possible chronic carriage. The next step is repeated coproculture, urinoculture and **biliculture** (by duodenal intubation). If confirmed, the questions of employment and monitoring are resolved by the set procedure.

### *Case 3*

A 1-year-old child has vomiting, up to 10 watery stools a day and a low-grade fever. No pathogenic bacteria were found on stool culture; the stool antigen rapid test was positive.

**Question:** What are the probable causative agent and the basis of treatment?

**Answer:** Rotavirus gastroenteritis. The basis of treatment is **rehydration** (oral rehydration solutions, intravenous if necessary). Antibiotics are not indicated. The rotavirus vaccine is important in prophylaxis.

## Quick review

**Table 22.3.** *Topic 22 — key concepts*

| Concept | Essence |
|---|---|
| *S.* Typhi | Gram−, peritrichous, lactose −, no gas, little $H_2S$ |

| Concept | Essence |
|---|---|
| Antigens | O (group), H (flagellar), **Vi** (virulence, carriage) |
| Pathogenesis | Intestine → bacteremia → parenchymatous → excretion → carriage |
| Week 1 | **Hemoculture** (Rappoport, 1:10) |
| Weeks 2–3 | Coproculture, urinoculture; Widal/PHA |
| Widal | O — acute; H — past/vaccination; Vi — carriage |
| Diagnostic criterion | Titer rise ≥4-fold in paired sera |
| Carriage | Biliculture is the most reliable (gallbladder) |
| Opportunistic pathogens | Citrobacter, Klebsiella, Enterobacter, Proteus (urease +, “swarming”) |
| Rotavirus | The main viral diarrhea in children; stool antigen; live vaccine |
| Adenovirus 40, 41 | Enteric types, longer-lasting diarrhea |

TOPIC 23

# Foodborne infections and food poisonings: salmonellosis, botulism

## 1. Learning objectives

In this class the student will master the following:

- To know the difference between foodborne infectious diseases (toxicoinfections) and intoxications;
- to explain the properties, pathogenesis and diagnosis of the causative agents of salmonellosis;
- to master the mechanism of action of the botulism causative agent and its toxin, the clinical expression of the disease and its diagnosis;
- Independent work: to know poisonings associated with Proteus, Campylobacter, *E. coli* and Listeria, and the serological identification of bacteria of the enteric group.

## 2. General classification: toxicoinfection and intoxication

**Table 23.1.** *Main groups of food-related conditions*

| Group | Meaning | Incubation period | Examples |
|---|---|---|---|
| **Foodborne toxicoinfection** | **Live microbes** enter with the food, multiply in the intestine and release toxin | 6–48 hours | Salmonellosis, *Proteus*, *C. perfringens*, opportunistic pathogens |
| **Food intoxication** | **Preformed toxin** enters in the product; a live microbe is not required | **Short** (30 min–6 hours), or longer in botulism | Staphylococcal enterotoxin, **botulism** |
| Infectious disease (by the food route) | A classic infection; food is only a transmission factor | Variable | Typhoid fever, shigellosis, listeriosis, campylobacteriosis |

**REMEMBER**

**The incubation period — the first indicator:** very short (1–6 hours), with vomiting predominating — **preformed toxin** (staphylococcal enterotoxin); 12–48 hours, with fever — **multiplication of live microbes** (a toxicoinfection of the salmonellosis type).

**Staphylococcal enterotoxin** (topic 16) is **heat-stable**: reheating a product kills the bacteria but does not inactivate the toxin. For this reason even “reheated” food can cause poisoning.

## 3. Salmonelloses

**Causative agents:** the many serovars of *Salmonella enterica*, the most frequent being **Enteritidis** and **Typhimurium**. (The typhoid–paratyphoid group was covered in topic 22.)

**Properties:** gram-negative, peritrichous (motile) rods; **lactose-negative**, they ferment glucose with production of **acid and gas**, **$H_2S$ positive** (the Kligler butt blackens), urease negative, indole negative.

**Epidemiology:** the main source is **animals and poultry**; transmission factors are undercooked eggs and egg dishes, poultry and beef, dairy products; in addition, the contact-household route (especially in children) and contact with reptiles.

**Pathogenicity factors:** adhesins, **invasion factors** (entry into the intestinal epithelium), enterotoxin and cytotoxin, **endotoxin (LPS)**; virulence plasmids in *S.* Typhimurium and others.

**Clinical forms:**

1. **Gastrointestinal (common)** — fever, abdominal pain, vomiting, watery or mucous diarrhea;
2. **Generalized** — typhoid-like and septic forms; in newborns, the elderly and the immunocompromised;
3. **Bacterial carriage** — acute or chronic.

**CLINICAL SIGNIFICANCE**

In ordinary gastrointestinal salmonellosis an antibiotic is **not indicated**: it does not shorten the course of the disease and may, on the contrary, prolong the shedding of bacteria. Antibiotics are used on a physician's prescription in generalized forms and in risk groups (newborns, the elderly, the immunocompromised). The basis of treatment is **rehydration**.

**Laboratory diagnosis:**

1. **Material:** stool, vomit, gastric lavage, remains of suspect food; in a generalized form — **blood** (hemoculture), urine.
2. **Enrichment:** selenite or magnesium medium (for stool).
3. **Inoculation:** Endo, Ploskirev, **bismuth sulfite agar** (black colonies).
4. **Identification:** Kligler (slant red, butt yellow, **gas +, $H_2S$ +**), the biochemical series, motility.
5. **Serological identification:** agglutination with O and H sera according to the **Kauffmann–White scheme** — first polyvalent, then group-specific and type-specific sera.
6. **Additional:** antibiogram, PCR; in epidemiological investigation — examination of food and environmental samples.

**MEMORY AID**

Salmonella in Kligler: **"red/yellow + gas + black"** (slant alkaline, butt acidic, gas and $H_2S$ present). In Shigella, neither gas nor $H_2S$ is present — **absent** (topic 21).

## 4. Botulism

**Causative agent:** *Clostridium botulinum* — a gram-positive, **obligate anaerobe**, **subterminal spore**-forming rod ("tennis racket" shape); motile.

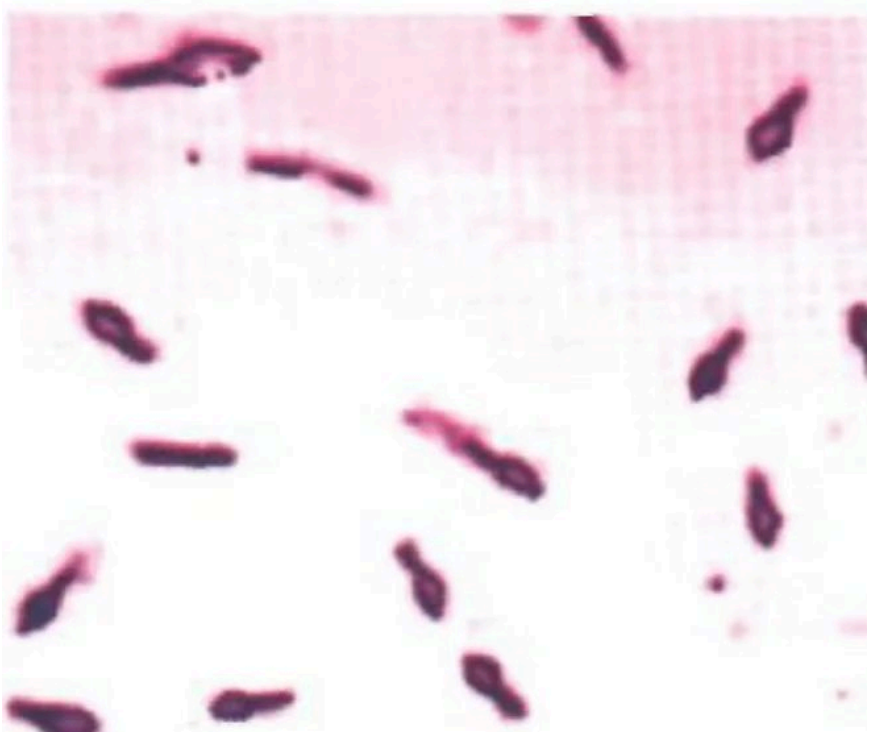


**Figure 23.1.** *The causative agent of botulism — gram-positive rods with subterminal spores ("tennis racket" appearance). Source: Muhamedov I.M. et al., 2019.*

**Botulinum toxin:** one of the most potent biological poisons in nature; **protein in nature**, **heat-labile** (broken down on boiling), but resistant to gastric acid and proteases. Serological types A, B, E, F (more common in humans) and others.

**Mechanism of action:** the toxin binds to peripheral cholinergic nerve endings and **stops the release of acetylcholine** → neuromuscular transmission is impaired → **flaccid (peripheral) paralysis** develops.

> **REMEMBER**
> Compare the two clostridial neurotoxins: **tetanus toxin — blocks inhibitory mediators → spastic (tonic) convulsions**; **botulinum toxin — blocks acetylcholine → flaccid paralysis**. In both, consciousness is preserved.

**Routes of transmission and forms:**

- **Foodborne botulism** — toxin accumulates in products in which anaerobic conditions have been created, such as home-canned sealed preserves, smoked fish and sausage;
- **Wound botulism** — multiplication of the pathogen in a wound;
- **Infant botulism** — germination of spores in the intestine in children under 1 year (therefore **honey is not recommended before age 1**).

**Clinical picture:** the incubation period is usually 12–36 hours (from 2 hours to several days, depending on the dose of toxin). At first nausea and abdominal discomfort may occur, then the typical **neurological signs**:

- **visual disturbance** — "a net before the eyes," double vision (diplopia), drooping of the eyelid (ptosis), dilated pupil;
- **swallowing and speech disturbance** — dysphagia, dysphonia, dry mouth;
- **symmetric, descending flaccid paralysis**; in severe cases — **paralysis of the respiratory muscles** (the main cause of death).

**Fever is usually absent and consciousness is preserved** — this distinguishes botulism from many infections.

**Laboratory diagnosis:**

1. **Material:** blood serum, vomit, gastric lavage, stool, **remains of suspect food**.
2. **The main method — detecting the toxin:** the classic variant is the **biological test (neutralization test)**: the material is mixed with type-specific antitoxic sera and injected into a laboratory animal; the serum that protects shows the type to which the toxin belongs. Modern variants — ELISA and other in vitro methods.
3. **Isolating the pathogen** — anaerobic culture (Kitt–Tarozzi and others); detecting toxin genes by PCR.
4. **In practice diagnosis rests on clinical and epidemiological data** — waiting for the laboratory answer must not delay treatment.

**CLINICAL SIGNIFICANCE**

The decisive treatment in botulism is giving **antitoxic serum** as **early** as possible: the serum neutralizes only toxin **in the blood that is not bound to nerve**. The effect of bound toxin cannot be reversed — so no time is lost. In severe cases breathing is supported. All prescriptions are made by a physician.

**Prophylaxis:** following the technology of home canning, not using suspect preserves (with a bulging lid or a changed smell), heating the product sufficiently before consumption (the toxin is heat-labile), not giving honey to children under 1 year.

## 5. Independent work: Other causative agents and serological identification

**Table 23.2.** *Other food-related causative agents*

| Causative agent | Transmission factor | Clinical feature |
|---|---|---|
| *Proteus* | Spoiled meat and fish products | Toxicoinfection; urease +, “swarming” growth |
| *Campylobacter jejuni* | Undercooked poultry, raw milk | Bloody diarrhea, severe pain; risk of Guillain–Barré (topic 20) |
| Diarrheagenic *E. coli* | Water, meat, vegetables | ETEC — watery; EHEC — bloody diarrhea, HUS (topic 20) |
| ***Listeria monocytogenes*** | **Refrigerated** ready-to-eat products, soft cheese, unpasteurized milk | **Psychrophilic**; in pregnant women (passes to the fetus), newborns and the elderly, **meningitis and sepsis**; a gram-positive rod, motile (“tumbling” motion at 22 °C), beta-hemolysis on blood agar |
| *Bacillus cereus* | Reheated rice and grain dishes | Two forms: vomiting (rapid) and diarrhea (slower) |

**CAUTION**

**Listeriosis** can be linked to products stored in the refrigerator, because the causative agent multiplies at low temperature. It is dangerous for pregnant women, newborns and the immunocompromised; in diagnosis, blood and cerebrospinal fluid culture is decisive.

**Serological identification of bacteria of the enteric group.** The isolated culture is tested with **diagnostic agglutinating sera**:

1. **A preliminary (orienting) reaction on a glass slide** — first with **polyvalent** serum: if it is positive, the group is determined;
2. Then it is clarified with **group** and **monoreceptor (type-specific)** O sera;
3. Then it is typed by the **H antigen** (in a motile culture);
4. Where necessary the **Vi serum** is used; if the Vi antigen interferes with O agglutination, the culture is heated;
5. The result is recorded as a serovar formula according to the **Kauffmann–White scheme**.

Serological identification is also important for epidemiological analysis: the same serovar shows that the cases spread from one source.

## 6. Practical skill: investigating a focus of food poisoning

1. **Epidemiological history:** who ate what and when; the number of cases; the **incubation period** and the dominant signs (vomiting or diarrhea; whether there are neurological signs).
2. **Collecting material:** from patients — vomit, gastric lavage, stool, blood (if a generalized form is suspected); **remains of the suspect product**, hand swabs from the people who prepared it.
3. **Choosing the direction:**

- short incubation + predominant vomiting → **staphylococcal enterotoxin**;

- 12–48 hours + fever + diarrhea → **salmonellosis** and other toxicoinfections;

- neurological signs (vision, swallowing, flaccid paralysis), no fever → **botulism**;

- bloody diarrhea → Shigella, Campylobacter, EHEC.

4. **Inoculation:** enrichment medium + Endo/Ploskirev/bismuth sulfite agar; salt egg-yolk agar for staphylococcus; when botulism is suspected — anaerobic culture and **detecting the toxin**.
5. **Identification and serological identification**, antibiogram.
6. **Conclusion:** the causative agent, the probable source and factor; sanitary and epidemiological measures.

## 7. Review questions

1. How do toxicoinfection and intoxication differ? What does the incubation period tell us?
2. What does the heat stability of staphylococcal enterotoxin mean in practice?
3. State the biochemical properties of Salmonella. What is the result in Kligler?
4. What are the clinical forms of salmonellosis? When is an antibiotic indicated?
5. Describe the stages of the diagnosis of salmonellosis.
6. What is the morphological feature of *C. botulinum*?
7. Explain the mechanism of action of botulinum toxin. How does it differ from tetanus toxin?
8. State the typical clinical signs of botulism. Why is there no fever?
9. How is the toxin detected in botulism? What is the basis of treatment?
10. What is special about Listeria, and who are the risk groups?

## Test questions

1. The main feature of food intoxication:
   A) Live microbes multiply in the intestine
   B) Preformed toxin enters in the product
   C) The incubation period is always long
   D) There is always fever
   E) Occurs only in children

2. The property of staphylococcal enterotoxin:
   A) Thermolabile
   B) Heat-stable — it persists even on reheating
   C) Formed only under anaerobic conditions
   D) Damages the nervous system
   E) Not a protein

3. The typical result of Salmonella in Kligler:
   A) Slant and butt yellow, no gas
   B) Slant red, butt yellow, gas +, $H_2S$ +
   C) Both red
   D) Only $H_2S$ +, no gas, lactose +
   E) No change at all

4. The selective medium for Salmonella:
   A) Bismuth sulfite agar
   B) Sabouraud
   C) Kitt–Tarozzi
   D) TCBS
   E) Bordet–Gengou

5. The scheme for the serological identification of Salmonella:
   A) Lancefield
   B) Kauffmann–White
   C) Runyon
   D) Bergey
   E) McFarland

6. The basis of treatment in ordinary gastrointestinal salmonellosis:
   A) An antibiotic
   B) Rehydration
   C) Antitoxic serum
   D) Bacteriophage
   E) A vaccine

7. The position of the spore of *C. botulinum*:
   A) Terminal
   B) Subterminal (“tennis racket”)
   C) Central
   D) No spore
   E) Exosporial

8. The mechanism of action of botulinum toxin:
   A) It stops the release of acetylcholine
   B) It stops the release of GABA and glycine
   C) It stops protein synthesis
   D) It raises cAMP
   E) It perforates the membrane

9. The type of paralysis that develops in botulism:
   A) Spastic
   B) Flaccid (peripheral)
   C) Convulsive
   D) Only unilateral
   E) Only in the facial muscles

10. The clinical signs characteristic of botulism:
    A) High fever and loss of consciousness
    B) Visual disturbance, dysphagia, flaccid paralysis; no fever, consciousness intact
    C) Bloody diarrhea
    D) Skin rash
    E) Jaundice

11. The relation of botulinum toxin to heat:
    A) Thermostable
    B) Heat-labile (broken down on boiling)
    C) Strengthened by heat
    D) Active only in the cold
    E) Temperature has no effect

12. The decisive part of treatment in botulism:
    A) Giving antitoxic serum early
    B) An antibiotic
    C) Bacteriophage
    D) A probiotic
    E) A vaccine

13. The recommendation to prevent infant botulism:
    A) Not giving honey before 1 year
    B) Not boiling milk
    C) Not giving vegetables
    D) Stopping breastfeeding
    E) Not giving eggs

14. The causative agent that multiplies in refrigerated products and is dangerous for pregnant women:
    A) *Listeria monocytogenes*
    B) *S. aureus*
    C) *Shigella sonnei*
    D) *V. cholerae*
    E) *C. tetani*

15. The causative agent of poisoning associated with reheated rice:
    A) *Bacillus cereus*
    B) *Listeria*
    C) *Proteus*
    D) *Campylobacter*
    E) *Yersinia*

## Clinical cases

### *Case 1*

3 hours after a wedding feast, several guests developed severe nausea and repeated vomiting; there was almost no fever and little diarrhea. Among the dishes was a creamy dessert.

**Question:** What is the probable cause, and what points to it?

**Answer:** Intoxication with **staphylococcal enterotoxin**: a very short incubation period, predominant vomiting, absence of fever and a product such as cream — the typical picture. In the examination, vomit, remains of the product and hand swabs of the people who prepared it are cultured; the toxin can be detected.

### *Case 2*

In a family, 20 hours after eating home-canned preserves, several people developed “a net before the eyes,” double vision and difficulty swallowing; no fever, consciousness intact.

**Question:** What are the diagnosis, the first measure and the laboratory tactics?

**Answer:** Botulism. The first measure is to give **antitoxic serum as early as possible** (by a physician) and to monitor breathing. Laboratory: **detecting the toxin** (neutralization test/ELISA) in blood serum, gastric lavage, stool and the remains of the product, anaerobic culture and PCR.

### *Case 3*

18 hours after eating an undercooked egg dish, a patient developed fever, abdominal pain and diarrhea. On stool culture lactose-negative colonies; in Kligler, gas and $H_2S$ positive.

**Question:** What are the probable causative agent and the final confirmation?

**Answer:** Salmonella (probably *S.* Enteritidis). The final confirmation is serological identification with O and H sera according to the **Kauffmann–White scheme**; the basis of treatment is rehydration.

## Quick review

**Table 23.3.** *Topic 23 — key concepts*

| Concept | Essence |
|---|---|
| Toxicoinfection / intoxication | Live microbe multiplies / preformed toxin |
| Short incubation + vomiting | Staphylococcal enterotoxin (heat-stable) |
| Salmonella | Lactose −, gas +, $H_2S$ +; bismuth sulfite agar |
| Serological identification | Kauffmann–White (O and H); polyvalent → group-specific → type-specific |
| Treatment of salmonellosis | Rehydration; antibiotic — generalized form and risk groups |
| *C. botulinum* | Subterminal spore, obligate anaerobe |

| Concept | Essence |
|---|---|
| Botulinum toxin | Acetylcholine blockade → flaccid paralysis; heat-labile |
| Clinical picture of botulism | Visual and swallowing disturbance, paralysis; no fever, consciousness intact |
| Treatment of botulism | Antitoxic serum as early as possible |
| Listeria | Psychrophilic; pregnant women, newborns; meningitis, sepsis |

TOPIC 24

# Zoonotic infections: anthrax and brucellosis. Plague and tularemia

## 1. Learning objectives

In this class the student will master the following:

- To know the concept of zoonotic infections and their epidemiological features;
- to explain the biological properties, pathogenesis and clinical forms of the causative agents of anthrax and brucellosis;
- to master the algorithm of laboratory diagnosis of these infections and the biosafety requirements;
- Independent work: to know the diseases caused by *Yersinia pestis* and *Francisella tularensis*, their diagnosis and prophylaxis.

## 2. General concept of zoonoses

**Zoonoses** are diseases whose source of infection is **animals**. A human is often the "final host," and person-to-person transmission is limited or absent.

**Routes of transmission:** direct contact (a sick animal, its skin and wool products), alimentary (meat, milk), airborne (dust, aerosol), vector-borne (arthropods).

**Occupational risk groups:** livestock breeders, shepherds, veterinary workers, workers at meat-processing plants and in the hide-and-wool industry, hunters, laboratory workers.

> **CAUTION**
> The causative agents of anthrax, plague and tularemia belong to the group of **especially dangerous infections**. When they are suspected, the material is collected in a special manner, transported in a prescribed container, and processed **in specially authorized laboratories** under a high biosafety regime; the health system is notified immediately. Such cultures are not handled in an ordinary diagnostic laboratory.

## 3. Anthrax — *Bacillus anthracis*

**Morphology:** a large gram-positive rod (1–1.5 × 3–10 µm) with blunt ("cut-off") ends; in smears it forms **chains** ("bamboo cane"); **nonmotile** (rare among bacilli); it forms a **capsule** (polypeptide) in the body and a **central spore** in the external environment.

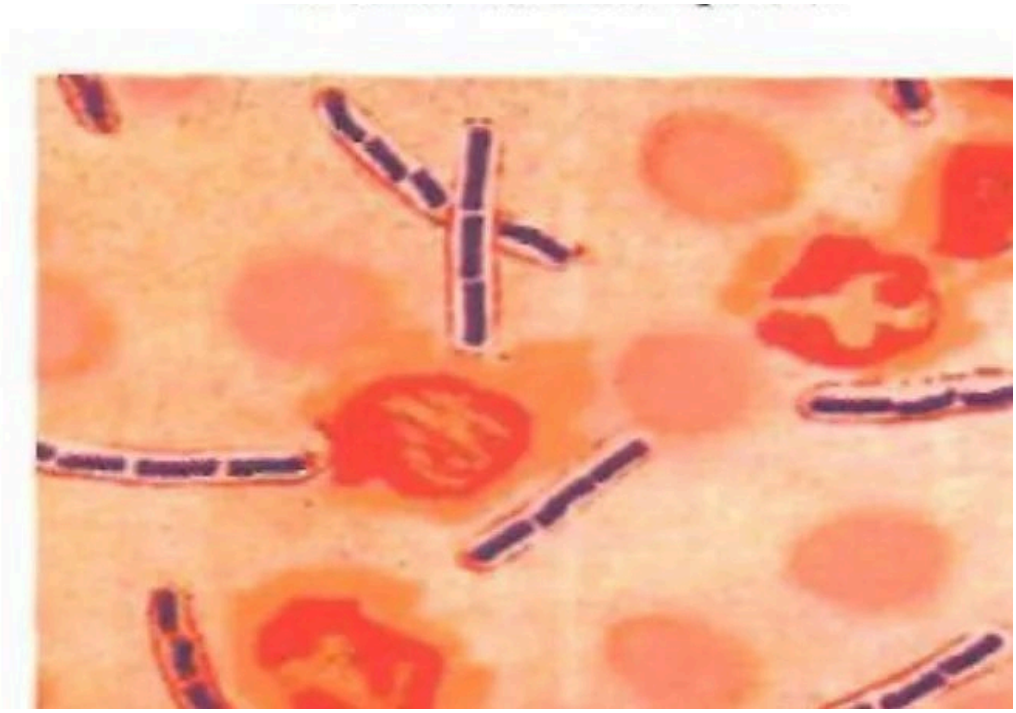


**Figure 24.1.** *The anthrax bacillus: large gram-positive rods, encapsulated chains in a blood smear. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties:** aerobe; grows on ordinary media; the colonies are **R-form** — rough, with thread-like projections at the edges ("**lion's mane**" or "Medusa head" appearance); in gelatin it grows in the shape of an **"inverted fir tree."**

**Pathogenicity factors:**

- **Capsule** (polypeptide) — antiphagocytic;
- **Exotoxin** — three-component: **protective antigen (PA)**, **lethal factor (LF)** and **edema factor (EF)**. Each alone is inactive; the combination of two acts: PA + EF → edema, PA + LF → tissue necrosis and death.

> **REMEMBER**
> Anthrax toxin is **three-component**: PA is the carrier and "door-opener," while LF and EF are the effector parts. The protective antigen (PA) is the main component of vaccines.

**Clinical forms:**

1. **Cutaneous form (the most frequent)** — a spot at the entry site → papule → vesicle → it bursts, leaving a **black eschar ulcer (carbuncle)** in the center; edema around it; **painlessness** is characteristic;
2. **Pulmonary form** — inhalation of spores; severe course;
3. **Intestinal form** — from eating infected meat; severe course.

**Laboratory diagnosis:**

1. **Material:** discharge from the ulcer and fluid under the eschar, sputum, stool, blood; material from dead animals; samples of skin, wool and meat.
2. **Bacterioscopy** — Gram stain and capsule stain: encapsulated, blunt-ended chains.
3. **Bacteriological** — culture, colony appearance (R-form), nonmotility, the "string of pearls" test (under the action of penicillin the cells become spherical), **susceptibility to bacteriophage**.
4. **Biological test** — in a laboratory animal (in special laboratories).

5. **Ascoli precipitation reaction** — detecting the heat-stable antigen in materials such as **skin, wool and meat**; the material is boiled (the antigen persists even if the bacteria are dead), then a ring reaction is set up with precipitating serum.
6. **Skin allergy test (anthraxin)** and PCR.

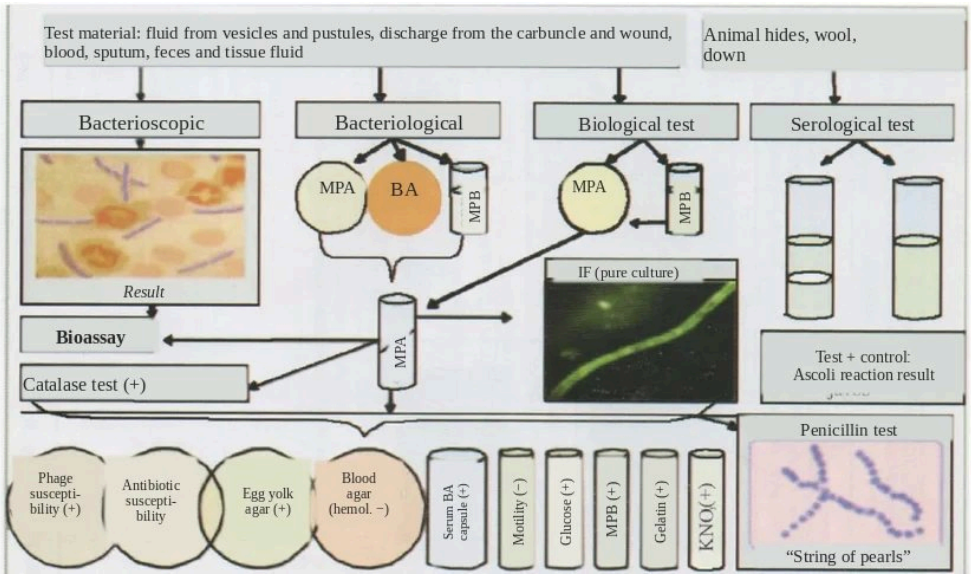



**Figure 24.2.** *Scheme of the microbiological examination in anthrax: bacterioscopic, bacteriological, biological and serological examinations. Source: Aliyev Sh.R. et al., 2013.*

**CLINICAL SIGNIFICANCE**

The **Ascoli reaction** is the classic way of making a diagnosis in material in which the causative agent cannot be grown or is dead (dried skin, wool, old meat): the antigen persists because it is heat-stable.

**Prophylaxis:** veterinary control and vaccination of livestock, destruction of material from sick animals by the prescribed procedure, vaccination of risk groups (live vaccine), occupational hygiene.

## 4. Brucellosis

**Causative agents:** *Brucella melitensis* (sheep, goats — the most pathogenic for humans), *B. abortus* (cattle), *B. suis* (pigs), *B. canis* (dogs).

**Morphology:** very small gram-negative coccobacilli; nonmotile, nonspore-forming; with a microcapsule.

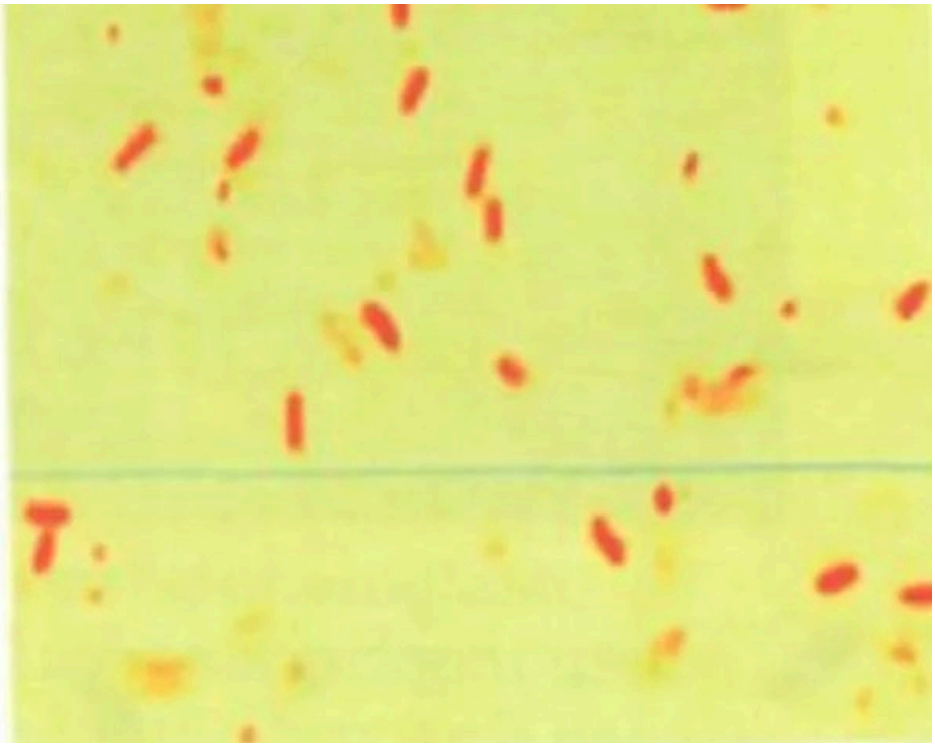


**Figure 24.3.** *Brucellae — very small gram-negative coccobacilli. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties:** aerobes, **fastidious and slow-growing** — growth may take 5–30 days; some species (*B. abortus*) require **additional $CO_2$ (5–10%)** for growth; they are grown in liver-broth and special media.

**Routes of transmission:** unpasteurized milk and dairy products (especially fresh cheese, cream), direct contact with a sick animal and its birth products, aerosol (in laboratories and on livestock farms).

**Pathogenesis:** brucellae spread through the lymphatic pathways and **live inside macrophages** (incomplete phagocytosis) → **granulomas** form in organs of the reticuloendothelial system (lymph nodes, liver, spleen, bone marrow) → repeated release into the blood produces **undulant fever**. Delayed-type hypersensitivity develops.

**Clinical picture:** prolonged **undulant fever**, profuse sweating, joint and muscle pain, enlargement of the lymph nodes, liver and spleen; when it becomes chronic — arthritis, spondylitis, damage to the nervous system and reproductive organs.

**Laboratory diagnosis:**

1. **Bacteriological** — blood and bone marrow aspirate are cultured; because growth is slow, it is observed for **up to 30 days**. Because of the high hazard it is done only in a special laboratory.
2. **Serological (the main one in practice):**

- **Wright reaction** — extended agglutination (in a tube), with a titer;
- **Huddleson reaction** — rapid screening on a glass slide;
- **Coombs reaction** — detecting incomplete antibodies (in the chronic form);
- ELISA (IgM/IgG) — a modern method.

3. **Skin allergy test (Bürnet test)** — with brucellin; it shows sensitization and is important in the chronic form.

4. **PCR** — fast and safer.

**MEMORY AID**
To remember brucellosis: **“milk + undulant fever + sweating + joint pain”**; in serology — **Wright** (tube, titer) and **Huddleson** (slide, rapid).

**CAUTION**
Working with a Brucella culture is **highly hazardous for laboratory staff**: it is transmitted by aerosol. Therefore when brucellosis is suspected the laboratory is warned in advance and the work is carried out under a special regime.

## 5. Independent work: Plague and tularemia

### *5.1. Plague — Yersinia pestis*

**Morphology:** a gram-negative, short, oval rod; it **stains bipolarly** (both ends darker — a “safety pin” appearance, clearly seen with Romanowsky–Giemsa); nonmotile, encapsulated.

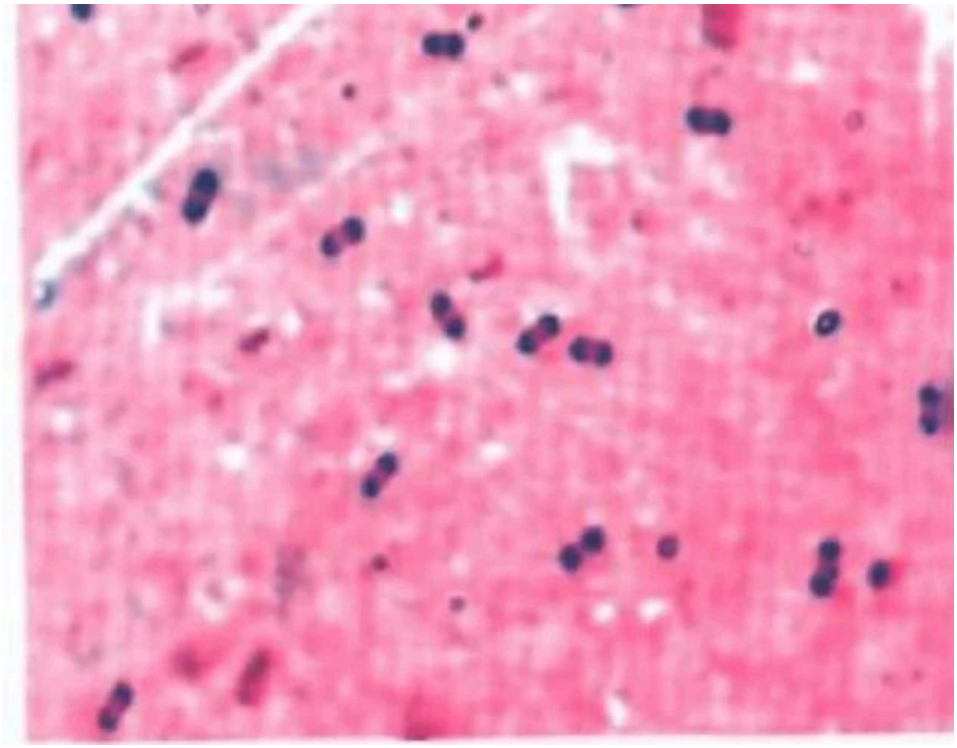


**Figure 24.4.** *The causative agent of plague — gram-negative rods that stain bipolarly. Source: Muhamedov I.M. et al., 2019.*

**Epidemiology:** a natural-focal, **vector-borne** infection; the reservoir is rodents, the vector is **fleas**; in the pulmonary form it spreads **from person to person by the airborne route**.

**Pathogenicity factors:** the F1 capsular antigen, the V and W antigens, “plague toxin,” plasminogen activator, LPS; they ensure resistance to phagocytosis and rapid spread.

**Clinical forms: bubonic** (the most frequent — severe inflammation of the lymph node nearest the entry site and a painful “bubo”), septic and **pulmonary** form (the most dangerous, spreading rapidly).

**Diagnosis:** bubo aspirate, sputum, blood, lymph nodes → bacterioscopy (bipolar rods) → culture → typing with phages → biological test; rapid methods — immunofluorescence, antigen tests, PCR. All work is done under the **especially dangerous infection** regime.

**Prophylaxis:** epidemiological surveillance in natural foci, measures against rodents and fleas, vaccination of risk groups, quarantine measures.

### *5.2. Tularemia — Francisella tularensis*

**Morphology:** a very small gram-negative coccobacillus; nonmotile, encapsulated.

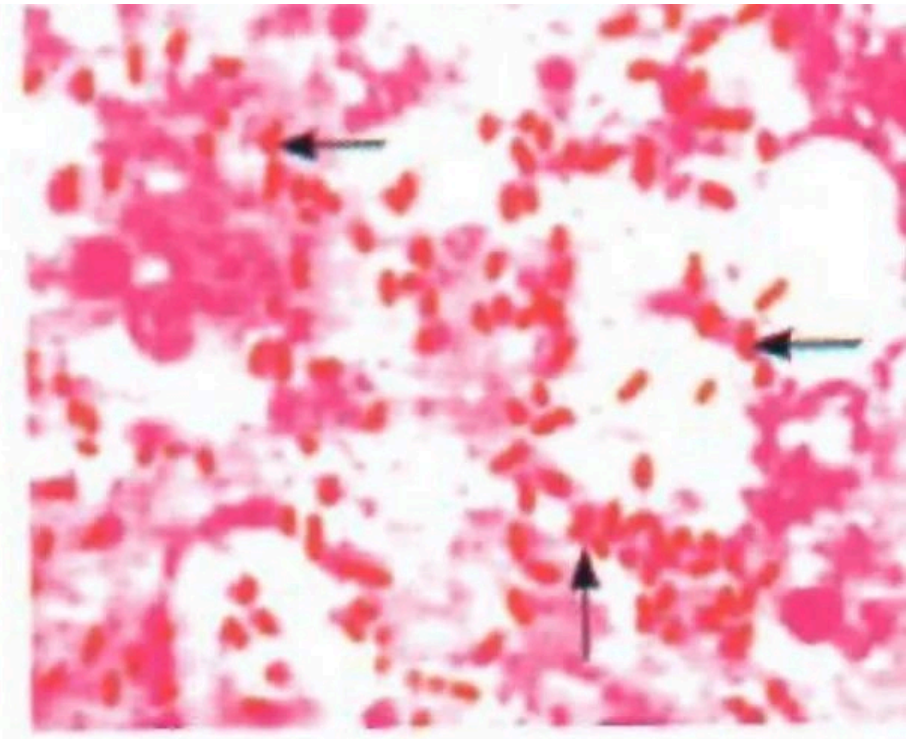


**Figure 24.5.** *The causative agent of tularemia (Francisella tularensis) — very small gram-negative coccobacilli. Source: Muhamedov I.M. et al., 2019.*

**Epidemiology:** the reservoir is rodents and hares; the routes of transmission are **very varied**: vector-borne (ticks, horseflies), contact (the skin of a sick animal), alimentary (contaminated water and food), airborne (dust). **The infectious dose is very low** — which is why it poses a high risk for laboratory staff.

**Features:** does not grow on ordinary medium — it requires a special medium **supplemented with cysteine/cystine** (a medium enriched with egg yolk); lives intracellularly.

**Clinical forms:** bubonic, ulceroglandular, oculoglandular, oropharyngeal (anginal-bubonic), pulmonary and generalized forms.

**Diagnosis (mainly by indirect methods):**

- **Serological** — agglutination reaction (Wright type), PHA, ELISA; the dynamics of the titer in paired sera;
- **Skin allergy test (tularin)** — becomes positive early (from day 3–5 of the illness);
- **PCR** and antigen tests;
- **Bacteriological method** — only in special laboratories.

**Prophylaxis:** surveillance in natural foci, protection of water sources, measures against rodents, vaccination of risk groups with the **live vaccine**.

**Table 24.1.** *Brief comparison of zoonoses*

| Disease | Causative agent | Source | Main transmission | Basis of diagnosis |
|---|---|---|---|---|
| Anthrax | *B. anthracis* | Cattle, sheep | Contact, alimentary, airborne | Bacterioscopy, culture, **Ascoli** |
| Brucellosis | *Brucella* spp. | Sheep, goats, cattle | Dairy products, contact | **Wright, Huddleson**, Coombs, ELISA |
| Plague | *Y. pestis* | Rodents | **Flea** (vector-borne), airborne in the pulmonary form | Bacterioscopy (bipolar), culture, PCR |
| Tularemia | *F. tularensis* | Rodents, hares | Highly varied | **Serology, tularin test**, PCR |

## 6. Practical skill: procedure when a zoonosis is suspected

1. **Establish the epidemiological history:** occupation, contact with animals, consumption of unpasteurized milk and fresh cheese, hunting, a tick or flea bite, travel to a natural focus.
2. **Assess the clinical syndrome:** a painless black-eschar ulcer → anthrax; undulant fever + sweating + joint pain → brucellosis; acute fever + a painful bubo → plague/tularemia.
3. **Warn the laboratory in advance** — this is essential for staff safety and helps to plan the examination correctly.
4. **Collect the material in the prescribed manner:** a sealed, unbreakable container, double packaging, a clearly completed requisition.
5. **Choose the examination method:** in especially dangerous infections preference is given to **serological and molecular** methods; culture work is done only in a specially authorized laboratory.
6. **Notify the epidemiological service** and begin the measures for the focus.

## 7. Review questions

1. What are zoonoses? What are their epidemiological features and occupational risk groups?
2. State the morphological and cultural features of *B. anthracis*.
3. What are the components of anthrax toxin and their actions?
4. State the clinical forms of anthrax and the characteristic sign of the cutaneous form.
5. Why and how is the Ascoli reaction performed?
6. What are the cultural features of Brucella? Why is growth observed for so long?
7. What is the role of macrophages in the pathogenesis of brucellosis?
8. Which reactions are used in the serological diagnosis of brucellosis?
9. What are the morphological feature and the routes of transmission of *Y. pestis*?
10. Why are indirect methods preferred in the diagnosis of tularemia?

## Test questions

1. The morphological feature of *B. anthracis*:
   A) Motile, noncapsulated
   B) Nonmotile, encapsulated in the body, forms chains
   C) Gram-negative coccus
   D) Spiral-shaped
   E) Small coccobacillus

2. The appearance of the anthrax colony:
   A) Smooth, shiny (S-form)
   B) Rough, with thread-like projections (“lion’s mane”)
   C) Mucoid, sticky
   D) Black, with a metallic sheen
   E) Blue-green

3. Not a component of anthrax toxin:
   A) Protective antigen (PA)
   B) Lethal factor (LF)
   C) Edema factor (EF)
   D) Erythrogenic toxin
   E) All are components

4. The most frequent clinical form of anthrax:
   A) Lung
   B) Intestinal
   C) Cutaneous (carbuncle)
   D) Septic
   E) Meningeal

5. The sign characteristic of the cutaneous form of anthrax:
   A) A severely painful ulcer
   B) A painless ulcer with a black eschar in the center
   C) Jaundice
   D) Paralysis
   E) Diarrhea

6. The material in which the Ascoli reaction is used:
   A) Only blood
   B) Materials such as skin, wool and meat
   C) Cerebrospinal fluid
   D) Urine
   E) Sputum

7. The type of the Ascoli reaction:
   A) Agglutination
   B) Precipitation
   C) CFT
   D) Hemolysis
   E) Neutralization

8. The Brucella species most pathogenic for humans:
   A) *B. abortus*
   B) *B. melitensis*
   C) *B. suis*
   D) *B. canis*
   E) *B. ovis*

9. The main transmission factor of brucellosis:
   A) Unpasteurized milk and dairy products
   B) Mosquito bite
   C) A contaminated syringe
   D) Sexual contact
   E) Soil

10. The main feature of the pathogenesis of brucellosis:
    A) Production of an exotoxin
    B) Living inside macrophages and forming granulomas
    C) Direct damage to the nervous system
    D) Breakdown of erythrocytes
    E) Entry into the intestinal epithelium

11. The extended agglutination reaction in brucellosis:
    A) Widal
    B) Wright
    C) Huddleson
    D) Ascoli
    E) Wassermann

12. The skin allergy test in brucellosis:
    A) Mantoux
    B) Bürnet
    C) Tularin
    D) Anthraxin
    E) Schick

13. The characteristic staining feature of *Y. pestis*:
    A) Bipolar staining (“safety pin”)
    B) Volutin granules
    C) Acid-fastness
    D) Metachromatic granules
    E) Spore

14. The main vector of plague:
    A) Louse
    B) Flea
    C) Tick
    D) Mosquito
    E) Fly

15. The substance added to the medium to grow *F. tularensis*:
    A) Cysteine/cystine
    B) Tellurite
    C) Bile salts
    D) Mannitol
    E) Optochin

## Clinical cases

### *Case 1*

A patient who works at a hide-and-wool plant has a painless sore on the hand: a black eschar in the center, vesicles around it and wide edema. The smear shows large gram-positive, blunt-ended, chain-forming rods.

**Question:** What are the probable diagnosis and the laboratory tactics?

**Answer:** Cutaneous anthrax. The laboratory is **warned in advance**; the material is collected in a special manner. The examination: bacterioscopy, culture and identification in a special laboratory, the **Ascoli reaction** on raw-material samples, PCR. The epidemiological service is notified.

### *Case 2*

A livestock breeder has had undulant fever for 2 months, profuse sweating and joint pain; the spleen and liver are enlarged. The patient ate cheese made from fresh (unpasteurized) milk.

**Question:** What should be considered, and which examinations are informative?

**Answer:** Brucellosis. In practice **serological methods** are the main ones: Huddleson (rapid screening), **Wright** (with a titer), Coombs and ELISA; the skin allergy **Bürnet** test; PCR. A blood culture is possible but slow and hazardous — only in a special laboratory.

### *Case 3*

A patient who hunted in a natural focus developed acute fever and a painful enlarged lymph node in the armpit.

**Question:** Which infections should be excluded, and what is the first measure?

**Answer:** Plague and tularemia should be excluded (both produce a bubo). The first measure is to **notify the laboratory and the epidemiological service immediately** and to collect the material in a special manner; the examination — bacterioscopy (bipolar rods), rapid antigen tests and PCR, and in tularemia additionally serology and the tularin test. Treatment is not delayed.

## Quick review

**Table 24.2.** *Topic 24 — key concepts*

| Concept | Essence |
|---|---|
| Zoonoses | Source — animals; occupational risk groups |
| *B. anthracis* | Large Gram+, nonmotile, capsule (in the body), spore (in the environment) |
| Anthrax toxin | PA + LF (death), PA + EF (edema) |
| Cutaneous form | Painless black-eschar carbuncle |
| Ascoli | Precipitation; heat-stable antigen in skin, wool, meat |

| Concept | Essence |
|---|---|
| *Brucella* | Small Gram−, slow-growing, $CO_2$; inside macrophages; granuloma |
| Brucellosis | Dairy products; undulant fever; Wright, Huddleson, Bürnet |
| *Y. pestis* | Bipolar staining; flea; bubonic and pulmonary forms |
| *F. tularensis* | Cysteine medium; very low infectious dose; serology and tularin |
| Safety | Especially dangerous infection regime, special laboratory |

TOPIC 25

# Dermatovenereal infections: causative agents of syphilis and gonorrhea. Chlamydiosis, mycoplasmosis, trichomoniasis, chancroid

## 1. Learning objectives

In this class the student will master the following:

- To know the biological properties of the causative agents of syphilis and gonorrhea;
- to explain the periods of the course of syphilis and the laboratory method suited to each period;
- to master the serological reactions (nontreponemal and treponemal tests) and their interpretation;
- to know the diagnostic tactics in the acute and chronic forms of gonorrhea;
- Independent work: to know the description and diagnosis of urogenital chlamydiosis, mycoplasmosis, trichomoniasis and chancroid.

## 2. Syphilis — *Treponema pallidum*

**Morphology:** a delicate spirochete; **8–14 uniform, regular coils**; length 6–14 µm, thickness 0.1–0.2 µm. It takes up ordinary dyes poorly (its name means “pale”); with **Romanowsky–Giemsa** it stains pale pink, with **Morozov’s method (silver impregnation)** black-brown. Its motility is varied: rotating, flexing, undulating.

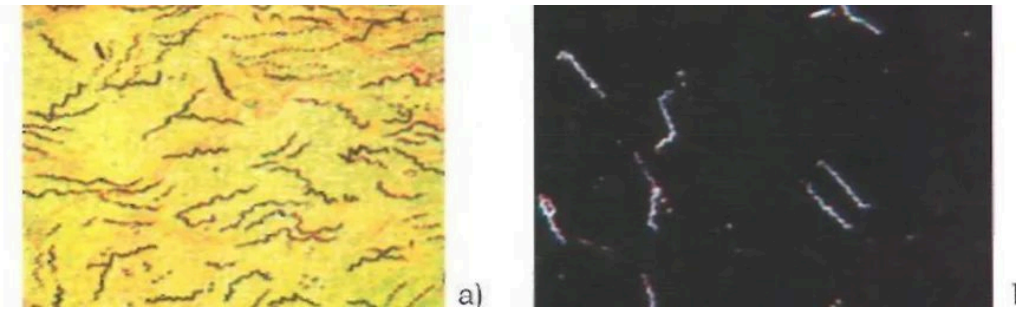



**Figure 25.1.** *T. pallidum in a smear from a hard chancre and in dark field: bacterioscopic methods in syphilis. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties: it does not grow on artificial nutrient media** (pathogenic strains). Therefore the classic bacteriological method is not used; diagnosis rests on **bacterioscopy and serology**.

**Resistance:** very unstable in the external environment — sensitive to drying, high temperature, antiseptics and soap; it survives briefly when moisture and temperature are maintained.

**Routes of transmission:** mainly **sexual**; also contact-household (rare), **transplacental (congenital syphilis)**, through blood (transfusion, syringe).

### *2.1. Periods of the disease and diagnostic tactics*

**Table 25.1.** *Periods of syphilis and the laboratory method*

| Period | Clinical picture | Preferred method |
|---|---|---|
| Incubation (3–4 weeks) | No signs | — |

| Period | Clinical picture | Preferred method |
|---|---|---|
| **Primary** | **Hard chancre** (a painless dense ulcer) + regional lymphadenitis | **Dark-field microscopy** (of the discharge); serology is initially negative (the “seronegative” stage), then becomes positive |
| **Secondary** | A generalized rash, a rash on the mucous membranes; an undulating course | **Serology — the highest titer**; treponemes can be found in rash elements |
| **Tertiary** | Gummas, damage to the internal organs, the cardiovascular and nervous systems | **Treponemal tests**; the titer of nontreponemal tests may fall |
| Innate | Infection of the fetus/newborn | Serology (child and mother), PCR, direct methods |

**REMEMBER**

**“In the primary period — the microscope; in the secondary and later periods — serology.”** This is the basic rule of syphilis diagnosis.

### *2.2. Serological diagnosis*

**Two groups of tests:**

1. **Nontreponemal (lipoidal) tests** — **MPR, RPR, VDRL**; cardiolipin is used as the antigen; they detect **reagins** in the body.

- *Advantage:* cheap, fast, **quantitative (with a titer)** — therefore used for **monitoring the effect of treatment** (the titer should fall);

- *Disadvantage:* **false-positive** results are possible (pregnancy, autoimmune diseases, some infections, old age).

2. **Treponemal tests** — **TPHA, ELISA, immunoblot, IF**; the antigen is the treponeme itself or its proteins.

- *Advantage:* high specificity — a **confirmatory test**;

- *Disadvantage:* they remain positive for many years even after treatment → **unsuitable for monitoring treatment**.

**CAUTION**

Rule of interpretation: **screening — nontreponemal; confirmation — treponemal; monitoring of treatment — the dynamics of the nontreponemal test titer.** If only the treponemal test is positive, it may also be a past (treated) infection.

**The Wassermann reaction (CFT)** — the historical classic method; it has now been replaced by modern tests.

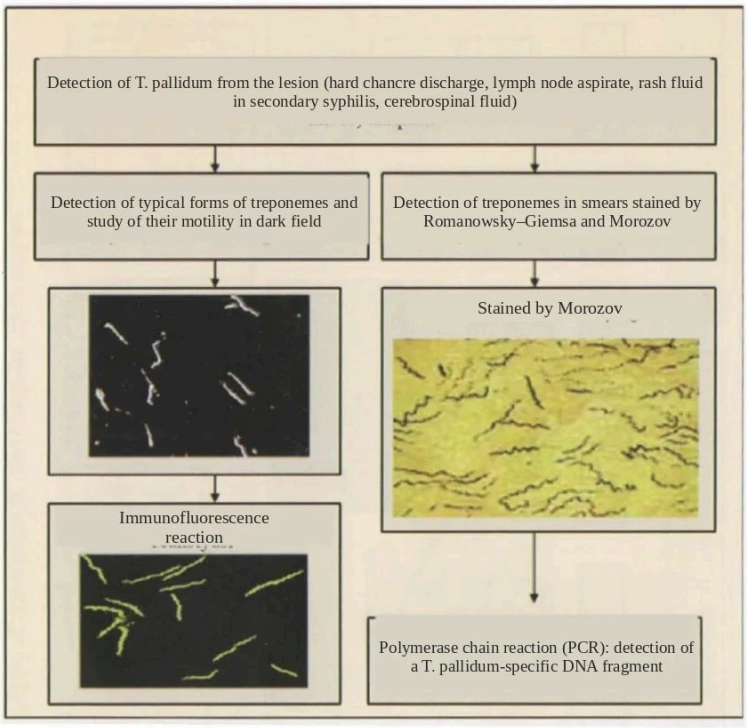



**Figure 25.2.** *Scheme of the bacterioscopic methods in syphilis: taking material from a hard chancre, dark field, the Romanowsky–Giemsa and Morozov methods, immunofluorescence, PCR. Source: Aliyev Sh.R. et al., 2013.*

## 3. Gonorrhea — *Neisseria gonorrhoeae*

**Morphology:** a gram-negative **diplococcus** with concave facing sides (“coffee bean”); in **acute gonorrhea it lies inside leukocytes in the pus** — a classic diagnostic sign. Nonmotile, nonspore-forming; it has pili and a microcapsule.

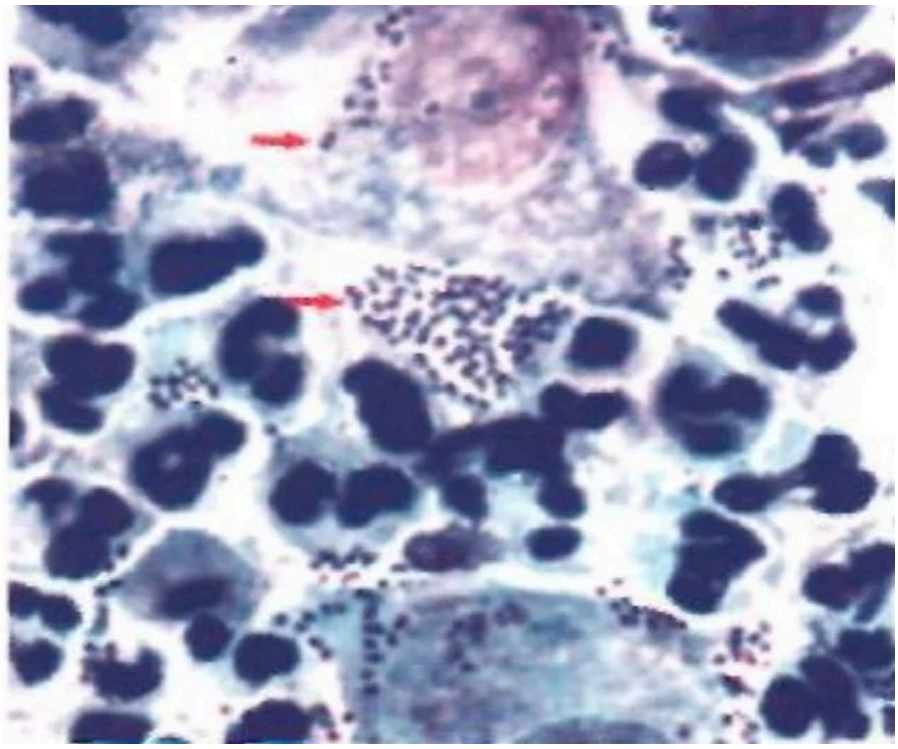


**Figure 25.3.** *Gonococci: gram-negative diplococci inside leukocytes in a pus smear. Source: Muhamedov I.M. et al., 2019.*

**Cultural properties:** a fastidious aerobe; **serum or chocolate agar**, **5–10% $CO_2$**; **oxidase positive**; it ferments **only glucose** (the difference from the meningococcus — topic 18). Very unstable in the external environment (sensitive to drying and cold).

**Pathogenicity factors: pili** (adhesion; because of antigenic variability immunity is not durable), **Opa proteins**, LPS (lipooligosaccharide), **IgA protease**, beta-lactamases (in some strains).

**Clinical picture:** in men — acute urethritis (purulent discharge, pain on urination); in women — often **with few or no symptoms** (cervicitis), so it is detected late and causes complications (pelvic inflammatory disease, infertility, ectopic pregnancy); in newborns — **blennorrhea** (conjunctivitis of the eye); in the disseminated form — arthritis, skin rash.

**Laboratory diagnosis:**

1. **Material:** a urethral, cervical canal, rectal or throat swab, conjunctival discharge.
2. **Bacterioscopy** — Gram and methylene blue: **gram-negative diplococci inside leukocytes**. Sensitivity is high in men with acute urethritis; low in women and in the chronic form.
3. **Bacteriological** — serum/chocolate agar + $CO_2$; identification: oxidase, carbohydrates (glucose only), **antibiogram** (important because resistance is widespread).
4. **Molecular (NAAT/PCR)** — currently the most sensitive method, especially in women, in the chronic form and in extragenital localization; it can also be done on urine.
5. **In the chronic form** — provocation methods (physiological, chemical, dietary) were used; in current practice **molecular methods** have largely taken their place.

**CLINICAL SIGNIFICANCE**
The gonococcus dies quickly after the material is collected, so inoculation is done **as quickly as possible** (at the patient's bedside or in transport medium). The material must not be cooled.

**MEMORY AID**
Two Neisseria: **Gonococcus — Glucose** (only); **Meningococcus — glucose + Maltose**. The gonococcus is noncapsulated, the meningococcus is encapsulated.

## 4. Independent work: Other urogenital infections

### *4.1. Urogenital chlamydiosis*

**Causative agent:** *Chlamydia trachomatis* (serovars D–K) — an obligate intracellular parasite (topic 4).

**Clinical picture:** often **asymptomatic or with few symptoms** — a "silent" infection; urethritis, cervicitis, pelvic inflammatory disease, infertility, reactive arthritis; conjunctivitis and pneumonia in newborns.

**The key to diagnosis:** the material must contain **epithelial cells** (a scraping, not the discharge itself). Methods: **PCR/NAAT (the main one)**, direct immunofluorescence, ELISA; cell culture — the gold standard, but rarely used in practice.

### *4.2. Urogenital mycoplasmosis*

**Causative agents:** *Mycoplasma hominis*, *M. genitalium*, *Ureaplasma urealyticum*.

**Features:** no cell wall → **naturally resistant to beta-lactams** (topic 4); some species also occur in the normal flora, so finding them does not always mean disease. *M. genitalium* — a definite pathogen

(urethritis, cervicitis).

**Diagnosis: PCR** (the main one), growth in special media (by urease and arginine), quantitative assessment.

### 4.3. Trichomoniasis

**Causative agent:** *Trichomonas vaginalis* — a flagellated protozoan; pear-shaped, with 4 free flagella and an undulating membrane; **it does not form cysts**.

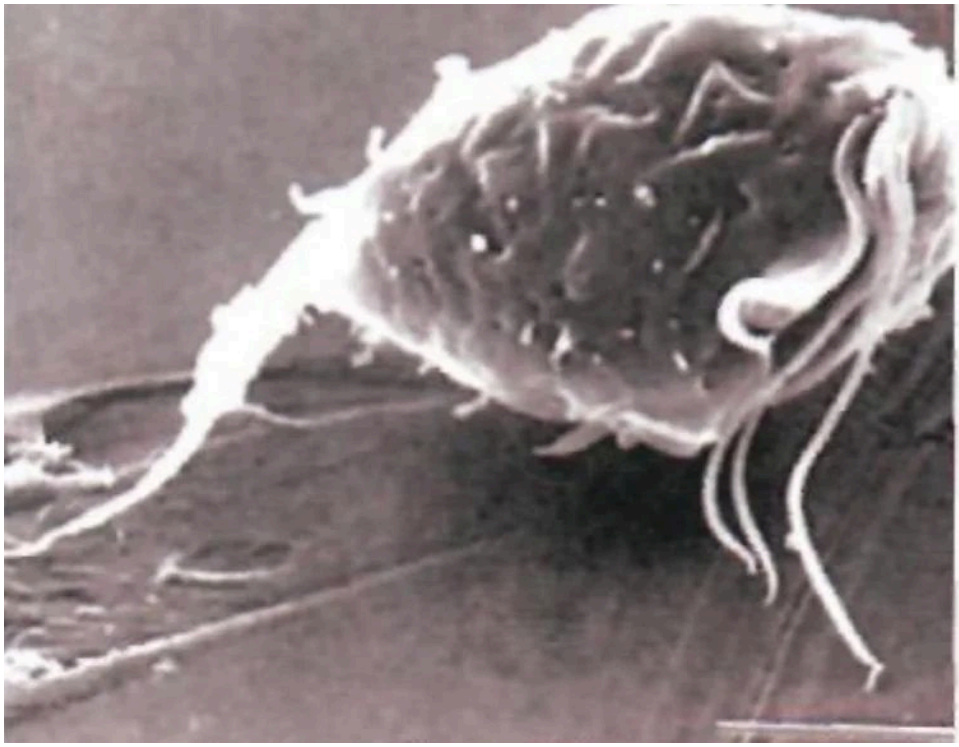


**Figure 25.4.** *Trichomonas vaginalis — a flagellated, pear-shaped protozoan (under the electron microscope). Source: Muhamedov I.M. et al., 2019.*

**Clinical picture:** in women — foamy, yellowish-green discharge, itching; in men often asymptomatic (carriage).

**Diagnosis: warm wet mount** — seeing the motile parasite immediately (topic 4); a stained smear (Romanowsky–Giemsa), culture in special medium, **PCR** (the most sensitive), antigen tests.

### 4.4. Chancroid

**Causative agent:** *Haemophilus ducreyi* — a gram-negative coccobacillus; in smears it lies as a **"school of fish" or in chains**.

**Clinical picture:** a **painful**, soft ulcer with a purulent base and painful suppurating lymphadenitis — these signs are decisive in differential diagnosis from the **hard chancre of syphilis** (painless, dense).

**Diagnosis:** a smear from the ulcer discharge (Gram), culture on specially enriched medium (fastidious, difficult), PCR.

**Table 25.2.** *Differential diagnosis of a genital ulcer*

| Feature | Syphilis (hard chancre) | Chancroid | Genital herpes |
|---|---|---|---|
| Pain | **None** | **Yes, severe** | Yes, burning |
| Edge/base | Dense, smooth, clear discharge | Soft, uneven, purulent | Grouped vesicles, erosions |

| Feature | Syphilis (hard chancre) | Chancroid | Genital herpes |
|---|---|---|---|
| Lymph nodes | Dense, painless | Painful, suppurate | Painful, soft |
| Causative agent | *T. pallidum* | *H. ducreyi* | Herpes virus (topic 30) |

## 5. Practical skill: the examination algorithm when syphilis is suspected

1. **Determine the clinical period:** ulcer (primary), rash (secondary), late signs (tertiary).
2. **In the primary period:** cleanse the ulcer with physiological saline and take tissue fluid from its base → **examine it immediately in dark field** (a live, unstained preparation). If necessary — PCR.
3. **Serology:** the **nontreponemal test (MPR/RPR)** — screening, **with a titer**; if the result is positive or the clinical suspicion is strong — confirm with the **treponemal test (TPHA/ELISA)**.
4. **Interpret the result:**

- nontreponemal (+) and treponemal (+) → active or past infection (assessed with the clinical picture and history);

- nontreponemal (+), treponemal (−) → probably a **false positive**;

- nontreponemal (−), treponemal (+) → past/treated infection or the late period.

5. **Monitoring of treatment:** only by the dynamics of the **nontreponemal test titer**.
6. **Epidemiological work:** examination and registration of sexual partners — by the set procedure.

## 6. Review questions

1. What are the morphological features and staining methods of *T. pallidum*?
2. Why is the bacteriological method not used in the diagnosis of syphilis?
3. State the periods of syphilis and the method suited to each.
4. What is the difference between nontreponemal and treponemal tests?
5. Which test is used to monitor the effect of treatment, and why?
6. State the morphological and cultural properties of the gonococcus.
7. Why is gonorrhea detected late in women, and which method is preferable?
8. What mistake must be avoided in transporting gonococcal material?
9. What is special about collecting material in urogenital chlamydiosis?
10. How do you distinguish a hard chancre from a soft one?

## Test questions

1. The coils of *T. pallidum*:
   A) 3–8, large and irregular
   B) 8–14, uniform and regular
   C) No coils
   D) Only 2
   E) More than 20, with hooked ends

2. The cultural property of the syphilis causative agent:
   A) It grows well on blood agar
   B) It does not grow on artificial nutrient media
   C) It grows on Sabouraud medium
   D) It grows on TCBS
   E) It grows in anaerobic medium

3. The most suitable bacterioscopic method in primary syphilis:
   A) Gram method
   B) Dark-field microscopy
   C) Ziehl–Neelsen
   D) Neisser
   E) Ozheshko

4. The characteristic sign of a hard chancre:
   A) Painful, soft, purulent
   B) Painless, dense, with a clear discharge
   C) A vesicular rash
   D) With a black eschar
   E) Many small ulcers

5. The nontreponemal tests include:
   A) TPHA
   B) MPR, RPR, VDRL
   C) ELISA (treponemal)
   D) Immunoblot
   E) IF

6. The test used to monitor the effect of treatment:
   A) Treponemal (TPHA)
   B) The titer of a nontreponemal test
   C) Immunoblot
   D) PCR
   E) Skin test

7. The disadvantage of treponemal tests:
   A) Very cheap
   B) They remain positive for a long time even after treatment
   C) Very low sensitivity
   D) Positive only in the primary period
   E) They give no titer

8. The morphological feature of the gonococcus:
   A) A gram-positive chain-forming coccus
   B) Gram-negative diplococcus, inside leukocytes
   C) Gram-negative rod
   D) Spirochete
   E) A gram-positive rod

9. The carbohydrate the gonococcus ferments:
    A) Glucose only
    B) Glucose and maltose
    C) Lactose
    D) Sucrose
    E) Mannitol

10. What is not done in transporting gonococcal material:
    A) Rapid delivery
    B) Using transport medium
    C) Cooling
    D) Inoculating at the patient’s bedside
    E) Providing $CO_2$ conditions

11. The most sensitive method in the diagnosis of gonorrhea in women:
    A) Bacterioscopy
    B) A molecular method (PCR/NAAT)
    C) Skin test
    D) The Ascoli reaction
    E) The Wright reaction

12. What the material should be in urogenital chlamydiosis:
    A) Only discharge
    B) A scraping containing epithelial cells
    C) Blood serum
    D) Urine sediment (only)
    E) Stool

13. The reason for the resistance of mycoplasmas to beta-lactams:
    A) They produce beta-lactamase
    B) They have no cell wall
    C) A thick capsule
    D) Efflux pump
    E) Absence of porin

14. The feature of *T. vaginalis*:
    A) It forms cysts
    B) It forms no cysts; only the trophozoite form exists
    C) It is a bacterium
    D) Has a cell wall
    E) Nonmotile

15. How chancroid differs from a hard chancre:
    A) Painful, soft, with a purulent base
    B) Painless and dense
    C) Vesicular
    D) With a black eschar
    E) No difference

## Clinical cases

### *Case 1*

A patient has a painless ulcer with firm edges on the genitals and painless enlargement of a regional lymph node. MPR is negative.

**Question:** Can the diagnosis be ruled out? What should be done?

**Answer:** No. At the beginning of primary syphilis serology may still be negative (the “seronegative” stage). **Dark-field** microscopy (or PCR) of the ulcer discharge is performed and serology is repeated at the set time.

### *Case 2*

On a screening examination RPR is positive (at a low titer) and TPHA is negative. The patient has no clinical signs of syphilis; she is pregnant.

**Question:** How should the result be interpreted?

**Answer:** Probably a **false-positive** nontreponemal result (pregnancy is one of the known causes). Because the confirmatory treponemal test is negative, a diagnosis of syphilis is not made; follow-up and, if necessary, repeat testing are carried out by the set procedure.

### *Case 3*

A male patient has acute purulent urethritis. Gram-negative diplococci were found inside leukocytes in the smear.

**Question:** What are the preliminary diagnosis and the next step?

**Answer:** Gonorrhea (gonococcal urethritis). The next step is culture (serum/chocolate agar, $CO_2$) and an **antibiogram** (because resistance is widespread), and PCR to exclude chlamydia; examination of sexual partners.

## Quick review

**Table 25.3.** *Topic 25 — key concepts*

| Concept | Essence |
|---|---|
| *T. pallidum* | 8–14 regular coils; does not grow on artificial media |
| Primary syphilis | Hard chancre (painless, dense); dark field |
| Secondary and later | Serology is the main method |
| Nontreponemal tests | MPR/RPR/VDRL — screening and **monitoring of treatment** (titer) |
| Treponemal tests | TPHA/ELISA/immunoblot — confirmatory; positive even after treatment |
| Gonococcus | Gram− diplococcus, inside leukocytes; only glucose; not cooled |
| Gonorrhea (in women) | Often asymptomatic; PCR/NAAT preferred |
| Chlamydiosis | Intracellular; material — a scraping with epithelium; PCR |
| Mycoplasmas | No wall → resistant to beta-lactams; PCR |
| Trichomonas | Warm wet mount; no cysts; PCR is the most sensitive |
| Chancroid | *H. ducreyi*; painful, soft ulcer |

TOPIC 26

# Vector-borne infections: rickettsiae, epidemic typhus, Q fever, relapsing fever borreliae. Lyme disease and leptospirosis

## 1. Learning objectives

In this class the student will master the following:

- To know the concept of vector-borne infections and the feature of natural focality;
- to explain the biological properties of rickettsiae and the pathogenesis and diagnosis of epidemic and endemic typhus and of Q fever;
- to know the characteristics of borreliae and relapsing fever and the mechanism of “relapse”;
- Independent work: to know the causative agents of Lyme disease and leptospirosis and their laboratory diagnosis.

## 2. General concept of vector-borne infections

**Vector-borne infections** are diseases in which the causative agent is transmitted by **arthropods (vectors)**. Vectors: lice, fleas, ticks, mosquitoes, horseflies.

**Two groups:**

- **Anthroponoses** — the source is a human (epidemic typhus, epidemic relapsing fever);
- **Zoonoses (natural-focal)** — the source is animals (endemic typhus, Lyme disease, tick-borne relapsing fever, Q fever).

**Natural focality** (the doctrine of E.N. Pavlovsky) — the causative agent, the vector and the reservoir animals circulate in a certain landscape even without human participation; a person falls ill on entering the focus.

## 3. Rickettsiae and typhus

**Rickettsiae** (topic 4) are small, pleomorphic gram-negative bacteria; **obligate intracellular parasites** (they multiply mainly in endothelial cells); they do not grow on artificial media.

**Table 26.1.** *Main rickettsial infections*

| Disease | Causative agent | Source | Vector/transmission |
|---|---|---|---|
| **Epidemic typhus** | *Rickettsia prowazekii* | **Human** | **Body louse**; not by the bite — the louse’s feces are rubbed into scratched skin |
| Brill–Zinsser disease | *R. prowazekii* (reactivation) | A person who previously had the disease | No vector, endogenous relapse |
| Endemic (murine) typhus | *R. typhi* | Rats | Flea |
| **Q fever** | *Coxiella burnetii* | Cattle, sheep, goats | **Aerosol/dust**, milk; ticks (among animals) |

**Pathogenesis of epidemic typhus:** the rickettsiae multiply in the **endothelium** of blood vessels → the vessel wall is damaged, thrombi and **vasculitis** develop → a skin rash (roseolar-petechial), damage to the nervous system and internal organs, severe intoxication.

**Clinical picture:** acute onset, high fever, severe headache, insomnia, excitement; from day 4–5 a **roseolar-petechial rash** (on the trunk and the inner surfaces of the limbs); cardiovascular and nervous system disturbances.

> **REMEMBER**
> **Brill–Zinsser disease** — the recrudescence of epidemic typhus after years: the rickettsiae persist latent in the body and become active when immunity falls. Therefore isolated cases may occur even in a louse-free area.

**The particularity of Q fever:** *C. burnetii* is **highly resistant** in the external environment (it forms a spore-like form); it is transmitted mainly by the **airborne route** (dust, animal birth products), a rash is usually absent; the clinical picture — fever, atypical pneumonia, hepatitis; in the chronic form, endocarditis.

**Laboratory diagnosis of rickettsioses:**

1. **Serological (the main one)** — CFT, **PHA**, **ELISA** (IgM/IgG); the dynamics of the titer in paired sera. The historical method — the **Weil–Felix reaction** (a cross-reaction with *Proteus* OX antigens; low specificity, no longer used in practice).
2. **PCR** — in blood and biopsy material; important in the early period.
3. **Bacterioscopy** — Romanowsky–Giemsa, the **Zdrodovsky** method (rickettsiae red, the cell cytoplasm blue).
4. **Isolating the pathogen** — chick embryo, cell culture, laboratory animal; because of the **high hazard** only in special laboratories.

> **CAUTION**
> In rickettsioses, diagnosis rests in practice on **serology and PCR**. The aerosol risk is taken into account when collecting material from the patient and when working in the laboratory.

**Prophylaxis:** louse control and sanitation, epidemiological surveillance in the focus; in Q fever — veterinary control of livestock, pasteurization of milk, occupational protection.

## 4. Borreliae and relapsing fever

**Morphology:** *Borrelia* — a group of spirochetes; **3–8 large, irregular coils**; stains **blue-violet** with Romanowsky–Giemsa (the spirochete that takes ordinary dyes best — so it is visible in a blood smear).

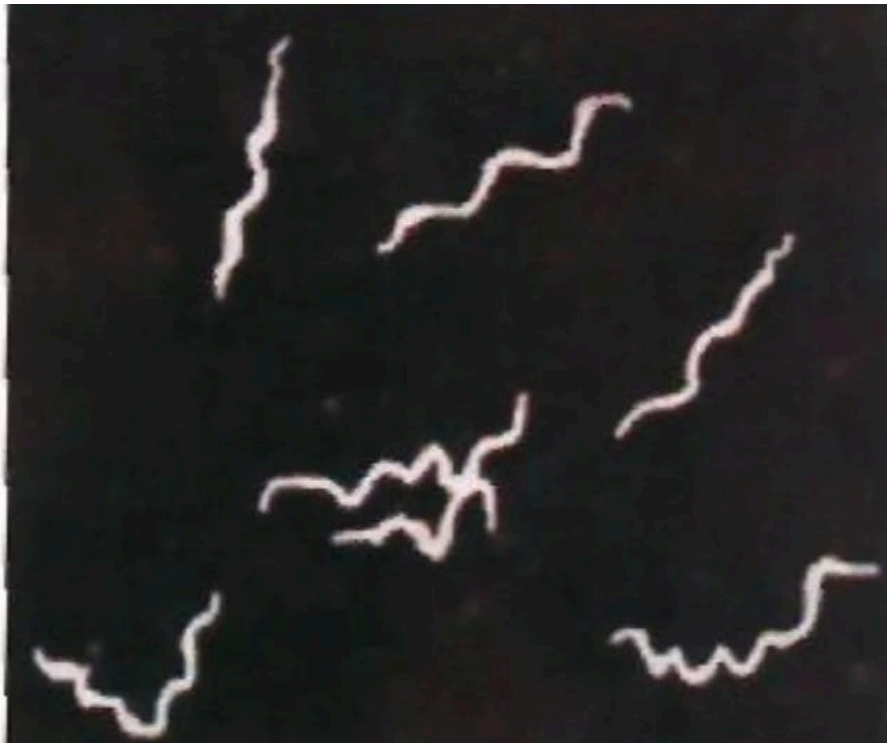


**Figure 26.1.** *Borreliae — large spirochetes with irregular coils (blood smear). Source: Muhamedov I.M. et al., 2019.*

**Table 26.2.** *The two forms of relapsing fever*

| Feature | Epidemic (lice) | Endemic (ticks) |
|---|---|---|
| Causative agent | *B. recurrentis* | *B. persica* and other species |
| Source | **Human** (anthroponosis) | Rodents (zoonosis, natural-focal) |
| Vector | Body louse | **Argasid ticks** |
| Spread | Epidemics (when sanitary conditions deteriorate) | Isolated cases, in foci |

**The “relapse” mechanism:** during the fever period borreliae are numerous in the blood; the body produces antibodies against them and they disappear (the apyrexia period). But a new population with a **changed antigenic structure (antigenic variation)** multiplies and produces the next wave of fever. For this reason the disease runs with **recurrent attacks of fever**.

> **REMEMBER**
> In relapsing fever the **blood smear is taken at the peak of fever** — exactly when borreliae are numerous in the blood. During apyrexia the smear is negative.

**Laboratory diagnosis:**

1. **Bacterioscopic (the main one)** — a **thick drop** and a thin blood smear taken **at the peak of fever**, stained with Romanowsky–Giemsa; live borreliae can also be seen in dark field.
2. **Biological test** — was used to distinguish the endemic form from the epidemic form (infecting mice).
3. **Serological and molecular** methods — additional.

## 5. Independent work 1: Lyme disease (Lyme borreliosis)

**Causative agent:** the *Borrelia burgdorferi* group; the vector is **Ixodes ticks**; the reservoir is rodents and large mammals; a natural-focal zoonosis.

**Clinical stages:**

1. **Early localized** — at the site of the tick bite a **migrating annular erythema** (a spot with a pale center and red edge that keeps expanding) — the most characteristic sign;
2. **Early disseminated** — after several weeks to months: the nervous system (neuroborreliosis — neuritis, meningitis), the heart (atrioventricular block), migrating joint and muscle pain;
3. **Late (chronic)** — chronic arthritis, acrodermatitis chronica atrophicans, chronic damage to the nervous system.

**Diagnosis:**

- **If there is a typical erythema — the diagnosis is made clinically**, and serology is not awaited (antibodies may not yet be present in the early stage);
- **In later stages — two-tier serology:** first **ELISA (screening)**, and with a positive or equivocal result **immunoblot (confirmation)** (topic 14);
- PCR — important in joint fluid and biopsy material;
- Culture — difficult and rarely used in practice.

> **CLINICAL SIGNIFICANCE**
> In Lyme disease **negative serology in the early stage is normal** — antibodies appear in 2–6 weeks. Therefore, with a typical erythema, treatment is started without waiting for serology (by a physician's prescription).

## 6. Independent work 2: Leptospirosis

**Causative agent:** *Leptospira interrogans* — a delicate spirochete; **very many fine coils**, with hooked ends (**S, C or "S" letter** appearance); highly motile.

**Epidemiology:** a natural-focal zoonosis; the reservoir is rodents (mainly rats), dogs, pigs and cattle. Transmission: **through water** (swimming, working in an irrigation ditch), and through soil and food contaminated with animal urine; the pathogen enters through **mucous membranes and damaged skin**.

**Clinical picture:** acute fever, severe **muscle (especially calf) pain**, redness of the conjunctiva; in the severe form — **jaundice, renal failure, hemorrhagic syndrome** (Vasiliev–Weil disease).

**Diagnosis — depends on the period:**

**Table 26.3.** *Choice of material and method in leptospirosis*

| Period | Material | Method |
|---|---|---|
| Week 1 (leptospiremia) | **Blood**, cerebrospinal fluid | Seeing in dark field, culture, PCR |
| From week 2 | **Urine** | Dark field, culture, PCR |
| From the end of week 1 | **Blood serum** | **Microscopic agglutination test (the gold standard)**, ELISA; paired sera |

**Additional:** leptospirae grow slowly in special liquid media (weeks), so culture gives its result late; for staining, Romanowsky–Giemsa and **silver impregnation (Morozov)** are used.

**Prophylaxis:** rodent control, protection of water bodies and water supply, occupational protective equipment, vaccination on indication (animals are also vaccinated).

## 7. Practical skill: examination when a vector-borne infection is suspected

1. **Epidemiological history:** travel to a natural focus, **a tick/louse/flea bite**, contact with livestock, swimming in a water body, occupation.
2. **Determine the syndrome:** fever + rash → rickettsiosis; **recurrent waves of fever** → relapsing fever; **migrating erythema** at the site of a tick bite → Lyme disease; fever + calf muscle pain + jaundice → leptospirosis; fever + atypical pneumonia + contact with livestock → Q fever.
3. **Collect the material at the right time:**

- in relapsing fever — **a thick drop and smear at the peak of fever**;

- in leptospirosis — **blood** in week 1, **urine** from week 2, **serum** in parallel;

- in rickettsioses — **paired sera** and PCR in the early period.

4. **Choose the method:** microscopy (borrelia, leptospira), serology (rickettsiosis, leptospirosis, Lyme disease), PCR (informative in all in the early period).
5. **Safety:** an aerosol and contact risk when working with rickettsiae and leptospirae; warn the laboratory.
6. **Epidemiological measure:** identifying the focus, vector control, monitoring of contacts.

## 8. Review questions

1. What are vector-borne infections? What is the essence of the doctrine of natural focality?
2. What are the source, vector and mechanism of transmission of epidemic typhus?
3. Which tissue is the main target in the pathogenesis of typhus?
4. What is Brill–Zinsser disease?
5. How does Q fever differ from other rickettsioses?
6. By which methods are rickettsioses diagnosed?
7. What are the morphological features of borreliae? What explains the waves of fever in relapsing fever?
8. When is the blood smear taken in relapsing fever, and why?
9. What are the stages of Lyme disease and the principle of early diagnosis?
10. How are the material and the method chosen according to the period in leptospirosis?

## Test questions

1. The vector of epidemic typhus:
   A) Tick
   B) Body louse
   C) Flea
   D) Mosquito
   E) Horsefly

2. How the causative agent enters the body in epidemic typhus:
   A) Through the saliva when a louse bites
   B) By rubbing louse feces into scratched skin
   C) With food
   D) Through the air
   E) Sexually

3. The main target in the pathogenesis of typhus:
   A) The endothelium of blood vessels
   B) Nerve cells
   C) Liver cells
   D) Renal tubules
   E) Erythrocytes

4. The rickettsiosis that relapses after years without a vector:
   A) Q fever
   B) Brill–Zinsser disease
   C) Endemic typhus
   D) Lyme disease
   E) Leptospirosis

5. The main route of transmission of *Coxiella burnetii*:
   A) Airborne (dust, aerosol)
   B) Only a tick bite
   C) Sexual
   D) Transfusion
   E) Through a wound, as in tetanus

6. The special method used to stain rickettsiae:
   A) Zdrodovsky
   B) Ozheshko
   C) Neisser
   D) Burri–Hiss
   E) CAMP

7. The main diagnostic method in practice in rickettsioses:
   A) Bacteriological culture
   B) Serological and molecular methods
   C) Coagulase test
   D) The Ascoli reaction
   E) Skin test

8. The morphological feature of borreliae:
   A) 3–8 large, irregular coils
   B) 8–14 regular coils
   C) Many fine coils with hooked ends
   D) Comma-shaped
   E) Coccobacillus

9. The cause of the waves of fever in relapsing fever:
    A) Antigenic variation (multiplication of new antigenic variants)
    B) Accumulation of toxin
    C) An allergic reaction
    D) Secondary infection
    E) The effect of drugs

10. When the blood smear is taken in relapsing fever:
    A) During apyrexia (no fever)
    B) At the peak of fever
    C) During recovery
    D) During the incubation period
    E) The timing is unimportant

11. The vector of Lyme disease:
    A) Ixodes ticks
    B) Louse
    C) Flea
    D) Mosquito
    E) Fly

12. The most characteristic early sign of Lyme disease:
    A) Migrating annular erythema
    B) An ulcer with a black eschar
    C) A roseolar rash
    D) Jaundice
    E) Bubo

13. The principle of serological diagnosis in Lyme disease:
    A) A single ELISA is enough
    B) Two steps: ELISA (screening) → immunoblot (confirmation)
    C) Only culture
    D) Only a skin test
    E) Only PCR

14. The morphological feature of leptospirae:
    A) Many fine coils, with hooked ends
    B) Large irregular coils
    C) No coils
    D) Comma-shaped
    E) A chain-forming coccus

15. The most informative material in week 2 of leptospirosis:
    A) Blood
    B) Urine
    C) Sputum
    D) Stool
    E) Throat swab

## Clinical cases

### *Case 1*

A patient had a high fever for 4 days, then 7 days of normal temperature, and the fever rose again. In a blood smear taken at the peak of fever, large spirochetes with irregular coils, stained blue-violet with Romanowsky–Giemsa, were found.

**Question:** What is the diagnosis, and why was the timing of the smear important?

**Answer:** Relapsing fever (borreliae). Borreliae are numerous in the blood **precisely at the peak of fever**; during apyrexia the smear is negative. The waves of fever are explained by antigenic variation.

### *Case 2*

A patient who worked in an irrigation ditch in summer developed acute fever and severe calf muscle pain 5 days ago; now jaundice and reduced urine output have been added.

**Question:** What are the probable diagnosis and the material suited to the period of the disease?

**Answer:** Leptospirosis (severe form). In week 1 — **blood** (dark field, culture, PCR), from week 2 — **urine**; in parallel, the **microscopic agglutination test** in serum (paired sera).

### *Case 3*

10 days after a rest in the forest, a patient developed at the tick bite site an expanding, pale-centered annular red spot. ELISA is negative.

**Question:** Can the diagnosis be ruled out?

**Answer:** No. In the early stage of Lyme disease antibodies may not yet have formed. A **typical migrating erythema** is sufficient basis for a clinical diagnosis; treatment is started by a physician without waiting for serology.

## Quick review

**Table 26.4.** *Topic 26 — key concepts*

| Concept | Essence |
|---|---|
| Vector-borne infections | Through arthropods; natural focality |
| Epidemic typhus | *R. prowazekii*, body louse, through feces; endothelium, vasculitis |
| Brill–Zinsser | Endogenous relapse after years |
| Q fever | *C. burnetii*, airborne, resistant, no rash |
| Diagnosis of rickettsiosis | Serology (CFT, PHA, ELISA) + PCR; Zdrodovsky |
| Borrelia | 3–8 large irregular coils; Romanowsky–Giemsa |
| Relapsing fever | Antigenic variation → waves of fever; smear at the peak of fever |
| Lyme disease | *B. burgdorferi*, Ixodes tick; migrating erythema; ELISA → immunoblot |
| Leptospira | Many fine coils + hooked ends; through water |
| Diagnosis of leptospirosis | Week 1 blood, from week 2 urine, serum — microscopic agglutination |

TOPIC 27

# RNA viruses — causative agents of respiratory disease: orthomyxoviruses, paramyxoviruses, coronaviruses

## 1. Learning objectives

In this class the student will master the following:

- To know the structure, antigens and mechanisms of variability of the influenza virus;
- to explain the properties of the paramyxoviruses (parainfluenza, measles, mumps, RS virus) and of coronaviruses;
- to master the algorithm of collecting material and of virological, serological and molecular diagnosis in respiratory viral infections;
- Independent work: to know the current problems of avian and swine influenza, RS virus and rhinoviruses.

## 2. Influenza virus (*Orthomyxoviridae*)

**Structure:** a medium-sized (80–120 nm), **enveloped (with a supercapsid)** virus; the genome is **single-stranded, negative-sense, segmented RNA** (8 segments in types A and B). The nucleocapsid has **helical** symmetry.

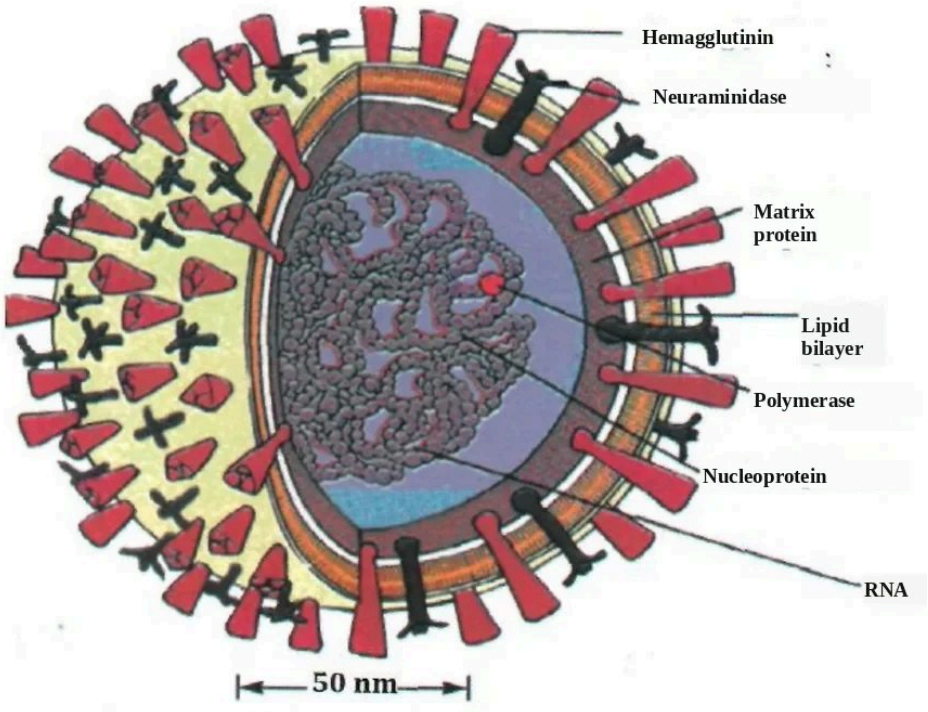



**Figure 27.1.** *The structure of the influenza virus: the envelope (supercapsid), hemagglutinin and neuraminidase spikes, matrix protein, segmented RNA and nucleoprotein. Source: Muhamedov I.M. et al., 2019.*

**Main antigens:**

- **Internal antigens** — nucleoprotein (NP) and matrix protein (M): they determine the **type (A, B, C)** and are stable;
- **Surface antigens (spikes):**

- **Hemagglutinin (H)** — provides binding to the cell receptor (sialic acid) and agglutinates erythrocytes; **the main protective antigen** — neutralizing antibodies form against it;

- **Neuraminidase (N)** — provides the release of new virions from the cell and their spread through mucus; the target of several antiviral drugs.

**Types:** A (in humans and animals; capable of pandemics), B (mainly in humans; epidemics), C (mild course).

### *2.1. Mechanisms of variability*

**Table 27.1.** *Variability of the influenza virus*

| Mechanism | Meaning | Consequence |
|---|---|---|
| **Antigenic drift** | Gradual change of H and N through point mutations | Seasonal **epidemics**; the vaccine composition is updated every year |
| **Antigenic shift** | Redistribution of segments (reassortment) when two strains infect one cell, owing to the **segmented genome** — entirely new H and/or N | Risk of **pandemic**: no immunity in the population |

**REMEMBER**

**Drift — a small change (epidemic), shift — a big jump (pandemic).** Shift occurs only in **type A**, because it also circulates in animals and creates conditions for reassortment.

**Pathogenesis:** the virus multiplies in the ciliated epithelium of the respiratory tract → epithelial necrosis and disruption of the clearance mechanism → viremia and intoxication (toxic effect, increased vascular permeability) → conditions for **secondary bacterial complications**.

**Clinical picture:** sudden onset, high fever, severe intoxication (headache, muscle pain), tracheitis (dry cough, retrosternal burning); catarrhal signs are relatively few. Complications: viral and **bacterial pneumonia** (often pneumococcus, *S. aureus*, *H. influenzae*), otitis, sinusitis, myocarditis, encephalopathy.

**CLINICAL SIGNIFICANCE**

The main danger of influenza is **secondary bacterial pneumonia**, especially in the elderly, in pregnant women and in people with chronic diseases. Therefore a renewed worsening after influenza (a second wave of fever) always demands attention.

### *2.2. Diagnosis of influenza*

1. **Material:** a nasal and throat swab/wash — **in the first 3 days** of the illness; in severe cases — material from the lower respiratory tract.
2. **Rapid methods:**

- **PCR (real-time)** — currently the main method: it determines the type and subtype;

- **Immunochromatographic rapid antigen tests** — fast, but less sensitive;

- **Immunofluorescence** — detecting viral antigen in epithelial cells.

3. **Virological method:** infection of a **chick embryo** (into the amniotic/allantoic cavity) or of a cell culture → indication by the **hemagglutination reaction (HA)** → identification by the **hemagglutination inhibition test (HI)** (topic 12). Mainly for epidemiological surveillance and for choosing vaccine strains.

4. **Serological method:** HI, CFT, ELISA in **paired sera** — a ≥4-fold rise in titer; of retrospective importance.

**Prophylaxis:** an annually updated **seasonal vaccine** (inactivated, subunit or live), vaccination of risk groups; nonspecific measures — hand hygiene, a mask, ventilation.

## 3. Paramyxoviruses (*Paramyxoviridae*)

**General features:** enveloped, **single-stranded, negative-sense, nonsegmented RNA**; helical nucleocapsid. Because it is nonsegmented, **there is no antigenic shift** — so the measles and mumps viruses are antigenically stable and the vaccines are long-lasting.

**Table 27.2.** *Paramyxoviruses of importance in medicine*

| Virus | Disease | Features |
|---|---|---|
| **Parainfluenza viruses** | ARVI, **croup** (laryngotracheitis) in children | Has hemagglutinin-neuraminidase (HN) |
| **Measles virus** | Measles | **Koplik spots**, rash; multinucleated giant cells; complications — encephalitis, panencephalitis |
| **Mumps virus** | Epidemic parotitis | Salivary glands; complications — orchitis, pancreatitis, meningitis |
| **RS virus (respiratory syncytial)** | **Bronchiolitis**, pneumonia in infants | Forms **syncytia** (multinucleated clumps) in cell culture; has no hemagglutinin |

**Diagnosis:** PCR (the main one), antigen tests and immunofluorescence (especially for RS virus), serology (IgM — a fresh infection; important in measles and mumps), isolation in cell culture (CPE — syncytium).

**Prophylaxis:** the **live combined vaccine** against measles, mumps and rubella — in the national immunization schedule; special immunoprophylaxis products exist against RS virus for risk groups.

> **MEMORY AID**
> **RS virus — "Syncytium"**: both its name and its main sign in cell culture. The main cause of bronchiolitis in infants.

## 4. Coronaviruses (*Coronaviridae*)

**Structure:** enveloped, **single-stranded, positive-sense RNA** viruses; the genome is the largest among RNA viruses. The **S (spike) protein** in the envelope forms crown-like projections — hence the name.

**Members:** seasonal coronaviruses (mild ARVI), **SARS-CoV**, **MERS-CoV** and **SARS-CoV-2** (COVID-19).

**Pathogenesis and clinical picture:** the S protein binds to the cell receptor; the epithelium of the respiratory tract and other tissues are damaged. The clinical picture ranges from mild ARVI to severe pneumonia and systemic complications; in severe forms, excessive activation of the immune response and disturbance of the coagulation system are important.

**Diagnosis:**

- **PCR (real-time)** — the gold standard; material: a nasopharyngeal swab;
- **Rapid antigen tests** — convenient for screening, with lower sensitivity;

- **Serology (IgM/IgG)** — for assessing past infection and the immune response; not the main method for acute diagnosis;
- **Isolation in cell culture** — only in special laboratories.

**Prophylaxis:** vaccination, a mask, hand hygiene, ventilation, isolation measures.

> **CAUTION**
> The clinical picture of respiratory viruses is largely similar; **only the laboratory determines the type**. Therefore in practice **multiplex PCR panels**, which cover several causative agents at once, are often used.

## 5. Independent work: Avian and swine influenza, RS virus, rhinoviruses

**Avian influenza (for example, A/H5N1, A/H7N9):** the natural reservoir is wild waterfowl; it is transmitted to humans mainly **by direct contact with sick birds**; person-to-person transmission is limited. It can run with severe lung damage. The main danger is the appearance, as a result of **reassortment**, of a new variant easily transmitted from person to person (antigenic shift).

**Swine influenza (A/H1N1pdm09):** the strain of the 2009 pandemic — the product of **redistribution** of segments of human, swine and avian influenza viruses. It now circulates among the seasonal influenza strains and is included in the seasonal vaccine.

**The pig — the "mixing vessel":** because pig cells carry receptors for both avian and human influenza viruses, the pig is a favorable environment for reassortment.

**RS virus: the main cause of bronchiolitis and pneumonia** in children under 1 year; it causes seasonal rises; immunity is incomplete — reinfection is possible. Diagnosis — antigen tests and PCR; special immunoprophylaxis is used in risk groups.

**Rhinoviruses (*Picornaviridae*): nonenveloped**, single-stranded positive RNA; there are more than 100 serotypes — so durable immunity does not form and a person falls ill several times a year. The optimal temperature for multiplication is **33–35 °C** — so they multiply mainly **in the upper respiratory tract** (the nasal mucosa). The most frequent cause of the common cold; diagnosis is usually not required.

> **REMEMBER**
> **Rhinovirus — many serotypes + loves low temperature → a frequently recurring "common cold"; a vaccine is hard to develop.**

## 6. Practical skill: collecting material for a respiratory virus

1. The material is taken **in the first days of the illness** (best on day 1–3) — viral shedding is highest in this period.
2. **Nasopharyngeal swab:** the swab is advanced along the nasal passage **to the posterior wall**, held for a few seconds and withdrawn with rotation; a throat swab is taken from the tonsils and the posterior wall.
3. The swab is placed in **viral transport medium** and its tip is broken off.
4. The sample is kept at **2–8 °C** and delivered quickly; if prolonged storage is needed, it is frozen by the set procedure (repeated freezing and thawing is prohibited).
5. On the requisition: the date of onset, vaccination status, travel/contact history.

6. In the laboratory: PCR (often a multiplex panel), an antigen test where necessary, and in epidemiological surveillance — virological isolation.
7. **Safety:** aerosols may form — protective equipment and the prescribed regime are observed.

## 7. Review questions

1. What are the structure of the influenza virus and the feature of its genome?
2. State the functions of the H and N antigens.
3. How do antigenic drift and shift differ? Why does shift occur only in type A?
4. What are the pathogenesis and main complications of influenza?
5. When is the material taken in the diagnosis of influenza, and which methods are used?
6. How are indication and identification performed in the virological method?
7. Why is there no antigenic shift in paramyxoviruses?
8. What are the sign of RS virus in cell culture and its clinical significance?
9. State the structure of coronaviruses and the diagnostic methods.
10. Why do rhinoviruses cause frequently recurring colds?

## Test questions

1. The genome of the influenza virus:
   A) Double-stranded DNA
   B) Single-stranded, negative-sense, segmented RNA
   C) Single-stranded positive RNA, nonsegmented
   D) Double-stranded RNA
   E) Single-stranded DNA

2. The main function of hemagglutinin:
   A) Release of the virion from the cell
   B) Binding to the cell receptor and agglutination of erythrocytes
   C) RNA synthesis
   D) Capsid formation
   E) Interferon production

3. The function of neuraminidase:
   A) Attachment to the cell
   B) Release and spread of new virions from the cell
   C) Doubling of the genome
   D) Protein synthesis
   E) Stopping apoptosis

4. The variability that causes seasonal epidemics:
   A) Antigenic drift
   B) Antigenic shift
   C) Transduction
   D) Conjugation
   E) Modification

5. The mechanism that leads to a pandemic:
   A) Antigenic drift
   B) Antigenic shift (reassortment of segments)
   C) Lysogeny
   D) Transformation
   E) Phage conversion

6. The type in which antigenic shift is observed:
   A) Type A
   B) Type B
   C) Type C
   D) The same in all types
   E) In no type

7. The classic system for propagating the influenza virus:
   A) Chick embryo
   B) Blood agar
   C) Sabouraud medium
   D) Kligler medium
   E) Löwenstein–Jensen

8. The reaction for identifying the influenza virus in the virological method:
   A) Hemagglutination (HA)
   B) Hemagglutination inhibition (HI)
   C) Coagulase test
   D) The Ascoli reaction
   E) The Wright reaction

9. The most dangerous complication of influenza:
   A) Secondary bacterial pneumonia
   B) Constipation
   C) Skin rash
   D) Arthritis
   E) Conjunctivitis

10. The reason there is no antigenic shift in paramyxoviruses:
    A) The genome is nonsegmented
    B) They are nonenveloped
    C) Contains DNA
    D) They occur only in animals
    E) Without a capsid

11. The virus that forms syncytia in cell culture:
    A) RS virus
    B) Rhinovirus
    C) Rotavirus
    D) Hepatitis A virus
    E) Adenovirus

12. The main viral cause of bronchiolitis in infants:
    A) RS virus
    B) Rhinovirus
    C) Measles virus
    D) Mumps virus
    E) Coronavirus

13. The protein that forms the crown-like projections of coronaviruses:
    A) The S (spike) protein
    B) The M protein
    C) NP
    D) Neuraminidase
    E) Hemagglutinin

14. The optimal temperature for multiplication of rhinoviruses:
    A) 33–35 °C
    B) 37 °C
    C) 42 °C
    D) 25 °C
    E) 4 °C

15. The current main method for diagnosing respiratory viruses:
    A) Real-time PCR
    B) Skin test
    C) Coagulase test
    D) The Ascoli reaction
    E) Oxidase test

## Clinical cases

### *Case 1*

A patient with influenza had his temperature return to normal on day 5, but on day 7 it rose again to 39 °C, the cough worsened with purulent sputum, and focal changes were found on X-ray.

**Question:** What has happened, and what examination is needed?

**Answer: Secondary bacterial pneumonia** (often pneumococcus, *S. aureus*, *H. influenzae*). Examination: sputum culture and **antibiogram**, blood culture; treatment is decided by a physician.

### *Case 2*

As part of epidemiological surveillance, the virus from a patient’s material was propagated in a chick embryo; the allantoic fluid agglutinated erythrocytes. When specific serum was added, the agglutination stopped.

**Question:** Which reactions were set up, and what does the result mean?

**Answer:** First the **hemagglutination reaction (HA)** — an indication that the virus is present; then the **hemagglutination inhibition test (HI)** — the type/subtype of the virus was determined (identification).

### *Case 3*

A 4-month-old infant has shortness of breath, wheezing and tachypnea; the season is winter. The antigen test on a nasopharyngeal swab was positive, and multinucleated cells (syncytia) formed in cell culture.

**Question:** Which causative agent, and what is its clinical significance?

**Answer: RS virus** — the main cause of bronchiolitis in infants. Treatment is mainly supportive; special immunoprophylaxis products are used in risk groups.

## Quick review

**Table 27.3.** *Topic 27 — key concepts*

| Concept | Essence |
|---|---|
| Influenza virus | Enveloped, negative-sense, **segmented** RNA; helical nucleocapsid |
| H / N | Receptor binding, protective antigen / release |
| Drift / shift | Mutation → epidemic / reassortment → pandemic (type A only) |
| Complication of influenza | Secondary bacterial pneumonia |
| Diagnosis of influenza | PCR (the main one); HA (indication) + HI (identification) |
| Paramyxoviruses | Nonsegmented → no shift; measles, mumps, parainfluenza, RS |
| RS virus | Syncytium; bronchiolitis in infants |
| Coronaviruses | Positive RNA, S protein; PCR is the gold standard |
| Rhinoviruses | Nonenveloped, 100+ serotypes, 33–35 °C; common cold |
| Material | Day 1–3, transport medium, 2–8 °C |

TOPIC 28

# Picornaviruses (poliomyelitis), rhabdoviruses (rabies), hepatitis viruses transmitted by the fecal-oral route: A and E

## 1. Learning objectives

In this class the student will master the following:

- To know the structure and members of the picornavirus family; to explain the pathogenesis and prevention of poliomyelitis;
- to master the properties of the rabies virus, its pathogenesis and the principles of emergency prophylaxis;
- to know the features of hepatitis A and E viruses and the interpretation of serological diagnosis;
- Independent work: to have an understanding of the Ebola, Lassa and Zika viruses.

## 2. Picornaviruses (*Picornaviridae*)

**General features:** very small (22–30 nm), **nonenveloped**, with an icosahedral capsid, **single-stranded, positive-sense RNA** viruses. Because they are nonenveloped they are **resistant** to the external environment, acid and bile acids — which is why most are transmitted by the **fecal-oral** route.

**Table 28.1.** *Main members of the picornaviruses*

| Genus | Representatives | Diseases |
|---|---|---|
| **Enteroviruses** | Poliovirus (types 1, 2, 3) | Poliomyelitis |
| | Coxsackie A and B | Herpangina, hand-foot-and-mouth disease, myocarditis, meningitis |
| | ECHO viruses | Meningitis, fever, rash |
| | Enterovirus 70, 71 | Conjunctivitis, damage to the nervous system |
| **Rhinoviruses** | 100+ serotypes | Common cold (topic 27) |
| **Hepatoviruses** | **Hepatitis A virus** | Hepatitis A |

### *2.1. Poliomyelitis*

**Causative agent:** poliovirus (serotypes 1, 2, 3).

**Transmission: fecal-oral** (the main route), less often airborne; the source is a sick person and a virus excretor.

**Pathogenesis (stages):**

1. **Alimentary stage** — the virus multiplies in the lymphoid tissue of the pharynx and intestine (tonsils, Peyer’s patches);
2. **Viremia** — it enters the blood; in most cases the process ends here;
3. **Neural stage** — the virus crosses the blood–brain barrier and infects the **motor neurons in the anterior horns of the spinal cord**.

**Clinical forms:** inapparent (asymptomatic, the most frequent), abortive (general infectious signs), meningeal (serous meningitis) and the **paralytic** form — **asymmetric, flaccid (peripheral) paralysis**, often in the legs; sensation is preserved.

**REMEMBER**

In poliomyelitis the paralysis is **asymmetric and flaccid**, and **sensation is not impaired** — this is related to the selective damage to the motor neurons.

**Diagnosis:**

1. **Material:** stool (the virus is shed longest there), throat wash, cerebrospinal fluid; paired sera.
2. **Virological method** — infection of a cell culture → **cytopathic effect** → typing by the neutralization reaction; **molecular typing** to distinguish it from the vaccine strain.
3. **PCR** — fast and sensitive.
4. **Serological** — the dynamics of the titer in paired sera by neutralization or ELISA.

**Prophylaxis — two kinds of vaccine:**

**Table 28.2.** *Poliomyelitis vaccines*

| A vaccine | Type | Advantage | Property |
|---|---|---|---|
| **OPV (oral, Sabin)** | **Live** attenuated | Also gives intestinal (local, IgA) immunity; cheap and easy to administer | In very rare cases vaccine-associated conditions may occur |
| **IPV (injectable, Salk)** | **Inactivated** | Safe; also used in the immunocompromised | Forms less intestinal immunity |

**CLINICAL SIGNIFICANCE**

In the poliomyelitis eradication program both vaccines are used: OPV interrupts the circulation of the virus, while IPV gives safe individual protection. They are used according to the national immunization schedule.

## 3. Rhabdoviruses: rabies (*Rhabdoviridae*, *Lyssavirus*)

**Structure:** a **bullet-shaped**, enveloped virus; the genome is **single-stranded, negative-sense, nonsegmented RNA**; helical nucleocapsid; the envelope bears **G glycoprotein** spikes (the main protective antigen, receptor binding).

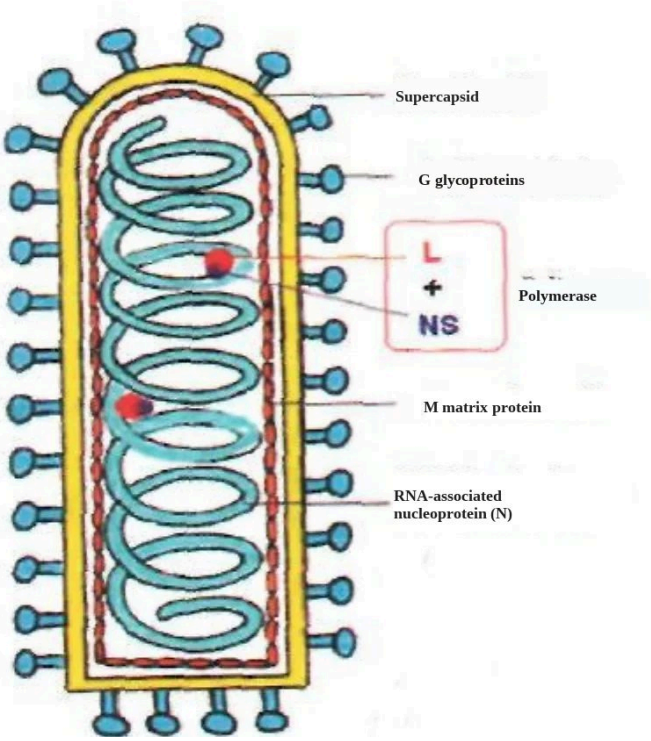



**Figure 28.1.** *The structure of the rabies virus: a bullet-shaped virion, supercapsid, G glycoprotein spikes, matrix protein and helical nucleoprotein. Source: Muhamedov I.M. et al., 2019.*

**Epidemiology:** a zoonosis; the source is wild and domestic animals (dog, cat, fox, wolf, bats). Transmission — through **a bite by a sick animal or saliva getting on a mucous membrane**.

**Pathogenesis:** the virus multiplies at the entry site → moves **along peripheral nerves to the center** → multiplies in the central nervous system (especially the hippocampus and brainstem) and causes encephalitis → then spreads **along nerves outward**, including to the **salivary glands** (which is why saliva is infectious).

**Incubation period** — usually 1–3 months; **it is shorter if the bite is near the head and deep**.

**Clinical picture:** a prodromal period (discomfort at the bite site, restlessness) → an excitation period (**hydrophobia** — fear of water and spasm on swallowing, aerophobia, photophobia, excitability) → a paralytic period. Once clinical signs have developed, the disease is considered practically untreatable.

**Morphological sign:** in the cytoplasm of nerve cells **Babes–Negri bodies** (topic 12) — viral inclusions; found on examining brain tissue after death.

**Diagnosis:**

- **In animals:** examination of brain tissue — immunofluorescence (the standard method), detection of Babes–Negri bodies, PCR;
- **In humans (during life):** immunofluorescence on a nuchal skin biopsy, **PCR** on saliva and cerebrospinal fluid, a smear from the cornea; the methods have limited sensitivity, so repeat samples are taken;
- After death — examination of brain tissue.

**Emergency prophylaxis (postexposure): wash the wound immediately with soap and plenty of water** and treat it with an antiseptic; then, as prescribed, the **vaccine** and, when indicated, **specific immunoglobulin** are used. The animal is placed under observation. All measures are determined by a physician on the basis of the current protocol.

**CAUTION**

In rabies **time is the decisive factor**: prophylaxis must begin before clinical signs appear. Therefore in any case of a bite it is necessary to seek medical help immediately.

## 4. Hepatitis viruses transmitted by the fecal-oral route

### *4.1. Hepatitis A virus (HAV)*

**Features:** the family *Picornaviridae* (hepatoviruses); **nonenveloped**, single-stranded positive RNA; **resistant** in the external environment (to acid, bile, drying and the usual disinfection regimes).

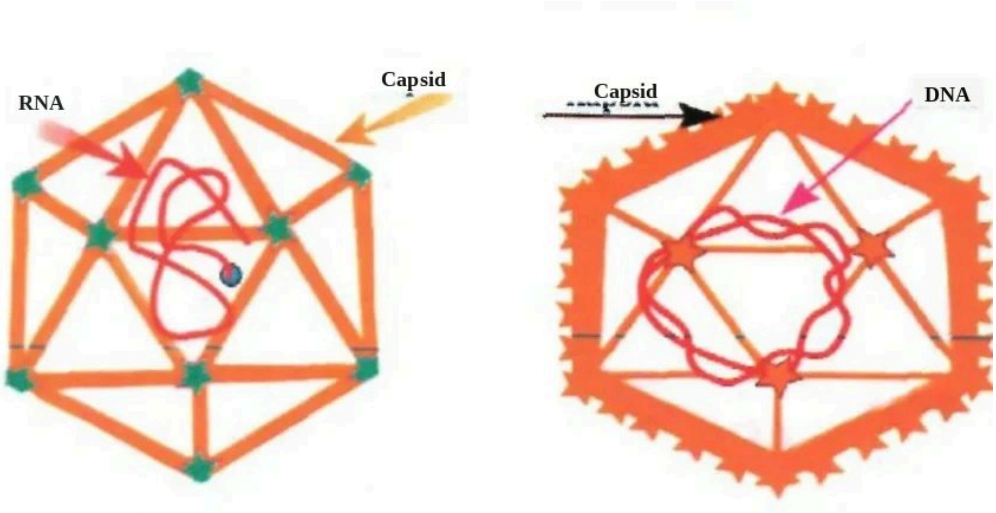



**Figure 28.2.** *The structure of the hepatitis A virus and the papillomavirus (schematic image). Source: Muhamedov I.M. et al., 2019.*

**Transmission: fecal-oral** — contaminated water and food (especially unwashed vegetables and fruit, seafood), contact-household. Viral shedding is **highest before jaundice appears** — so the patient is most infectious until jaundice begins.

**Clinical picture:** the incubation period is 2–6 weeks; a pre-icteric period (fever, loss of appetite, nausea), an icteric period, recovery. In children it often runs **asymptomatically**. It **does not become chronic**, carriage does not develop; acute liver failure is rare.

**Diagnosis: anti-HAV IgM** — a sign of acute infection (the main marker); **anti-HAV IgG** — past infection or immunity after vaccination; PCR (stool, blood) — in epidemiological investigation.

**Prophylaxis:** sanitary and hygienic measures (water supply, food control, hand hygiene), the **inactivated vaccine**; immunoglobulin for contacts on indication.

### *4.2. Hepatitis E virus (HEV)*

**Features:** *Hepeviridae*; nonenveloped, single-stranded positive RNA; transmitted by the fecal-oral route, mainly **through water**; some genotypes are zoonotic (pigs and others, via undercooked meat).

**Clinical picture:** resembles hepatitis A; usually does not become chronic (an exception may occur in the severely immunocompromised). **The main feature is a severe course in pregnant women** and a high risk of acute liver failure.

**Diagnosis:** anti-HEV IgM and IgG, PCR (blood, stool).

**Table 28.3.** *Comparison of hepatitis A and E*

| Feature | Hepatitis A | Hepatitis E |
|---|---|---|
| Family | *Picornaviridae* | *Hepeviridae* |
| Transmission | Fecal-oral (water, food, contact) | Fecal-oral (mainly **water**), some genotypes zoonotic |
| Chronicity | Absent | Usually none |
| Risk group | Children (often asymptomatic) | **Pregnant women** (severe course) |
| A vaccine | Available (inactivated) | Available in limited regions |
| Marker | anti-HAV IgM / IgG | anti-HEV IgM / IgG |

**MEMORY AID**
**"A and E — Alimentary/Enteral"**: both are transmitted through the intestine and do not become chronic. **B, C, D** are transmitted by blood and sexual routes and may become chronic (topic 29).

## 5. Independent work: Ebola, Lassa and Zika viruses

**Table 28.4.** *Some especially dangerous and emerging viruses*

| Virus | Family | Reservoir/transmission | Clinical picture |
|---|---|---|---|
| **Ebola** | *Filoviridae*; filamentous virion, negative-sense RNA | Bats; person to person by **direct contact with body fluids** | **Hemorrhagic fever**: high fever, bleeding, multiple organ failure; high mortality |
| **Lassa** | *Arenaviridae* | **Rodents** (food and dust contaminated with their feces and urine) | Fever, pharyngitis, hemorrhagic signs; a complication — **deafness** |
| **Zika** | *Flaviviridae* | **Aedes mosquitoes**; also the vertical and sexual routes | Usually mild (fever, rash, conjunctivitis); the main danger is, in pregnancy, **damage to the fetal nervous system (microcephaly)** |

**General principles:** work with these viruses requires laboratories of the **highest biosafety level**; diagnosis is made mainly by **PCR and antigen tests**; care and isolation measures are carried out according to strictly set protocols. When infection is suspected the health system is notified immediately.

**CLINICAL SIGNIFICANCE**
In a patient who presents with fever and who has **traveled to an epidemic area** in recent weeks, establishing the travel history is of decisive importance: it makes it possible to consider malaria, hemorrhagic fevers and other imported infections in good time.

## 6. Practical skill: the sequence of examination when hepatitis is suspected

1. **Clinical and epidemiological information:** jaundice, loss of appetite, darkening of the urine; a link with water/food, travel, parenteral interventions, sexual history.
2. **Biochemical tests:** transaminases and bilirubin — they confirm liver damage (but do not show the causative agent).
3. **Marker panel: anti-HAV IgM**, **HBsAg** and **anti-HBc IgM**, **anti-HCV**, and if necessary **anti-HEV IgM** (to be viewed together with topic 29).
4. **Interpret the result:** IgM — an **acute** process; IgG — past infection or immunity.

5. **Confirmation and assessment:** PCR (viral RNA/DNA), and viral load where necessary.
6. **Epidemiological measure:** in A and E — examining the source of water and food, monitoring contacts, hygiene rules; reporting by the set procedure.

## 7. Review questions

1. What are the general features of picornaviruses? How does being nonenveloped affect their route of transmission?
2. State the stages of the pathogenesis of poliomyelitis.
3. What are the characteristic features of paralysis in poliomyelitis?
4. Compare OPV and IPV.
5. What are the structure of the rabies virus and the role of the G glycoprotein?
6. How does the virus move in the pathogenesis of rabies?
7. What are Babes–Negri bodies and where are they detected?
8. What is the first step of emergency prophylaxis in rabies?
9. What is the route of transmission of hepatitis A, and when is the patient most infectious?
10. Who is the main risk group for hepatitis E? How are the markers interpreted?

## Test questions

1. The structure of picornaviruses:
   A) Enveloped, contain DNA
   B) Nonenveloped, single-stranded positive RNA
   C) Enveloped, segmented RNA
   D) Double-stranded DNA, nonenveloped
   E) Nonenveloped, double-stranded RNA

2. The main route of transmission of poliovirus:
   A) Vector-borne
   B) Fecal-oral
   C) Parenteral
   D) Sexual
   E) Vertical

3. The structure damaged in poliomyelitis:
   A) Motor neurons in the anterior horns of the spinal cord
   B) Sensory neurons in the posterior horns
   C) The sheath of a peripheral nerve
   D) The cerebellum
   E) The pituitary

4. The feature of paralysis in poliomyelitis:
   A) Symmetric, spastic
   B) Asymmetric, flaccid; sensation is preserved
   C) With loss of sensation
   D) Only in the facial muscles
   E) With loss of consciousness

5. The vaccine that better forms intestinal (local) immunity:
   A) OPV (live, oral)
   B) IPV (inactivated)
   C) A toxoid
   D) Recombinant vaccine
   E) Polysaccharide vaccine

6. The shape of the rabies virus:
   A) Spherical
   B) Bullet-shaped
   C) Filamentous
   D) Icosahedral
   E) Oval

7. The route by which the rabies virus reaches the central nervous system:
   A) Through the blood
   B) Along peripheral nerves
   C) Through the lymph
   D) Through the intestine
   E) Through the air

8. The inclusions detected in nerve cells in rabies:
   A) Guarnieri bodies
   B) Babes–Negri bodies
   C) Volutin granules
   D) Druses
   E) Clue cells

9. The first step of emergency prophylaxis in rabies:
   A) Washing the wound immediately with soap and plenty of water
   B) Taking an antibiotic by mouth
   C) Waiting and observing
   D) Suturing the wound
   E) Applying a cold compress

10. The clinical sign characteristic of rabies:
    A) Hydrophobia (fear of water)
    B) Diarrhea
    C) Jaundice
    D) Rash
    E) Bubo

11. What the resistance of the hepatitis A virus is related to:
    A) Its lack of an envelope (it resists acid and bile)
    B) Its envelope
    C) Its DNA content
    D) Its segmented genome
    E) Its lack of a capsid

12. The period in which the patient is most infectious in hepatitis A:
    A) Before jaundice appears
    B) In the middle of jaundice
    C) After recovery
    D) After a year
    E) Only in the chronic stage

13. The marker indicating acute hepatitis A:
    A) anti-HAV IgM
    B) anti-HAV IgG
    C) HBsAg
    D) anti-HCV
    E) anti-HBs

14. A feature of hepatitis A:
    A) It does not become chronic, carriage does not develop
    B) It often becomes chronic
    C) It is transmitted only parenterally
    D) It has no vaccine
    E) It occurs only in adults

15. The main risk group for hepatitis E:
    A) Pregnant women
    B) Newborns
    C) Athletes
    D) The elderly (only)
    E) There is no risk group

## Clinical cases

### *Case 1*

A child developed asymmetric flaccid paralysis of the leg after a fever; sensation is preserved.

**Question:** What are the probable diagnosis and the material to be examined?

**Answer:** The paralytic form of poliomyelitis. The material is **stool** (the virus is shed longest there), throat wash, cerebrospinal fluid and paired sera. Methods: virological isolation, **PCR** and molecular typing (to distinguish a wild strain from a vaccine strain). The case is reported by the set procedure.

### *Case 2*

A patient was bitten on the street by an unknown dog; the wound is in the facial area and is deep.

**Question:** What is the first measure, and why is it urgent?

**Answer: Wash the wound immediately with soap and plenty of water** and treat it with an antiseptic; then the vaccine and, when indicated, specific immunoglobulin are used by the procedure set by a physician. The reason for urgency: with a wound near the head and a deep one, **the incubation period is shortened**, and once clinical signs appear the disease is practically untreatable.

### *Case 3*

Several children in a group have jaundice; the common feature is that they drank water from the same place. anti-HAV IgM is positive.

**Question:** What are the diagnosis, the route of transmission and the measures?

**Answer:** Acute hepatitis A; fecal-oral (waterborne) transmission. Measures: examining and disinfecting the water source, a hygiene regime, monitoring of contacts, vaccination/immunoglobulin on indication, reporting of cases.

## Quick review

**Table 28.5.** *Topic 28 — key concepts*

| Concept | Essence |
|---|---|
| Picornaviruses | Nonenveloped, positive RNA, resistant, fecal-oral |
| Poliomyelitis | Alimentary → viremia → neural; asymmetric flaccid paralysis |
| OPV / IPV | Live, intestinal immunity / inactivated, safe |
| Rabies virus | Bullet-shaped, negative RNA, G glycoprotein |
| Pathogenesis of rabies | Along nerves to the CNS, then to the salivary glands |
| Babes–Negri | Inclusions in nerve cells |
| Emergency prophylaxis | Wash the wound + vaccine (± immunoglobulin), on time |
| Hepatitis A | Nonenveloped, resistant; most infectious before jaundice; does not become chronic |
| Marker of hepatitis A | anti-HAV IgM — acute; IgG — past/immunity |
| Hepatitis E | Through water; **severe in pregnant women** |
| Ebola / Lassa / Zika | Contact / rodents / Aedes mosquito and vertical |

TOPIC 29

# Retroviruses. Human immunodeficiency virus. Hepatotropic viruses B, D, C. Fungal diseases

## 1. Learning objectives

In this class the student will master the following:

- To know the structure of retroviruses and the particularity of their replication cycle;
- to explain the pathogenesis, stages and algorithm of laboratory diagnosis of HIV infection;
- to master the properties of the hepatitis B, D and C viruses and the interpretation of their serological markers;
- Independent work: to know the causative agents, pathogenesis and laboratory diagnosis of fungal diseases.

## 2. Retroviruses and HIV

**Retroviruses** (*Retroviridae*) are enveloped viruses containing **two copies of single-stranded positive-sense RNA**; their main feature is the enzyme **reverse transcriptase (revertase)**: it synthesizes DNA from RNA, and this DNA is then **integrated** into the host chromosome (**provirus**).

**HIV (human immunodeficiency virus)** is the causative agent of AIDS (acquired immunodeficiency syndrome); there are two types: **HIV-1** (widespread worldwide) and **HIV-2** (mainly West Africa, with a slower course).

**Structure:**

- **Genome** — two copies of RNA; the main genes: *gag* (internal proteins — p24 and others), *pol* (enzymes: reverse transcriptase, integrase, protease), *env* (envelope glycoproteins — **gp120 and gp41**);
- **Conical capsid** (p24 — an important diagnostic antigen);
- **Supercapsid** — from the host cell membrane; **gp120** binds to the receptor, **gp41** provides membrane fusion.

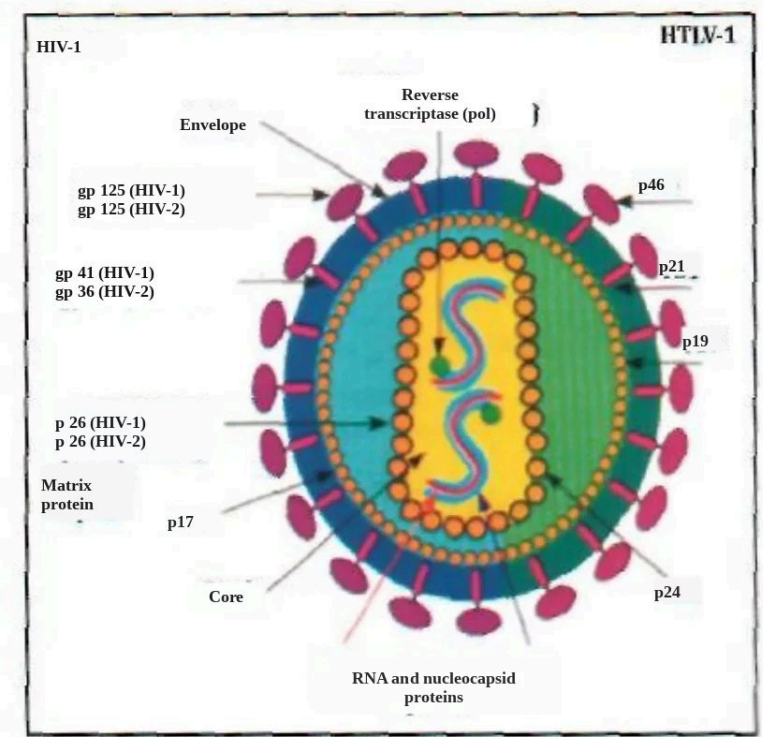



**Figure 29.1.** *The structure of HIV (HIV-1 and HIV-2): the envelope, gp120 and gp41 glycoproteins, matrix protein, p24 capsid protein, RNA and reverse transcriptase. Source: Muhamedov I.M. et al., 2019.*

**Target cells: $CD4^+$ T lymphocytes** (the main ones), macrophages, dendritic cells, microglia. The virus binds to the **CD4 molecule** (the main receptor) and to **chemokine coreceptors** (CCR5, CXCR4).

**Replication cycle:** adsorption (gp120 + CD4 + coreceptor) → membrane fusion (gp41) → **reverse transcription** (RNA → DNA) → formation of the provirus with the help of **integrase** → transcription and translation → maturation of proteins by **protease** → release by **budding**.

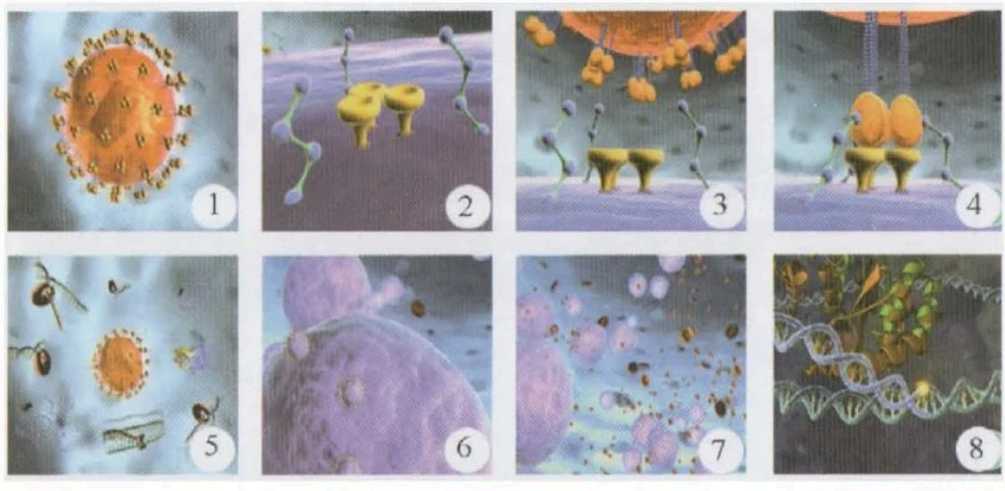



**Figure 29.2.** *The replicative cycle of HIV: entry into the cell, reverse transcription, integration, formation of new virions. Source: Aliyev Sh.R. et al., 2013.*

**REMEMBER**

Modern antiretroviral therapy targets exactly these stages of the cycle: there are groups of drugs that stop **entry**, **reverse transcriptase**, **integrase** and **protease**. Treatment is given in a combination regimen, determined by a physician.

**Routes of transmission:** sexual; parenteral (blood, syringe, medical and nonmedical manipulations); **vertical** (pregnancy, delivery, breastfeeding). It is **not transmitted** through household contact, air or insects.

**Pathogenesis:** the virus gradually destroys $CD4^+$ cells → cell-mediated immunity is deeply impaired → **opportunistic infections** and tumors develop. Stages of the disease: acute retroviral syndrome → the **latent (asymptomatic) period** (lasting years) → persistent lymphadenopathy → the **AIDS stage**.

**AIDS indicator conditions:** Pneumocystis pneumonia, toxoplasma encephalitis, cryptococcal meningitis, disseminated forms of candidiasis, tuberculosis and atypical mycobacterioses, cytomegalovirus retinitis, Kaposi sarcoma and others.

## *2.1. Laboratory diagnosis of HIV infection*

**Table 29.1.** *Diagnostic methods and their place*

| Method | What it detects | When |
|---|---|---|
| **ELISA (4th generation: antibody + p24 antigen)** | Antibodies and early antigen | **Screening** |
| **Immunoblot (Western blot)** | Antibodies against individual proteins | **Confirmatory test** |
| **PCR (viral RNA/DNA)** | Viral nucleic acid | Early diagnosis, the **"window period,"** infants, measuring viral load |
| **$CD4^+$ cell count (flow cytometry)** | Immune status | Assessing the stage and follow-up |
| **Viral load** | The amount of viral RNA in the blood | Monitoring the effect of treatment |

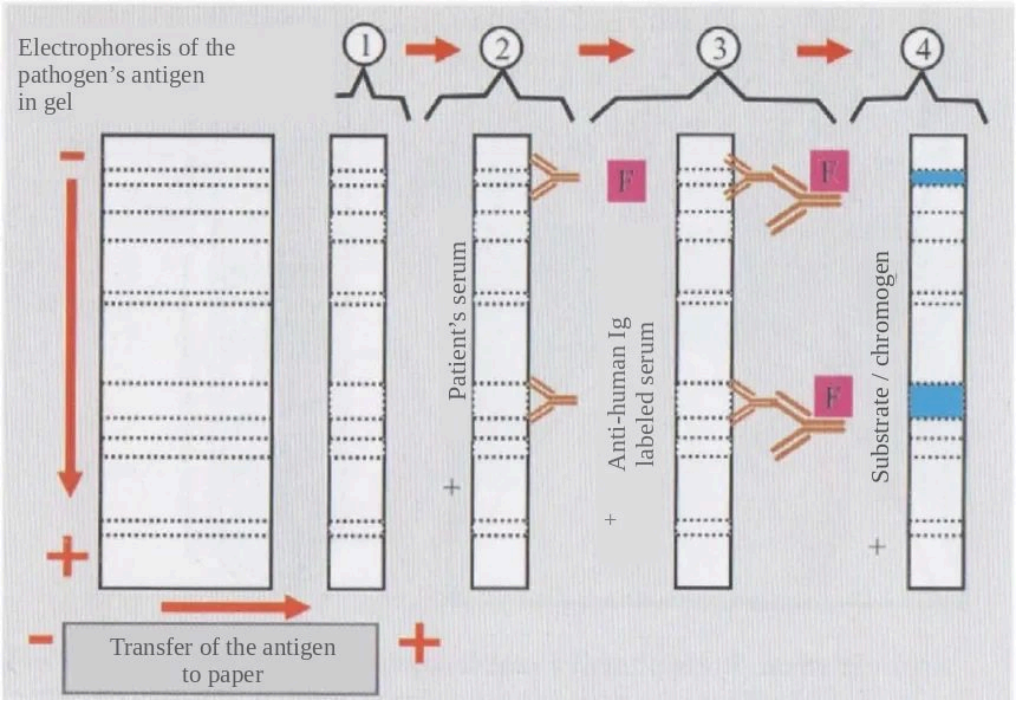



**Figure 29.3.** *Scheme of performing the immunoblot (Western blot) reaction: separating antigens by electrophoresis, transfer to a membrane and detection of antibodies by bands. Source: Aliyev Sh.R. et al., 2013.*

**CAUTION**

The **“window period” (serological window)** is the period after infection when antibodies cannot yet be detected. During this period **ELISA may be negative**, but the person is infectious. Therefore after a risky contact the test is **repeated** at set intervals; PCR is used for early detection.

**CLINICAL SIGNIFICANCE**

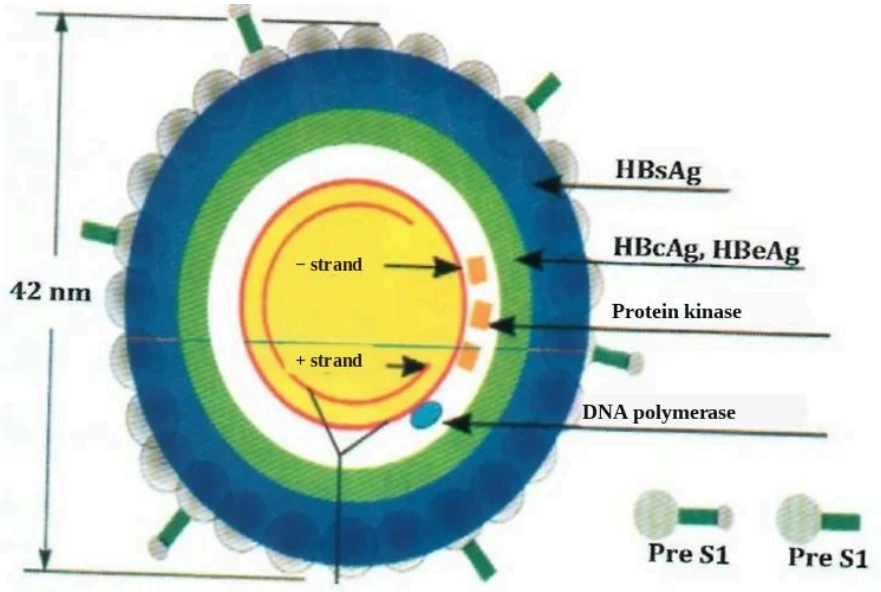
In an infant born to an HIV-positive mother, **the mother’s IgG antibodies** cross the placenta and may persist up to 18 months. Therefore up to this age the diagnosis is made **not by antibodies but by PCR**.

**Prophylaxis:** safe sexual behavior, disposable medical instruments and blood safety, programs for preventing mother-to-child transmission, postexposure and preexposure prophylaxis (by a physician’s prescription), early detection and treatment (a person on treatment whose viral load is undetectable does not transmit the infection).

## 3. Hepatotropic viruses: B, D, C

### *3.1. Hepatitis B virus (HBV)*

**Features:** *Hepadnaviridae*; **partially double-stranded DNA**; enveloped; its distinctive enzyme is **DNA polymerase with reverse transcriptase activity**. The complete virion is the **Dane particle**.




Schematic structure of the hepatitis B virus

**Figure 29.4.** *The structure of the hepatitis B virus: HBsAg (surface antigen), HBcAg and HBeAg, DNA polymerase, partially double-stranded DNA. Source: Muhamedov I.M. et al., 2019.*

**Antigens:**

- **HBsAg** — the surface (Australia) antigen; **the main marker of infection**; the vaccine is made from this antigen;
- **HBcAg** — the core antigen; not detected free in the blood, only in liver cells;
- **HBeAg** — a marker of active replication and **high infectivity**.

**Transmission:** parenteral (blood, medical manipulations, syringe), **sexual**, **vertical**. The infectious dose is very low, and the virus is resistant in the external environment.

**Clinical picture:** acute hepatitis (often with a gradual onset); in adults in most cases complete recovery, but in a proportion **chronicity**; in newborns and young children the likelihood of chronicity is much higher. In a chronic course — the risk of cirrhosis and hepatocellular carcinoma.

**Table 29.2.** *Interpretation of hepatitis B markers*

| Marker | Meaning |
|---|---|
| **HBsAg** | Infection is present (acute or chronic) |
| **anti-HBc IgM** | **Acute** infection (or a flare of chronic infection) |
| anti-HBc total (IgG) | Has met the virus (past or chronic) |
| **HBeAg** | Active replication, high infectivity |
| anti-HBe | Replication has decreased |
| **anti-HBs** | **Immunity**: recovery or **after vaccination** |
| HBV DNA (PCR) | Viral load, monitoring of treatment |

**MEMORY AID**
**Only anti-HBs positive → vaccinated. anti-HBs + anti-HBc positive → recovered from the disease.** Because the vaccine consists only of HBsAg, core antibodies are not formed after vaccination.

### *3.2. Hepatitis D virus (HDV)*

**Features:** a **defective** virus — it cannot multiply independently and uses **HBsAg** as its envelope. Therefore it occurs only together with HBV:

- **coinfection** — HBV and HDV are acquired at the same time;
- **superinfection** — HDV is added to a chronic HBV carrier; it usually runs more severely and increases the risk of cirrhosis.

**Diagnosis:** anti-HDV (IgM/IgG), HDV RNA (PCR) — tested in HBsAg-positive patients.

**Prophylaxis:** vaccination against hepatitis B — indirectly also protects against D (D does not develop without B).

### *3.3. Hepatitis C virus (HCV)*

**Features:** *Flaviviridae*; enveloped, **single-stranded positive-sense RNA**; high genetic **variability** (many genotypes and quasispecies) — therefore **there is no vaccine** and immunity does not protect.

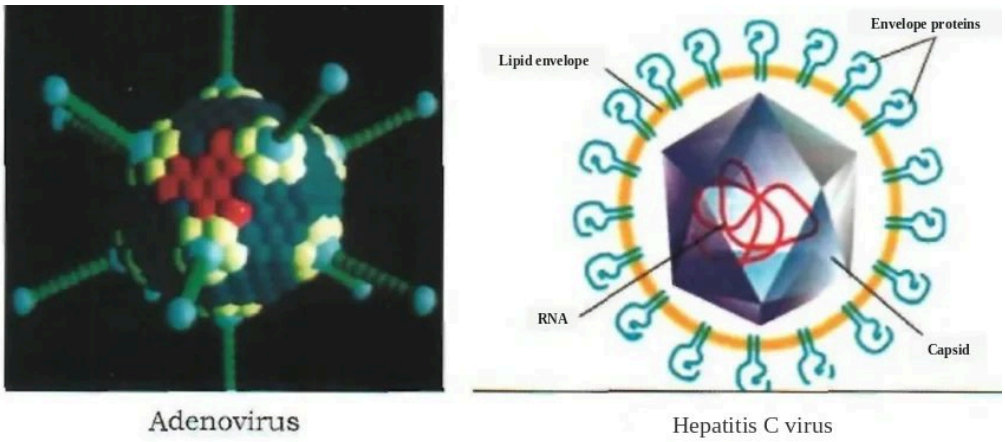



**Figure 29.5.** *The structure of adenovirus and hepatitis C virus (schematic image). Source: Muhamedov I.M. et al., 2019.*

**Transmission:** mainly **parenteral** (blood, syringe, medical manipulations); the sexual and vertical routes are relatively less common.

**Clinical picture:** the acute phase often runs **asymptomatically**; **the likelihood of chronicity is high** → the risk of cirrhosis and hepatocellular carcinoma. Therefore HCV is often found incidentally, during screening.

**Diagnosis: anti-HCV** (screening, ELISA) → confirmation with **HCV RNA (PCR)** and determination of active infection; genotyping — to choose the treatment regimen.

> **CLINICAL SIGNIFICANCE**
> **A positive anti-HCV does not prove a current infection**: it also stays positive in a past infection that has cleared spontaneously or as a result of treatment. Active infection is confirmed only by **HCV RNA**.

**Table 29.3.** *Brief comparison of hepatitis B, C and D*

| Feature | B | C | D |
|---|---|---|---|
| Genome | Partially double-stranded DNA | Single-stranded positive RNA | Single-stranded RNA (defective) |
| Transmission | Parenteral, sexual, vertical | Mainly parenteral | Together with HBV |
| Chronicity | In a proportion (more in children) | In many cases | Often severe |
| A vaccine | **Yes** (recombinant, HBsAg) | Absent | Indirectly through the hepatitis B vaccine |
| Main marker | HBsAg, anti-HBc IgM | anti-HCV + HCV RNA | anti-HDV, HDV RNA |

## 4. Independent work: Fungal diseases (mycoses)

**Classification:**

**Table 29.4.** *The main groups of mycoses*

| Group | Site of damage | Causative agents |
|---|---|---|
| Superficial mycoses | Skin surface, hair | *Malassezia* (tinea versicolor) |
| **Dermatomycoses** | Skin, hair, nails | *Trichophyton*, *Microsporum*, *Epidermophyton* |
| Subcutaneous mycoses | Subcutaneous tissue | Sporotrichosis, etc. |
| **Deep (systemic) mycoses** | Internal organs | *Histoplasma*, *Coccidioides* (dimorphic) |
| **Opportunistic mycoses** | When immunity is reduced | *Candida*, *Aspergillus*, *Cryptococcus*, *Pneumocystis* |

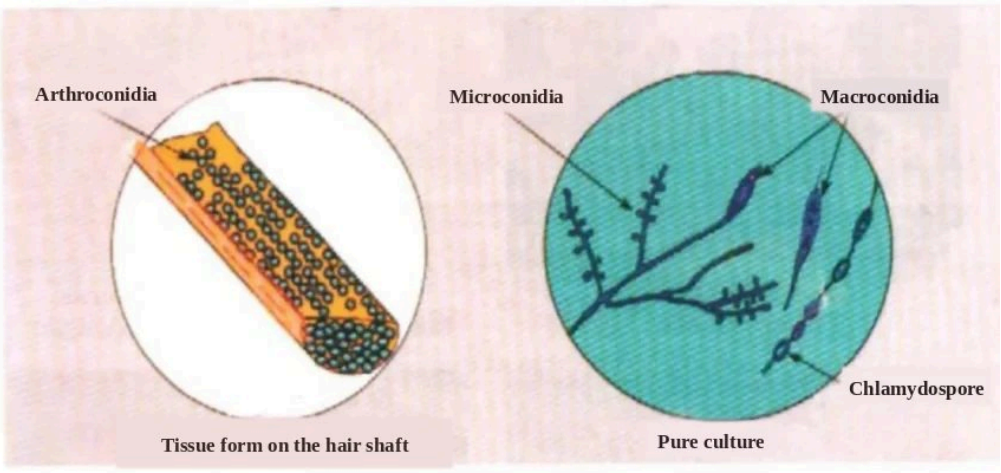



**Figure 29.6.** *Trichophyton tonsurans: its location on the hair shaft and the microscopic appearance of a pure culture (arthroconidia, microconidia and macroconidia). Source: Muhamedov I.M. et al., 2019.*

**Important mycoses associated with HIV infection:** Pneumocystis pneumonia (*Pneumocystis jirovecii*), cryptococcal meningitis (*Cryptococcus neoformans* — an encapsulated yeast-like fungus; viewed in the cerebrospinal fluid **with India ink**, and the **capsular antigen** test), esophageal and disseminated forms of candidiasis.

**Laboratory diagnosis (see also topic 3):**

1. **Microscopic:** the material (skin scrapings, hair, nails, sputum, cerebrospinal fluid) is cleared in **10–20% KOH**; staining — Gram, Romanowsky–Giemsa, PAS, calcofluor; for cryptococcus in the cerebrospinal fluid, an **India ink preparation**.
2. **Culture:** inoculation on **Sabouraud** medium, at 22–28 °C and 37 °C; identification by colony appearance and microscopic structure; for dimorphic fungi both temperatures are important.
3. **Antigen detection:** cryptococcal capsular antigen (cerebrospinal fluid, blood), galactomannan in aspergillosis.
4. **Molecular:** PCR, sequencing; MALDI-TOF.
5. **Antifungal susceptibility** — in disseminated cases and when resistance is suspected.

**REMEMBER**

The fungal membrane contains **ergosterol** (absent in bacteria) — this is the target of most antifungal drugs; the cell wall consists of **chitin and glucan**. Therefore antibacterial antibiotics do not act on fungi and may, on the contrary, create conditions for their growth.

## 5. Practical skill: interpreting the markers of parenteral hepatitis and HIV

1. **Screening panel:** HBsAg, anti-HCV, HIV (4th-generation ELISA).
2. **If HBsAg is positive:** anti-HBc IgM (is it acute?), HBeAg and anti-HBe (degree of replication), **HBV DNA**; in an HBsAg-positive patient **anti-HDV** is also tested.
3. **If anti-HCV is positive: HCV RNA (PCR)** — to confirm active infection; if positive, genotyping.
4. **If the HIV screening is positive:** a **confirmatory test** by the set algorithm (immunoblot and/or a molecular method); after confirmation, the **CD4⁺ count** and **viral load**.

5. **If a negative result was obtained after a risky contact:** taking the “window period” into account, the test is repeated at set intervals.
6. **At all stages:** confidentiality, counseling (before and after the test), observing safe care rules.

## 6. Review questions

1. What is the main enzyme of retroviruses and what does it do?
2. What are the target cells and receptors of HIV?
3. Describe the replication cycle of HIV step by step.
4. What are the stages of HIV infection? Give examples of AIDS indicator conditions.
5. What is the “window period” and how does it affect diagnosis?
6. Why is the diagnosis in a newborn of an HIV-positive mother made by PCR?
7. State the main antigens of hepatitis B and their meaning.
8. How do you distinguish, by markers, a vaccinated person from one who has recovered from the disease?
9. Why can hepatitis D not multiply independently?
10. What does a positive anti-HCV prove, and what does it not prove?
11. Into what groups are mycoses divided? Give examples of opportunistic mycoses.

## Test questions

1. The main enzyme of retroviruses:
   A) DNA polymerase
   B) Reverse transcriptase (revertase)
   C) Neuraminidase
   D) Hemagglutinin
   E) Protease (only)

2. The main target cell of HIV:
   A) B lymphocytes
   B) $CD4^+$ T lymphocytes
   C) Erythrocytes
   D) Neutrophils
   E) Platelets

3. The glycoprotein that provides the binding of HIV to the cell receptor:
   A) gp41
   B) gp120
   C) p24
   D) p17
   E) HBsAg

4. The enzyme that provides the integration of the HIV genome into the cell chromosome:
   A) Protease
   B) Integrase
   C) Neuraminidase
   D) Ligase
   E) Catalase

5. Not a route of transmission of HIV:
   A) Sexual
   B) Parenteral
   C) Vertical
   D) Household contact
   E) Through breastfeeding

6. The HIV screening test:
   A) Immunoblot
   B) ELISA (4th generation)
   C) PCR
   D) Flow cytometry
   E) Skin test

7. The confirmatory test in the diagnosis of HIV:
   A) ELISA
   B) Immunoblot
   C) Agglutination
   D) The Ascoli reaction
   E) The Wright reaction

8. What the “window period” is:
   A) The period when antibodies are not yet detectable, but the person is infectious
   B) The recovery period of the disease
   C) The end of the latent period
   D) The period of immunity after the incubation period
   E) The period after the end of treatment

9. How HIV is diagnosed in a child up to 18 months of age:
   A) By antibodies
   B) By PCR
   C) By a skin test
   D) By immunoblot
   E) By blood culture

10. The main surface antigen of hepatitis B:
    A) HBsAg
    B) HBcAg
    C) HBeAg
    D) anti-HBs
    E) anti-HBc

11. The indicator of active replication and high infectivity:
    A) anti-HBs
    B) HBeAg
    C) anti-HBe
    D) anti-HBc IgG
    E) HBsAg (only)

12. If only anti-HBs is positive, this means:
    A) Acute infection
    B) Immunity after vaccination
    C) Chronic infection
    D) A flare of the disease
    E) Hepatitis D

13. The feature of the hepatitis D virus:
    A) Multiplies independently
    B) Defective; uses HBsAg as its envelope
    C) Contains DNA
    D) Transmitted only by the fecal-oral route
    E) Its vaccine was developed independently

14. The test that confirms active hepatitis C infection:
    A) anti-HCV
    B) HCV RNA (PCR)
    C) HBsAg
    D) anti-HBs
    E) Bilirubin

15. The main target of antifungal drugs:
    A) Peptidoglycan
    B) Ergosterol
    C) 70S ribosome
    D) DNA gyrase
    E) Mycolic acid

## Clinical cases

### *Case 1*

A patient who came 2 weeks after a risky contact had a negative ELISA for HIV.

**Question:** How should the result be interpreted?

**Answer:** This may be the **“window period”** — antibodies have not yet formed. The result does not rule out infection. The test is repeated at set intervals; **PCR** is used for early detection. Counseling is given to the patient.

### *Case 2*

On a preventive examination: HBsAg negative, anti-HBs positive, anti-HBc negative.

**Question:** What does this picture mean?

**Answer: Immunity after vaccination**. The vaccine consists only of HBsAg, so core antibodies (anti-HBc) are not formed; in a person who recovered from the disease, anti-HBc would also be positive.

### *Case 3*

A patient is anti-HCV positive, but HCV RNA was not detected; liver tests are normal.

**Question:** What is the conclusion?

**Answer:** Probably a **past infection** (cleared spontaneously or as a result of treatment). anti-HCV persists for a long time and does not prove active infection. The result may be repeated at a set interval and assessed together with clinical information.

## Quick review

**Table 29.5.** *Topic 29 — key concepts*

| Concept | Essence |
|---|---|
| Retroviruses | Reverse transcriptase → DNA → integration (provirus) |
| HIV genes | *gag* (p24), *pol* (enzymes), *env* (gp120, gp41) |
| Target | $CD4^+$ T lymphocytes + coreceptors (CCR5, CXCR4) |
| Diagnosis | ELISA (screening) → immunoblot (confirmation) → CD4 and viral load |
| Window period | No antibodies, infectivity present → PCR |
| HBV | Partially double-stranded DNA; HBsAg, HBcAg, HBeAg |
| Vaccinated / recovered | Only anti-HBs / anti-HBs + anti-HBc |
| HDV | Defective, HBsAg envelope; coinfection/superinfection |
| HCV | Positive RNA, variable, no vaccine; anti-HCV → HCV RNA |
| Mycoses | Superficial, dermato-, subcutaneous, deep, opportunistic |
| Antifungal target | Ergosterol (membrane) |

TOPIC 30

# Herpes and pox viruses. Monkeypox. Nosocomial (hospital-acquired) infections

## 1. Learning objectives

In this class the student will master the following:

- To know the classification, general properties and subfamilies of the herpesvirus family;
- to explain the mechanism of latency and recurrence of herpes infections;
- to master the structure and diagnosis of poxviruses (smallpox, monkeypox);
- Independent work: to know the causative agents, causes, diagnosis and preventive measures of hospital-acquired (nosocomial) infections.

## 2. Herpesviruses (*Herpesviridae*)

**General properties:** these are large (150–200 nm), **enveloped** viruses with **double-stranded linear DNA**; an icosahedral capsid, with a layer of **tegument** between the capsid and the envelope. They multiply in the nucleus.

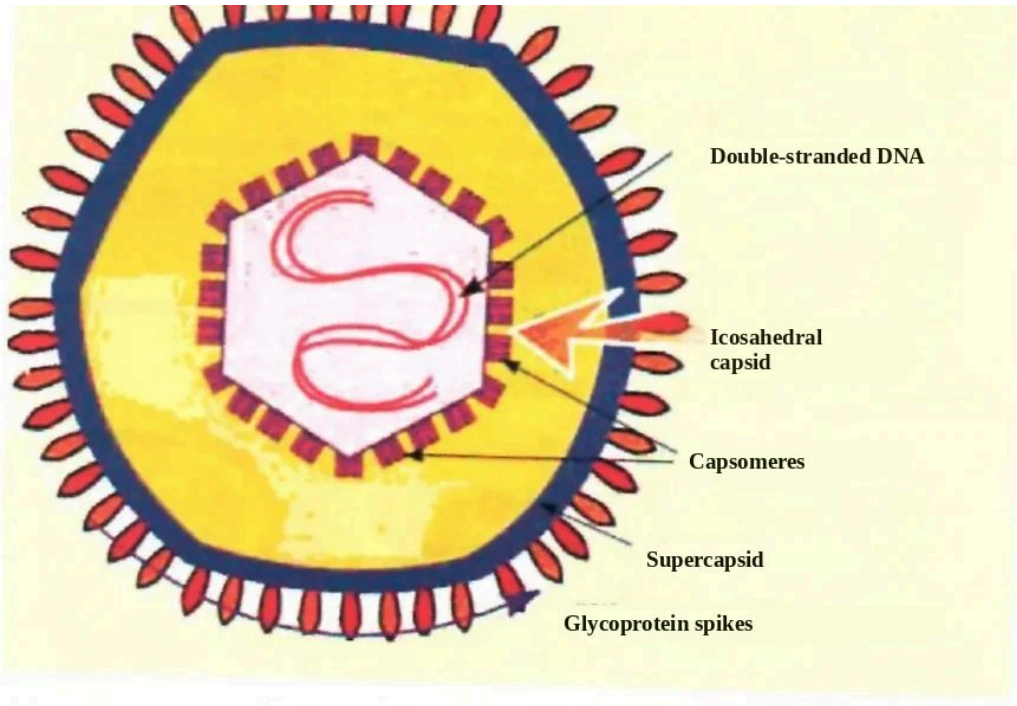



**Figure 30.1.** *Scheme of the structure of the herpesvirus: double-stranded DNA, icosahedral capsid, capsomeres, supercapsid and glycoprotein spikes. Source: Muhamedov I.M. et al., 2019.*

**The main biological property — latency and recurrence:** after the primary infection the virus does not leave the body completely, but is kept in certain cells in a **latent** state and **is reactivated** when immunity falls (stress, another illness, immunosuppression, UV light, menstruation).

**Table 30.1.** *Human herpesviruses*

| Subfamily | Virus | Site of latency | Main diseases |
|---|---|---|---|
| **α (alpha)** | **HSV-1** | Trigeminal ganglion | Herpes labialis, gingivostomatitis, keratitis, encephalitis |

| Subfamily | Virus | Site of latency | Main diseases |
|---|---|---|---|
| | **HSV-2** | Sacral nerve ganglia | Genital herpes; neonatal herpes |
| | **VZV (type 3)** | Ganglia of the spinal cord and cranial nerves | **Chickenpox** (primary), **shingles** (reactivation) |
| **β (beta)** | **CMV (type 5)** | Monocytes, salivary glands, kidney | Congenital CMV infection, severe forms in transplantation and AIDS, retinitis |
| | HHV-6, HHV-7 | Lymphocytes | A sudden-onset exanthem in children |
| **γ (gamma)** | **EBV (type 4)** | B lymphocytes | **Infectious mononucleosis**; association with Burkitt lymphoma and nasopharyngeal carcinoma |
| | HHV-8 | Lymphocytes | Association with Kaposi sarcoma |

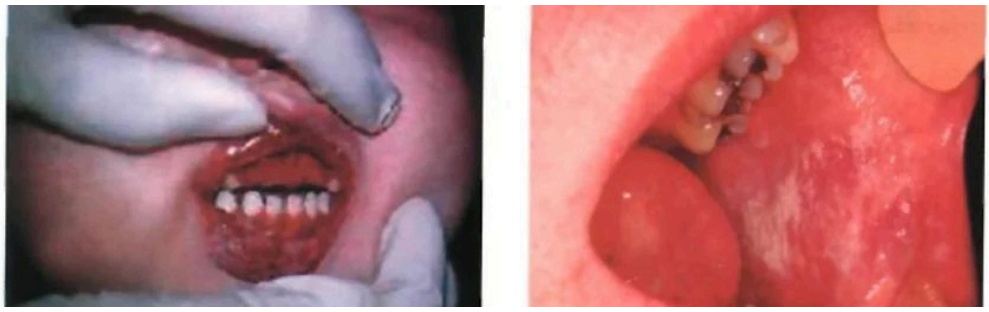


**Figure 30.2.** *Clinical manifestations of herpes infection: eruptions around the mouth and on mucous membranes. Source: Muhamedov I.M. et al., 2019.*

**REMEMBER**

Remember the herpesviruses by their site of latency: **α — in nerve ganglia** (HSV, VZV), **β — in monocytes/secretory cells** (CMV), **γ — in lymphocytes** (EBV, HHV-8).

**Routes of transmission:** direct contact (saliva, vesicle fluid), sexual (HSV-2), airborne (VZV), vertical and transfusion (CMV), through saliva — the “kissing disease” (EBV).

**Clinically important conditions:**

- **Neonatal herpes** — infection with HSV-2 in the birth canal; it may give a severe, disseminated form;
- **Congenital CMV infection** — hearing loss, damage to the nervous system;
- **Infectious mononucleosis (EBV)** — fever, tonsillitis, lymphadenopathy, splenomegaly, **atypical mononuclear cells** in the blood;
- **Shingles (reactivation of VZV)** — a painful vesicular rash along a nerve, on one side.

**Laboratory diagnosis:**

1. **Material:** vesicle fluid and scrapings from the base, saliva, cerebrospinal fluid, blood, biopsy; in newborns — from several sites.
2. **PCR — the main modern method**: especially in encephalitis (HSV in the cerebrospinal fluid), neonatal infection and CMV; **quantitative PCR** to monitor viral load in transplant patients.
3. **Cytologic method** — the Tzanck preparation: in a smear taken from the base of the rash, **multinucleated giant cells** and intranuclear inclusions.
4. **Serologic** — IgM (new infection), IgG (past infection); in EBV, the heterophile antibody test (for mononucleosis) and a panel of specific antibodies.

5. **Virus isolation** — in cell culture (CPE: rounding, syncytia, inclusions); rarely used in practice.
6. **Immunofluorescence and antigen tests** — rapid methods.

**Prophylaxis and control: vaccines against chickenpox and shingles** are available; there is no widely used vaccine against HSV and CMV. Antiviral drugs control reactivation but do not eliminate the latent virus — treatment is prescribed by a physician.

> **CLINICAL SIGNIFICANCE**
> **Suspected herpes encephalitis** is an emergency: **HSV PCR** is done immediately on the cerebrospinal fluid; treatment may be started on clinical suspicion, without waiting for the result, because treatment started in time significantly improves the outcome.

## 3. Poxviruses (*Poxviridae*)

**General properties: the largest viruses** (250–300 nm), visible under the light microscope with special staining; **double-stranded DNA**; **complex symmetry**, brick-shaped. Unlike other DNA viruses, they **multiply in the cytoplasm** (because they have their own enzymes).

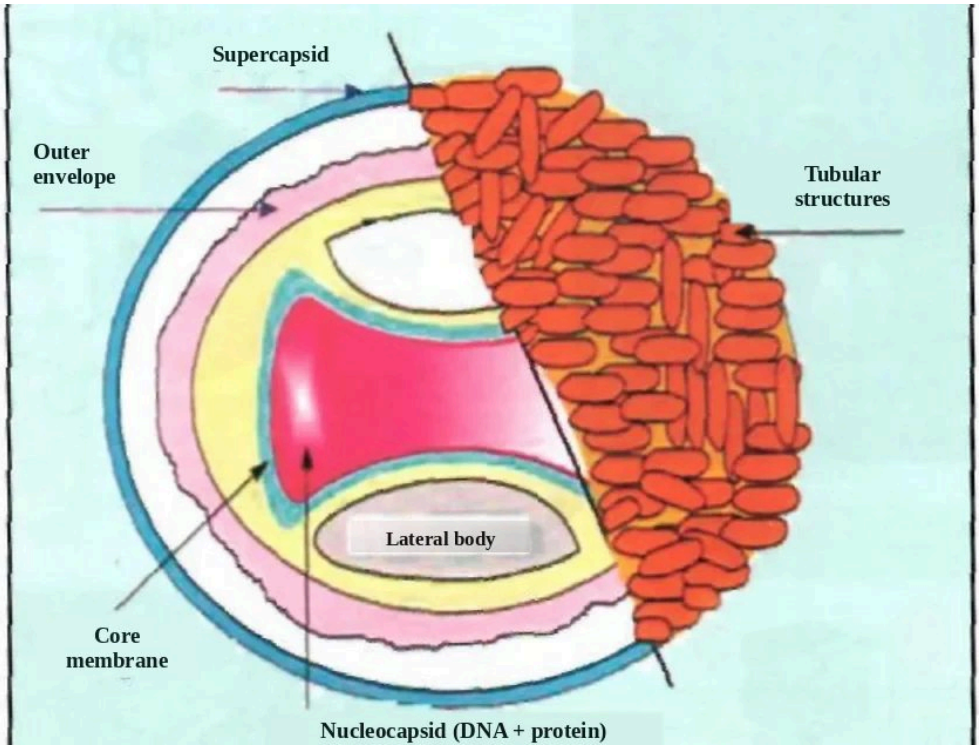



**Figure 30.3.** *The structure of the smallpox virus: the outer envelope, tubular structures, lateral bodies and the nucleocapsid (DNA + protein). Source: Muhamedov I.M. et al., 2019.*

**Smallpox (variola):** historically one of the most severe infectious diseases; **declared eradicated worldwide by the World Health Organization in 1980** — the first and only complete eradication in the history of medicine. The main reasons: the virus circulates only in humans, a stable antigenic structure, an effective vaccine and a mass vaccination strategy.

**Monkeypox (mpox):** a zoonotic infection of the genus *Orthopoxvirus*; the reservoir is rodents and primates (West and Central Africa). Transmission: contact with a sick animal, from person to person by **close direct contact** (rash lesions, body fluids, prolonged face-to-face contact), contaminated objects; the vertical route is also possible.

**Clinical picture:** fever, headache, muscle pain and **enlargement of lymph nodes** (a sign that differs from smallpox), then a rash: macule → papule → vesicle → pustule → scab. The rash lesions in a

given area are usually **at the same stage**.

**Diagnosis:**

- **PCR (the main method)** — scrapings from a rash lesion, vesicle fluid, scab;
- electron microscopy (brick-shaped virions), antigen tests — supplementary;
- serology — of limited value because of cross-reactivity among orthopoxviruses;
- work is carried out under the set biosafety regime.

**Control:** isolation of the patient, monitoring of contacts, protective equipment; smallpox vaccines give cross-protection against orthopoxviruses and are used in risk groups on indication.

> **MEMORY AID**
> Tell poxviruses apart: **the largest + brick-shaped + multiply in the cytoplasm** — other DNA viruses (herpes, adeno, papilloma) multiply in the **nucleus**.

## 4. Independent work: Nosocomial (hospital-acquired) infections

**Definition: healthcare-associated infections** — infections that were not present in the patient at hospital admission and were not in the incubation period, and that were acquired in a medical institution (also the occupational infections of medical staff).

**Main causes:**

- **patient factor** — reduced immunity, severe underlying disease, age, diabetes mellitus;
- **invasive procedures** — catheters, artificial ventilation, operations, drains (breach of natural barriers);
- **microbial factor** — **multidrug-resistant** "hospital strains"; biofilm formation;
- **organizational factors** — breaches of hand hygiene, breaches of the sterilization and disinfection regime, crowded rooms, unjustified use of antibiotics.

**Table 30.2.** *The main forms of nosocomial infections and their typical causative agents*

| Shape | Typical causative agents |
|---|---|
| Urinary tract infection (often catheter-associated) | *E. coli*, *Klebsiella*, *Proteus*, enterococci, *Candida* |
| Postoperative wound infection | *S. aureus* (including MRSA), enterobacteria, anaerobes |
| Pneumonia (ventilator-associated) | *P. aeruginosa*, *Klebsiella*, *Acinetobacter*, *S. aureus* |
| Bloodstream infection (catheter-associated) | Coagulase-negative staphylococci, *S. aureus*, *Candida* |
| Intestinal infection (antibiotic-associated) | *Clostridioides difficile* |
| Viral | Hepatitis B and C, HIV (parenteral), respiratory viruses, rotavirus |

**Features of "hospital strains":** multidrug resistance, resistance to disinfectants, long survival in a moist environment, biofilm formation, circulating in a ward for a long time.

**Diagnosis and control:**

1. **Microbiological monitoring** — identification of the isolated strains, following the **antibiogram** and the resistance profile;
2. **Epidemiologic typing** — phage typing, molecular typing: whether the strains come from one source or are independent;

3. **Control of the environment** — samples of swabs, air, instruments and equipment, solutions (topic 8);
4. **Examination of staff** — detecting carriage (for example, *S. aureus* in the nose);
5. **Control of sterilization and disinfection** — biological indicators (topic 9).

**Preventive measures:**

- **hand hygiene** — the most effective and cheapest measure;
- strict observance of the rules of asepsis and antisepsis;
- placing invasive devices only on indication and **removing them as early as possible**;
- isolation and contact measures;
- **rational antibiotic use** programs;
- staff training, vaccination and the work of the infection control service.

**CLINICAL SIGNIFICANCE**

In catheter- or prosthesis-associated infections, bacteria inside the biofilm are much more resistant to antibiotics. Therefore treatment is often ineffective without **removing the foreign body** or replacing it.

## 5. Practical skill: collecting material from a vesicular rash

1. The **newest, still unruptured vesicle** of the rash is chosen (it contains the highest amount of virus).
2. The skin is wiped with an antiseptic and one waits **until it is dry** (so that the antiseptic does not get into the sample).
3. The vesicle is opened with a sterile needle or the tip of a scalpel; the fluid is collected with a swab, and a **scraping from the base of the vesicle** is also taken (cells are needed).
4. The material is placed in **virological transport medium**; for PCR a dry swab may also be used (according to the laboratory's requirements).
5. The sample is stored at **2–8 °C** and delivered quickly.
6. In parallel, a **Tzanck preparation** can be made (to see multinucleated giant cells).
7. The requisition states: the date the rash began, its stage, its spread, vaccination and travel history, and immune status.
8. **Safety:** gloves and protective equipment; when monkeypox is suspected — the set procedure for isolation and packaging.

## 6. Review questions

1. What are the general structure and the main biological property of herpesviruses?
2. Name the subfamilies of herpesviruses and their sites of latency.
3. What disease does VZV cause in primary and in recurrent infection?
4. In which cells does EBV persist, and what disease does it cause?
5. Which method is decisive in the diagnosis of herpes encephalitis?
6. What is seen in the Tzanck preparation?
7. State how poxviruses differ from other DNA viruses.
8. What factors made the eradication of smallpox possible?
9. How is monkeypox transmitted, and how is it detected?

10. What is a nosocomial infection? State its main causes and preventive measures.

## Test questions

1. The genome of herpesviruses:
   A) Single-stranded RNA
   B) Double-stranded DNA
   C) Segmented RNA
   D) Partially double-stranded DNA
   E) Single-stranded DNA

2. The main biological property of herpesviruses:
   A) Antigenic shift
   B) Latency and reactivation
   C) Multiplication only outside the cell
   D) Not passing through a bacterial filter
   E) Spore formation

3. The site of latency of HSV-1:
   A) Trigeminal ganglion
   B) B lymphocytes
   C) Monocytes
   D) Sacral nerve ganglia
   E) Bone marrow

4. The virus that causes chickenpox and shingles:
   A) HSV-1
   B) VZV (type 3)
   C) CMV
   D) EBV
   E) HHV-8

5. The site of latency of EBV:
   A) Nerve ganglia
   B) B lymphocytes
   C) Erythrocytes
   D) Epithelial cells (only)
   E) Neutrophils

6. The sign characteristic of the blood picture in infectious mononucleosis:
   A) Atypical mononuclear cells
   B) Basophilia
   C) Eosinophilia
   D) Thrombocytosis
   E) Reticulocytosis

7. The decisive method in the diagnosis of herpes encephalitis:
   A) HSV PCR in the cerebrospinal fluid
   B) Blood culture
   C) Skin test
   D) Coagulase test
   E) The Ascoli reaction

8. What is detected in the Tzanck preparation:
   A) Multinucleated giant cells and intranuclear inclusions
   B) Druses
   C) Volutin granules
   D) Clue cells
   E) Bipolar rods

9. The site of multiplication of poxviruses:
   A) Nucleus
   B) Cytoplasm
   C) Mitochondrion
   D) Outside the cell
   E) On the ribosome

10. The size and shape of poxviruses:
    A) Smallest, spherical
    B) Largest, brick-shaped
    C) Bullet-shaped
    D) Filamentous
    E) Spiral

11. When smallpox was declared eradicated:
    A) In 1960
    B) In 1980
    C) In 1995
    D) In 2005
    E) Not yet eradicated

12. The clinical sign that distinguishes monkeypox from smallpox:
    A) Enlargement of lymph nodes
    B) Absence of a rash
    C) Absence of fever
    D) Occurrence only in adults
    E) No scab formation

13. The main method of diagnosing monkeypox:
    A) PCR
    B) Coagulase test
    C) The Wright reaction
    D) Mantoux test
    E) Oxidase test

14. The most effective measure in preventing nosocomial infection:
    A) Hand hygiene
    B) Giving prophylactic antibiotics
    C) Painting the room
    D) Admitting more patients
    E) Keeping catheters in for a long time

15. An important condition of treatment in catheter- and prosthesis-associated infections:
    A) Only increasing the antibiotic dose
    B) Removing or replacing the foreign body
    C) Only wiping with an antiseptic
    D) Delaying treatment
    E) Giving a probiotic

## Clinical cases

### *Case 1*

A patient suddenly developed fever, confusion and seizures; MRI showed changes in the temporal lobe.

**Question:** Which causative agent should be excluded first, and which test is needed?

**Answer: HSV-1 encephalitis**. **HSV PCR** is done urgently on the cerebrospinal fluid. If clinical suspicion is strong, treatment may be started by a physician without waiting for the result, because early treatment significantly improves the outcome.

### *Case 2*

A 60-year-old patient developed a painful vesicular rash along a nerve on one side of the chest. As a child he had chickenpox.

**Question:** What are the diagnosis and the mechanism?

**Answer: Shingles** — reactivation of VZV. After the primary infection the virus stays latent in the nerve ganglia and, when immunity falls, is reactivated and produces a rash along the nerve.

### *Case 3*

In the intensive care unit, a patient on a ventilator developed pneumonia; the isolated *Acinetobacter* proved multidrug resistant. A similar strain was found in two more patients in the same unit.

**Question:** What conclusion and what measures are needed?

**Answer:** A hospital-acquired infection; the strains were probably spread from one source — this is determined by **molecular typing**. Measures: contact isolation, **hand hygiene**, checking the disinfection regime, monitoring instruments, equipment and solutions, rational antibiotic use, involving the infection control service.

## Quick review

**Table 30.3.** *Topic 30 — key concepts*

| Concept | Essence |
|---|---|
| Herpesviruses | Enveloped, double-stranded DNA, tegument; multiply in the nucleus |
| Main property | Latency + reactivation |
| α / β / γ | HSV, VZV (nerve ganglia) / CMV (monocyte) / EBV, HHV-8 (lymphocyte) |
| VZV | Chickenpox → shingles |
| EBV | Infectious mononucleosis; atypical mononuclear cells |
| Diagnosis of herpes | PCR (main), Tzanck preparation, serology |

| Concept | Essence |
|---|---|
| Poxviruses | The largest, brick-shaped, multiply in the **cytoplasm** |
| Smallpox | Eradicated in 1980 |
| Monkeypox | Zoonosis; close contact; lymphadenopathy; PCR |
| Nosocomial infection | Invasive procedures + resistant strains + breaches of the regime |
| Prevention | Hand hygiene, asepsis, early removal of devices, rational antibiotic use |

# Answers to the test questions

**Topic 1:** 1-C; 2-B; 3-C; 4-B; 5-B; 6-C; 7-B; 8-C; 9-C; 10-A; 11-B; 12-B; 13-B; 14-B; 15-B.

**Topic 2:** 1-B; 2-C; 3-B; 4-C; 5-C; 6-B; 7-C; 8-B; 9-B; 10-B; 11-B; 12-B; 13-C; 14-B; 15-B.

**Topic 3:** 1-B; 2-B; 3-C; 4-C; 5-B; 6-B; 7-B; 8-D; 9-B; 10-B; 11-B; 12-B; 13-C; 14-B; 15-B.

**Topic 4:** 1-B; 2-C; 3-C; 4-B; 5-C; 6-B; 7-A; 8-B; 9-B; 10-B; 11-B; 12-B; 13-B; 14-B; 15-B.

**Topic 5:** 1-C; 2-C; 3-B; 4-A; 5-B; 6-B; 7-C; 8-C; 9-C; 10-B; 11-B; 12-B; 13-B; 14-C; 15-B.

**Topic 6:** 1-D; 2-D; 3-C; 4-B; 5-B; 6-C; 7-B; 8-B; 9-B; 10-B; 11-A; 12-B; 13-C; 14-B; 15-B.

**Topic 7:** 1-B; 2-C; 3-B; 4-B; 5-B; 6-B; 7-B; 8-B; 9-B; 10-B; 11-D; 12-A; 13-C; 14-B; 15-B.

**Topic 8:** 1-B; 2-B; 3-B; 4-B; 5-B; 6-B; 7-C; 8-B; 9-B; 10-A; 11-C; 12-A; 13-B; 14-A; 15-A.

**Topic 9:** 1-B; 2-B; 3-C; 4-A; 5-B; 6-B; 7-C; 8-B; 9-A; 10-C; 11-B; 12-B; 13-B; 14-B; 15-C.

**Topic 10:** 1-B; 2-B; 3-B; 4-B; 5-B; 6-A; 7-A; 8-B; 9-B; 10-A; 11-B; 12-B; 13-B; 14-A; 15-B.

**Topic 11:** 1-B; 2-B; 3-B; 4-B; 5-B; 6-C; 7-B; 8-B; 9-B; 10-B; 11-A; 12-B; 13-B; 14-B; 15-B.

**Topic 12:** 1-B; 2-B; 3-B; 4-A; 5-A; 6-B; 7-C; 8-B; 9-B; 10-A; 11-B; 12-B; 13-B; 14-B; 15-B.

**Topic 13:** 1-B; 2-C; 3-B; 4-B; 5-A; 6-B; 7-B; 8-B; 9-C; 10-B; 11-B; 12-A; 13-B; 14-B; 15-C.

**Topic 14:** 1-B; 2-B; 3-B; 4-B; 5-B; 6-B; 7-B; 8-A; 9-A; 10-B; 11-B; 12-B; 13-B; 14-B; 15-B.

**Topic 15:** 1-B; 2-B; 3-B; 4-A; 5-B; 6-B; 7-B; 8-B; 9-B; 10-B; 11-B; 12-C; 13-B; 14-B; 15-B.

**Topic 16:** 1-B; 2-B; 3-B; 4-B; 5-A; 6-B; 7-B; 8-B; 9-B; 10-A; 11-B; 12-B; 13-B; 14-C; 15-B.

**Topic 17:** 1-C; 2-B; 3-B; 4-B; 5-B; 6-B; 7-B; 8-A; 9-B; 10-B; 11-B; 12-B; 13-A; 14-B; 15-A.

**Topic 18:** 1-B; 2-B; 3-B; 4-A; 5-B; 6-A; 7-B; 8-B; 9-B; 10-B; 11-B; 12-B; 13-B; 14-B; 15-A.

**Topic 19:** 1-B; 2-B; 3-B; 4-C; 5-A; 6-B; 7-B; 8-A; 9-B; 10-A; 11-B; 12-A; 13-B; 14-A; 15-A.

**Topic 20:** 1-B; 2-B; 3-B; 4-A; 5-D; 6-B; 7-A; 8-B; 9-B; 10-B; 11-A; 12-B; 13-A; 14-B; 15-C.

**Topic 21:** 1-A; 2-B; 3-B; 4-B; 5-B; 6-B; 7-A; 8-B; 9-B; 10-B; 11-A; 12-A; 13-B; 14-B; 15-B.

**Topic 22:** 1-B; 2-C; 3-B; 4-C; 5-B; 6-A; 7-B; 8-A; 9-C; 10-B; 11-C; 12-A; 13-B; 14-A; 15-B.

**Topic 23:** 1-B; 2-B; 3-B; 4-A; 5-B; 6-B; 7-B; 8-A; 9-B; 10-B; 11-B; 12-A; 13-A; 14-A; 15-A.

**Topic 24:** 1-B; 2-B; 3-D; 4-C; 5-B; 6-B; 7-B; 8-B; 9-A; 10-B; 11-B; 12-B; 13-A; 14-B; 15-A.

**Topic 25:** 1-B; 2-B; 3-B; 4-B; 5-B; 6-B; 7-B; 8-B; 9-A; 10-C; 11-B; 12-B; 13-B; 14-B; 15-A.

**Topic 26:** 1-B; 2-B; 3-A; 4-B; 5-A; 6-A; 7-B; 8-A; 9-A; 10-B; 11-A; 12-A; 13-B; 14-A; 15-B.

**Topic 27:** 1-B; 2-B; 3-B; 4-A; 5-B; 6-A; 7-A; 8-B; 9-A; 10-A; 11-A; 12-A; 13-A; 14-A; 15-A.

**Topic 28:** 1-B; 2-B; 3-A; 4-B; 5-A; 6-B; 7-B; 8-B; 9-A; 10-A; 11-A; 12-A; 13-A; 14-A; 15-A.

**Topic 29:** 1-B; 2-B; 3-B; 4-B; 5-D; 6-B; 7-B; 8-A; 9-B; 10-A; 11-B; 12-B; 13-B; 14-B; 15-B.

**Topic 30:** 1-B; 2-B; 3-A; 4-B; 5-B; 6-A; 7-A; 8-A; 9-B; 10-B; 11-B; 12-A; 13-A; 14-A; 15-B.

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**Sources of figures.** The photomicrographs, diagrams and diagnostic algorithms in the manual were taken mainly from sources [1] and [2]; the source is shown under each figure. The figure showing the structure of the light microscope was taken from open internet sources, and its labels were translated.